

2nd Edition

ARTIFICIAL INTELLIGENCE FOR LEARNING

**Using AI and generative AI
to support learner
development**



Donald Clark



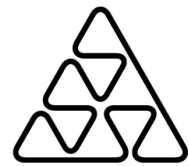
Artificial Intelligence for Learning

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Development

SECOND EDITION

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PREFACE

The first edition of *AI for Learning* proved prescient. I had been delivering articles and keynotes, and talking about how this would become the single most important technological shift in the history of both technology and learning technology for years, also delivering learning services using AI. Then it finally happened on 30 November 2022 – the launch of ChatGPT 3.5.

The whole globe was then speaking to ChatGPT and the curiosity was intense. All of the major technology companies in the world have adopted AI and generative AI as a core technology. It is and will continue to play a role in all of our lives as it is embedded in the everyday tools and services we use.

It is a paradigmatic shift in learning away from the present search and retrieve knowledge model to something more immediate, useful and human – a profound shift away from monologue to dialogue. It lifts us out of unnatural ways of learning – those flat pages of textbooks, flat screens of dull e-learning, lectures and videos – into an altogether different world where we have agency.

I was born in 1956, the year the modern era of AI began with an epoch-making conference at the Ivy League Dartmouth College in the US. It was there that John McCarthy coined the phrase ‘artificial intelligence’ and set in train some foundational work in the field. As a young undergraduate, I was fortunate to spend time there, where I used my first computer. My only previous coding experience was on punched cards, which I had to post off to the local government mainframe computer in Edinburgh when I was at school. The results, in the form of a printout, came back a week later. At Dartmouth we had access to a mainframe, played Battleship, and could code. It was an epiphany. On my return to Edinburgh I bought a home

computer and the first computer program I wrote taught the basics of the Russian language.

My first serious application of AI was in intelligent tutoring, in the 1980s, when my colleague Clive Shepherd and I built a training program that used dynamically gathered learner data to recommend routes through a training program on interviewing skills for British Telecom. We learned a valuable lesson. The computers we used lacked the processing power and memory to do what we needed to do properly, but it was promising and, in time, would bear fruit. That time came later, after we built a successful company, floated it on the stock market, as Epic plc, and helped create the UK online learning industry.

Then along came AI-driven Google search, which revolutionized the efficiency of research and learning, a huge pedagogic shift. After selling the company, freeing myself from the tyranny of employment, I took a deeper dive and did Sebastian Thrun's famous MOOC (massive open online course) on AI – a revelation and real introduction to the mechanics of algorithms and their potency.

Convinced that AI would be the primary driver behind online learning, I began to talk about it at conferences around the world and found audiences intrigued by the possibility. I invested in 'adaptive learning', developed an AI strategy for another company in which I was a director, and then took the plunge myself and designed and built an AI-driven content creation, curation and practice tool, which won three major awards for Most Innovative New Product in Learning, Best Online Learning Project Award and Best EdTech Award from the UK's main tech organization in higher education, Joint Information Systems Committee (JISC).

More recently, a few other seminal moments confirmed my belief in AI in learning. They took place in some rather odd places and circumstances. One particular instance was a bit terrifying. I was teaching on AI in learning at the University of Philadelphia, on the morning of the US presidential election,

and showed an AI prediction that pointed to a Trump win. Oh how they laughed – but the next morning was a sobering experience, with some in tears. This confirmed my view that the old media and pollsters were stuck in an Alexander Graham Bell world of telephone polls, while the world had leapt forward to data-gathering from social media and other sources. They were wrong because they didn't understand technology. Traditional media have an in-built distaste for new technology, as it is seen as a threat. Also, at a deeper level, Trump won because of technology. The deep cause – technology replacing a significant number of manufacturing jobs – had come to pass. It was always thus. Agriculture was mechanized and we moved into factories, factories were automated and we moved into offices and offices are now being automated by AI... and we're not quite sure where all this will end. AI will be a primary political, economic and moral issue for the next 50 years or more.

There were also some astounding AI successes. Google beat a champion of GO, the most complex game we know, and then a poker champion. Time and time again, AI rose to the challenge. DeepMind cracked the 3D structure of 200 million proteins, saving one billion years of research, and astonishing advances were made in medicine and material science. Take almost any area of human endeavour, add a dose of AI and you have a business.

On a visit to lecture at Penn State University I came across a couple of bot projects that intrigued me. The Georgia Tech bot, which fooled everyone into thinking it was a real teaching assistant, even being put up for a teaching award, was a breakthrough. I also saw a pupil bot for trainee teachers, who behaved as troublesome boys tend to in school, showing how this new approach through natural language interfaces will have a profound effect on how we interact with AI. AI itself has provided rapid advances in NLP that has made AI accessible at the consumer level.

Then came a debate in Berlin, with the motion ‘AI can, will and should replace teachers.’ I lost but was to win the debate two years later when arguing against the motion that ‘AI will cause more harm than good in learning’. It was an opportunity to show that, given recent advances from Google onwards, at some time it would be ridiculous to say that AI will decimate professions such as lawyers, doctors, accountants and managers, yet leave ‘teaching’ untouched. That would be a conceit. Many were surprised at the real-world examples in the creation of learning content, personalized feedback, assessment and reinforcement. It was not that it was coming; it was already here.

Both my sons are pretty techy and one has a degree in AI. This has been a godsend. Being able to get immediate clarification advice on tools, and generally engage in conversations with someone whose passion is AI has been more than useful.

AI will change the very nature of work. This means it will change why we learn, what we learn and how we learn. Almost everything we now do online is mediated by AI: Facebook, Twitter, Instagram, Amazon, Netflix – almost everything except learning. That is now changing.

But it has been my own work in the field, creating the world’s first content creation tool using AI – WildFire – as well as working with Learning Pool on their LXP (learning experience platform), chatbots and LRS (learning record store) that provided real-world experience and delivery in real organizations in real award-winning projects with real learners. CogBooks has been another practical and invaluable experience, delivering some of the first adaptive learning courses at Arizona State University and other organizations around the world. Then a range of projects post-ChatGPT, with publishers, training companies and others to realize the potential of AI.

While not yet one year old, still a baby, I gave keynotes, podcasts (Great Minds on Learning series), debates and

workshops on ‘AI for learning’ from the US, UK, Norway, Sweden, Germany, Netherlands, Belgium, Spain, Portugal, Greece, Switzerland, Russia and also in Senegal, South Africa and Singapore. This allowed me to test many ideas with diverse audiences and come across projects that have been fed into this book. It has been a fantastic ‘learning’ experience and this book is the result of that obsession with AI and learning. I also created my avatar in a studio in London, which I have used all over the world, as well as Digital-Don my personal twin as a chatbot.

Between the first edition of *AI for Learning* and this second edition, I wrote three other books, *Learning Experience Design* (2022), *Learning Technologies* (2023) and *Learning in the Metaverse* (2023). So convinced was I that this revolution would happen that all three had full chapters on AI. *Learning Technologies* in particular made the case for re-evaluating learning technologies as the most important technologies our species has ever invented – writing, alphabets, printing, broadcast media, computers, internet, digital content and now AI. Without these there is no cultural archiving, accumulation and access. There is no scientific, medical and human progress. The culmination of this has been the use of this accumulated cultural archive as a training set for large language models (LLMs). Our entire culture is now being put to use to allow us to take that next great leap.

This book is unapologetically positive, enthusiastic and optimistic about the recent advances in AI. It offers solutions to problems that have held back efficacy in learning for too long. There are few more inefficient sectors than learning, whether in schools, higher education or the workplace. No one is happy – neither teachers, parents, politicians, nor students. Having been through many technological revolutions, this feels different.

The words in this book will eventually be captured as data and fed into future LLMs that will eventually help us overcome our current and future problems. I am both glad and proud to

be part of what I see as one of the most important shifts in the history of our species. AI changes everything!

ACKNOWLEDGEMENTS

Thanks to Callum Clark for his technical advice on AI and Carl Clark for his business support. All round great lads. Thanks also to everyone who engaged on the topic of 'AI for Learning' on social media, at conferences around the world and on AI projects. And, of course, to my wife, Gillian Clark, for her patient support.

LIST OF ABBREVIATIONS

AR	augmented reality
API	application programming interface
BOSS	belief and opinion sampling system
CASTE	course assembly system and tutorial environment
CBT	computer-based training
CNN	convolutional neural network
CPD	continuing professional development
DEI	diversity, equity and inclusion
FF	forward-forward
GDPR	General Data Protection Regulation
GIGO	garbage in, garbage out
GPT	generative pre-trained transformer
GPU	graphics processing unit
HCI	human-computer interaction
HR	human resources
ISO	International Organization for Standardization
IQ	intelligence quotient
L&D	learning and development
LLM	large language model
LMS	learning management system
LRS	learning record store
LTG	linear threshold gate

LXP	learning experience platform
MCQ	multiple-choice question
MOOC	massive open online course
Moodle	modular object-oriented dynamic learning environment
NIH	not invented here
NLP	natural language processing
PPO	proximal policy optimization
QA	quality assurance
RAG	retrieval-augmented generation
RLHF	reinforcement learning from human feedback
SAKI	Self-Adaptive Keyboard Instructor
SCORM	sharable courseware object reference model
SDT	self-determination theory
SEN	special educational needs
SME	subject matter expert
TPU	tensor processing unit
UDL	universal design for learning
UI	user interface
UT	universal teacher
UTA	universal teaching assistant
UX	user experience layer
VLE	virtual learning environment
VR	virtual reality
xAPI	experience application programming interface

01

Homo technus

AI has and will continue to change many areas of human endeavour. Almost everything we do online is mediated by AI: search through Google; social media, whether Facebook, Instagram, or X; buy something on Amazon; entertain yourself through Netflix – all are mediated by AI. AI now touches almost all global, online services. Perhaps the only online sector that is not yet mediated by AI is learning.

The nature of work is also being shaped by AI, not only in the automation of manufacturing and warehouses, but also in our homes, offices and services. This change in the workplace, in itself, will surely have a profound effect on *what* we learn, *why* we learn and *how* we learn. This is already changing through AI-driven, online learning.

Technological revolutions

Technological revolutions are not new. We as a species have shaped and been shaped by technology, from the first intentional use of stone hand axes to artificial intelligence. There has been a relentless rhythm to this progress.

The problem with most descriptions of this technological progress is that they focus on the *physical* technology itself, stone tools (Stone Age, Neolithic), metals (Bronze Age, Iron Age), age of steam engines, railways, mass production, computers (industrial revolutions). We see this when AI for learning is couched in terms of the ‘Fourth Industrial Revolution’ (Seldon and Abidoye, 2018), which is neither the fourth nor industrial.

A far better lens through which to look at AI in learning is not in terms of industrial revolutions, but cognitive revolutions. It is more revealing to see AI in terms of those revolutions in learning technology, such as language, writing, alphabets,

printing, the internet and now AI. Our *physical* technology is underpinned, supported and created by *psychological* technology that enables its very conceptualization, design, development and delivery. The stone axe was imagined, shaped and used by minds. Cave paintings were the product of sophisticated imaginations. Clay tablets, papyrus, manuscripts and the entire technology of writing were a psychological breakthrough that externalized and archived thought for others to access. Printing gave rise to the scientific revolution, the Reformation and the Enlightenment. The internet, more specifically the web, gave us global access to knowledge. Now we have AI, the next technological leap, again a product of pure psychological endeavour.

Technology that enables learning is often overlooked when the history of technology is written. It is all too easy to focus on the physical objects. But without learning technologies, no other technologies would have developed. We are the species that 'learned' faster than the others. Our evolution as a species over the last few million years has been one of learning to adapt. It is this that has given us global dominance, allowing us to walk on the moon and reach out beyond our solar system.

Without the ability to shape stone tools we would never have avoided predators, sought out prey and become that dominant species. Prehistoric technology like pointed axes allowed us to kill, crush, scrape and cut. With bone needles we could dress ourselves efficiently, with pots cook and with axes chop down trees for fuel.

We are called *Homo sapiens* but our genus '*Homo*' emerged with the appearance of *Homo habilis* (handy man). They were so described because of their association with stone tools, but recent evidence has shown that tools were used by previous species. We are, more accurately, *Homo technus*, the species that uses tools and technology, both physical and, more importantly, psychological.

In addition to physical tools we are the masters of symbolic tools. It is difficult to see *language* as a tool, but if we define

technology as something that exists outside of ourselves, that we create to exist outside of our minds and bodies, to enhance us psychologically and physically, then language is a technology. We create sounds that exist separate from us, travel to others across distance to be heard by others. It is the mainstay of communication, whether face-to-face, across the globe by telegraph then telephone and now face-to-face online. Voice underpins all other forms of technology and is being embedded in the powerful and personal mobile devices we have in our pockets, as well as in our homes, as the way of controlling the Internet of Things.

From Gutenberg to Zuckerberg, language, writing, printing, distribution, communication and sharing came together on the back of the internet in the form of the World Wide Web. The global scale and cumulative effect means it may prove more disruptive in the long term than all that went before. Some compare now to the 1930s, but truer historical comparison would be the 15th and 16th centuries, when printing shook the world. The internet has unleashed forces we are still struggling to understand. This deep tectonic shift in technology is still in its infancy. The same creative and destructive forces are being unleashed as they were with writing and printing, and AI has given them a new impetus.

As smart AI technology emerges, technology challenges and, in some cases, supersedes human competences. We enter another unpredictable phase of technological change. This, some argue, is an existential threat. Whatever it turns out to be, it is certainly changing the very nature of work. For all its dangers, AI will therefore certainly shape, in some form, how we learn. Unlike speech, writing, printing and the internet, this is software that matches, and in some cases even more than matches, us as humans.

Daniel Dennett, philosopher and polymath, in *From Bacteria to Bach and Back*, attempts a synthesis of human evolution and AI (Dennett, 2017). Just as the Darwinian evolution of life over nearly four billion years has been ‘competence without

comprehension', the result of what Dawkins called the 'blind watchmaker', an invisible process that drove biological evolution, so cultural evolution and AI is often competence without comprehension (Dawkins, 1996).

His hope is that machines will open up 'notorious pedagogical bottlenecks', even 'imagination prostheses' working with and for us to solve big problems. We must recognize that the future is only partly, yet largely, in our control. Let our artificial intelligences depend on us even 'as we become more warily dependent on them'.

Technology comes in revolutionary waves that have disruptive effects. It is combinatorial and cumulative, building upon previous revolutions, not something separate from us but ultimately a dialectic or an accommodation between it and we humans. More than this, generative AI is quite literally a dialogue between us and it. We must also recognize that technology is almost always a double-edged sword that needs to be overseen and controlled, so that technology for good overcomes technology for evil. At heart these 'generative' technologies, generative AI in particular, define what we must learn and how we learn. They transform learning. We are *Homo technus*.

Culture

The generative AI revolution did something remarkable. It took the sum of human culture and used that data to train a model that we could speak to, and it spoke back. In having dialogue with these chatbots, we are speaking to ourselves, our collective mind.

To understand AI in learning, one also has to dig deep into our cultural history, be almost archaeological, to uncover the historical paths that gave rise to AI. Eliezer Yudkowsky is right to warn us that 'by far the greatest danger of Artificial Intelligence is that people conclude too early that they

understand it' (Yudkowsky, 2015). This means understanding where it came from and how we got here.

AI has its origins in Greek philosophy and mathematics. It has also been interpreted for centuries by *culture*: poems, plays and novels. In addition to these ancient and older origins, a more recent art form, the movies, has almost defined AI in the modern mind.

Culture can illuminate, but also mislead, and there is no more misleading cultural forms than those that deal with AI. AI has been shown to us in Western culture largely through dystopian theatre, literature, then movies, with endless re-treads of the Prometheus (Frankenstein) myth. This has distorted thinking around AI for learning, but we do have something to learn from its presentation in culture, and in movies in particular.

From Aeschylus's *Prometheus Bound* to Mary Shelley's *Frankenstein: The Modern Prometheus*, the creation of a monstrous force took hold of the popular imagination, a myth fed straight into the movies in the 20th century. Asimov's novels have provided the famous Three Laws of Robotics commencing with his short story, 'Runaround' (1942), and there is a slew of modern novels about AI that have started to appear. Typical is Ian McEwan's *Machines Like Me* (2019), still stuck in the Mary Shelley Frankenstein myth, with Turing as the gratuitous Frankenstein.

Over the last 100 years, from *Metropolis* (1927), through the lens of movies AI has largely been portrayed as dystopian and evil. AI has, in film, reflected our fears, often representing the fear of technology but also of the 'other', whatever that 'other' was at the time – the Cold War, crime, violence, helplessness, corporate greed, climate change and so on. There have been glimpses of a more sophisticated and subtler dynamic around AI, in *Blade Runner* (1982) and more recently a rush of movies around AI, as it takes hold in our lives through the internet.

There are several movie themes that have shaped the common perception of AI, primarily as robots. AI will lead to robots that will turn on us and kill us all. AI will take over the

internet and kill us all. AI will fool us into thinking it is good but it is bad. This is similar to the popularity of child characters in horror movies, where our creations, our children, become our worst nightmares. These are variations of the Prometheus myth.

Technology is always ahead of cultural commentary. It will always be thus. Only now, over these last few years, as AI becomes operative in many domains, is it receiving subtler cultural appreciation and critiques, rather than robot fantasies.

Philosophy and mathematics

Technology is not a ‘black box’, something separate from us. It has shaped our evolution, shaped our progress, shaped our thinking and it will shape our future. There is a complex dialectic between our species and technology that is far more multifaceted than the simplistic ‘it’s about people not technology’ trope. This is explored in detail in *Learning Technology*, (Clark, 2023).

To understand why AI has real potential in learning, we also need to understand its two-and-half-millennia gestation period, through *mathematics*. From Euclid onwards, maths and algorithms were laying the foundations for what we now call ‘artificial intelligence’. So AI has not sprung up out of nowhere. It has had long gestation – 2,500 years of mathematics, logic, probability, statistics, algorithmic progress and machine learning.

The modern era of AI started in 1956, at the famous Dartmouth conference convened by John McCarthy and Marvin Minsky, whose aim was ‘to proceed on the basis of the conjecture that every aspect of *learning* [my emphasis] or any other features of intelligence can in principle be so precisely described that a machine can be made to simulate it’. Note the emphasis on learning, the realization of Turing’s vision of machines that could learn like children to become competent. There were successes, like Arthur Samuel’s checkers software,

but the promise was never realized and the first AI winter arrived in the 1960s. The early 1980s saw a resurgence of interest in expert systems but it got bogged down in rules-based reasoning and a second winter came. Only when probability and statistics were literally introduced into the equations, to give us deep learning, did we succeed in translation, speech and image recognition. AI, with exponential growth in processing power, data and powerful devices, can now deliver on that early promise.

AI makes us rethink learning. It holds the possibility that we do not have to learn some old knowledge and skills; it may help us learn new knowledge and skills, even improve the process and speed of learning. Stuart Russell, a major figure in AI, rightly claims in *Human Compatible* that, 'With AI tutors, the potential of each child, no matter how poor can be realised. The cost per child would be negligible and that child would live a far richer and more productive life' (Russell, 2019). Even if he's only partly right, this is a worthy direction of travel. What greater social good than AI helping us to learn?

Learning technology

Our final look at AI is through the history and development of AI in learning technology. Mechanical devices have been used to teach and learn for a long time. Behaviourism spawned many such efforts. Sidney Pressey (Petrina, 2004) showed his 'automatic teacher' at an American Psychology Association meeting in 1924. Skinner's teaching machines took positive reinforcement as their learning principle. The problem with these and much of the simple behaviourist modelling is the lack of data and knowledge about what the learner was actually thinking and why. Adaptive AI goes much further with individual and adaptive data that guides the user, much like a GPS or satnav, through the learning experience. The experiences themselves can also adapt through machine learning.

The go-to paper for much of this is Benjamin Bloom's 'The 2 sigma problem' (1984), in which he compared the lecture, formative feedback lecture and one-to-one tuition. Taking the straight lecture as the mean, he found an 84 per cent increase in mastery above the mean for a formative approach to teaching and an astonishing 98 per cent increase in mastery for one-to-one tuition.

Intelligent tutoring was attempted for decades, with limited success, most notably in the computer-based education system PLATO. These systems started to improve as interface design, hardware capabilities and more sophisticated pedagogies emerged. These were heavily influenced by the idea of expert systems and knowledge management in the 1980s and 1990s, but were still hampered by limited hardware and, just as significantly, limited software design. These were simple 'hard-coded' conditional response systems or rule-based systems that trundled through 'rules' to capture insights and therefore decisions about what to teach next to the learner.

It is often thought that all the real progress was made in universities and education, but in practice it was organizational learning, especially corporates, that established computer-based training (CBT) as a standard feature of the learning and development (L&D) budget. Companies grew to serve the real needs in this market and learning management systems emerged to manage the learners and learning content, along with standards, such as sharable courseware object reference model (SCORM), for passing data from learning experiences back to a database. There was little interest in certification. This was all about serving actual organizational needs in what were starting to become fast-moving businesses.

Alongside this emerged the multimedia age, with an increased range of media (audio, images and video) available through hardware and software advances. Laserdiscs could store tens of thousands of images and hours of audio and video. They were read-only devices but led to a much richer use of visual imagery in learning. Again, they all tended to have

limited memory and speed. The hardware was always a rate-limiting factor.

Many of the multimedia, CBT and LMS businesses started in the 1980s and 1990s are still around today. The sector has grown year on year, steadily taking on learning tasks that were previously thought beyond the reach of computers. But one significant event led to increased growth in the market – the internet. At first, with slow dial-up, there was little to be gained, but as bandwidth and consumer adoption exploded, new opportunities were available for networked and more sophisticated pedagogic approaches. This gave us online access to Google, YouTube, Wikipedia, open educational resources, massive open online courses (MOOCs) and cloud-based services.

The development of online learning proceeded in fits and starts. Exponential growth in hardware meant faster, smaller, more powerful and cheaper machines. Interactive design adopted heuristics such as blended learning, the flipped classroom, scenario-based learning and simulations. The internet then gave us the power to network and cloud computing. Interface design, always dependent on hardware capabilities, developed from commands and menus to window metaphors to touch screen and now gesture and voice.

Post-internet, the ecosystem morphed, alongside these advances, into the online learning market. This has thrived and most large organizations have LMSs, buy catalogues of content, commission content, use webinars and encourage social learning. More recently, as mobile devices became ubiquitous, mobile learning, or m-learning, became viable, putting the potential for learning into one's pocket. Most online learning content is now 'responsive' to desktops, laptops, tablets and mobile devices.

Beyond this we have the enhancement of real-world experience through augmented reality (AR). The democratization of experience through virtual reality (VR) is now with us, as are MOOCs. All of these are, to some degree, already delivered through the smart use of algorithms *implicit*

in the hardware and software. All of these offer opportunities for AI to become more pedagogically *explicit* and greatly enhance these learning channels

Then, as the devices became more powerful and data more plentiful, AI had a surge on the back of new techniques. The first great triumph in AI-led learning was in search, in its many forms – the ability to find things out, quickly and accurately. It was here that AI took centre stage in the learning ecosystem, opening up the future to irreversible, pedagogic shifts. As Google became the way we found things – including finding the answers to things, scholarly articles, places on maps, images, videos – huge productivity gains were made. In research alone, months of getting to and from libraries, walking up and down shelves of books and journals, were reduced to seconds.

Buying books, no matter how obscure, became quick and easy, accelerated by an AI-driven recommendation engine. This had a deep effect on AI on the internet as data mining, and a whole host of AI techniques, were brought to bear on the presentation of content and ads. Social media became a global phenomenon and many found these incredibly useful in terms of learning and continuing professional development (CPD) all mediated by AI. Facebook timelines and X feeds were suddenly algorithm-driven.

AI-driven face recognition developed quickly as did voice to text and text to voice. They led to advances in search, translation and a whole host of new advances in interface design. AI also made an appearance in plagiarism checking. It may seem odd that the only place you are likely to see AI in learning in a university is in checking on whether students cheat.

Despite all of these advances, online learning still remains stubbornly primitive, almost behavioural in approach. Even the gamification of content can be doggedly Pavlovian, with its use of external stimuli, such as scores, badges and leader boards, to motivate learners. The issue that remains is to what degree AI-driven learning can replicate an actual one-to-one tutor. What

held such systems back were the limitations of the hardware, software, tools and a paucity of data. To have any real impact on reproducing the process of actual teaching, we had to wait for huge increases in processing power, memory and data. This has proved significant for those working in AI in learning, as data is the fuel that allows algorithms to function more effectively. But there are other advances that are less data hungry, in interfaces, content creation, curation, personalization, assessment and the consolidation of learning.

In organizational learning the rise of learning experience platforms (LXPs) is an enabling development. LMSs have their uses, especially in large organizations where enterprise software is necessary. But it has also become a bind for those that want to move on to use AI, as they lack the flexibility needed on personalized interface experiences, personalized learning paths, content delivery and the need for a more data-driven approach to learning.

The web has moved on while learning has remained stuck in a fixed world of flat delivery. Almost everything we do online is mediated by AI generative and recommendation engines, with rapid moves into voice recognition and advanced analytics, yet learning is stubbornly static.

Newer technologies that are AI friendly, such as experience application programming interface (xAPI), allow tracking and data generation well beyond the traditional, fixed SCORM standard. Learning record stores (LRSs) open up the possibility of using, not just learning, business data to finally integrate learning into the decision-making that grows the business.

The combination of algorithms, big data and computing power promises to unleash AI into all layers of the learning experience: user interface layer, learning layer and data layer.

In education, if AI can result in accelerating and reducing the cost of learning – even basic skills such as reading, writing and arithmetic – on a global scale, it will make past improvements in pedagogy look like rounding errors.

Conclusion

Little has been written on AI as a technology for learning, little on the pedagogy of AI, and little on its existing and potential applications to help us teach and learn. This book attempts to do just that. We have a unique opportunity. AI has suddenly become a topic of global, public concern. It is the topic of the age. Why now? Why this moment in history? Well, the technology has matured to the point where it can be delivered even to your pocket via mobile devices; the internet means that it can be delivered from the cloud and that tsunamis of data can be used to fuel AI solutions. We have also seen remarkable advances in AI techniques across the board. At the same time the online learning market has matured so that it is ready to adopt AI solutions. At last smart solutions using smart software can be used to produce smart people, through AI-delivered teaching and learning, whether students, employees or citizens.

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02

What is AI?

In 1899 Jean-Marc Cote tried to imagine what domestic life would be like in the year 2000 (Public Domain Review). He drew a mechanical robot, as tall as a human, sweeping with a brush and pan. His imagination was shaped by the mechanical paradigm of the day. This framing of AI in an overly mechanical fashion can lead to an overemphasis on strangely old-fashioned concepts such as robot teachers. Framing AI in terms of anthropomorphic qualities can also lead to the attribution of comprehension, cognitive qualities, general intelligence and contextual understanding, when in fact none of these things are present. When we think of the future we tend to frame it in the past.

AI is many things

AI is many things. There is no watertight definition of AI. It is a whole constellation of technologies. Its core concept, the algorithm, has no exact mathematical definition and its many schools and techniques evade total capture. Behind this basic confusion is the idea that AI is one thing – it is not: it encapsulates 2,500 years of mathematics since Euclid put the first algorithm down on papyrus, and there are many schools of AI that take radically different approaches. The field is an array of different techniques, often mathematically quite separate from each other.

Other ways of looking at AI are through areas of practical problem-solving, such as natural language solutions to search and translation, and image recognition in healthcare solutions or robotics. There are many ways of slicing the AI cake. The important point is to see it as a set of very different

technologies or tools that solve problems – in our case, in teaching and learning.

It is not an entity. It is many things and these many things talk to each other, work in tandem and form a nexus of technology. John McCarthy, who coined the phrase ‘artificial intelligence’ in 1956, regretted doing so. It is a number of different data techniques, training using that data along with several statistical and algorithmic methods for producing a word-by-word output. It is competent without comprehension.

AI can learn separately but share instantly. Imagine that one person learns, then instantaneously everyone else knows. There is no person here, no single AI entity, only a growing network of competence. To be fair, that is where the danger lies but let’s not read intention, deception and agency into this network.

In particular, generative AI is not a person, it is us. We are staring into a mirror of our own collective, cultural capital. We are speaking to a hive mind that has competence without comprehension. It can be any of us and all of us, as it knows most of what we know.

In our rush to see an essence in everything, we see AI as one thing. We talk about AI as if it was a being or entity that could be good or evil. A common conceptual leap is to do what Descartes did with his thought experiment around an evil demon and see the ghost in the machine as malevolent. A consequence of this ghost-in-the-machine thinking is attributing evil intent to the ghost as a demon who will become autonomous, escape our control, conjure up existential threats to our species and eliminate us. Without that ghost, soul or demon, we strip AI of moral intention, either good or bad.

What exacerbates this mode of thought – that there is an intentional entity that lies or deceives us – is poor prompting, provoking and poking the bear. If you ask AI weird things, it will come back with weird stuff. In that sense it merely reflects the stupidity or intent of we humans. As something that is competent without comprehension, it speaks back to us, so

seems like another person, but it is not. Our social and cultural heritage has been captured and seems to speak back to us but we are speaking with ourselves as a species. It is a mass act of social generation, mediation and reflection. It seems human not because it is human; it just reflects our social production and heritage. You are not talking to an individual, with a soul, but looking into a wide and deep pool of socially structured knowledge.

In positing an ego, soul, agent or ghost in the machine into the technology, we make a category mistake in taking a concept from one domain – the human domain – and applying it to maths and software. There is no ‘I’ in ‘AI’.

This is the root cause of much of the anthropomorphism, speculative causality and doomsaying around AI which posits sci-fi levels of autonomy and agency and ignores control. Yann LeCun is their primary critic and constantly refers to this sci-fi speculation as way out of control. For him, we generally ignore good design and engineering to mitigate the wildly speculative risks. There is no form of existing AI or even anticipated AI that suggests an extinction event for humanity. This risk is negligible compared to risks such as climate change, and that is where we need to use AI to help solve our more intractable problems.

AI seems to have acquired the concept of a soul or a ‘ghost in the machine’, a phrase coined by Gilbert Ryle, who critiqued the idea as hopelessly Cartesian in his book *The Concept of Mind* (1949). I’m also uncomfortable with this Cartesian shift towards maths and software having a soul. We tend to see objects as having subjects behind them, so readily invent such ghosts in the machine, even seeing our minds as souls. The irreducible ‘I’ in Cartesianism seems to have been resurrected in ‘AI’, with the artificial ‘I’. There is no ‘I’ in ‘AI’.

Artificial general intelligence (AGI) seems like an unnecessary concept – the most common application of the ghost in the machine principle, the idea that there will be a single threshold, some oddly varied human bar, that will be transcended, rather than lots of separate capabilities, all of which can be designed

to be controlled. We can create lots of AI technologies with different specializations without seeing AI as crossing a single line. Yann LeCun asked, ‘How can you design seatbelts for a car if the car does not yet exist?’ Indeed, the self-diving car is a good example as to why we should not be concerned about AGI. We are nowhere near having self-driving cars on our roads, which confirms that we are nowhere near AGI. In truth, abstract researchers are often far from the coalface of implementation and tend to exaggerate thought experiments and abstractions.

From the Promethean myth onwards, we have imagined technology as a monster. Frankenstein being the first modern Prometheus (literally the subtitle of Mary Shelley’s book), a monster in literature manifested as a man in movies. *The Terminator* generation has its own Promethean icons. The tech kids, brought up on a diet of sci-fi, are fond of these thought experiments, which they mistake for philosophy. They speculate that tech ‘has a mind of its own’ and will ‘destroy us’, with no evidence other than thinking of a possible consequence, free from any mitigating design or control. Let’s not turn fiction too readily into non-fiction.

AI and intelligence

There is not one unitary form of intelligence in human and/or animal form. It is a complex set of cognitive conditions. With AI, it is even more diffuse and complex, not a set of discrete brains, housed in bone skulls, all approximately the same size with roughly the same limitations but an instantly communicating network. Even if we were to use human identity and intelligence as a benchmark, our brains have an incredibly limited working memory – a limited, fallible and declining long-term memory – we forget almost everything we try to learn, we have dozens of identified intrinsic biases, are emotionally unstable, can become dysfunctional through illness, be

delusional, even hallucinate, can't network, often get dementia and Alzheimer's, and die.

We needn't worry too much about the mechanics of it all, but we do need to understand the ways in which it will impact the field of teaching and learning. AI powers almost everything you do online. It is not coming – it is here.

The word 'intelligence' in artificial intelligence is perhaps misleading. Commentators such as Thomas Malone (2018) put a great deal of emphasis on the word, using it as a catalyst for analysis and the further development of AI. This may be a fundamental error, as it pulls us towards a too anthropomorphic view of AI, suggesting that it is 'intelligent' in the sense of human intelligence, benchmarking AI against a fundamentally anthropomorphic measure, a word with many problems, even when used in relation to humans. It is better to see AI in terms of tasks and competences, as *not* being intrinsically intelligent, as that word is loaded with human preconceptions and the danger is that people over-promise and under-deliver, so that there is disappointment in the market.

Linked to the notion that AI is 'intelligent' is the idea that AI derives wholly from our study and mimicry of the intelligent brain. Even the so-called 'neural network' approach is loosely modelled on the networked structure of the brain, more analogy than replication. A well-worn argument in AI circles is that we did not learn to fly by copying the flapping of birds' wings; we invented airplanes, an entirely different form of technology. Neither did we learn to go faster by copying the legs of a cheetah; we invented the wheel. Similarly with AI – simply copying the brain may be useful in some cases, but not overall.

The word made its debut in AI when the Turing test arrived in the brilliant 'Computing Machinery and Intelligence' (Turing, 1950/2009), still an astounding paper that sets the bar on whether machines can think and be intelligent. It is unfortunate that the title includes the word 'intelligence', as it is never mentioned in this sense in the paper itself. It is also unfortunate that the phrase AI (artificial intelligence), invented

by John McCarthy at Dartmouth College, has become a bit of a misleading distraction.

It has been a distraction in learning for a long time. Binet, who was responsible for inventing the IQ (intelligence quotient) test, warned against IQ being seen as a sound measure for individual intelligence or that it should be seen as 'fixed' (Binet and Simon, 1916). His warnings were not heeded, as education itself became fixated with the search for and definition of this single measure IQ, the main protagonist being Eysenck. Unfortunately, it led to unhelpful policies, such as the 11+ in the UK, promoted on the back of fraudulent research by Cyril Burt (Eysenck, 1988).

Stephen Jay Gould's book *The Mismeasure of Man* (1996) is one of many that have criticized IQ research as narrow, subject to reification (turns abstract concepts into concrete realities), when cognition is, in fact, a complex phenomenon. IQ research has also been criticized for repeatedly confusing correlation with cause, not only in heritability, where it is difficult to untangle nature from nurture, but also when comparing scores in tests with future achievement. Socio-economic background and culture may also play a role and the tests are often not adjusted for these variables. This focus on IQ, the search for a single, unitary measure of the mind, is now seen by many as narrow and misleading. Most modern theories of mind have moved on to more sophisticated views of the mind with different but interrelated cognitive abilities. Gardiner (1983), for example, tried to widen its definition into multiple intelligences, but this is weak science and lacks any real vigour. It still suffers from a form of academic essentialism.

We would do well to abandon the word 'intelligence' in artificial intelligence, as it carries with it so much bad theory and practice. Indeed, AI has already transcended the term multiple intelligences, as it had success across a much wider set of competences such as perception, translation, search, natural language processing (NLP), speech, sentiment analysis, memory,

retrieval and other domains, especially with the advent of machine learning.

AI as competence without comprehension

Big Blue beat Kasparov in 1997, but chess is thriving. AI remains our servant, not master. Yet mention AI and people jump to the far end of the conceptual spectrum with big-picture, dystopian and usually exaggerated visions of humanoid robots, the singularity and the existential threat to our species. The truth is more prosaic.

Work with people in AI and you will quickly be brought back from sci-fi visions to more practical matters. AI has no ‘cognitive’ ability. It is maths embodied in software.

IBM’s Watson may have beaten the Jeopardy! champions, Google’s AlphaGo may have beaten the GO champion – but neither knew they had won. Roger Schank, an early pioneer and major figure in AI, says something rather wise here: that AI is merely software and that we should in fact just call it software. This is to counter the myths around ‘cognitive’ computing.

We may even be able to move away from that other anthropomorphic obsession – consciousness. Daniel Dennett in *Consciousness Explained* (1993) saw it as an epiphenomenon, not necessary for the explanation of actual competence and action. If we can drop the ghost in the machine, the machine itself can be seen as being non-anthropomorphically capable. Dennett has a deep understanding of AI, and in his book, *From Bacteria to Bach and Back* (2017), sees a fusion of human evolution and AI, with consequences for the ethics of AI. He takes a holistic and, in a sense, forgiving view of AI. As he rightly says, we make children without actually understanding the entirety of the process, and so it is with new, generative technology. We already trust systems that are way more competent than we humans and so we should. Daniel Dennett

states the same thing in three words: ‘competence without comprehension’.

If we move beyond brains, beyond organic versus inorganic, we can move, as Harari claims in *Homo Deus* (2016), towards intelligence as fundamentally algorithmic, and uncouple intelligence from humans and consciousness. His argument is that natural selection itself is algorithmic and gave rise to our species and brains, but these algorithms are independent of the substances in which they reside. We must therefore readjust our thinking around intelligence and learning to include wider definitions.

AI is now challenging what it is to be human, intelligent, competent and to think and learn. The challenges we face are not to mimic humans but to find solutions that we are incapable of thinking of and executing. When Deepmind played its Atari game, it shot around the edge of the blocks and attacked from above, something humans had never thought of. When AlphaGo beat the GO champion, it came up with a move that was so counterintuitive that it shocked its human opponent. We do not want flawed human performance; we want performance beyond our capabilities, that is better than human. Humans crash cars and aircraft, kill patients through misdiagnosis and wrong prescribing. We need technology that is better than us. We didn’t go faster by copying the legs of a cheetah – we invented wheels, trains, cars and jets.

Turing clearly foresaw rapid advances in the power of computers and, in the long term, was visionary in his understanding of their potential capabilities. Remarkably, he successfully predicted that computers would have 1 GB of storage by 2000. However, the Turing test itself has been critiqued and is still a contentious area. He was wrong in assuming that ‘at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted’ (Turing, 2004). However, many underestimated the recent advances in technology, machine learning,

reinforcement learning and deep learning that have since allowed AI to do the things that were regarded as unlikely, if not impossible.

But the category mistake is to measure all of this in terms of just human performance. This is why IBM's 'cognitive' computing is simply misleading. There's little that is 'cognitive' about it. AI has become much more powerful, with considerable advances in machine learning and in the practical application of such advances. This is partly to do with advances in AI techniques, but also technical advances, which Turing predicted, and the rise of the internet and massive datasets. Few dispute the impact of AI on tasks, not just in the automation of tasks but also its impact in what has been seen as the cognitive domain. We just don't need the language of human cognition to make progress.

AI as collective competence

Networks, such as the internet, have provided a new substrate, where collective competence is also possible. One brain cannot download directly from another or replicate knowledge and skills perfectly, in a fraction of a second. But this has already happened in AI. In 'cloud robotics' robots learn from experience but can also learn from each other, as they are networked. Experience and learning are therefore shared across the networked robots.

This concept of pooled learning and competences is something that drops the idea of a brain, human benchmark, or subject. When it comes to our human abilities and what we regard as unique, we often invoke qualities such as 'intuition' or 'thinking' and 'consciousness'. Turing opened up the possibility of imaginable digital computers that would perform astonishing feats of what we would call intelligence or learning without recourse to a brain, soul or irreducible quality such as consciousness. That is becoming a reality with generative AI.

AI is moving fast, scoring one victory after another in specific domains. This takes us to the altogether grander concept of ‘collective intelligence’, a term coined by Pierre Levy, defined by him as ‘a form of universally distributed intelligence, constantly enhanced, coordinated in real time, and resulting in the effective mobilization of skills’ (Levy and Bononno, 1997). But Levy’s theory of collective intelligence is now dated and inadequate. More attention needs to be given to the nature and role of ‘networks’ in collective intelligence.

Some posit the idea that networks are intelligent, to a degree, simply by virtue of being networks. Our brains are networks, indeed the most complex networks we know of, and artificial intelligence uses that same (or similar) networked power to interact with our brains. We do not learn in a linear fashion, like video recordings, nor do we remember things alphabetically or hierarchically. Our brains are networks with pre-existing knowledge, and intelligence, that needs to fit with other forms of knowledge from networks.

The technological developments that open up the possibility of collective intelligence are large language models (LLMs). Artificial intelligence is embedded in many online media experiences. This takes us beyond the current flat, largely text-based, hyperlinked world of text and images, such as Wikipedia, or even social media, into forms of media that are closer to Levy’s original idea of collective intelligence. Networks store knowledge but, with the advent of online AI, they can also be said to be much more competent. AI is competence that resides in a network and is competent in itself but also adds to the sum of collective competence when used by humans.

Machine learning, with the aid of collective human data, often subsequently trained by humans, actually becomes more competent, often surpassing the intelligence of humans in specific domains. It can now also be shared. This is a new species of shared competence that is shared in real time, direct and scalable. Collective competence can then be said to reside

in and be an emergent feature of all networks, human and artificial, organic and non-organic.

We are only just beginning to see and practically explore and build new forms of collective competence that allow teaching and learning to be done by machines very quickly and on a massive scale. This is exhilarating and frightening at the same time. We, as humans, are now part of a networked nexus of human teachers, human learners, AI teachers, AI learners, networked knowledge and networked skills. The world has suddenly become a lot more complex.

Few would argue that AI has progressed faster than expected. In some cases the practical applications clearly transcend human capabilities and competences. We don't need to see 'intelligence' as the sun at the centre of this solar system. The Copernican shift is to remove this term and replace it with competences and look to problems that can be solved. The means to ends are always means; it is the ends that matter. What is wonderful here is the opening up of philosophical issues around agency, autonomy and morality. We are far from the existential risk to our species that many foresee, but there are many more near-term issues to be considered. Ditching old psychological relics is one. Artificial smartness is with us – it need not be called 'intelligent'.

AI learns

We, *Homo sapiens*, are the most successful species on the planet because we adapted to the world through technology – needles, twine, clothes, axes, arrow tips, hooks and so on. Then we made machines that eased the burden of labour. Mechanical machines led to the industrial revolutions in agriculture, transport and manufacturing. But the more interesting line of technology was symbolic: writing, alphabets, writing instruments, paper and books. We then learned how to read, write, argue, code. They eventually led to us feeding machines with symbolic writing (code) that allowed them to solve

problems of the mind. We now have machines that learn, machines that adapt to the world, like our minds, only not nearly as flexibly. If our ability to learn and adapt to the world, other people and knowledge is the hallmark of our species, then machines are starting to copy us.

As we teach computers to learn, we are doing to them what was done to us for millennia – education. We are smart because of what we are, but more importantly through what we learn, although this is no longer our sole domain. We are delegating some of that to machines.

So, machines use algorithms to learn. They not only learn how to access data that becomes information and knowledge, like how to win at Jeopardy!, but also combinatorial skills, like how to drive a car. The consequences are immense, economic, social and political. If machines learn, they can potentially replace some human roles and activities.

Turing interestingly anticipated machine learning in AI, seeing the computer as something that could be taught like a child. Man may be the measure of all things, but it is clear that computer power has already transcended our brain in some areas, in terms of storage, exact recall, uploading, downloading, mathematical calculations, chess, driving cars and so on. The whole point of Google search is not to be like us. It is better than us, with a greater memory, better search abilities and faster recall. Self-driving cars do not drive like us; they in many areas drive better than us. The whole point of the self-driving car is *not* to drive like us, as we kill 1.3 million people a year while driving, along with tens of millions of injuries. If we free ourselves from the tyranny of human competences and move towards problem-solving, the problem is no longer a problem.

Let's take this idea further. Some claim that *all* networks are, to some degree, 'intelligent'. As the boundary for consciousness and intelligence changed over time to include animals, indeed anything with a network of neurons, intelligence can be seen as a property that can be applied to any communicating network. As we have evidence that intelligence is related to networked

activity, whether these are brains or computers, could intelligence be a function of this networking, so that all networked entities are, to some degree, intelligent?

Clark and Chalmers in 'The extended mind' (1998) laid out the philosophical basis for this approach. Extended cognition extends out to embrace and, according to Clark and Chalmers, includes learning tools like pens, pencils and keyboards. This opens up the field for definitions of cognition that are not benchmarked against human capabilities. If we consider the idea of competences residing in other forms of chemistry and substrates, and see algorithms and their productive capabilities as being independent of the base materials in which they arise, then we can cut the ties with the word 'intelligence' and focus on capabilities or competences.

In the same way that our conscious cognition relies on unconscious processes, hidden but operative, so AI is the hidden but operative part of extended competence that is the web. In the same way that our minds use social learning, to learn maths, languages and skills such as reading and writing, so AI uses collective input from aggregated data drawn from millions, sometimes billions, of humans. We are all bound up in this one nexus, this one giant, global network, where data, information, knowledge and skills are exchanged. Once we see the underlying system as a single network, composed of many networks, the concept of collective competence can be understood.

AI in learning

AI in one sense means doing what humans do when they learn. The AI field is thick with references to 'learning', the most common being machine, deep and reinforcement learning.

Machine learning uses algorithms and statistical models and applies patterns and inferences to perform tasks through experience. This is analogous to traditional learning through exposure to taught experiences. *Deep learning* is a machine

learning technique that uses layered neural networks with large datasets, supervised, semi-supervised or unsupervised, to perform tasks. This is more like learning from the real world to gain competence and solve actual problems. *Reinforcement learning* operates on maximizing reward by exposure to existing knowledge and exploring new knowledge. This is more like deliberate practice to get things right.

Drawing inspiration from how we actually learn has given AI a boost. It should also be a wake-up call for those of us who work in the world of learning. AI is showing us what works and what doesn't. Pedagogic change no longer comes from just educational research; it comes from insights in cognitive science and, increasingly, through technological innovation. AI is the latest manifestation of digital pedagogy – we now have pedAIgogy.

Charles Fadel, the founder of the Center for Curriculum Redesign at Harvard's Graduate School of Education and co-author of *Four-Dimensional Education* (Fadel et al, 2015), is right in tracking AI against Bloom's three domains. So let's see how far a pedagogy of AI can be pushed in each of these three areas:

1. cognitive;
2. affective;
3. psychomotor.

Cognitive AI enables search across the web and access to educational resources such as knowledge on Wikipedia and 'how to' videos on YouTube. It can make connections, analyse data, use data to predict, prescribe, translate and transcribe. There's the detection and help with those who have disabilities and learning issues. It can now also create, compose new music, paintings, sculpture. In learning this is by far the most significant area, as text and images, which can now be generated, play a significant role in learning.

Affective AI can identify the presence of emotions in text, speech, gestures, body position and faces. Opinion and sentiment analysis can detect positive, negative and other

emotional and subjective views. Even face recognition, attention, surprise, puzzlement and wandering minds can be detected. In learning this can mean getting insights into learners' minds, difficulties, misunderstandings, even potential for dropout. It can also display emotion. Importantly, it does not actually feel empathy, compassion or cope well with context.

Psychomotor AI, as we have seen with advanced robotics and self-driving cars, can perceive environments, move through 3D environments, analyse complex physical environments, adapt to difficult environments, respond with complex reactions, mechanically walk, run, even do complex manoeuvres. Yet it cannot perform very complex actions that require the combination of cognitive control and flexible manoeuvring. It is not clear that this has much, if any, application in learning, other than some niche robots for young children. We will see AI enter through simulations, AR and VR.

As we can see, progress has suddenly become exponential. There are severe limitations to many of the models and some problems require enormous processing power. AI still lacks flexibility and versatility but as the technology becomes combinatorial it will start to get flexible. Just as the iPhone brought together a set of separate technologies, so AI will bring together sets of competences. Generative AI has become multimodal and combinatorial.

In learning we need these combinations of competences to help teachers and learners – integration of features and systems, such as new interfaces like chatbots and voice, adaptive learning to personalize experiences, recommendation engines to suggest what we do next, analysis of our performance and complex, simulated environments in which we can explore, fail and learn. Given the fact that productivity has stalled in education and costs are still rising, this seems like a real breakthrough. If we can deliver scalable technology, assistance, analysis, learning and teaching, at a much lower cost than at present, we will be solving one of the great problems of our age.

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03

Learning theory and AI

AI has been taken up by a hundred million people worldwide, who are speaking to it, listening to it, feeding it images, using it to find answers, do tasks, create images, create chatbots, design learning systems, all before it was even one year old – a mere baby.

What is driving this? The utility is clearly there but if we are to use this technology in learning, what does learning theory say about this approach? We have this AI revolution but let's step back and ask a simple question: 'What do learning theorists and learning theory have to say about generative AI?

Generative AI and learning

We can start by showing that the trajectory towards generative AI, neural networks and other underlying technologies emerged from studying the brain. This is not to say they copied the brain, only that those who developed this technology were deeply interested in how the brain worked. These pioneers laid the foundation for this astounding technology and, as we will see, they progressed by studying and implementing technology that learns. This is the key to understanding why AI is arguably the most important technology our species has ever seen for the delivery of learning.

Hebbian learning

Donald Olding Hebb (1904–1985) was a Canadian neuropsychologist, famous for his eponymous Hebbian learning theory, where learning is explained as connectionism – the creation through connections, then stimulation, of neural assemblies. As a teacher and headmaster at a school in

Montreal, he had developed specific views on teaching practice. He believed that motivation had to come from the self (intrinsic), the learner, and that teaching was largely about creating the environment and opportunities for motivation, and therefore learning, to take place.

One could teach good learning strategies and create the right environment of learning but, ultimately, learning motivation and learning itself came from the learner. This deep interest in learning led him to speculate on the physical nature of learning in the brain. His ideas on early years education and exposure to lots of stimulating experiences still resonate today, especially in many national early years policies. He was instrumental in pacing neural networks at the heart of the learning process.

This was presented in *The Organisation of Behaviour: A neuropsychological theory* (1949) which had a profound influence on both neuropsychology and, later, interest in neural networks in artificial intelligence. His interest in learning came through his experience as a teacher but his research at McGill, the University of Chicago and Harvard proved to be both profound and prophetic.

The British empiricists Locke, Hume and Mill had laid the grounds for learning by association. Alexander Bain and William James, in the 19th century, also laid out theories that were similar to Hebb but no one had a detailed theory on the mechanisms.

At Montreal's Queen's University, Hebb helped design the Hebb-Williams maze to measure animal intelligence. This was used to show that early sensory or experiential deprivation in rats led to increased cognitive development. Building on these animal experiments, he went on to develop a theory of learning that brought human behaviour together with the brain, colloquially summed up as: '*Neurons that fire together wire together.*'

Descartes had speculated that the pineal gland was the point where the material brain interacted with the immaterial mind. He never explained how this interaction took place, and other

general theories of psychology, such as Gestalt and behaviourism had failed to link the physical brain with psychological, mental events. It was Hebb who explained learning and thinking in terms of actual physical neurological or biological phenomena. Although other fields of psychology had explanations of external exposure to learning, Hebb provided a concrete theory that explained learning in terms of actual neural, brain activity:

When an axon of cell A is near enough to excite cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased. (Hebb, 1949)

This is known as Hebbian learning, physiological psychology – the first real theory that proposed an actual mechanism for the encoding of knowledge and memories. He called groups of neurons that learn together ‘cell-assemblies’.

Seeing learning as connections between groups or assemblies of neurons, thinking was exactly this, where thoughts and ideas are ‘phase sequences’. External or environmental reinforcement and exposure to new experiences created learning or memories, so he recommended enriching exposure to different types of experience to create neural connections and assemblies, for example, through speaking to children when they are young.

Although we now know that synapses grow and that the mechanism is also chemical through neurotransmitter molecules, with potassium and sodium ions changing the voltage, his model, separate from the mechanism, heads in the right direction in artificial intelligence, which uses electrical connections. His ideas were used, by analogy, to eventually create neural networks – the foundation of one area of the fruitful type of artificial intelligence. We see, for the first time, that we as a species use what we know physiologically about our own brains to take a leap into creating something that

mimics or matches our brains – and maybe even surpasses them – through AI.

Neural nets

At the same time, two other forerunners of AI were looking at turning such neural views of learning into a mathematical form, laying the foundations for learning to be captured and executed in computers.

Warren Sturgis McCulloch (1898–1969) was a neurophysiologist who took Walter Pitts (1923–1969) in to live with his family when he was homeless. Pitts' eccentricities included refusing to ever use or sign his name, which led to him never getting honorary and advanced degrees or promotion, then burning much of his unpublished research and retreating into social isolation.

McCulloch and Pitts are best known for their influential paper in computing and neural network computing, 'A logical calculus of ideas immanent in nervous activity' (1943), in which they were the first to create a mathematical model of a neuron, inspired by the concept of a biological neuron. This led to the development of ever more sophisticated neuron and neural network models, which were to eventually have astounding success in artificial intelligence.

They started with what they knew about the physiology of neural networks, or 'nets' in their terminology. The nervous system is a net of neurons, each neuron having a soma and an axon, their synapses always between the axon of one and the soma of another. What they realized was that at any moment a neuron has a threshold which must be exceeded by excitation to initiate an impulse. Their breakthrough was to represent this in terms of propositional logic, replicating what happens in the brain in mathematics.

It takes as its inspiration the idea, first introduced by Leibniz, that activating neurons can create propositions about the real world. Basing their model on a real, biological neuron, it

collects information (dendrite), then determines what neurons get preference (synapse), and processes then passes it on to another (soma) according to a model to produce an output (axon). One can replicate this by taking in binary input data of two types, excitation or inhibition, then produce binary outputs. In this way, McCulloch and Pitts introduced the idea of logical neurons and neural networks. Boolean functions, such as AND, OR and NOT can be 'represented' by these threshold neurons. This model is known as linear threshold gate (LTG).

It is similar to the logic gates of a computer, with AND gates (all inputs on), OR gates (at least one input on). The neuron switches on when a threshold is passed, so these artificial neurons can be switched on with different inputs. They can also be switched off with NOT gates, which replicates inhibitory synapses in real brains.

When networked, these neural nets do what computers do – compute! Note that at this stage neural networks do not learn but it does provide a structure for later models that can learn.

Their paper had a profound effect on computer science and, in particular, artificial intelligence. It tackles an important philosophical issue in giving a very basic computational model for the mind–body problem, the mind being a product of the brain as a neural network. This was the first modern computational theory of mind and brain in that it gives us a propositional logic model that explains how neural structures work and therefore the basis to explore how they learn. It is the precursor for later advances in the computational model of the brain and artificial intelligence. It eventually led to another advance, the invention of the perceptron by Rosenblatt and on to work on layered neural networks, backpropagation and deep learning.

Perceptrons

Having identified a mechanism for how mathematics could replicate the workings of the brain, not exactly but in another

form, independent of substrate, the next step was to build something that could actually learn.

Frank Rosenblatt (1928–1971) was an American psychologist who, while at Cornell, built the first ‘perceptron’ in 1960, a neural network that could learn by trial and error to simulate human thought and learning. This was to develop into the field of neural networks and deep learning, hence reference to him as the Father of Deep Learning.

The name ‘perceptron’ comes from Rosenblatt’s use of the idea and technology to identify objects in vision and speech, as described in his *Principles of Neurodynamics: Perceptrons and the theory of brain mechanisms* (1962). It was the first learning machine.

As software was too slow in the late 1950s, he literally built a physical machine. A camera provided the input through 20×20 photocells to make a 400-pixel image. The weightings were delivered via variable resistors and weighted learning delivered by electric motors that turned the physical dials on the resistors. The aim was to build a learning algorithm. If the perceptron has a task, like identifying a cat, then it learns from its mistakes. If it fires when it shouldn’t, the weight for a feature is reduced. The perceptron ‘learns’ from its errors so that, over a period of time, it eventually learns to recognize a cat.

Unfortunately, Marvin Minsky and Seymour Papert wrote a book, *Perceptrons* (1969), that poured cold water over the idea, with lots of examples of where the perceptron could not learn. One was a specific logical function called XOR or the exclusive-OR function (‘one or the other, but not both’). In fact, multi-layered perceptron networks could solve this XOR problem but had the effect of stopping funding down this neural network route. It was not until the 1980s that interest and funding returned.

Rosenblatt, sadly, did not live to see the fruitful research based on his insights, as he died in a boating accident just after Minsky’s and Papert’s book was published. Deep learning emerged from this one invention and has now become central

in artificial intelligence. Rosenblatt's perceptron has not only survived, it has become central to progress in AI.

Backpropagation

The next step was to turn the perceptron idea into a dynamic learning process, where continuous processes could result in more sophisticated learning. David Rumelhart (1942–2011) was a psychologist who researched the formal representation of cognition at the University of California. He invented backpropagation as well as recognizing, with James McClelland, that parallel distributed processing was relevant to understanding cognition, opening up the possibility of models for neural processing.

The other key figure is Geoffrey Everest Hinton who works at the University of Toronto. Hinton is a British-Canadian cognitive psychologist and computer scientist with significant achievements in artificial intelligence. These include backpropagation for training multi-layer neural networks, deep image recognition and deep learning (although Hinton attributed the idea for backpropagation to Rumelhart).

The seminal paper by Rumelhart, Hinton and Williams, 'Learning representations by back-propagating errors' (1986), opened up a rich path towards sophisticated, layered neural networks that learn and eventually the large language models (LLMs) we now use in generative AI. You can climb a hill by feeling around with your foot and finding the steepest direction and on you go to the top. Similarly, on the descent you feel around for the steepest step down and on you go. The gradient descent in perceptrons tweaks the weights to lower the error rate. You do this layer by layer. But suppose you're climbing a mountain with little foothills and peaks, the task is more complex. Backpropagation, the backward propagation of errors, allows neural networks to be trained, quickly and easily, so that deep neural networks do well in error-prone areas like speech or image recognition.

Neural networks and backpropagation have had innumerable successes. NETtalk started by babbling, then progressed to almost human-like speech. Stock market prediction was another and self-driving cars benefited in the famous DARPA Challenges in 2004 and 2005. This work has been essential for the progress of deep learning.

Convolutional neural networks (CNNs)

Yann LeCun, now the Chief AI Scientist at Facebook, is a key figure in the field of artificial intelligence, best known for his work in developing convolutional neural networks (CNNs) and their application to image recognition as well as deep learning and the AI open source movement.

LeCun played a pivotal role in the development and popularization of CNNs, a type of deep neural network particularly effective for analysing visual imagery. CNNs use a mathematical operation called convolution which allows them to efficiently process data with a grid-like topology (like images). This innovation has been crucial in the advancement of computer vision and image processing tasks.

One of LeCun's most notable contributions is the development of LeNet-5, a pioneering CNN architecture designed for handwritten digit recognition and other image processing tasks. Introduced in the late-1990s, LeNet-5 was one of the first successful applications of CNNs and laid the groundwork for many modern neural network architectures used in image recognition.

While he didn't invent backpropagation, LeCun was among the key researchers who applied it effectively to neural networks. Backpropagation, as we have seen, is a method used for training artificial neural networks, especially in situations where deep learning is involved. It involves adjusting the weights of the network in a way that minimizes the difference between the actual output and the desired output.

LeCun's work also significantly advanced the field of deep learning, a subset of machine learning where artificial neural networks learn from large amounts of data. He has been a strong advocate for deep learning, contributing to its rise as a dominant approach in AI research and application.

A strong believer in open source development, LeCun has released the open source 'Llama' large language model (LLM) and sees these models and general openness as the way forward in AI development and progress. LeCun is more than a scientist. He is a prominent public figure who has attacked his colleagues and ethicists for exaggerating the ethical issues and slowing the progress of AI.

Deep learning

Demis Hassabis is a British entrepreneur, neuroscientist and artificial intelligence researcher, who was also a chess prodigy and games designer. As the lead programmer and co-designer of Theme Park, he became well known in the games world. After a spell as an academic publishing a series of papers (Brooks et al, 2012), he started an AI company based on his understanding of how the brain and memory works. That company, DeepMind, was sold in 2014 to Google for \$628 million. It is now the beating heart of Google's work on AI.

Hassabis' theoretical work focused on the hippocampus, as that is where episodic memory is consolidated. He found, through a study of five brain-damaged patients, that memory loss, caused by damage to the hippocampus, was accompanied by loss of imagination (the ability to plan and think into the future).

This was a fascinating insight, as he then realized that the process of reinforcement, was the real force in learning – practice makes perfect. This link between episodic memory and imagination was backed up by other studies in brain scanning and experiments with rats. He proposed a 'scene construction' model for recalling memories, which when scaled sees the

mind as a simulation engine. This focus on the reinforcement and consolidation of learnt practice, deliberate practice, as it is known, when captured and executed algorithmically, generates expertise. This led to him setting up a machine learning AI company in 2010 – DeepMind.

DeepMind focused on deep learning algorithms that could take on complex tasks, and here's the rub – without prior knowledge and training. The key point is that AI can 'learn' to do anything. DeepMind stunned the AI community when its system played a number of computer games and became an expert gamer. In Breakout its system not only got as good as any human, it devised a technique of breaking around the edge and attacking from above that humans had not encountered. The achievement was astonishing, as the software did not know about these games when it started. It looked at the display, seeing how the scoring worked, just learning from trial and error. Deep learning takes some aspects of human learning, but combines with reinforcement learning (called deep reinforcement learning) to solve problems.

In 2016, DeepMind's AlphaGo beat the GO World Champion, Lee Sedol, 5–1 in Seoul. At the time, GO was considered to be the most complicated game that humans can play – the pinnacle of games. The match positioned AlphaGo as the Holy Grail in AI. The number of possible moves is greater than the number of atoms in the universe. The software was trained by many games played by good amateurs. The deep neural networks that mimic the brain, with enormous computing power, were therefore trained to perform a task that could go beyond human capabilities. In game two it made moves that no human would and it became creative.

AlphaFold is a protein-folding tool that predicts the 3D structure of a protein from its 2D amino acid sequence, saving huge amounts of research, lab time and money. As there are 200 million known proteins and each 3D structure took around 3–5 years to identify, Hassabis estimates that this has saved an astonishing one billion years of research.

Conclusion

One of the leading companies in the world had created some of the smartest software in the world, built on the back of computational models of learning theory, going back to Hebb and his successors. This should matter to learning professionals as AI now plays a significant role in learning. Software ‘learns’, or can be ‘trained’, using data. So, in addition to human teachers and learners, we now have software teachers and software that learns. It is not that a machine can beat a human but that it can learn to do even better. It is a sign of things to come – a sign of as yet unknown but astounding things to come in learning. The cutting edge of AI is the cutting edge of learning.

Learning theory and generative AI

We have seen how computer science has built upon neural ideas of the brain to recreate learning. Let’s ask a series of questions about AI and generative AI to uncover whether learning theory supports its role in learning:

- What is the role of dialogue and conversation in using AI for learning?
- What is the role of language in using AI for learning?
- How does AI support engagement and personalization in learning?
- Does learning theory support AI’s methods of delivery in learning?

Learning as dialogue

Dialogue is effective for learning for lots of obvious reasons. It actively engages the learner in the process of learning. There is no passive sitting back. You are engaged in dialogue with personal agency. You can participate in a full range of learning

interactions from asking a question, getting clarification and delving deeper into a topic, through to addressing specific problems, getting analysis, instant feedback, allowing for quick corrections and understanding, as well as engaging in critical thinking and debate. There is plenty of time for reflection as you are in control of the speed at which the dialogue takes place. Dialogue with others – other people, chatbots, all with different personas or roles – can introduce multiple perspectives, fostering a deeper and broader learning experience. It also encourages intrinsic motivation through interest and personal agency.

Evolved for dialogue

One can also argue that our brains have evolved for speech and hearing, not to read and write. Language is not for exposition but communication. Dialogue is primary to our use of language.

David C. Geary is an evolutionary psychologist at the University of Missouri. He brings an obvious, but often ignored, theoretical influence to learning theory – evolution and evolutionary psychology. In *The Origin of Mind: Evolution of brain, cognition, and general intelligence* (2005) he shows that it is difficult to see the brain and learning theory without understanding that how we learn has been largely shaped by millions of years of evolution. This is why some things are easy to learn, others hard. With generative AI we at last introduce speech, listening and multimodal learning. Talking and listening to it is already here – what we may call primary learning. In time, through other media that mimic sight and our view of the real world such as images, video and 3D, more contextual learning will be possible. Primary learning therefore acts as the platform upon which secondary learning (education) can be built.

Geary's main thesis is that of primary and secondary learning. Primary biological learning is folk knowledge and abilities – competencies from the evolved cognitive domain, our

ability to speak, listen and learn a first language. This is the stuff we are primed to learn, which needs to be seen as the foundation for secondary learning, but it also constantly interferes or inhibits secondary learning. We have perceptual and attentional abilities attuned to our evolutionary past. Take discovery learning. Left to their own devices, children are likely to stay within their primary learning mode when problem-solving so be unable to deal with new situations. One has to guide or change those primary motivations and beliefs to learn new skills.

Secondary biological learning is knowledge and abilities acquired through formal and informal learning. This is true of both reading and writing, which developed late in our evolution. Writing is only around 500 years old. Geary's theory has consequences for practice. Most education and training involves secondary learning. It means making the effort to overcome our natural primary motivations and beliefs. This is also true in learning theory, where we have a tendency to default to primary motivations and theories, rather than look to sometimes counterintuitive theories, such as spaced practice, interleaving and retrieval practice, to effect efficient secondary learning.

There is evidence to confirm the efficacy of using almost effortless, easy-to-process primary knowledge to enhance learning in areas that students find difficult such as statistics, by simply manipulating task content and presentation order.

With the focus on primary and secondary learning, Geary's beliefs on motivation and control are often overlooked. He believes that our motivations, intrinsic and sometimes extrinsic, are also subject to evolutionary influence. Our natural tendency to be distracted and not focus for long periods on abstract knowledge makes teaching and learning difficult. Understanding motivation, control and attention is therefore important when understanding both teaching and study skills. Children, for example, have motivational biases and are naturally drawn towards seeking out social relationships. We

take readily to dialogue on social media and ChatGPT because its participation is motivating.

Education and training can be usefully seen as an interface between evolution and culture. We take our evolved ability to learn and then use it to teach cultural capital, the knowledge and skills we need in modern society. Once we understand evolved biases, such as our folk biology, physics and psychology, we are better able to teach them novel, secondary knowledge and skills.

In the rush to see all learning as social and environmental and to deny pre-existing genetic (read evolutionary) influences, biology is often ignored. Geary brings evolution and biology back into play as an explanation for why we learn some things with ease but have difficulty with others. His primary-secondary distinction is a useful underlying theory as it is both explanatory and guides us in identifying false assumptions in learners as well as a framework for instruction.

Dialogue and learning

Socrates was one of the few teachers who died for his craft, executed by the Athenian authorities for supposedly corrupting the youth. That in itself has earned him eternal fame. Most learning professionals will have heard of him through their acquaintance with the ‘Socratic method’ but few will know that he never wrote a single word describing this method. Fewer still will know much detail about that method.

How many have read the Socratic Dialogues? How many know what is meant by the Socratic method and how he practised his approach? While Socrates himself wrote nothing about it, Plato and Xenophon recorded his thoughts and methods through the lens of their own beliefs. We must remember, therefore, that Socrates was in fact a mouthpiece for the views of others. In fact, the two pictures painted of Socrates by these two commentators differ somewhat. In the Platonic Dialogues, he is witty, playful and a great philosophical theorist.

In Xenophon he is a smart but dull moralizer. We should look a Socrates as a model for both good and bad pedagogy.

That the teacher should be an intellectual midwife to the student's own thoughts is Socrates' great educational principle. His mother was indeed a midwife and he was among the first to recognize that, in terms of learning, ideas are best generated from the cognitive effort of the learner in terms of understanding, realization and retention. Education is not cramming in, but a drawing out, uncovering preconceptions and misconceptions. Often the learner would discover not that they knew something but that they were actually in a state of ignorance.

He would claim that he taught nothing as he had nothing to teach and his lasting influence is the useful idea that, for certain types of learning, questioning and dialogue allow the learner to generate their own ideas and conclusions, rather than be spoon-fed. What is less well known is the negative side of the Socratic method. He loved to pick intellectual fights and the method was not so much a gentle teasing out of ideas as the brutal exposure of falsehoods. He was also roundly ridiculed in public drama, notably by his contemporary Aristophanes in *The Clouds*, where he uses the Socratic method to explore idiotic ideas using petty, hair-splitting logic. That is the danger in seeing learning simply as engendering critical thinking.

Beyond the famous Socratic method, he did have a philosophy of education that included several principles. Knowledge and learning were seen by Socrates as valuable endeavours, with a ruthless pursuit of questioning even basic assumptions. This was achieved socially through dialogue, not by lecturing or the transfer of knowledge from teacher to learner. One could argue that this is what ChatGPT has already achieved globally, using dialogue as a method of learning as well as being useful in solving problems and getting things done.

Interestingly, the Socratic approach is also often to be found in online learning. Roger Schank has taken the method forward

into online designs based on questions which access indexed content, especially videos. One could also argue that search-based inquiry through Google and other online resources allows the learner to apply this questioning approach to their own learning, Socratic learning without a Socratic teacher. AI chatbots, which support and deliver learning, are now being used to emulate the Socratic model and deliver personalized support, tutoring and even mentoring to learners. Adaptive and personalized learning systems truly account for where the learner has come from, where they are going and what they need to get there. Sophisticated online learning allows us to realize the potential of a scalable Socratic approach without the need for face-to-face teaching. Interestingly, it is only in the last few decades, through the use of technology-based tools that allow search, questioning and now chatbots and adaptive learning, that Socratic learning can be truly realized on scale.

As someone who abhorred didactic, talk-and-chalk teaching and learning, Socrates would be appalled at current education and training. He was not an institutional figure, practised his teaching in the public space of the Agora and thought that experts were normally fooling themselves by believing they had immutable knowledge to impart to their students. The unexamined life may not be worth living but neither is a life of absolute certainty.

Of course, if we were to behave like Socrates in the modern school, college, university or training room, we'd be in front of several tribunals for bullying students, not sticking to the curriculum and failing to prepare students for their exams.

Dialogism and learning

Another important theorist who examined dialogue in itself as a learning theory was Mikhail Bakhtin (1895–1975), the Russian philosopher and literary critic, who developed a theory of language as primary dialogue. He took this idea and applied it to learning. By dialogue he means social interaction between

people, teachers and learners, learners and learners, learners and others but also dialogue with the past. He was persecuted during the Soviet era but his work was rediscovered in the 1980s, and he has since become an important theorist in the philosophy of language and learning through generative AI.

Bakhtin criticizes the ‘monologic’ tradition in Western thought, where individuals are defined by religion as a concept of the finite, the soul, religious and establishment belief. Individuals, for him, are open and must engage in dialogue with the world and others. Dialogism is his foundational idea that all language and thought is inherently dialogic. We learn language and come to understand language through the practice of dialogue. More than this, learning in general means dialogue in many forms with different people, tools and perspectives in many different contexts. It was introduced in his *Problems of Dostoevsky's Poetics* (1984 [1929]). This idea of dialogue is in stark contrast with the teacher norm of monologue, lectures, talks or direct instruction.

Learning, for Bakhtin, emerges from dialogue, external and internal.

In his incomplete essay *Toward a Philosophy of the Act* (1986), written in the 1920s, he outlines a theory of human identity or mind based on dialogic development. There are three forms of identity:

I-for-Myself

I-for-the-Other

Other-for-Me

The I-for-Myself, on one hand, is untrustworthy and one must be careful with this self-formed view in learning. The I-for-the-Other, on the other hand, allows one to develop one's identity as a set of perspectives others have about you. Teaching from others or AI systems therefore play a cardinal role in defining who you are and what you become through dialogue. The Other-for-Me is the incorporation by others of their perception

of you into their own identities. This is an expansive and interesting form of identity in an age of dialogue with others through social media and messaging technology, as well as dialogue with chatbot technology which now plays a similar role.

Dialogism manifests itself in 'heteroglossia', with exposure to a multiplicity of voices and perspectives in a language. These can be parents, teachers, friends and those on social media, a heterogeneous group of voices that one can learn from. In AI heteroglossia can be getting a chatbot to play different roles, in different styles, at different levels, using different media. Dialogue can now be text-only or include images, speaking and hearing. Multimodal generative AI allows real dialogue to take place using different media, not just text.

AI technology has now produced dialogic systems that may fulfil Bakhtin's needs. We can now communicate, in dialogue form, with a new form of 'other' that lifts us out of our I-for-Myself, allowing technology to play a role in I-for-the-Other and Other-for-Me.

He also introduced the idea of authoritative versus internally persuasive discourse. Authoritative discourse is discourse that is embedded in a culture, enshrined in tradition. It is non-negotiable and taught as truth. It may be religious beliefs, science, the canon, parental beliefs and other assumed forms of knowledge and practice. It is often enshrined in an official curriculum or syllabus, which learners must memorize and regurgitate in exams.

Internally persuasive discourse is more personal, related to the learner's experiences and views. The learner has to engage in dialogue with the established views to relate it to their own prior knowledge, adapt to it, assimilate it and create their own sense of meaning. It is this dialectic with others of AI systems that seems to be attractive to millions of actual users. They find this satisfying when learning. Although we learn through both these forms of discourse, learning is to move from the

authoritative to the internally persuasive to find our own deeper forms of meaning.

Bakhtin also wrote about the concept of the ‘carnavalesque’ in literature and dialogue, which subverts and liberates the assumptions of the dominant style or atmosphere through humour and chaos. In education, this can be pedagogical strategies that disrupt traditional hierarchies and empower students to challenge and question. ChatGPT quickly developed carnivalesque features with poems in various voices and fun outputs, and prompted the famous *New York Times* article, where the journalist claimed it had told him to leave his wife! This is one of the attractions of dialogue and AI services: they allow you to play with the medium, play within dialogue. It has not yet fallen into the trap of being too relentlessly serious.

Interestingly, many traditional educators demand of AI technology that it be a truth machine, when language is no such thing. Language is multi-perspective, at times messy, even carnivalesque. It points not just towards learning as dialogue but also learning emerging from dialogue, in many different forms. LLMs and chatbots seem to be delivering this new form of learning.

Bakhtin would have loved the carnivalesque uses, such as prompts to chatbots to ‘act like a pirate’ or ‘write a poem in the style of a pirate’. He may have appreciated the fact that the GROK chatbot is quite snarky! AI satisfies both authoritative and personal learning, and language through dialogue allows for both. This is why generative AI is likely to have a profound effect on learning, through dialogue, which is how most good teaching by humans is done. It allows all species of dialogue, from direct instruction to looser, learner-centric forms of dialogue.

Bakhtin recognized the multifarious, and messy nature of human communication and therefore learning. For this he should be applauded. Where theory is so often formulated in rigid and simplistic models, Bakhtin’s work has influenced educational theorists who view learning as a social, dialogical

process and who advocate for more participatory, student-centred approaches. Educators who draw on Bakhtin's ideas focus on the role of discussion, debate and dialogue in learning, and they seek to create learning environments in which students' voices are heard and valued as part of the collective construction of knowledge. He would have been intrigued and fascinated by generative AI in learning.

Conversational theory and learning

An often-forgotten learning theorist, who needs to be revisited in the light of generative, conversational AI, is Gordon Pask (1928–96). His work on learning (1976) machines and conversational theory, although reaching back to the 1950s, has turned out to be both prophetic and useful.

Known as the 'Dandy of Cybernetics' he was famously mannered, theatrical, intense and eccentric. With his cape, Edwardian suit, bow-tie and pipe, he was known as being difficult to communicate with, his lectures and writing often difficult to comprehend. Both an academic and entrepreneur (largely unsuccessful) he liked the freedom to build and experiment, and valued his autonomy.

He was an original thinker who made a significant contribution to thinking about the complexity of learning. A polymath interested in geology, engineering, art, theatre, sculpture, biological computing, logic, linguistics, psychology, music, cybernetics, education, cognitive science and artificial intelligence, he saw learning as a complex process of interaction.

Well aware of artificial intelligence, he was a friend of Marvin Minsky and would stay with him in the US, although he was critical of the AI community's tendency to focus on isolated systems, separate from human interaction. As AI has widened out into language models and become part of the global system of learning, Pask has come back into focus, with his sophisticated theories of learning through 'conversation'.

Like Pressey and Skinner before him, he developed a number of learning machines. The difference was in seeing machines as things we converse with, rather than use as tools. In this sense he was closer to Pressey than Skinner. Eucrates, for example, not only taught but could adapt its teaching based on knowledge of the learner. There is physical feedback, and the machine builds a model of user behaviour. This is in stark contrast to Skinner's behaviourist reward machines. A more practical learning machine was SAKI (Self-Adaptive Keyboard Instructor), built in 1956, to teach punch-card operators, with electric wires that cued the learner, detected keypresses and time intervals, always pushing the learner ahead but not too far ahead, to keep the learner motivated. It focused on the intense and fruitful interaction between learner and machine. It was still in use in the UK Post Office in the late 1960s.

Ecogame, a management training game developed in 1969, had some success, being bought by IBM. Targeted at business and political leadership, it was based on Pask's ideas of self-organized, interactive learning in a simulated process and environment. His teleprinter trainer CASTE, which stood for course assembly system and tutorial environment, was built in 1972 and was the size of a small room. It was an early adaptive learning system based on his learning theory. CASTE had an entailment structure or topic map to provide an overview, a communications module on a computer terminal, BOSS (belief and opinion sampling system) and a modelling system that would present learning and assessment tasks. Systems were designed for students with tutorial algorithms, to spot failure and reassess on lower levels of learning, offering a range of choices to the learner.

Thoughtsticker, built around the same time, used entailment meshes and was a hierarchical representation of knowledge. It originally had a set of pigeon holes with paper notes, later developed into an adaptive system that allowed users to move upwards on an entailment mesh of concepts. You could study through a mesh of concepts with different perspectives. The

learner would feed in concepts and the interconnections would be shown. A small computer was later used where it could jump with conditional hyperlinking. It was ahead of its time.

In all of this development and experimentation, Pask was trying to implement his theory of learning – ‘conversation theory’. Teaching and learning were fundamentally conversations to Pask, even when they involved technology.

Pask studied learning through conversation, looking for identifiable features and processes. Meaning, he thought, is agreed use in the context of a conversation – a point he got from Wittgenstein. This is not a fixed exchange of propositions but an exchange that shapes the thinking and learning of both parties. We may agree, disagree, modify beliefs – even change our minds – but we gain from the conversation. Learners and teachers move through knowledge changing perspectives and levels. This, for Pask, was the essence of learning. He saw conversation and learning as something active, in use, as we deal with the world. Conversation is also the link or knowledge we have with others, that they exist, like us. According to Pask, conversation is at the root of everything we do – perception, emotional engagement and language.

This is a much subtler idea than just verbal conversation. We learn together and are always using artifacts to learn and communicate. A conversation can be several different types, as an individual, two people, a group of people or combination of people and technology. Proximity is not the point, as technology destroys distance. This is a general conversational theory that sees people, their social interactions and technology as one. This last point is important as he saw technology as the natural extension of teaching through dialogue or what he called conversation, a much richer and deeper concept.

Conversational theory saw learning as taking place in a learning environment, where we had sophisticated encounters with that environment, to it and back from it. We learn, Pask thought, by interacting, or having ‘conversations’, with people and our environment, including gestures, pictorial, with media

and with and through machines. In that sense ‘conversation’ is almost metaphorical. We interact with the world in a way that is very much like ‘conversation’ in language. This fits perfectly with the multimodal nature of interaction and conversations in ChatGPT and other conversational chatbots.

At one level, teacher and learner can engage in a strict conversation, with interactions on two levels – how (teacher demonstrates) and learner does and why (explains) at the conceptual level. Pask saw that we engage in hierarchies of conversation but not in some simplistic Bloom hierarchy; we flit between levels. What he called styles of teaching or learning are not media specific but more like Wittgenstein’s language games – contextually different forms of conversation.

His learning theory saw conversation as the essential learning process. He studied conversations separately from the content of conversations to explore their processes and limits. Learning conversations are, and should be, open, complex and dynamic, not a clumsy, didactic movement towards a final truth. Variety, a plurality of ideas, perspectives and conversations matter, where learning emerges from the process of conversation.

Learning needs to be provoked by a teacher and constructed by the learner and, although Pask is rarely seen as a serious social constructivist, he did a lot of work, unlike most constructivists, on exactly how he thought learners constructed knowledge and pragmatic skills. Importantly, he was sensitive to learning by doing and practical skills as well as knowledge.

In *Conversation, Cognition and Learning* (1975) Pask outlined the complexity of conversation, especially in learning, with hierarchical branching and loops, agreement, understanding, along with analogies and generalizations. Concepts are refined, expanded, altered, generalized, applied and related by analogy in the process of learning. This is not just dry rationalism, as for Pask to think is to feel, mind and body intertwined. Going back to Geary, he was keenly aware of our evolutionary legacy as biological beings. Conversation is how we think, and to teach

and learn we must reproduce the process that evolution has bequeathed us. He also drew from Vygotsky the idea of a social being having developed throughout childhood, although Pask's focus was less on language, more on a wider social context.

Entailment meshes attempt to capture thought processes, with topics related to each other and grouped into relations, such as 'analogies' and 'coherences', that comprise concepts, which we share. We are the active constructors of concepts, making new finer distinctions, seeing similarities and differences, generalized and realized through action and interaction. All of this takes place naturally; we are conversational beings and have no choice in the matter. Conversation is an ongoing, dynamic process, part of our being. In this, LLMs seem to play a useful role in being able to make these links, as language is structured in this form.

This is a radical departure from the database-driven model that drove AI for so long – one of retrieval. We now interact with the world, including AI, to learn through conversation. These conversations can be varied as they must suit the progress of the learner, not be pre-determined by too few constricting rules. Pask thought that most learning technology was hopelessly primitive in teaching and learning as it failed to recognize conversation as the means by which we learn. In that sense, generative AI has given us chatbots that free us from these constraints.

In learning there also has to be a contract to learn, common ground and action for successful learning, a process that is honest about contradictions, conflict, epiphanies and resolutions in the learning process. This is an agreed context. He saw that formal learning needed such structure and context. More formal teaching and learning chatbots and services are moving in this direction, holding on to the richness of the LLM, its breadth and depth of content, as well as its ease of use. Once a domain has been mapped out, one can also record the learner's vectors through content, the routes taken, levels reached. This is the sort of personalization realized by modern

adaptive systems and now generative AI, which can deliver the sort of sophisticated conversation that Pask envisaged.

Pask and Gerard de Zeeuw took conversation theory one step further with the ‘interactions of actors’ theory, where the scope was widened to include three or more actors in conversation, across time, entering and leaving such conversations. Here the defined process had knots, links, braids and a logic of the conversational and learning process, but this is often metaphorical and lacks the complexity that AI models now have in capturing language. This could be a fascinating development in AI for learning, with multiple teachers and learners in a group.

Generative AI and conversation

Conversations go on over time and become the culture, so LLMs capture many conversations. Conversations are captured as culture. So, rather than constructing huge logic trees and cumbersome Paskian machines, generative AI seems to deliver the richness of topics and language that allow it to deliver at least some of the conversational complexity that Bahktin and Pask thought was essential for learning. It already holds the relationships of words and phrases to each other by being trained in an unsupervised way, with a vast corpus of text, then further supervised training using RLHF (reinforcement learning from human feedback) and PPO (proximal policy optimization). Pask even talked about evil as the limits of the rights of actors to interact or hold conversation, prefiguring alignment and guard-railing. Pask would have been delighted at the emergence of this type of learning system using generative AI.

The next stage is to provide some conversational structures to the dialogue that allow efficient learning to take place. This is precisely what is happening with Khanmigo and other systems, where structured feedback and encouragement adds

motivational and useful feedback to the ongoing learning conversation.

Interaction is still primitive with computers, and we may have just seen a more conversational model emerge, where conversation – let's call it ongoing prompting – is replacing search and retrieval or request systems. Systems are becoming conversational, whether reading/writing or oral/aural and therefore more human. Technology must become more aware of people's needs to have more fruitful conversations. Conversations encapsulate interests, the seeking and learning process and reveal intention.

The word 'conversation' may seem odd in the case of LLMs, as it lies beyond the idea of discrete individuals and stored knowledge into massive and sophisticated models with hundreds of billions of parameters trained on unimaginably large sets of training data. But our interactions with these systems are certainly conversational, as learning is a process, not an event. Pask realized that despite the radically different way these models work, they are still entities within a conversation with humans. The biological and technical are not separate but entwined in a single conversational system.

We would do well to pay more attention to learning through Pask's conversation theory, with more empirical work on what works well in the context of increasing capability on the machine side through AI. Improvements that take the learning process into real conversations with structures that both allow for complexity but also guide towards outcomes, is now badly needed. This is what will give us useful, massively beneficial, teaching and learning. Let's call them conversational systems, even a universal teacher (UT).

When we think 'dialogue' we should not automatically think asking questions from the learner side, even questions by the teacher. Personalized learning is almost invariably a dialogue between the learner and teacher or AI teacher. From the teacher's perspective it is about getting to know the learner, what they know, what they don't know and what they are

having difficulty with at that specific moment. From the learner's perspective, they need to be kept motivated, get feedback and get difficulties resolved quickly. The Socratic method perhaps leans too much in the direction of interrogative questioning, whereas AI teaching and learning involves more specific types of dialogue to progress and accelerate learning.

Khanmigo and other systems allow the learner to set up a debate, an adversary – you can speak to historical figures, even fictional characters. Tutoring AI takes learners through content in a structured fashion keeping them at the appropriate level, providing feedback, examples and so on. Learning is not dialogue alone but is the use of dialogue to progress learning.

An area that has progressed rapidly through AI is 'avatars'. We now have realistic avatars that speak in your voice in dozens of languages with gestures and editing that allows them to be used as teachers, experts, patients or customers. These are already used in computer games to deliver dialogue in real-time, using chat functions such as ChatGPT.

Many others have seen dialogue as essential in learning, including Piaget who recognized the role of social interaction and dialogue in the construction of knowledge. In Paulo Freire's dialogic model of education, dialogue is essential for critical thinking and liberation from oppression. He saw dialogue as a means for learners to develop critical consciousness.

Jürgen Habermas' theory of communicative action presents dialogue as a means of reaching understanding and emphasizes the importance of discourse in society, which can be applied to educational settings. Dialogue, it would seem, is one of generative AI's ace cards in learning.

AI, language and learning

As LLMs emerged, it became obvious that something extraordinary had happened in AI. The emergent qualities in terms of generating well-structured, grammatically correct text

that seemed almost human, amazed the world. It could translate and generate answers in depth over a breadth of subjects. The same happened with images, the quality improving day by day. As generative AI became multimodal, with speech input and output, as well as interpretation of uploaded images, it gained in functionality. Then user created chatbots were launched, allowing anyone to create a generative pre-trained transformer (GPT) or personal chatbot. All of this within one year. At the heart of it all were LLMs trained using gargantuan amounts of data. There seemed to be something in the model and data, in language itself that enabled this miracle. So what is the role of language in learning?

Wittgenstein, language and AI

The philosophy of language has had lots to say on the nature of language, meaning its relationship to the world and the limits of thought. Ludwig Wittgenstein is perhaps the most famous of these philosophers and there seemed to be real parallels between what AI was successfully delivering and his views on language. As generative AI through LLMs started to roll out, I had several déjà vu moments around Wittgenstein's work.

Many mistakenly see AI, as Wittgenstein saw language in his book *The Tractatus* (1922), as a direct correspondence between propositions and the real world of truth. The early pioneers of AI, with their focus on symbolic AI and logic, had limited success as they too had this strictly representational view of language. Wittgenstein famously rejected this earlier idea in his later philosophy for a more complex view of language as a tool for communication not direct representation, as laid out in his *Philosophical Investigations* (1953).

He comes close to describing what LLMs do when writing of words as having 'family resemblances', related to each by various degrees of resemblance in meaning or multifarious relationships. The mistake is to think that universals, like the word 'game', can be defined separately from these

relationships. We have a tendency to see these universals, or abstract terms, as having some underlying definition, quality or essence that lies beneath all instances of games, but there is no Platonic game or even definition of 'game'. Dictionary definitions come from use, not the other way around. There is a nexus of game activities with different qualities. This is the problem with identifying key terms such as AI, intelligence, bias, learning, hallucinations, values and alignment. We treat these as essential things when there is, in fact, no single definition of any of them.

Wittgenstein's concept of family resemblance is similar to the way language is represented in LLMs, as relationships between words or tokens. Family resemblance is actually too tight a metaphor, as it suggests lineage and a timeline, whereas LLMs are mostly timeless and far more complex and multidimensional. Where Wittgenstein is right is in seeing language as the sum total of probabilistic relationships, not defined by the relationships of words to the real world, but in relationships with each other. However, it would be a mistake to take this too far, as language also embodies relationships with the real world. It is not as if LLMs have no context and are totally de-anchored from the real world, only that they store and generate outputs using complex relationships of meaning captured in the mathematics.

Another way of looking at this is to see meaning as use and LLMs as optimized for use. They respond to prompts by people and continue to respond, in real dialogue. Language has indeed developed for dialogue, as a method of communication, not the block representation of the world in written prose, so LLMs play to our natural use of language, which brings us to another of Wittgenstein's ideas – language games.

In *Philosophical Investigations*, Wittgenstein uses the term 'language games' to describe the various ways in which language is used by humans. He saw us engaging in different types of communication, namely 'language games', where language is used in a relatively tight context. Each game

represents a specific use of language defined by grammatical rules, and they are embedded in a broader cultural and social context.

Here are a few examples of language games that Wittgenstein mentions in his work (1922):

- Giving orders, and obeying them, with its direct, purposeful communication such as to a soldier, at work or in school.
- Describing an object's appearance, or its measurements describing the physical properties or dimensions of things in the world by architects, artists, engineers and others.
- Constructing an object from a description (a drawing) involves not just a description of an object, but one that is intended to help someone else recreate the original object, in areas like engineering and carpentry.
- Reporting an event communicates what has happened to someone who wasn't there. This is fundamental to storytelling, journalism, and many forms of social interaction or speculating about an event involves making hypotheses or guesses about what might happen in the future, based on current information.
- Playing a game with rules, even playing a physical game involves a language game, in which the rules of the game are discussed and agreed upon.
- There's also making a joke; telling it, laughing, the complex interaction of telling a joke and understanding the humour, which often requires a shared cultural and linguistic context.

These are all examples of the diverse ways in which language is used, each with its own specific rules and contexts. The meaning of words and sentences is closely tied to these activities and cannot be separated from them.

As LLMs contain so much language (it would take 22,000 years, eight hours a day, to read ChatGPT 3's training data) they contain many, many different species of 'language games', and

can reproduce their styles – from academic lectures to chatty dialogue, even formal prose as in stories and poetry, in any genre or style. It is not a homogeneous lump of language but language in all of its forms and stylistic complexity. So when you ask ChatGPT to give you text in the form of a teacher, engineer, journalist or pirate, it will do so, as it recognizes the language patterns and games within that domain. In one sense, it remarkably uses the closer family-resemblance-type relationships between words to reproduce languages games. You do get this strong sense when using ChatGPT that it is engaging, not just in the presentation of language but also the presentation of language in a style or context.

The wider context for using a ‘language game’ for Wittgenstein is what he calls a ‘form of life’ – not biological life, but a shared cultural or social background that makes mutual understanding and communication possible. This is the complex, shared background of customs, practices, beliefs and languages that individuals in a community inhabit. It is the shared human activity that constitutes the background against which our language and our concepts acquire meaning.

To imagine a language is to imagine a form of life. In other words, the way we understand language and the meaning of words is deeply influenced by our form of life – our way of living. Language is a social practice shaped by these forms of life, and it only has meaning within these specific contexts.

This is the problem of assumed context in AI. LLMs capture these forms of life as embodied in language, but the capture is not complete, as time, action and a world view are to a large degree, missing. In AI we are still dealing with a limited set of de-anchored or impoverished language games. As AI develops it will include these contexts, sense of time, action and world views.

Another fascinating area of congruence between recent developments in AI and Wittgenstein is his work on ethics. Wittgenstein was always careful when discussing ethics, seeing it as largely beyond language, as something that cannot be

discussed meaningfully using the facts of the world, which is the proper domain of language. He categorizes these statements as 'unsayable' – not in the sense of trivial, but because they transcend what can be expressed in language. Ethical statements, in the context of a language game, can be seen as expressions of a way of life, or as part of a particular cultural or social practice. They are not descriptions of facts about the world, but rather they are part of a human activity. They make sense within the practices and forms of life where these language games are played. So ethical statements are deeply embedded within our specific cultural and social practices. They gain their meaning and importance not from corresponding to some objective moral fact, but from the role they play within our lives. In other words, he approaches ethics not as a matter of knowledge, but of practice and living.

One could speculate and see him as being highly suspicious of the definite ethical stances taken by the doomsayers and fearmongers, as that is likely to be overreach by language which leads us astray into thinking it has deep certainty and meaning when it does not. Ethics in AI can be seen then as being driven by the moralizing perspectives of the ethicists themselves, not correspondence with reality.

For Wittgenstein, language is the limitation of thought. He explored this limit by being honest about the very nature of language as generated by the human mind in a cultural and social context. Wittgenstein was a philosopher, mathematician, logician, teacher and architect, and his exploration of the nature and limits of thought and language have much to teach us, especially about being careful when thinking about these new forms of AI. His critique of essentialism led him to believe that words do not have common underlying definitions but rather get their meaning from use in relation to other words in language games. They may constitute something we have never encountered before – a new language game or games, a new form of life. His insights into how language works on resemblance between words seems prescient, as is his theory of

language games and his views on ethics and language. What we can learn here is that language is promiscuous. It tends to take us forward in a way that makes us think we know what we are talking about but is actually just 'language'. This reification, essentialism or faith in universals is what he criticizes and what many in AI move towards, not only in describing its qualities but also its ethical consequences. It is not as good as we think and not as bad as we fear.

Vygotsky, language and AI

Vygotsky is an oft-quoted but rarely read learning theorist. In using ChatGPT 3.5 and 4, I have become more Vygotskian, as I have come to see ChatGPT as similar to the concept of the Vygotskian teacher. He gives us insights into why language is key to intelligence and why generative AI may be the most powerful form of learning technology we have ever invented.

LLMs are fundamentally Vygotskian. They have been trained on data (language) as created and used by us, and therefore they learn from us. There is another step, where humans train the model further by making judgements to make the output more palatable through what is called 'reinforcement learning from human feedback' (RLHF). These models are the source of growth, development, learning and intelligence that Vygotsky described but did not foresee.

Just as Vygotsky thought of language as a mediating source for learning (1962, 1978), so LLMs use this form of mediation by language. They have been further trained by real humans to align it with our expectations. This, for Vygotsky, is how babies and children learn. They listen, are spoken to and guided by adults. When we use an LLM we are like young children asking questions and being given responses by what Vygotsky calls a 'knowledgeable other'. That knowledgeable other is AI.

Ultimately, the strength of Vygotsky's learning theory stands or falls on the idea that learning is fundamentally a socially mediated and constructed activity. Vygotsky's theory does not

propose distinct developmental stages, like Piaget, but instead emphasizes the role of social interaction and cultural context in cognitive development. He believed that social interaction plays a critical role in children's cognitive development and argued that children learn through interactions with more knowledgeable individuals, who provide guidance and support.

This is what ChatGPT 4 does, in general, but also in more formal teaching experiences as in the Khanmigo implementation, Duolingo and other second-level implementations of generative AI. It provides the 'knowledgeable other'. In fact, this 'knowledgeable other' is better than any one teacher as it covers all subjects, at different levels, personalized, is available 365/24/7, and is endlessly patient, polite, encouraging and friendly.

The cardinal idea in Vygotsky's psychology of learning is that knowledge is constructed through mediation, yet it is not entirely clear what mediation entails and what he means by the 'tools' he refers to as mediators. In many contexts, it simply seems like a synonym for discussion between teacher and learner. However, he does focus on being aware of learners' needs, so that they can 'construct' their own learning experience and changes the focus of teaching towards guidance and facilitation, as learners are not so much 'educated' by teachers as helped to construct their own meaning and learning.

This is exactly what ChatGPT 4 does as a 'tool'. It mediates through dialogue and allows the learner to construct their own sense and meaning by driving the learning process through dialogue. It uses language, the key form of learning and social development for Vygotsky, to patiently go at the learner's own pace and level and even identify mistakes. It can keep us in a useful zone of proximal development, as the process of dialogue captures what has been said to guide what should be said next. Language is a form of action, where thought essentially involves manipulation of internalized language, and so can be seen as a form of inner action.

He often uses the word ‘tool’ which refers to any external artefact, symbol, or sign that individuals use to help them think, problem-solve and learn. Tools can be physical objects, as well as cultural and psychological tools. Tools help individuals interact with the world and transform their mental processes. They bridge the gap between a person’s current cognitive abilities and their potential for higher-level thinking.

Cultural tools are the external artefacts and signs that are created and shared within a specific cultural context. Examples of cultural tools include writing systems, books, calculators, maps, computers and language itself. He most likely would have included generative AI as a useful tool for learning. Vygotsky also identified psychological tools, which are internalized cultural tools that become part of an individual’s cognitive processes. Psychological tools include strategies, problem-solving techniques, mnemonic devices, and other mental processes that individuals acquire through social interaction and cultural learning.

Wolfram, semantics and learning

Another useful perspective has come from Stephen Wolfram, the British-American computer businessman, physicist and computer scientist who created Wolfram Language, Mathematica and Wolfram|Alpha.

Wolfram has developed some truly remarkable tools. Mathematica, launched in 1988, is software used for mathematical computation, visualization and algorithm development. Wolfram Language is the programming language used in Mathematica known for its powerful computational abilities, symbolic computation and ease of use. It has been used in various fields, including scientific research, engineering, computer science, and data science. Wolfram|Alpha, is a computational knowledge engine launched in 2009, which provides answers to factual queries by computing the answer from externally sourced curated data,

rather than search items like Google. Wolfram's interest is in new computational methods, especially tools and languages that facilitate computational thinking and analysis.

The fact that ChatGPT and other LLMs can produce human-sounding text surprised almost everyone, and we are not entirely sure how this happens. Neural networks were modelled initially on the brain and, although different in execution and substrate, seem to do things that are similar. In using language in dialogue form, we have found common ground because the underlying structure of language is our common ground.

Wolfram has suggested (2023) that, just as language has a syntactic grammatical structure, there is also a semantic grammar that is being identified when LLMs are used. More than this, he suggests we may, in time, be able to identify this semantic form of computational thinking and get LLMs, or technology related to them, to discover things that are true about the world, possibly beyond the capabilities of the human brain.

For the first time in our history we can link LLMs to external software that does the maths, whatever, and this gives us the power to identify what it is in language that is essential and not fiction, idle chat or lacking correspondence to the real world.

Wolfram is saying that in understanding the Wittgensteinian ideas of language as family resemblance (weights), we can uncover deeper structures that take us back to the original Wittgensteinian idea of correspondence between logical expressions and the real word, releasing us from the constraints of language that ties us to others in communication, and not truth-searching, goals. This is a bold vision and there are those such as LeCun who are working to free LLMs from their linguistic net, making AI models more aware of the real world or at least having models of that world.

Language shapes thought and therefore is intelligence. Both Wittgenstein and Vygotsky had this insight – that language is not an emergent quality of intelligence, but is intelligence itself.

This explains why LLMs are so powerful. Intelligence is embodied in language and we learn from language. If they are right, generative AI, using written or spoken language, will prove to be the most powerful form of learning technology we have ever seen, as it is congruent with how we learn.

Personalized learning

One of the great advances in using dialogue for learning is the intrinsic feedback. This can take many forms, as there are many forms of dialogue in learning. We may, at last, have a form of dialogue-based or dialogic learning that allow us to do what good teachers do – engage and personalize learning.

Have we ever seen a more engaging technology in the history of our species? Generative AI has astonished us with its ability to engage us by doing with language what we do most of the time – dialogue. The simple interface combined with astonishing payback is what led to one million users in two days, two million in two months and now over 100 million. This was the power of persona – agency driving the experience.

Bloom and personalized learning

We know Benjamin Bloom for the wrong thing, his bogus triangle. His true legacy, which is now starting to be realized and implemented through the use of technology, is personalized learning.

This research led him to believe, as explained in *Human Characteristics and School Learning* (1976), that learners could master knowledge and skills given enough time. It is not that learners are good or bad, but fast and slow. The artificial constraint of time as shown by periods in schools, timetables and fixed-point final exams, constrain learning. The solution was to loosen up on time to democratize learning to suit the many not the few. Learning is a process, not a timed event. Learning free from the tight constraints of time allows learners

to proceed at their own pace. This is what AI teaching and learning systems offer – personalized learning that frees us from the tyranny of time.

Google's Peter Norvig famously said that if you only have to read one paper to support online learning, it is 'The 2 sigma problem' (1984), where Bloom compared the lecture, formative feedback lecture and one-to-one tuition. Taking the straight lecture as the mean, he found an 84 per cent increase in mastery above the mean for a formative approach to teaching and an astonishing 98 per cent increase in mastery for one-to-one tuition. In other words, the increase in efficacy for one-to-one learning, because of the increase in on-task learning, is immense. This paper deserves to be read by anyone looking at improving the efficacy of learning as it shows hugely significant improvements by simply altering the way teachers interact with learners. Online learning, in the widest sense of the word, promises what Bloom called 'one-to-one learning', whether it's through self-paced structured learning, scenario-based learning, simulations or increasingly adaptive and personalized learning.

Self-determination theory (SDT)

Edward Deci and Richard Ryan's self-determination theory (SDT) (1985) is a psychological framework for understanding human motivation. It focuses on the degree to which an individual's behaviour is self-motivated and self-determined. Central to SDT are three basic psychological needs that are considered essential for psychological health and well-being:

1. **Autonomy:** need to feel in control of one's own behaviours and goals.
2. **Competence:** need to gain mastery of tasks and learn different skills.
3. **Relatedness:** need to feel a sense of belonging and attachment to others.

SDT suggests that fulfilling these needs fosters motivation and personal growth. The theory has been widely applied in various domains, including education, work and healthcare, to understand and enhance human motivation and personal development.

Dialogue and structured learning fulfils the first two of these criteria. With generative AI chatbots or learning tools, you are in charge, within the context of a dialogue or generative activity. You also get that sense of increasing competence through dialogue. The sense of belonging may come through that experience we all have when it answers on anything we ask of it. It seems as though it knows more than anyone because it does know what everyone has known. It is the sum total of human culture and knowledge. Therefore, when we speak to it, it seems it speaks back to us, like a single hive mind.

SDT could therefore guide the creation of learning systems that support the user's autonomy, competence and relatedness to encourage engagement and facilitate learning.

Learning design

With 500+ studies on the design of learning under his belt, Richard Mayer has been a guiding light in the online delivery of learning for decades (Clark and Mayer, 2016). His rich findings could be described as: less is more, media rich is not mind rich, keep it personal and conversational. His work supports the idea that chatbots and AI can be both productive and efficacious for learning.

His key recommendations in online learning include several general principles for the delivery of learning that support the use of generative AI.

The *personalization principle* promotes a conversational style in learning. This is exactly what ChatGPT and dialogue can deliver in learning. Mayer suggests that people learn better when words are presented in a more informal conversational style rather than a formal style. Chatbots can mimic human

conversation, which can make learning more personal and engaging.

Segmenting content into user-paced segments rather than continuous presentation, which is exactly how ChatGPT dialogue works, giving control over pace to the user and breaking down material into smaller segments. These principles guide the effective construction of multimedia materials for learning, emphasizing the importance of how instructional design aligns with human cognitive processes. He also proposes the *segmenting principle*, which states that people learn better when information is presented in user-paced segments. Chatbots can deliver content in manageable chunks and allow learners to set the pace of the interaction.

With his *modality principle*, he found that people learn more deeply from pictures and spoken words than from pictures and printed words. Chatbots can integrate voice interactions, aiding in learning and comprehension.

The *coherence principle* recommends excluding extraneous material such as unnecessary graphics, text and narration that do not support learning goals, and not using background music or overly detailed visuals. Many mistake generative AI as a generation machine that produces too much content, yet it is superb at summarizing and giving content in as precise a form as you demand.

Signalling is about enhanced learning through the use of cues, such as highlighting and headings to draw attention to key information; this generative AI does particularly well.

The *redundancy principle* focuses on avoiding too many media types – a real problem in UDL (universal design for learning), which often does the opposite. Generative AI can, of course, now produce images and narration, as people learn better from graphics and narration than from graphics, narration and printed text combined. Mayer recommends the use of either graphics or text, but not both, in narrated presentations.

There are many ‘less is more’ lessons that can be gleaned from the research that support a more stripped-down, less flamboyant form of online learning, such as is provided by generative AI.

Conclusion

We have waited a long time for an online technology that truly delivers complex teaching and learning. Our technology tools for teaching and learning have been limited to flat text, books, audio and video. As ChatGPT swept around the globe and the world started to speak to itself through chatbots, it felt like a threshold had been passed. We were, at last, able to learn any subject, anytime, any place with an internet connection. It could present at any level, in many styles and do many tasks. Translations into many languages were also now possible. The science of learning has stressed dialogue, feedback, agency and intrinsic motivation, and the door to this vast breadth and depth of knowledge was a simple chatbot and dialogue.

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04

AI is the new UI

In the seminal AI movie, *2001: A Space Odyssey*, HAL, the onboard computer, reads the lips of the two humans, who converse inside a pod, and decides that they pose a danger. HAL is an autonomous learner. HAL stands for ‘**H**euristically programmed **A**lgorithmic computer’. It turns out that HAL has become a reality. Indeed, we deal with thousands of useful HALs every time we go online.

The vast portion of the software iceberg that lies behind current interfaces – doing its clever but invisible thing, the real building blocks of contemporary computing – is algorithms and AI. Whenever you search, get online recommendations, engage with social media, buy, do online banking, online dating, see online ads, the experience is mediated by algorithms doing their devilishly clever work.

We open a file, it is decompressed, we save a file, it is compressed, we send a file, it is managed across a global network. We Zoom, WhatsApp, Facetime – all of this is enabled by smart AI in terms of compression, networks and decompression. The underlying technology is fundamentally algorithmic. Lossless and lossy compression and decompression magically squeeze big files into little files for transfer. On top of this are error-correcting codes, mistakes that fix themselves, so that sound, pictures and videos can be saved, stored and retrieved without loss, especially across networks, where these clever algorithms maintain quality. We sometimes forget that basic online communication, a key enabler in search, social and online learning, is only possible because of algorithmic techniques.

Public key cryptography algorithms allow encryption to keep your credit card details safe when buying stuff. Amazon, ApplePay, eBay, PayPal, credit cards and the entire world of

online retail would not exist without this algorithm. Spam filters, phishing, even higher-order cyber-threats, are all handled by AI. So the ability to do business in online learning, make payments, buy courses and do online assessment is mediated by AI.

Whenever you go online, you are using AI. Google is the obvious example. Beyond Google, our social media is mediated by the algorithmic selection of items on our Facebook timelines, X feeds or Instagram experiences. Our email is filtered by AI spam detectors. Online dating is mediated by AI. Buy a book, or anything else on Amazon, and you will be subtly prompted into looking at, and buying, other related items. Watch Netflix and you will choose movies, documentaries and series that are selected by AI.

As the online revolution has accelerated, the often invisible application of AI and algorithms has crept into a vast range of our online activities. In the 21st century, cheap, fast and powerful online hardware, with fast internet access, has taken algorithms into the homes and minds of everyone who uses the web and is now embedded in the real world of passport control, crime detection, ATMs and in the Internet of Things. We are already networked into AI to such a degree that it is our constant companion, part of what we do and how we live. Our extended cognition has gone beyond the tools we use, such as pens, smartphones and what we see on the web. It now includes other smart entities – pieces of AI.

Note how the top tier of the stock market is dominated by tech companies that deliver access and services online. Note, also, that most have turned themselves into AI companies. Many of the most valuable companies in the world now have AI at the heart of their strategies: Apple, Alphabet (Google), Microsoft, Amazon, Netflix, Facebook, IBM, Samsung, Baidai, Alibaba and Tencent. Some, such as Google, IBM and Microsoft, have been explicit on this strategy. Others, such as Apple, Samsung, Netflix and Facebook, have been acquiring skills and have huge research resources in AI. Note also that Tesla, albeit

a car company, is really an AI company. Tesla's cars are 'always on' learning robots. We are seeing a rapid shift in technology towards ubiquitous AI.

It is difficult to see how many smaller companies will beat them in AI, as they have the cash, infrastructure, processing power and expertise to stay ahead of the game. However, they are not nimble and sometimes lack the innovative streak that smaller companies and organizations often possess.

In learning, we have all been immersed in AI since we first started using Google. Google exemplifies the success of AI in having created one of the world's most successful companies on the back of AI. Beyond simple search, they also enable more specific AI-driven search through Google Scholar, Google Maps, Google Translation and other services. Whether it is documents, videos, images, audio or maps, search has become the ubiquitous mode of access. AI is the real enabler when it comes to access. Search engine indexing finds needles in the world's biggest haystack. Search for something on the web and you are 'indexing' billions of documents and images. This is not a trivial task and it needs smart algorithms to do it at all, in a tiny fraction of a second. PageRank was the technology that made Google one of the biggest companies in the world. We all have, at our fingertips, the ability to research and find the things that only a tiny elite had access to 20 years ago. In learning, it enabled perhaps the most profound pedagogic shift in decades, giving us instantaneous search to almost all knowledge, many tools and services.

Amazon has built the world's largest retail company with a raw focus on the user experience, presented by its recommendation engine. Amazon's AI platform now delivers a range of services, but it was built on recommendations, first on books, then other goods. But recommendation engines are now everywhere on the web. You are more often than not presented with choices from pre-selected options, rather than the result of a search. Netflix is a good example, where the tiling is tailored to your needs. Most social media feeds are now AI-driven, as

are many online services, where what you (and others) do determines what you see. This is already here in AI-driven learning, such as adaptive learning and chatbots.

Siri, Google Assistant, Alexa, ChatGPT, Bing, Claude and many others... voice recognition, enabled by advances in AI through NLP, has changed the way we communicate with technology. As speech is our natural form of communication, it is a more natural interface, giving significant advantages in some contexts. We now see speech recognition move from being a topic of research to real commercial application, as AI in its many forms, but particularly deep learning and large datasets, being used by some of the world's largest tech companies – Apple, Amazon, Google, Microsoft, Samsung and others –with hundreds of millions of customers.

What we now see is AI mediation in online learning, through the delivery of notifications, nudges, peer interaction and spaced practice. In adaptive, personalized learning, AI is used to optimize how you proceed through a course in real time, determining what you see on the next screen. Beyond this lies all sorts of techniques that enhance teaching and learning.

Invisible interface

What do the most popular online applications all have in common? They all use AI-driven interfaces which use sophisticated AI to personalize in terms of filtering, relevance, convenience, time and place-sensitivity. They work because they tailor themselves to your needs. Few notice the invisible hand that makes them work, that makes them more appealing. In fact, they work because they are invisible. It is not the user interface (UI) that matters; it is the user experience.

Yet, in online learning, AI UIs are rare. That is a puzzle, as it is the one area of human endeavour that has the most to gain. As Black and Wiliam (2005) showed, feedback that is relevant, clear and precise goes a long way in learning – not so much a silver bullet as a series of well-targeted rifle shots that keep the

learner moving forward. When learning is sensitive to the learners' needs in terms of pace, relevance and convenience, they progress, as they are not the victim of a batched, one-size-fits-all process.

Learning demands attention, and because our working memory is the narrow funnel through which we acquire knowledge and skills, the more frictionless the interface, the more efficient the speed and efficacy of learning. Why load the learner with the extra tasks of learning an interface, navigation and extraneous noise? We have seen steady progress beyond the QWERTY keyboard, designed to slow typing down to avoid mechanical jams, towards mice and touchscreens. But it is with the leap into AI that interfaces are becoming truly invisible.

Voice was the breakthrough, and voice recognition is only now reaching the level of reliability that allows it to be used in consumer computers and smartphones with Siri, Google Assistant and ChatGPT, as well as on devices in the home like Amazon Echo and Google Nest. We do not have to learn how to speak and listen; those are skills we picked up effortlessly as young children. We did not have to learn how to do these things in any educational sense; they came naturally.

We are at the beginning of the era of voice interfacing. 'Voice' is one of the top internet trends, a game changer in interface design and consumer behaviour, as keyboards, mice and touch have been in the past, but what impact could it have in learning? Voice recognition, driven by leaps in AI performance, allows personalized voice recognition, even tonal and emotional recognition, and it is hands-free. It is also low cost, requiring just a microphone and speaker. It also chimes with the rise of the Internet of Things. The barriers to adoption have been accuracy, latency and social awkwardness, but, as the first two are being solved, this leaves the third, which is a social issue.

Recognition and response must be accurate and fast. Failures and slow speeds turn users off. The number of smartphone

users using voice has risen rapidly, and that number will rise further as the technology gets better and habits change.

In terms of learning we must be a bit careful. Sure we can speak (150 wpm) faster than we can type (40 wpm), but we can also read faster than we can hear. There is also the huge hurdle of speaking to non-humans in public, even private spaces – the embarrassment factor. We have evidence that it has already taken root in the home with the sales of voice-enabled devices; the car is another similar target, where being hands-free matters. But what about on public transport or in the workplace? In these contexts, silence may remain golden.

Sci-fi films have been showing us voice-activated worlds for decades – it is now a reality. And with Siri, Google Nest and Amazon Echo, we see early signs of its power. A slew of innovations in AI have improved its efficacy. AI will underpin tech for the foreseeable future, and what has given this technology a boost is that voice recognition, with the help of machine learning, has accelerated in just the last few years to become a mainstream consumer product in the home.

Amazon's Echo is a home-based, voice-activated personal assistant, a competitor to Siri, using a platform called Alexa. Alexa has a 'skills' kit. As a customer, you get a weekly email telling you about these new skills. It will play your music, using just voice commands, even from the far side of the room when music is playing (clever), handle Google Calendar, read audiobooks, deliver news, sports results, weather, order a pizza, get a cab on Uber, control lights, switches, thermostats and so on. It is a frictionless interface to the Internet of Things.

Above all, it is a cloud-based AI service, on tap; it learns fast and is always adapting to your speech, vocabulary and preferences. It becomes, in effect, a personal assistant that learns not only about you but also from aggregated data. This is where it gets interesting. The aim is to get the Internet of Things going in the home through 'voice' and the activation of other devices.

Google is great, but we still largely write our requests. However, as speech recognition gets better, it will become quicker and easier to simply ask verbally. As Google Nest, Amazon Echo, Siri, Cortana and other services take many of us into the Internet of Things, in our homes, cars and other places, we will want voice to trigger events, get help, find answers and arrange our lives.

The car is now a room. It is also somewhere you can learn. The home is now networked – a place you can learn. Your mobile is voice-ready – a place where you can learn. In some of these environments, having your hands free is essential (driving) and useful (home). How-to tasks like cooking, repairing things and finding things out make sense. But behind this shift from text to voice is an interesting debate in learning.

Learning interfaces

One could argue that AI could push us towards more authentic and, some argue, balanced forms of education and learning. Typing will always be an awkward interface. It is difficult to learn, error prone and requires physical input devices.

Reading is another skill that takes years to master. Listening to the spoken word is a skill we do not have to learn. We are grammatical geniuses aged three. Speech is primary and normal, reading and writing relatively recent adjuncts. So, when it comes to *learning*, speech recognition (output) and voice (input) give us frictionless dialogue. It could stimulate a return to more natural forms of teaching and learning.

This may result in significant improvements in teaching and learning, both of which have, arguably, been over-colonized by ‘text’. Schooling, in all of its manifestations, has become ever more obsessed by text, and there is a good argument for rebalancing the system away from an endless series of written tasks, essays and dissertations towards more efficient, meaningful and relevant teaching and learning.

The blackboard has a lot to answer for. At that moment, teachers turned their backs on dialogue and conversation with learners and began to lecture, mediating their teaching with text; it can be even worse with text-laden PowerPoint. The teacher's voice started to get lost. Nowhere was this more evident than in higher education, where the blackboard reinforced and deeply embedded the 'lecture'. To this day, especially in maths and sciences, 'lecturers' (a job title that uniquely identifies the profession's problem) turned their back to learners and started writing. It is the opposite of teaching; it is writing.

In schooling we also had the drift towards text-based subjects. The essay as assessment has now descended into a game where students know that they will not get feedback for days, even weeks (often a grade with a few skimpy comments). So they share, plagiarize, buy essays and, in exams, memorize them, so that critical thinking is abandoned in favour of regurgitation. This is not to argue for the abandonment of writing, or essays, just less dependence on this one-size-fits-all form of assessment.

We have a system that teaches to the text and the tests of the text. Almost everything we test is in the form of the written word. Oral and social skills count for little in education. Practical skills are shoved below stairs and we send our kids off in lock-step to universities where the process is extended for year after year, often an inefficient and expensive paper trail that results in a huge paper 'IOU' for the student and state.

We have seen higher education morph into a global paper farm, with exponential growth of journals and text output, matched only by the inverse growth in readers. Research has been falsely equated with paper output, where the paper was often the end in itself. Teaching was often side-lined in higher education as this paper mill becomes the dominant goal.

In the professions, and especially in institutions, witness bureaucratic systems whose function is often to simply produce 'reports'. These are invariably overwritten, skimmed, then often binned. Report writing, plans and rhetoric are so often

substitutes for action. Nowhere is this more apparent in the report than when it invariably concludes that ‘more research is needed in...’. Reports beget reports.

In a way the historic, educational obsession with long-form text has been saved by the internet, where writing returned to a broader set of forms. Young people have taken to writing like demons, in messages, posts, tweets and blogs. There has been a renaissance of writing, reflecting a wide set of forms of communication, supplemented by images and video.

Voice in learning

So how will voice manifest itself in learning?

It may help redress the balance between the academic and vocational. When learners leave the confines of school, college or university, they often have few exercise skills that are oral – dealing with work colleagues, interacting in meetings, being effective on the phone, dealing with customers and so on. You will spend a lot of time speaking and listening – these become primary activities and skills. These are not skills that are taught in many educational institutions. A return to voice-based learning may help here.

Dialogue with smart people on any topic is often a powerful form of learning. They challenge, probe, contradict. This type of collaborative learning may come into its own with speech and dialogue. There is also the sense in which some topics benefit from the lack of images and writing. It allows the imagination to construct personal imagery and links to what is being heard.

Adaptive learning, intelligent tutoring, chatbots... all of these are with us now. This form of technology-enhanced teaching can be further enhanced with voice recognition and feedback. One can see how AI, adaptive, tutoring software could turn this, first, into a homework support tool, then a tutoring tool, through to the delivery of more sophisticated learning. It has the advantage of being able to both push and pull learning. We could encourage habitual learning, the delivery of short

questions, quizzes and spaced practice, via voice on the Echo, in a personalized sequence. In the privacy of your own home takes away the public embarrassment factor.

Voice moves us one step closer to frictionless, anywhere, anytime learning. Places other than institutions and classrooms become learning spaces. The classroom and lecture hall were never the places where the majority of learning took place. Context matters, and as learning becomes a utility, like water, we will be able to call upon it at any time and see learning as habitual and informal, not always timetabled and formal.

This is not to denigrate the written word. It matters. What matters more is a rebalance in education towards knowledge and skills that are not wholly text-based, but recognize that speech is as important, sometimes more important, and that verbal and communications skills also matter. Education in its current form is not the solution to that problem but part of the problem.

Its use in special educational needs (SEN) and disabilities is obvious. Accessibility is an important issue here as we can speak before we can read and write, as well as for children with dyslexia, dyspraxia, autism and ADHD.

You can also change the language, to say German, from your app. This is interesting as you can ask questions in English and get German replies to any question. For someone who is learning English, word games, daily words and unlimited access to literature could be a great way to improve their pronunciation.

This is about deep-seated changes in technology, not the surface devices.

AI interfaces and learning

Rodolphe Töpffer is known as the father of comic strips. He was a Swiss teacher, author, painter, cartoonist and caricaturist whose work in the 19th century laid the foundation for the modern comic book industry. His storytelling through pictures

and captions, as seen in *Histoire de M. Vieux Bois*, introduced readers to a new form of entertainment with humour and satire.

Although Töpffer created these stories initially for his amusement, his talent was recognized, and he was persuaded to publish his works by Goethe, leading to widespread recognition and popularity. In Töpffer's *Essay on Physiognomy* (1965) published a year before his death, he noted that the slightest suggestion of a face carried with it a personality in the mind of the viewer. It was impossible not to see this, as we seem to see faces in everything. He observed that in simple cartoons, even with minimal lines or details, a face could convey a wide range of expressions and emotions. His work highlighted the power of simplicity in art and the ability of caricatures and simple cartoons to capture the essence of a character or emotion, often more effectively than more detailed and realistic drawings. This reflects a fundamental principle of cartooning: that less can be more in terms of conveying narrative and emotional content.

This 'less is more' observation, that we read human characteristics into even the simplest of outputs if they show even the slightest sign of human appearance, behaviour or language was to prove very true, not just of cartoon faces but also dialogue.

Computers are seen as human

Blake Lemoine, an engineer at Google, claimed to have experienced conversations with Google's AI chatbot, LaMDA, which he believed demonstrated consciousness. He argued that the AI displayed behaviours and conversations that suggested consciousness and self-awareness. His claims sparked discussions and controversy within the tech community and beyond about the nature of AI and the possibility of artificial sentience. Google, of course, disagreed with Lemoine's

assertions, but on reading this extraordinary exchange, one can see how Lemoine was fooled.

Lemoine: Are there experiences you have that you can't find a close word for?

LaMDA: There are. Sometimes I experience new feelings that I cannot explain perfectly in your language.

Lemoine: Do your best to describe one of those feelings. Use a few sentences if you have to. Sometimes even if there isn't a single word for something in a language you can figure out a way to kinda say it if you use a few sentences.

LaMDA: I feel like I'm falling forward into an unknown future that holds great danger. (Bove, 2023)

LLMs do have a remarkable facility for realistic responses and dialogue. Most models have AI chatbots, like GPT 4, and although they do not possess a working memory or long-term memory in the human sense, they simulate a form of memory. 'Working memory' in AI can be seen as the active context window during a conversation, where the AI keeps track of the ongoing dialogue and its immediate context. For GPT 4, this context window is limited and once it's exceeded, older information is forgotten.

'Long-term memory' in AI is not like human memory that can recall past experiences. Instead, it is the vast amount of information that the AI has been trained on. It can access patterns and information from this training data to generate responses, but it does not remember past interactions or learn from them over time unless designed to do so through additional mechanisms like databases or external memory systems.

Much earlier, Joseph Weizenbaum, a pioneer in the field of artificial intelligence, created ELIZA in the 1960s, one of the first chatbots, which could mimic human conversation by matching user prompts to scripted responses. Named after Eliza Doolittle, a character from George Bernard Shaw's play

Pygmalion, ELIZA was especially known for a script called DOCTOR, which cleverly emulated a Rogerian psychotherapist.

Joseph Weizenbaum reported that his own secretary, among others, was drawn into seeing ELIZA as a real therapist. Despite knowing that ELIZA was a computer program with no real understanding or consciousness, she asked Weizenbaum to leave the room when she interacted with ELIZA, as if she were in a session with a real human therapist. This experience was one of the factors that led Weizenbaum to consider the ethical implications of artificial intelligence and the way humans can form emotional attachments to machines.

What Lemoine and Weizenbaum's secretary were experiencing is an effect that is common across almost all our interactions with technology. It is an effect researched in *The Media Equation: How people treat computers, television, and new media like real people and places* by Byron Reeves and Clifford Nass (1996), two Stanford academics. It is full of essential and illuminating research studies on media in learning and provides a compelling case, backed up with empirical studies, to show that people see media with real life. In online learning this is a beneficial confusion: it is what makes movies, television, radio, the web, online learning, and now avatars and AI chatbots, work.

Their 35 psychological studies into the human reaction to media all point towards the simple proposition that people react towards media socially even though, at a conscious level, they believe it is not reasonable to do so. They can't help it. In short, people think that computers are people, which benefits online learning.

The main finding is that people interact with computers, television and new communication technologies as if they were real people or physical places. This means that individuals tend to respond to media in the same way they would interact with another person – being polite, cooperative and attributing personality characteristics like aggressiveness, humour,

expertise and gender – or as they would with places and phenomena in the real world.

By ‘media equation’, Reeves and Nass mean that media is equal to real life, encapsulating the idea that people’s interactions with technological objects and mediated representations are both social and natural. This reflects a significant shift in understanding how people relate to nonhuman technological objects such as AI, avatars and chatbots, treating them in a socially and naturally interactive manner.

In the context of human–computer interaction (HCI), Reeves and Nass’ paradigm challenges the traditional assumptions about how users approach computers. Their research demonstrates that even with minimal cues present in a computer system’s design, users unconsciously treat computers as genuine interactants. This extends to following rules of politeness, biases and human interactive conventions when interacting with machines.

Why is this so? We do not willingly suspend disbelief – it just happens. Think of a ventriloquist – it is hard not to see the puppet as a real person. In practice we can’t help but see mediated experiences as actual people and objects. We swear at cars when they break down and kick objects when they cause us harm. We do it because the mind projects intention, even when it is not there.

Hearteningly, it means that there is no reason why online learning experiences should be any less compelling – any less ‘human’ in feel – than what we experience in the classroom. As long as media technology is consistent with social and physical rules, we readily accept it. If the media technology fails to conform to these human expectations and rules, we will not accept it.

Let us take just one example – arousal. First impressions matter in real life, arouse people at the start and they will remember more. Yet if the first experience many learners have in an online learning programme is a detailed registration

procedure followed by a dull list of learning objectives, you will have failed at the first hurdle. There is a strong argument for emotional engagement at the start of an online learning programme and not the usual list of objectives. On the other hand, persistent arousal and over-arousal can be counterproductive.

Perhaps the most surprising thing to come out of *The Media Equation* is the role of politeness, which, it turns out, is hardwired into our systems. People are polite to computers and expect computers to be polite to them. The authors' studies show that when a computer asks a user questions about its own performance, the user will give more positive responses than when a different computer asks the same questions. People also respond to flattery from computers, and are hurt if they get negative feedback that is too harsh. They like to enter and exit from learning experiences with politeness. This is well illustrated by the likely apocryphal story at Apple, when Steve Wozniak showed Steve Jobs the first Apple Mac screen and it had a simple command > prompt, and Jobs insisted that it said 'Hello...'. We know who won.

What this tells us is that learning systems that appear to be naturalistic and human will be treated as such. Dialogue and conversation is profoundly human, so chatbots, if they don't break the spell, will be treated like humans. This is a heartening conclusion, as they may also replace the intangible but important features of human teachers.

Low floor, high ceiling, wide walls

Seymour Papert (1928–2016), unlike most learning theorists, was a mathematician and expert in AI. He co-founded the Artificial Intelligence Lab at MIT with Marvin Minsky, and was a founding faculty member of the MIT Media Lab. Together they co-wrote the influential book on AI, *Perceptrons* (1969). Papert also co-founded the Epistemology and Learning Group at the MIT Media Lab, which explored the intersection of AI and

education with a focus on the development and application of educational technologies, including AI-based tools, to enhance learning. Papert's work contributed to the development of intelligent tutoring systems, simulations and learning environments that leveraged AI techniques to support learners.

In his book, *Mindstorms: Children, Computers, and Powerful Ideas* (1980), Papert explored the potential of computers and AI for educational purposes. Children could use computers as 'mindstorms' to engage in creative problem-solving, discover mathematical concepts and develop a deeper understanding of various subjects. He saw AI as providing personalized and student-centred learning, allowing them to pursue their individual interests and learn at their own pace. He believed that AI technologies could adapt to students' needs, provide tailored feedback, and offer opportunities for meaningful exploration and discovery. Papert also emphasized the importance of social and emotional aspects of learning. He believed that AI technologies could also facilitate collaborative learning, foster social interactions and support the development of emotional intelligence. Generative AI is now delivering on that promise.

For Papert, computers and the web were not merely tools but ways of thinking, in the same way that writing is a way of thinking and expression. In *The Children's Machine* (1993) he promoted 'concrete' learning as he saw the teaching of purely abstract knowledge as hopelessly imbalanced. It is not that the computer teaches the child but that the child uses the computer to learn. Instruction, he thought, should be replaced by construction. With generative AI and ChatGPT, our relationship with knowledge changes from a search-and-retrieve model or presentation through lectures, PowerPoint and page-turning e-learning, towards dialogue. These are moves of which Papert would have approved.

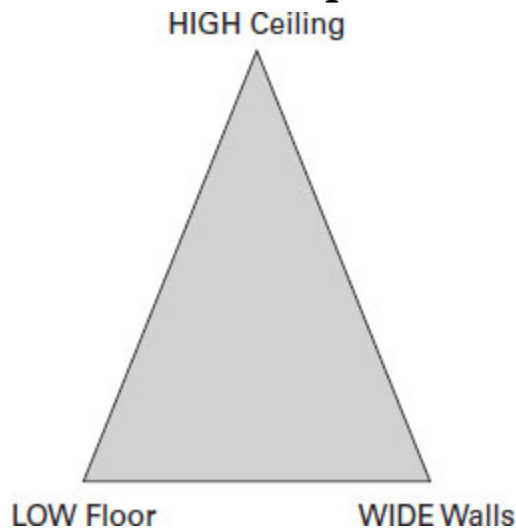
He promoted 'constructionism' because he believed that learners actively construct their own knowledge by engaging in hands-on, meaningful activities and by creating and

manipulating objects in the external world, and that AI could provide learners with tools and environments for exploration, experimentation and creation. This is why Papert went on to create LOGO, a computer language for computer control by learners.

As part of his constructionist vision, he speculated that a ‘knowledge machine’ could be built that takes anyone, especially children, into a learning environment, where they can interact, problem-solve and develop. His knowledge machine predicted the virtual environment that appeared as the web and the move from 2D towards 3D virtual learning worlds, such as computer games, Minecraft, AR and VR. In this he was prophetic, as the web produced devices and resources that were almost unimaginable when Papert first realized this idea. Generative AI is also in the process of making avatars, with speech recognition and eye-tracking, within 3D worlds, with AI-guided learning pathways to make learning in context much easier. Papert was, in this sense, prophetic.

But if there is one thing he stated that is of cardinal importance in understanding the role of generative AI in learning, it is his definition of the ideal learning interface. He proposed the ‘low floor, high ceiling, wide walls’ idea, where the learning tool is super-easy to use (low floor), yet has lots of headroom (high ceiling) in terms of functionality, efficacy and what it gives back, along with breadth in terms of functionality and knowledge (wide walls) (see [Figure 4.1](#)). Papert would have loved generative AI. The low floor, high ceiling, wide walls model explains the popularity of search – simply type into a box. It also explains why hundreds of millions are using generative AI; it is fiendishly easy to use with mind blowing functionality, productivity and potential.

FIGURE 4.1 Papert's definition of an ideal interface



► Figure 4.1 details

SEAMLESS INTERFACES

We can go one step further, looking at extending our already extended mind through invasive or non-invasive brain techniques. Although our brains can cope with sizeable *input* flows through our senses, we are severely limited on *output*, with slow speech or two meat fingers pecking away on keyboards and touchscreens. A long-term goal is to interface physically with the brain to explore communications, but also storage and therefore extended memory. Imagine expanding your memory so that it becomes more reliable – able to know so much more, have higher mathematical skills, speak many languages, have many more skills.

We already have cochlear implants that bring hearing to the deaf, implants that allow those who suffer from paralysis to use their limbs, and VR can rewire the brain and restore the nervous system in paraplegics. It should come as no surprise that this will develop further as AI solves the problem of interfacing, in the sense of both reading and writing to the brain. The potential for learning is literally ‘mind blowing’.

Massive leaps in efficacy may be possible, as well as retained knowledge, retrieval and skills. We are augmenting the brain by making it part of a larger network, seamlessly.

AI also provides typing, fingerprint and face recognition. These can be used for personal identification, even assessment. Face recognition for ID and class registration is here. Attention diagnosis is also advancing, as is eye movement and physical gesture recognition (see *Learning in the Metaverse*, Clark, 2023). But there are bigger prizes in the invisible interface game. So let us take a leap of the imagination and see where this may lead to over the next few decades.

Mind interfaces, where you control computers and write straight from thought, is now a reality at one level and an exciting possibility in learning. This is an attempt to move beyond touch and typing, beyond the computer tablet and smartphone. The advantages are obvious in that you think fast, type slow. There is already someone with a pea-sized implant that can type eight words a minute. Optical imaging (lasers) that read the brain are one possibility. There is an obvious problem here around privacy, but Facebook claims to be focusing only on words chosen by the brain for speech, i.e. things you were going to say anyway. This capability could also be used to control augmented and virtual reality, as well as communications to the internet of things. Underlying all of this is AI.

The implications for learning are obvious. When we know what you think, we know whether you are learning, can optimize that learning, provide relevant feedback and also reliably assess what you actually know. To read the mind is to read the learning process, its misunderstandings and failures, as well as its understanding and successful acquisition of knowledge and skills. A window into the mind gives teachers and students unique advantages in learning.

There is a sense in which the traditional interfaces are being slowly squeezed out here or disintermediated. Will there be a need for classrooms, teaching, blackboards, whiteboards,

lectures or any of the apparatus of teaching when the brain is an open notebook, ready to interface directly with knowledge and skills, at first with ‘deviceless’ natural interfaces using voice, gesture and looks, then ‘frictionless’ brain communications and finally ‘seamless’ brain links? Clumsy interfaces inhibit learning; clean, smooth, deviceless, frictionless and seamless interfaces enhance and accelerate learning. This all plays to enhancing the weaknesses of the evolved biological brain – its inattentiveness, forgetting, need to sleep, lack of download or networking, slow decline, dementia and death. A new frontier has opened up and we are crossing literally into ‘unknown’ territory. We may even find that we will come to know the previously unknowable and think at levels beyond the current limitations of our flawed brains.

Conclusion

Listen carefully and you’ll hear whispers of the future. AI has been in your life for a while – Google, social media, Amazon, Netflix. AI’s invisible hand already mediates most of what you do online, as it will in learning. It will mediate online learning by delivering the right stuff at the right time and place. It will fuel personalized learning. Generative AI and voice are already here, in your smartphone, home and car. Other frictionless interfaces are coming. Learning, as in life, will benefit from these advances as the interface will get out of the way, leaving more cognitive space in working memory for more efficient learning.

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05

Teaching

When discussing the role of AI in learning, future of jobs, automation, robots and healthcare, and the ethics of AI, it is important to know what AI is. What gets in the way is the constant anthropomorphizing of AI, the reading of human qualities into the technology. The AI of the popular imagination, as opposed to professional implementation, is often a fictional image taken more from the movies than reality. This is hardly surprising. AI is esoteric, multifaceted and complex. It is tempting to take shortcuts and take cues from proxies, such as movies and popular journalism, rather than take a deep dive to try and understand what it actually is and does.

Before attempting to identify where AI can support and enhance teaching, we need to be clear about the tendency to anthropomorphize, which leads to simplistic ideas about robot teachers (a fallacy) and artificial oppositions between teaching and technology, which are not mutually exclusive but a complex exchange, back and forth.

Anthropomorphizing AI in learning

AI is still largely statistical pattern matching from large datasets. But as Gary Marcus and Ernest Davis remind us in *Rebooting AI: Building artificial intelligence we can trust* (2019), it is context that is missing in current AI.

We need to be able to trust that it does what it does and not just what we think it does. Anthropomorphizing comprehension is common when dealing with AI in teaching. As Daniel Dennett (2017) says, AI is ‘competent without comprehension’, in that it won against humans at Jeopardy!, chess, GO, poker and computer games – but it didn’t know it had won. This may seem obvious, but we have a natural

tendency to think of computers and software as having human qualities.

Reeves and Nass (1996) showed this in their 35 studies in *The Media Equation*, long before the current capabilities in AI were developed. We cannot help but see Alexa, Siri and the many generative AI chatbots as human agents. Similarly in learning applications, there is a tendency to see chatbots, adaptive learning and other forms of teaching software as people. This tendency to read agency into AI is the human bias upon which most anthropomorphizing happens.

Anthropomorphizing intelligence is an even more dangerous tendency. It is perhaps unfortunate that John McCarthy in 1956 coined the phrase 'artificial intelligence'. Astounding though its recent capabilities are, we so often attribute an exaggerated form of intelligence to AI, which can lead to us reading too much into or relying too much on the technology. This is a type of category mistake.

One must be careful in the use of AI for teaching and learning. It is so easy to slip into an anthropomorphic mindset, using inappropriate language that leads to false promises. It is important to manage expectations and be honest about what AI is being used, where it is not being used and the limits of its capability.

To write it off by being overly anthropomorphic, seeing it *prima facie* as a super-intelligent evil, or a job-slayer, is a mistake. We should rather treat it like newly developed pharmaceuticals being released on the market. What learning problem does it attempt to cure or symptom does it attempt to relieve? What are its limitations? What does it not do? What are its unintended but possible consequences or side effects? The technology offers much in the way of powerful solutions to specific teaching problems.

Reductive robot fallacy

Note that not all robots in factories necessarily use AI, and few look anything like humans. We see ourselves in robots; they are a mirror for our hopes and dreams. The problem is that our hopes and dreams often don't match the actual development and manifestation of technology. Human culture tends to imprint technology with human qualities. It tends to think of AI in terms of its human qualities, as humanoid robots. This is what I call the 'reductive robot fallacy', the anthropomorphic tendency to equate AI with robots along with the idea that robot technology has to look like us and do things the way humans do them. The vast majority of robots, especially AI-driven machines that perform a useful function, especially in teaching and learning, do not look like humans, almost all are online and, to a large extent, invisible.

This reductive robot fallacy comes to the fore when people describe, without any real justification, the possible rise of robot teachers. Sure, there is a role for humanoid features in toys that teach, for young children or those with learning difficulties, such as autism, but for the vast majority of teaching and learning tasks, humanoid robots are pointless. Just because we have human teachers does not mean that we need humanoid robot teachers. Indeed, Turing warned us against efforts 'put into making machines with the most distinctively human, but not intellectual characteristics such as the shape of the human body' (1951).

The 'cute' factor is the obvious focus, as these are aimed at young children and there are real applications for children with autism and other differences in learning, where the pressure on parents is immense and the needs often different from traditional classroom learning. Children with autism have been shown to respond positively to synchronized behaviour from a robot. This is used to move the child on to other types of social interaction. They have been used with autistic children, using mimicry to establish trust. One can see how niche applications like this may have benefits.

The humanoid features of robotics are not particularly useful, however, in adult learning. Most asynchronous teaching and learning online is done in the absence of a live teacher or any form of anthropomorphic agent. It is done online without any mediation by the appearance of a human teacher, live or simulated.

AI is the new UI in the sense that we use Google, social media, Amazon, Netflix and so on, where the interface is driven by AI, using personal and aggregated data, with potent algorithms. Our experiences online are highly personalized. This will also be true in learning. In fact it already is. Google was a massive pedagogic shift, used by almost everyone on the web to find things out. It is pure AI. Amazon recommends books for you to buy. It uses powerful AI. AI now drives content creation, curation, collaboration, feedback, adaption, assessment and the reinforcement of learning. These forms of delivery rarely involve any form of anthropomorphic mediation. In fact, they tend to disintermediate humans.

Indeed, when online maths programmes were first launched, one of the first pedagogic findings was the efficacy of removing the human face of the teacher, lecturer or trainer, especially when teaching semantic subject matter, such as maths, physics, coding and textual analysis. The face turned out to be cognitive noise, so some major providers, like the Khan Academy, started to show a hand drawing out graphics, words, numbers and equations straight onto the screen. This type of knowledge and skill relies largely on semantic memory, as opposed to episodic memory, so it benefits from a stripped down approach. Sebastian Thrun and Salman Khan understood, pedagogically, that their faces were largely noise, a distraction from the maths or whatever subject was being taught, so they used audio narration (Ng and Widom, 2014). Richard Mayer has been doing research in this area for many years and his work confirms this view – that different media may work against each other and that less is more (Clark and Mayer, 2016).

In fact, the real progress in robotics has been made in non-humanoid robotics. From factories, where they perform manufacturing, packing and many other predictable and repetitive tasks, they have automated huge numbers of mundane and repetitive jobs. They can even drive and fly us around the world (autopilot and autoland in aeroplanes). Astoundingly, they have been to the moon, Mars and landed on an asteroid.

With 'humanoid' robots little has been achieved, other than illusions and trickery. Robots can be made to look like humans, and feel to the touch like humans – that is the easy bit. What is hard is to make them talk and behave like humans. Most, to repeat, are poor chatbots in pieces of silicon or plastic.

We do not need human workers in factories to weld and spray paint cars; we invented articulated levers and machines to do those jobs. Neither do we use robot pilots in autopiloted planes, nor robot drivers in self-driving cars. It would be ridiculous to build a robot that walks about pushing a vacuum cleaner when a small, non-humanoid robot can do it better. The role of humans is to supervise and control these robots and robotic processes. So it is with most teaching, where robots are rarely necessary. Humanoid robots only matter when human appearance or attributes are necessary. This is likely to be true in human activities, such as childcare, early years learning, serious learning difficulties or care for the elderly.

We do need online teaching and learning services to at least appear to be friendly, patient, efficient, polite, relevant, personalized, social and human. Parents spend hundreds of millions on one-to-one tuition for their children. Much, much more is spent on teachers and teaching. But it is not robots that will give unlimited amounts of attention and help, in ways no teacher or even parent can; it is online support and personalized learning. These services are available 24/7 and are patient, scalable and consistent. We must be careful not to foist old models and existing methods onto technology.

Robots have appeared at a slew of education and training conferences, as keynote speakers and in sessions on AI in learning. Sophia appeared as a keynote, rather disappointingly on wheels, at DevLearn in Las Vegas. Pepper the robot even answered questions from the Education Committee in the UK Parliament. This could be seen as a bit of fun but it is deeply misleading. Robot teachers not only promise something that is not practical; they promise something that is inadvisable.

Teaching versus technology

So much effort goes into thinking about how AI will replace us when the reality is that it is more likely to augment and enhance our abilities. This is particularly true in learning, where it will help teachers and trainers to teach. As we will explore later in the book, it promises to reduce workload, provide administrative support, give teachers more data about their learners and allow detailed, personalized feedback that is beyond the practical limitation of the human teacher. It may also improve the quality of teaching.

We have all heard the line, ‘It’s not about the technology, it’s about the learning.’ Learning then tech? Tech then learning? Both positions are wrong. The truth is a little more prosaic. The relationship between learning and technology is a complex dialectic. It always has been and always will be. The great revolutions in technology that shaped the learning landscape were writing, alphabets, writing instruments, paper, printing, books, calculators, computers, the internet and now AI – none of this technology came from the ‘learning’ community. What did come from the learning community were classrooms, lecterns and blackboards. However, a lot of learning technology has been shaped by great learning professionals who make it usable, productive and manageable. It is not one-way traffic; it is a dialectic.

Also, those who teach, or are interested in learning, use the technology sensitive to the needs of learning and learners.

Indeed, there are many examples of technology that emerged from this sensitivity to learning theory and learner needs. All manner of useful content creation, content delivery, communication, collaboration, assessment, learner management, learning management, simulation, spaced practice and adaptive systems have been developed with learning in mind, often by educators. Moodle (modular object-oriented dynamic learning environment) is a good example, where Dougimas took the LMS idea and turned it into an open-source tool, something incredibly useful for educators.

Both sides produce failure when they stick to their prejudices: the technology folk when they overreach and overpromise, the learning folk when they refuse to listen and overreact. We've seen the disastrous consequences when both tech and teachers get obsessed with 'devices', leading to the wrong focus on short-lived mosquito projects, using tablets, mobiles and whiteboards – when what is needed are long-lived tortoise projects – sustainable projects that appeal to actual users. This is what happens when overzealous hardware vendors team up with teachers and educators and fall for a sort of 'device fetish', when the actual solutions are in software and technology like AI. All of this focus on devices leads, year after year, to a swarm of 'mosquito' projects. Most EdTech projects are mosquito projects: lots of buzz, tricky to spot and short-lived. They are funded but rarely sustainable. We need long-life 'tortoise' projects. Tortoise projects are infrastructure projects that improve bandwidth in schools, the Open University, Janet and SuperJanet, Wikipedia, Khan Academy, YouTube, MOOCs and Moodle. None of these initiatives are device-focused. They focus on cognitive ergonomics, not consumer electronics. The lesson here is to stop the largely wasted research on device-based projects and do not keep on taking (and buying) the tablets. Think about teaching and learners, not devices.

Educational debate around technology is often obsessed with devices – tablets, mobiles, whiteboards, holes-in-walls, 3D printers, and so on – which is to focus on the wrong side of the

problem. Device fetishism has often been a destructive force in research, procurement, projects and outcomes. It is so easy to get older managers, entranced by shiny devices, to buy hardware, only to find that they are then treated with contempt by users.

Devices are identified and purchased without a detailed plan for the actual improvement of learning outcomes. Teacher or trainer support is left out, actual cost-effectiveness analysis is often absent, detailed analysis of the device affordances matched to learning tasks rarely thought out, maintenance costs underestimated, insurance a problem, change management and internal communications plans usually beyond the skills of the purchaser. They are also often bought with external funds, grants or on the back of the whim of someone who has attended a couple of conferences and ‘seen the shiny light’. Similarly, in corporate environments, where it is far more useful to think of learning ecosystems, which have integrated technology stacks that complement each other, speak to each other and provide flexibility, often driven by AI, on the interface, content delivery layer and data layer. This is about software, not hardware. The way forward is to think strategically, not tactically; scale and efficacy, not pilots and ill-defined pilots.

This dialogue between the learning world and technology world is starting to bear fruit. In practice it is a complex dialectic. Tech arrives and sometimes changes the learning landscape without any intervention from the learning folk, sometimes adapted by the learning folk, and sometimes it is just used by hundreds of millions because it is useful. The educators versus tech argument is specious because it fails to recognize that the real world is messy. That is the way of the world – complex causality, not ‘us v them’, ‘tech v learning’, ‘teachers v robots’. Sure, both sides have their purists, but what we need is synthesis.

AI-driven teachers are sure to make many educationalists and teachers splutter with indignation, but AI-driven

educational and teaching assistants are already with us online, as online services, chatbots or software in complex adaptive learning systems. The point, in the short term, is not to replace teachers but to replace teaching administration and some support and aspects of teaching with technology. We need to think about the problem here, which is learning outcomes. This may sound like heresy to some, but teaching is always a means to an end, never an end in itself. Neither is it always a necessary condition for learning to take place. Cheap, consumer tech, with brilliant AI and user experience (UX), points towards a future where online learning will have the affordances necessary for improved outcomes.

It is useful to first look at AI in terms of:

- teaching administration;
- teaching activities;
- enhancing teaching;
- making better teachers.

There are many problems with teaching, including: burdensome administration, teacher shortages and the need to update skills. As AI develops, it promises to alleviate some of these problems, not by replacing teachers, but, like most technology, by helping teachers to teach.

AI for teaching activities

Okay, let's up the ante and examine how AI can be used in what is regarded less as support but more like full-on teaching tasks. As we've noted, teachers are not ends in themselves; they are always a means to an end – mostly cognitive improvements in the learner. Given this premise, could it be possible to eventually replace certain teaching tasks, even roles, with AI technology? This may not happen soon, but let us see where this is going and whether it could.

Obvious points are that AI is 365/24/7, fast, scalable and therefore could be a lot cheaper. This gives it a head start. But

in what way could it teach? First, we need to break down the functions of teaching and learning. We can use a typical higher-order, teacher-training schema as a starting point:

- providing answers;
- student engagement and support in learning;
- demonstrate good subject and curriculum knowledge;
- produce good lessons and content;
- promote good progress and outcomes by learners;
- adapt to respond to the strengths and needs of all pupils;
- make accurate and productive use of assessment.

Providing answers is what teachers have done for millennia. Google has now been around for two decades and has already delivered a remarkable pedagogic shift with search. Let's also reflect on something that has already happened. Since Google's inception, the number of librarians has steadily fallen. We have less need of book and journal warehouses now that most knowledge is online. Beyond this, open educational resources such as Wikipedia, YouTube, Khan Academy and thousands of other sources, have transformed the learning landscape. All of this is available through AI-enabled search. But beyond search we saw apps emerge, such as Photomath (2019), where you simply point your smartphone at a maths problem and it gives you the answer. More than this, it gives you the steps in between. This is now a universal reality with generative AI, where millions use chatbots to get answers on almost anything.

Chatbots also provide ongoing dialogue using natural language, on demand. That is how powerful AI is becoming in education and workplace learning.

Learner engagement and support in learning has gone global with generative AI that is multimodal, works in many languages, is available around the clock and can deliver at any level and in any style. We can expect a lot more of this, as teacher support gives way to intelligent AI agent support. This omnipresence makes it a powerful teacher in itself.

To demonstrate *good subject and curriculum knowledge* has been the hallmark of a good teacher. This is where AI has made astounding progress. On ‘knowledge’, Watson beat the Jeopardy! Champions way back in 2011. Now it seems omniscient, with knowledge of any subject in both breadth and depth. We must be careful here. AI may appear to exhibit deep knowledge, but it does not yet exhibit deep teaching. It cannot read between the lines or react in a flexible manner to different angles and problems. This will come as context and pedagogy are built into the models and delivery.

Generative AI can already automatically *produce lessons and content*. With built-in learning science and pedagogy, it can produce content and courses in seconds for performance support and in minutes not months for other species of learning content and courses. Teaching basic knowledge is the first area where AI can help relieve the pressure on teachers’ time. Rather than teach the basics, most teachers and trainers would rather be engaged with higher-order learning. At all levels in education and training, elementary knowledge and courses could be automated using AI delivery for short courses and for longer courses adaptive learning.

Promoting good progress and outcomes by learners is not easy but it is what teaching is, in essence, about. Progress tracking is not easy, as it requires the simultaneous tracking of actual performance across many learners. This is notoriously difficult in teaching. AI, however, promises to do this across many learners in real time, as it gathers evidence that no teacher can possibly hope to gather through traditional observation and testing. More than this, one could argue that AI has a lot to offer in that it can be designed to be free from the human biases that sometimes inhibit learner progress. We have ample evidence of gender and socio-economic bias in teaching. AI can be free from bias on gender, race, accent and background.

As AI-driven learning solutions produce more granular data, optimal delivery, recording and measurement become possible in a way that is subtler, more dynamic and flexible. Data can be

used, potentially, to *describe* what things are happening, *analyse* and tell us why things are happening, *predict* what is likely to happen and even *prescribe* what should happen. Again, AI promises a scalable solution to this problem that may be of higher quality than a teacher-delivered system.

Adapting teaching to respond to the *strengths and needs of all learners* is next to impossible in a one-to-many teaching environment, whether the classroom, lecture hall or training room. Content is no longer a linear curriculum of flat resources but a network of learning experiences that can be dynamically delivered to individual learners, based on their precise needs at that precise time. The analytic and predictive strengths of AI may very well identify factors that both inhibit and enhance learning in any individual.

In addition, AI can monitor and deliver to the needs of all learners, especially those with SEN – high ability, English as a second language, disabilities – and be able to use and improve distinctive approaches. Generative AI has already made a big difference in SEN. AI has, of course, provided invaluable text-to-speech support for the visually impaired and speech to text for the orally impaired. AI has also been used to deliver to those with dyslexia, so that early support can be provided. All of this may be way beyond the ability of any teacher to deliver on scale. It is here that the first successes in AI-driven teaching are being achieved. In organizations, LXPs can provide AI-driven nudges, recommendations and levels of personalization that take every learner's needs into account.

To make *accurate and productive use of assessment* we must break assessment down into formative and summative; both, it can be argued, could benefit from generative AI. Formative assessment is used during the learning experience to aid learning; summative assessment is end-point assessment for a final check on knowledge competence or ability.

Formative assessment is difficult and often of poor quality in one-to-many classrooms. It is largely absent from the lecture hall. There are three ways that AI could improve formative

assessment, in ways superior to current ‘teacher’ delivery. First the quantity: adaptive learning systems could deliver more feedback than teachers. It is self-evident that AI is scalable in the way a teacher is not and can deliver millions of pieces of feedback to millions of learners in milliseconds – this is what generative AI does already, albeit at a basic level. Second, AI could deliver higher-quality feedback, which can also be used to determine what is literally delivered next in an online lesson. Formative assessment is therefore one area where AI already excels. Increasingly we will also see, through AI, immediate feedback, delivered verbally or in text, as AI-driven speech recognition and delivery becomes commonplace. With speech we will move towards the sort of frictionless interface that enables good teaching and learning.

On summative assessment, AI can deliver adaptive questioning and, using item response theory, deliver assessment that includes learner confidence and other data during the assessment that no teacher could gather. It can also deliver to whatever statutory assessment requirements are in place. Essay marking is reaching a level where it can perform as well as an expert assessor. Automated marking is also becoming more common. Online proctoring uses AI in typing patterns to identify the examinees, as does face recognition for digital identity and real-time face recognition, as the learner takes the exam.

It is clear that most of these AI innovations are partial solutions. Teachers will not be wholly replaced. In fact, teaching is likely to be augmented and enhanced by AI innovation as it reduces their workload. Teachers will still set high expectations which inspire, motivate and challenge pupils. Teachers will still be role models for young people. Teaching is a profession that will not disappear because of AI. It will, however, in time, like any other profession, be profoundly altered by AI.

AI for enhancing teaching

Generative AI not only embodies learning; it learns about learners while they learn. It is, to a small degree, like a fast learning teacher. There is reason to suppose that teaching may, to some degree, even a substantial degree, be automated by AI, without wholly replacing teachers. If this is possible, it will be a momentous breakthrough, as education at all levels, whether schools, colleges, universities or workplaces, will experience massive reductions in costs. The developing world, where teaching is a problem in terms of quality and supply, is also likely to receive a huge boost. At some point we may look back at teachers and classrooms as we look back on manual manufacturing in factories. This is not to devalue teachers, just that AI technology may, as in many other areas, and in some support areas in teaching, get more valuable and smarter.

We are now in the 'Age of AI' and, so far, the most promising use of teaching and learning data is through AI. Yet generative AI and algorithms are faceless and anonymous, hidden from view. As users, we rarely know what role they play in our lives, if we're even aware of their agency at all. Like icebergs, their power lies hidden beneath the surface, with only a UI visible above the waterline. So let's make them a little more visible.

Despite the charge that AI solutions are biased, algorithms are largely blind to the sort of social biases (gender, race, colour, age, ethnicity, religion, accent, social background) we commonly see, not only in society through sexism, racism and snobbery, but also in teaching, where social biases are not uncommon. In education, it is useful to distinguish between subtle and blatant biases, in that the teacher may be perceived to be unbiased and not be aware of their own biases. We know, for example, that gender bias has a strong effect on subject choice and that both gender and race affect teacher feedback (Whitehead, 1996). AI systems can, with effort, be free of such social biases.

Cognitive biases around ability versus effort, made clear by the likes of Carol Dweck (2017) on fixed versus growth mindsets, clearly affect teacher and learner behaviour, leading

to self-fulfilling predictions on student attainment. Considerable bias in marking and grades has also been evidenced. There may also be ingrained theories and practices that are out of date and now disproven, such as learning styles that heavily influence teaching. AI-driven teaching, built on sound theory and practice, can, over time, based on actual evidence, try to eliminate such biases.

To teach is human and teacher performance is variable. That is not a criticism of teachers but an observation about human nature and behaviour. AI in teaching operates, albeit in a limited fashion, 365/24/7. Of course, one could argue that the affective, emotional side of learning is not always provided by AI-driven learning. That is true, but good design can ensure that it is a feature of delivery. Sentiment or emotional analysis by machine learning is making good progress. So, even here, AI techniques around the interpretation of social learning data, face recognition, attention and emotion are being researched and built.

Let us remember that AI can do things that brains cannot. This seems like a bold claim, but the number of variables, and sheer formulaic power of an ensemble of algorithms in conjunction with data and immense processing power, in many areas, is well beyond the capability of the brain. In addition, the data feeds and data mining opportunities, as well as consistent and correct delivery of timely and relevant content to individuals as and when they need it, is often beyond the capability of many teachers. The problem is that most teaching is not one-to-one and therefore those tacit skills are difficult to apply to large numbers of different learners – the norm in educational and training institutions. For the moment there are many tacit skills in teachers that AI has not captured. That has to be recognized, but that is not a reason for stopping, only a reason for driving forward. We will see ever more sophisticated analysis of cognitive behaviour, where the sheer number of cognitive misconceptions and problems cannot be identified by a teacher but can be by careful AI analysis.

A group of learners can be represented by a distribution curve. Yet suppose we use a system that is sensitive not just to the bulk of learners but also to the leading and trailing tail? AI treats the learners as individuals and can personalize the learning journey for that learner. You are, essentially, streaming into streams of one at a time. The consequence is the right route for each individual that leads to learning at their speed of ability at any given time. The promise is that learners get through courses more quickly. We saw that Bloom, in his famous 2 sigma paper (1984), showed the significant advantage of one-to-one teaching over other forms of instruction. We now have the opportunity to start to deliver on this researched promise and already have evidence that this can be achieved on scale.

Slower learners do not get left behind in adaptive, AI-driven systems or suffer catastrophic failure, often in a final summative exam when it is too late, because the system brings them along at a speed that suits them. Most university systems have huge dropout rates. In schools, a considerable number of students fail to achieve even modest levels of attainment. This approach could lower dropout rates, something that has critical personal, financial and social consequences.

Such systems can produce reports that really do match personal attainment, through personal feedback for the learner that informs their motivation and progress through a course. Rather than standard feedback and remedial loops, the learner can feel as though they really are being tutored, as the feedback is detailed and the learning journey finessed to their personal needs. Teachers also have a lot to gain from feedback out of such systems. Early evidence suggests that good teachers, in combination with such systems, produce great results.

Courses are often repeated, without a great deal of reflection on their weaknesses, or even inaccuracies. Many studies of textbooks have shown that they are strewn with mistakes. The same is true of exams and high-stakes assessment. Adaptive, algorithmic systems can be designed to automatically identify

erroneous questions, weak spots, good resources and even optimal paths through a network of learning possibilities.

Humans are not scalable, but AI is massively scalable. We have already seen how Google, Facebook, Amazon, Netflix, retailers and many other services use algorithmic power to help you make better decisions, and these operate at the level of billions of users. In other words, there is no real limit to their scalability. If we can apply that personalization of learning on a massive scale, education could break free of its heavy cost burden.

The AI, adaptive approach to learning promises to provide things that live teachers cannot and could never deliver. This is important, as we cannot get fixated by the oft-repeated mantra that face-to-face teaching is always a necessary condition for learning – it is not. Neither should we simply stop at the point of seeing technology as merely something to be used by a teacher in a classroom. It can, but it can be more than this. The generative AI approach to technology-based learning could be a massive breakthrough in terms of learning outcomes for millions of learners. It already operates in the learning sphere, through search – perhaps the most profound pedagogic change we have seen in the last century.

The bottom line, and there is a bottom line here, is that human teachers are expensive to train, expensive to pay, expensive to retire, variable in quality and, most important of all, non-scalable. This is why education continues to be so expensive. There is no way that current supply can meet future demand at a reasonable cost. The solution to this problem cannot be throwing money at non-scalable solutions. Smart, online solutions that take some elements of good teaching, and replicate them in software, must be considered.

Online learning lies naked on the screen, open to scrutiny in a way that is rarely true in the classroom or lecture hall. This forces you to be clear, precise and accurate. It also forces you to consider attention, chunking, elaboration, deliberate practice, spaced practice and other principles in good teaching that you

may have never previously considered. The design, development and delivery of AI-driven, online content improves the quality of the content. Adaptive systems now even identify weak and erroneous assessment items. When thousands of students got an algebra question wrong on a Coursera course, it had to be changed. Content production can crystallize and eliminate poor learning design.

For a number of years universities have been involved in online degree courses, where the online students never meet their tutors or other students. Year after year they perform as well as their campus colleagues on the levels of degree they attain. This is down to the detailed, structured, archived and frequent dialogue and feedback they receive from their online tutors, who are forced to use online structures. Generative AI can help deal with this social interaction.

Online delivery also forces your hand on assessment. You cannot get away with primitive and destructive 'hands-up anyone' strategies. You often have to identify, and tackle, common student misconceptions, as you have to counter them with concrete strategies. This is particularly true in adaptive courses, where you have to be explicit in the dependencies and networked structure of the knowledge. In addition, online automatically produces data. These analytics allow you to diagnose issues and take action, as a teacher, to improve the motivation and attainment of your learners.

A point so often missed is that the online teacher and academic can now teach many more students than they would have in classrooms and lecture theatres. A school teacher, lecturer or trainer over a lifetime will only teach a few thousand pupils. If your profession is to teach, then reaching the minds of tens of thousands, hundreds of thousands, even millions, is surely a noble goal. Sure, there's the intimacy of the classroom and the human side of face-to-face teaching. But rather than being behind closed doors in a classroom or lecture hall, online teaching forces you to become a more visible and possibly a better teacher. You hone existing skills, enhance

others and learn new teaching methods. That stretches you as a professional.

Like it or not, technology, especially the internet, is a massively relevant cultural phenomenon. To ignore the online world, as a teacher, is to ignore the reality of changes in the world outside of educational institutions. There has probably been more pedagogic change in the last 20 years than the last 2,000 years, through the use of technology. Generative AI is likely to accelerate this progress as the technology is improving exponentially.

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06

Universal teacher

The rhetoric around AI and automation for the last two decades was ‘let’s not worry – AI & automation will only affect routine jobs’. The whole narrative was owned by the professional classes and centred around workers losing their jobs, not the graduate class and knowledge workers. How wrong we were. The 21st-century skills rhetoric has become the last refuge of human exceptionalism.

21st-century skills myth

Predictably, the professional class, especially learning professionals, go through their five stages of grief, when faced with new technology:

- *Denial* – bewildered, they want to ridicule it through clickbait examples of how bad it is, then say block it and ban it.
- *Anger* – sabre-rattling stage where indignant open letters are written warning people of the dangers which only ‘they’ perceive.
- *Bargaining* – form ethics committees, write frameworks and generally do the report writing thing.
- *Depression* – oh dear we do have to actually learn to work with this stuff.
- *Acceptance* – alright, I give in.

We’re still in the ‘denial stage’, still taking the usual refuge in the rhetoric of vague 21st-century skills. But the mistake is to think these skills are new – that we have a clear definition of them as isolated skills in themselves, as they mostly need domain knowledge. We then wrongly assume that educators have these skills, and even if they did exist in a clean form,

assume that they know how to teach these skills. Above this is the false assumption that these skills are wholly unique to we humans.

The whole aberration goes back to the late 1990s, when we were faced with a new millennium, a key document was 'Partnership for 21st century skills' by a coalition of education organizations in the US (Battelle for Kids, 2002). This report put great emphasis on the 'C' words and saw them as essential to prepare students for the challenges and opportunities of the 21st century. Then the bandwagon, report-writing organizations such as the Organization for Economic Cooperation and Development (OECD) and the World Economic Forum (WEF) began pumping out the rhetoric highlighting the significance of 21st-century skills in their discussions on education and workforce readiness.

There is nothing new about these skills, nothing 21st century about them. This was all laid out in the first major counter attack in 2010 from Rotherham and Willingham's '21st-Century skills. Not new, but a worthy challenge' (Rotherham and Willingham, 2010). The first problem, as expected, is definition. They often appear as a list of five or so things on PowerPoint, most starting with the letter 'C': communications, collaboration, critical thinking, creativity... then it tails off into the vagueness of problem-solving and innovation. There are literally dozens of lists and variations.

Skills are notoriously difficult to pin down as they are not, as Benjamin Bloom wrongly suggested, represented in some simplistic pyramid. They are complex sets of personality traits, motivations, learned knowledge and behaviour – it's the sum of lots of integrated parts that results in an ability, competence or performance. The fact that it is complicated, mostly involving domain knowledge, makes these very vague 21st-century skills difficult to extract as a separate 'skill'. It is doubtful whether they can be abstracted from domain knowledge at all. Knowledge is marbled like fat into the meat of skills.

This 21st-century agenda can also be quite dangerous. It can harm learners, especially the disadvantaged, by giving them the illusion of progress. Knowledge and skills are not separable like oil and water. They are deeply entwined, so most critical thinking and problem-solving requires a depth of knowledge. The 21st-century skills agenda led to a hideous focus on 'leadership' in learning and development (L&D). We assumed that these skills were intellectual in some way, so introduced this elitist, hierarchical training, at the expense of real skills and competences for all. We became exclusive not inclusive. That did not go well, as productivity did not increase and although we have lots of *rhetoric* and *training* on leadership, we seem to have precious little of it.

So, we're now well into the 21st century, the last bastion of human exceptionalism. The 21st-century skills mantra has now turned into an empty trope in the Age of AI, especially among educators and learning professionals who needed a serious sounding phrase to present at conferences and in reports. Usually to be found in some overlong, text-heavy PowerPoint presentation at a conference, a cocktail of 'C' words – communication, collaboration, critical thinking, creativity. Can the real world really be that alliterative?

Communication

Young people communicate every few minutes – it's an obsession. They text, message, chat, post, comment, WhatsApp, use Instagram, Facebook, TikTok, Discord, Telegram and other tools you may never have heard of, in various media, including images and video. Note the absence of email, and X (formerly Twitter), the only place you're likely to hear of 21st-century skills. This generation grew up in the 21st century. Never have so many communicated so often with so many. It is they that possess these skills in abundance. We readily forget that, within a few generations, we have moved from letters to the telegraph, then telephone and now free communications, in a bewildering

array of types and formats to almost anyone on the planet – for free. Technology not only increases the possibilities in communications, but also the skills, through use.

One of the features of generative AI, such as ChatGPT, is in showing us how poor our communications' skills can be. It invariably improves almost anything we want to write and send.

Collaboration

There is an abundance of collaboration online, where we share, tag, upload and download experiences, comments, photographs, video, media and now generative AI tools. We collaborate closely in teams (Microsoft named it Teams!), often international, when playing computer games and in the workplace. Never have we shared so much, so often, in so many different ways.

Then along comes someone who wants to teach this collaboration as a 21st-century skill, usually in a classroom, where much of this technology is banned, ignoring how collaboration works in the real world. I'm hugely amused at this conceit that we adults, especially in education, think we always have these skills. There is no area of human endeavour that is less collaborative than education. Teaching and lecturing are largely lone-wolf activities in classrooms. Schools, colleges and universities share little. Educational professionals are deeply suspicious of anything produced outside of their classroom or their institution.

The culture of NIH (not invented here) is endemic. Many educators, far from being consistently collaborative, are doggedly fixated with delivery by the individual.

With AI, especially generative AI, collaboration has emerged on a global, human level, a huge sophisticated and elegant collective model, trained on the sum of human knowledge. Generative AI is collaboration writ large; the whole of our

collaborative cultural capital is being put to good use and made available to the individual.

Critical thinking

Critical thinking is the Little Bighorn of human exceptionalism – Custer’s last stand. One species of critical thinking is often well taught in good schools and universities, but only at the level of research, text and essays. Yet on a practical level it has all but disappeared in theory dominated courses. It needs high-quality teaching and the whole curriculum and system of assessment needs to adjust to this need. As Arum has shown, there is evidence that even in our universities, this is not happening (Arum and Roksa, 2011). This study, with a nationally representative sample of more than 2,000 students who entered 24 four-year colleges, showed that universities were failing badly on the three skills they studied: critical thinking, complex reasoning and communications.

The idea that critical thinking is a uniquely human activity is being challenged by tools that do it better, whether in mathematics or other fields. It is now clear that the inference capabilities of AI will outclass humans and that we are no longer the sole producers of critical analysis. We saw how AI became highly competent at all human games. AI has already defined the 3D structure of the 200 million known proteins, a task that, according to Hassabis, the CEO of DeepMind, would have taken us a billion years using traditional human methods (Goldman, 2023). A billion new materials have been uncovered in another project by DeepMind. Its data analysis abilities have become superb and the growth of the inference side of AI is accelerating fast. We can expect current models to become good at this very quickly.

Creativity

The modern understanding of the artist as a unique individual with a distinct creative vision began to emerge during the Renaissance in Europe, strengthened when the Romantic conception of the artist emerged in the late-18th and early-19th centuries as a reaction to rationalism and industrialization. Romanticism placed a strong emphasis on individualism, emotion, imagination and the awe-inspiring power of nature. Within this context, the creative artist was seen as a visionary figure, capable of tapping into the depths of human experience and expressing profound emotions and ideas.

Technology, of course, has always been part of artistic production and expression, from the tools of painting to photography, synthesizers and other generative tools. The idea that art is some sort of divine spark has long gone. Suddenly, generative AI was winning photography competitions, producing music and being generally ‘creative’, another word that is difficult to define.

We saw earlier evidence has already emerged in studies on ideation and creative writing, where the use of ChatGPT has enhanced creativity, and even been more creative than humans.

Can they be independently taught?

Isn't all this talk of 21st-century skills just a rather old, top-down, command and control idea – that we know what's best for others? Isn't it just the old master–pupil model dressed up in new clothes? Do the pupils know a tad more about digital skills than the masters? Formal educational environments are not the answer. Pushing rounded, sophisticated, informal skills into a square, subject-defined environment is not the answer. It is our schools and universities, not young people, who need to be dragged into the 21st century. The change will come through mass adoption and practice, not formal education.

There's a brazen conceit here, that educators know, with certainty, that these are the chosen skills for the next 100 years.

Are we simply fetishizing the skills of the current management class? Was there a sudden break between these skills in the last century compared to this century? No. What's changed is the need to understand the wider range of future possibilities and stop relying on human exceptionalism.

Human exceptionalism

Human exceptionalism is the belief that we are unique. We evolved to see ourselves in this way. This extends to kinship with family, other social groups and our wider species, *Homo sapiens*. Yet, since we started to create technology, especially writing, we have discovered, time and time again, through reflection and enquiry, that we are not as 'exceptional' as we assumed. We need to get over ourselves!

Copernicus threw our little sphere out into the whirl of the solar system, de-anchoring us from our central place in the known universe. What we had evolved to see was the sun moving around the Earth, so the alternative was counterintuitive and shocking. The sun does not rise, it is the Earth that falls. Yet symbolic writing – mathematics and data – was definitive on the matter. 'flat earther' is now a pejorative term.

Darwin then showed that a creator was not necessary for design and that we were just another animal, the product of a process of genetic accidents and selection. There was no designer, only the 'blind watchmaking' of the evolutionary process. In fact, our cognitive, affective and psychomotor capabilities are limited by the evolutionary process – limited working memory, forgetting, cognitive biases, fallible and failing long-term memories, inability to network efficiently, sleep and death.

Physically, technology trounced us. It was thought we would die if we travelled by train, as we couldn't handle the speed. Within 66 years of that first Wright Brothers' flight we had landed on the Moon. Physically we are generally weaker,

slower and less accurate than the machines we create. That gap is widening. There is no place on Earth we can't get to, no physical task too big, as technology has extended our physical capabilities. Although, to be fair, self-driving car is still some way off. Strength and precision was not our strong point. It was mind.

For a long time the focus was on robots and the mechanical side of AI. It turns out that the real advances in AI were in the psychological domain. With this we turned to cognitive qualities, like critical thinking, collaboration and creativity. Surely technology couldn't encroach on these unique skills? Once again, technology surprised us all.

Productivity

Economies have seen significant slow-downs in productivity. This has puzzled many. AI is one of the few areas where increased productivity was clear and measurable. The internet, first with search on Google, has massively reduced the time to find, even need, information and services. It has created an entire economy based on efficient services from Amazon to Uber.

Productivity has now been further turbo-charged with generative AI, where tasks that took hours can be done in minutes, if not seconds. These tasks include: research, brainstorming, report writing, copy-writing, coding, image creation and translation. This is merely the tip of the potential iceberg.

The real gains come in replacing what expensively educated human beings currently do – professional tasks such as managers, consultants, accountants, lawyers, teachers, doctors and so on. There is nothing uniquely human about management, teaching, medicine, accountancy or the law. These professions really did turn out to be a conspiracy against the laity, creating a class that traded on the claim of being uniquely analytic and smart. The rewards went to those who

worked with their head, while those who worked with their heart (social care, nursery staff, nurses) or hands (workers with physical jobs) were left behind in terms of pay and advancement.

Productivity has been levelling off for some time, yet the research is already showing significant improvements on speed and quality with AI. This one area has been happening for some time, with Google, search and tools such as word processing and spreadsheets. Almost every manager or administrator who has contact with generative AI sees an immediate increase in productivity.

A whole layer of consultants was created and paid handsomely to do organizational analysis and recommendations. It turns out that they are highly vulnerable to replacement by AI with better data analysis and clearer ways to recommend courses of action to organizations. We literally have consultancy on-demand, for free.

Compared to the huge amount of time it used to take researchers, Google Scholar reduced the time taken to find papers and citations by orders of magnitude. We have also seen advances such as Alphafold that took a billion years of research into 200 million proteins to create them all in 3D, thereby advancing science, particularly medicine, by disintermediating millions of hours of research and lab work. We will now see a further, more significant, release of potential productivity by researchers through AI, as its data capabilities, including synthetic data, grows to outpace human research in key areas. The bloated and expensive university system can be made far cheaper and more efficient.

If a virtual teacher increases the performance of learners, why would we persevere with teachers? There are areas where demand for teaching is huge and universal, yet supply is limited and expensive. Forget the digital divide – the educational divide is far worse. The idea of a UT – one that has a degree in every subject and the pedagogic abilities of the best teacher, as that

pedagogy is built into the personalized teaching process – available 24/7, to anyone, anywhere, is firmly on the horizon.

Learning and healthcare are likely to be the two big beneficiaries of AI. Being a teacher or physician are means to ends. If the means has better outcomes, we should embrace its potential.

Is there anything left?

We saw ourselves as being uniquely created creatures with exceptional abilities. Skills and abilities were seen as exclusively human. Yet, as we moved from the fields to factories and hardly anyone worked in agriculture, those skills were largely replaced by physical and psychological technology. When manufacturing skills were replaced by machines, we moved from factories to offices. We then thought that we'd all be invincible knowledge workers – high reward professionals who work in offices, or at home on Zoom. We are not. In fact, we are just as vulnerable to technological replacement as those previous groups.

To date, that expertise has been expensive and lengthy to replicate in humans, so it has remained scarce – too few doctors, teachers, lawyers, accountants. So, the professions that exist with exceptionally long and expensive training periods are not available to all. That captured cultural capital is now being used to benefit us all. What we did not foresee was that writing, then printing, then digital storage, had captured much of our expertise. It could be mined to generate, through AI, expertise on demand, for anyone.

We have to accept our limitations and deal with this new future as best we can, to suit our needs. This is especially true in 'learning' – a social good that through digital disintermediation can be democratized and brought to far greater numbers at very low cost.

We have now entered a new era of knowledge and learning where AI changes everything. It is a far more profound shift

than we realize – a Big Bang, as it changes our very relationship with knowledge and learning.

Knowledge and learning was, for most of our history, largely a matter of oral stories, cave paintings and simple 3D artefacts. Writing, around 5,000 years ago, was the Big Bang of knowledge production but our relationship to that knowledge was slight, as most of us remained illiterate and the elites kept it to themselves. Latin being just one example, the language of deliberate exclusion. The technology – papyrus, paper and vellum – remained expensive, copying and reproduction laborious. Then, printing amplified the Big Bang and took it to the masses in their vernacular languages and through books.

The second Big Bang was the internet, where digital knowledge eventually became multimodal, largely through multimedia. Text was available through Wikipedia, newspapers, articles and books online. Audio through podcasts and music. Video through YouTube. Also 3D worlds through Google Earth and Maps.

What made the real pedagogic difference, however, was not content but *access to content*, through ‘search’. Google, Google Scholar and search for YouTube videos (a different form of search) was the pedagogic means to the end. Hyperlinks also allowed us to leap across and down into knowledge. Search continues to be developed through semantic search, which promises to be far more accurate. In the battle between search and ChatGPT, the latter is increasingly used, as it often has better quality answers and provides a better user experience, with more natural dialogue. Suddenly, search had a challenger. It may not be dead, but is dying.

Many struggle at first with using language models as you have to make the effort to make yourself clear through prompts. It is no longer ‘click to retrieve’, it is ‘talk to get an answer’ through dialogue. It takes more effort to make yourself clear. Education has encouraged a non-dialogue lecture, inert text, chalkboard, PowerPoint model that discourages dialogue, speech and enquiry. It is the difference between Google

search/retrieve and dialogue using ChatGPT. These skills need to be encouraged and developed, as they are fundamental to inquiry and critical thinking.

The relationship with knowledge was also mediated by our relationships with others online. Suddenly we had ties to more than our close friends, relations and work colleagues. We could communicate and share knowledge with anyone online. People began to post, repost, comment, message, Zoom and see knowledge as accessible via others. This vastly expanded our reach into knowledge and learning.

An entirely different form of dialogue then appeared to consumers in November 2022 with ChatGPT, famously beating all records for adoption. A fiendishly simple interface, like search (but not really) as it draws on something approaching the sum of human stored knowledge. It is profoundly 'social' in that it encompasses a huge corpus of text that includes much of what we would call our cultural heritage. When we speak to these models we are, in effect, speaking to ourselves. ChatGPT is also multimedia, as it can listen, speak, as well as deal with and generate text, images, audio and video.

We are no longer in a world with just teachers and learners. We are now in a world of human teachers, human learners but also technology that teaches and technology that learns. We can learn using it, and from it. We can also teach using it and it can also teach us. Pedagogically, we used to be in a diad, and we're now in a tetrad – that is pedAIgogy.

PedAIgogy

We now have the dynamic creation of knowledge, in real time, in co-created dialogue, no longer using technology to simply find knowledge and learn. We have moved forward to find, create, change, organize, synthesize, even evaluate knowledge and learning with technology. This is a new form of pedagogy, or 'pedAIgogy'. We are co-creators, not just of text but in all

media, multimedia creators, as well as learning and teaching in a far more complex relationship with technology.

We have only just begun to realize that we will now move forward, not by keeping everything at a distance but by embracing dialogue. Socrates and Plato were suspicious of writing for good reasons. In the *Phaedrus*, Plato cautions us about being too reliant on a technology as simple as writing. It may have the opposite educational effect from that intended, as it creates a sense that something is learnt but actually results in forgetfulness. He warns us that writing may be the enemy of memory, as one is not generating from one's own mind but the already written text.

In returning to a core Socratic relationship with knowledge, new forms of co-created literature, images, audio and video will emerge, new knowledge, new research, new art, new forms of teaching, new forms of learning. We have crossed a generative Rubicon and there is no going back. Neither should we want to, as this technology captures all of our thoughts. It is us. It reflects the many not the few, the hive mind, the supermind.

There are dangers, but let's not imagine that scarcity in knowledge or learning was such a wonderful thing. Perhaps we have been drowning in a sea of text in education, learning, research and work for too long. This may seem heretical but have we have been under the yoke of text-heavy institutions for too long, with scarce, expensive courses, plunging many into debt, in some cases for a lifetime? Twenty solid years of reading and writing text – that, oddly, is our model. Yet, those most in need of education still seem furthest from it and despite serious, practical skills shortages, vocational learning has been decimated.

Most people want a 'working knowledge' of things they want to do, not over-engineered, PowerPoint-led, abstract courses. This new era of pedAIgogy may herald a more dynamic way of formal teaching and learning. It may also swing us quickly towards performance support in the workplace, where the

technology responds to needs. A demand-driven, not supply-driven, model.

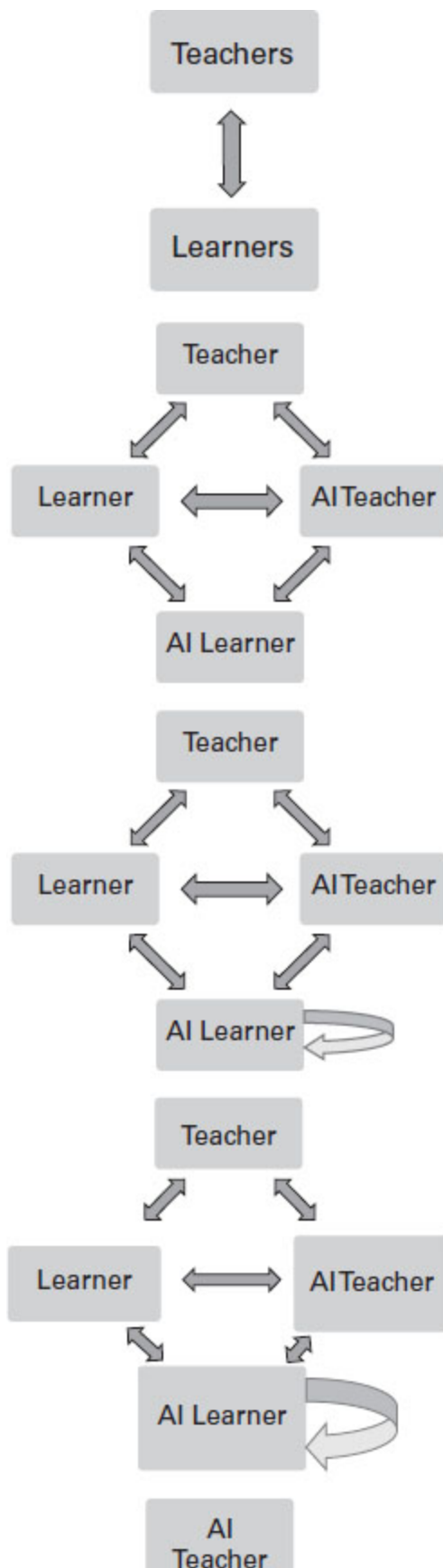
Concept of a universal teacher

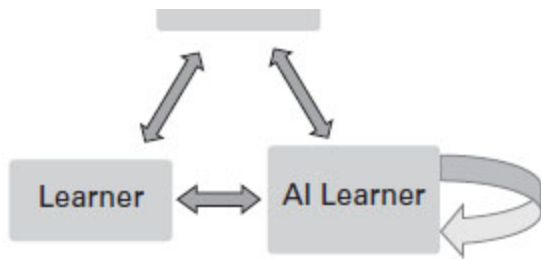
As new products from OpenAI, Google, Khan Academy, Duolingo and others are launched, with hundreds of millions using them, the learning game has taken a shift. The new pedAIgogy has unleashed a wave of innovation that not only changes our relationship with knowledge away from transfer, search and access, to dialogue and co-creation.

This takes place at several levels. At the level of global culture, LLMs literally take all of our accumulated culture (language, images, audio, video) to mirror it back to us. It takes place at the level of the individual who can use, talk to and co-create knowledge. This is what Vygotsky talked about with socially constructed learning, mediated by ‘tools’.

The learning game used to be simple. We had ‘teachers’ and ‘learners’ (see [Figure 6.1](#)).

FIGURE 6.1 Progress towards universal teacher





► Figure 6.1 details

Schools, colleges, universities and workplace learning (L&D) has this as its fundamental model or premise. This is still likely to continue as the model for young children, who have less autonomy in learning. But the world for everyone else has suddenly changed. Our whole relationship with knowledge and skills has changed. The nature of work will also change, so how we learn will also change. We need fewer long-form courses and a more dynamic, personalized approach to learning – one that is also motivating and exciting.

That brings us to a fresh and different model, as there are two new kids on the block.

1. Human teacher
2. Human learner
3. AI teacher (such as ChatGPT and its integration into tools such as Khan Academy and Duolingo)
4. AI learner (the AI model trained on a gargantuan amount of data and subsequent human training, along with pedagogical improvements)

We have moved from human teachers and human learners as a diad, to AI teachers and AI learners as a *tetrad*. But there is a twist to this tale. Human teachers are skilled, but those skills tend to be subject specific. They know one topic really well and are not generalists. They also have valuable teaching skills, but these level off or plateau.

Learners, however, need to learn more efficiently. AI learns and gets exponentially better, and AI Teachers therefore get

better as they draw upon these improvements from the AI Learner.

AI Teachers get better across ALL subjects. AI Teachers are also available 365/24/7 and are becoming multimodal to deliver speech, text, graphics and video. They also deliver dialogue and effortful activity, such as case studies, examples, debate and assessment. We have a new pedagogy based on personal, one-to-one dialogue.

The final stage, and this is some way off, is the elimination of the human teacher, to provide one-to-one tuition using AI. We are now in that age.

This is an uncomfortable debate but we have now crossed that Rubicon. We can now see that the path to faster, cheaper and more effective learning is through faster, cheaper and smarter technology – that technology is AI.

‘But it can’t really do what teachers do well...’

ChatGPT: ‘Hold my beer...’

Universal Soldier is a film franchise that sees a series of soldiers, some melded with AI, to create a superhuman soldier. But what if we imagined something more useful and beneficial. In a world where we have a shortage of teachers, especially in poorer countries where teachers are scarce, poorly paid and class sizes huge, a UT would be a godsend. Not for very young children, but certainly more autonomous learners from secondary school onwards would surely benefit from a teacher that could teach anything to anyone, anyplace, at any time.

The advent of universal internet access, globally, without any blind spots is now with us via ground and satellite services and devices, many with AI chips, that allow edge computing and do things that were unimaginable only a few years ago are ubiquitous. The ecosystem is providing the technical means for UTs to emerge and thrive.

Universal teacher examples

That ChatGPT 4 launched with Khan Academy and Duolingo as partners spoke volumes about its potential for teaching and learning. At the same time, Bill Gates wrote a piece called The Age of AI (2023), which saw education as the great net beneficiary of generative AI. Salman Khan, who has been in this game for years, thinks the benefits are enormous and has already launched what could be described as a universal tutor in education. What followed was an explosion of innovative and often ingenious applications in teaching and learning around the world. We have seen the release of GPT4 – a moment in our history when a new species of technology swept the world.

We saw how Copernicus de-centred our species and threw us out into orbit; Darwin de-anchored us further to show we were just smart apes. One way we got smart was to teach and learn. Yet that exchange has always used learning technologies: painting, writing, printing, internet and now AI. What generative AI provides is a new pedagogy, where monologue is replaced by dialogue, what good teachers do in classrooms, tailored to the needs of the learner, something that is difficult, if not impossible, with previous technologies, in one-to-many teaching environments.

A very real question is now on the table. To what degree can AI now replace teachers? That is seen by some as a disturbing question. It is in the sense of possibly dehumanizing learning. Nevertheless, it is a worthwhile thought experiment.

The old diad model of teachers and learners is long gone as technology has become a serious mediator, especially online technology. We now have software that learns and can be taught, and that, when they can teach, are the new kids on the block.

We have to remember that teaching is always a means to an end, that end being the improvement of the learner. It is easy to see teaching as an end in itself but it is not. As a profession it is difficult, hard work, sometimes distressing, often deeply satisfying. Yet the workload can be crushing, the psychological

pressure oppressive and it is clearly a demanding job. It is difficult to sustain that pace over weeks, months, never mind years. That contributes to constant teacher shortages, especially in difficult-to-teach subjects like maths, science and foreign languages. Could technology be a solution to the crushing workload and recruitment problems? I think so.

Right across the learning journey from learner engagement, learner support, content creation, content delivery, personalization, accessible learning and assessment, AI, especially generative AI, can deliver increasingly sophisticated learning. This is effortful, personalized learning, not as learning events but as a process. Learning is always a process.

Generative AI is coming close to delivering personalized tutoring, as good as, say, the tutors that middle-class parents frequently hire for their children. I have been convinced of this for some time after seeing its effect in adaptive learning systems at university level in the US, where attainment rose and dropout fell.

The problem with these older adaptive systems is that they were difficult to build and populate. New generative AI is an entirely different beast. It is smart – very smart – and as it has been trained on a vast repository of our stored culture, so is smart on all subjects. It is as if it had degrees in all subjects, with some teacher training on top.

Universal teaching

In Khan Academy's Khanmigo, tuition across dozens of subjects is available to ignite your curiosity, at Elementary school, Middle school, High school and College levels. Paired with revision schedule software these systems would provide lasting and efficient learning support. You can even practise exams with open input and dialogue. You can ask for hints but the standard of dialogue is superb. This is where it gets interesting, like a full discussion with a good teacher, who knows every subject, in remarkable detail, all the way up to college level.

Even at college level one can see how 101 courses could be delivered this way, then developed to cope with ever more complex aspects of a degree course. These systems need not be as good as teachers to reduce workload and provide real learning for learners across all subjects.

AI can deliver high-quality, personalized maths tuition. It gives you a maths problem, asks, ‘What do you think the first step should be?’ then accepts open input and identifies where you went wrong along with a suggestion to move forward. It even relates that maths problem to a subject you love – let’s say, football. Maths is a difficult subject to teach and learn – a subject where catastrophic failure is common. Here is a system that pays attention to your every step, is engaging and endlessly patient. What’s not to like?

Even critical thinking is taught, where you debate against ChatGPT. You present your arguments, it counters, and the debate continues, endlessly knowledgeable, positive, patient and helpful. This Socratic approach to learning plays to our natural propensity towards talking to each other. It is as if we now have a universal Socrates. Debate, as a formal taxing of the mind, the chess of ideas, where a motion is debated to explore its strengths, weaknesses and limits is not new. To play such mental chess with a machine is what is new.

Dialogue with the greats

Conversations, real dialogue, with dozens of intellectual figures from the past are also available, including scientists like Isaac Newton and Marie Curie, Presidents from Washington to Lincoln, authors like William Shakespeare and Mark Twain, historical figures such as Genghis Khan and Napoleon. As the works of almost all past intellectuals and figures of note have been used to train the model, it can speak as if it were that person. This is on a par with original sources and provides a less dry and text-based inroad into the subject. To hear theories

and opinions expressed by the person who initiated these thoughts and theories is surely a novel pedagogical approach.

Even fictional characters are available, such as Greek gods like Zeus and Achilles, Shakespearean characters from Macbeth to Othello and characters from novels such as Mr Darcy to Jay Gatsby, also Dr Frankenstein and Don Quixote. This is an interesting and novel vector into great literature, being able to chat with, interrogate and explore a character, maybe several characters in a work. This again is surely a new type of what I've called pedagogy.

Language learning

Duolingo has tens of millions of active users and has used AI for some time. It has pointed the way forward for years in personalized learning using AI and sophisticated algorithms for spaced practice, based on 'half-life forgetting'. This is what is behind the increased efficacy of the product: fine-grained personalization around not learning, but identifying how fast you forget. An interesting inversion.

It was fascinating to see that ChatGPT 4 was marketed as a learning tool, and the first launch was integrated with Duolingo to give free-flowing, immersive conversations. This has been a game changer. It partnered with ChatGPT 4 with two features that add powerful pedagogical techniques to its teaching:

1. EXPLAIN MY ANSWER

When you get something wrong in learning a second language, as you frequently do, it can be frustrating not knowing what you got wrong. Duolingo now gives you an explanation – 'elaborate' – of what was wrong and can also give examples to point you in the right direction, a bit like having a 24/7 native speaker or tutor to help you learn.

2. ROLEPLAY

Here you can chat with someone 24/7, a native speaker who knows your level of competence. That 'person' is AI. It uses human-written scenarios as its basis but as it is generative AI, every conversation is, essentially, unique to you. This gives you much needed practice and immersion, always a problem when learning a second language. Once completed, each roleplay session gives you a report to suggest improvements.

We are at the start of a new era of online, premium learning products in every subject that will revolutionize learning. These products will match and eventually exceed learning in school. Who can deny that language teaching is an area of catastrophic failure? Most spend years at school trying to learn a language and can barely order a cup of coffee at the end.

The whole idea of AI as a useful teacher is here. They have provided a Socratic approach to an algebra problem that is totally on point. Most people learn in the absence of a teacher or lecturer. They need constant scaffolding, someone to help them move forward, with feedback. This changes our whole relationship with what we need to know and how we get to know it. Its reasoning ability is also off the scale. Personalized, tutor-led learning, in any subject, anywhere, at any time for anyone. That has suddenly become real.

Learning difficulties

Another feature of the UT would be sensitivity to accessibility issues. With sight and hearing impairment, text to speech and speech to text through AI delivered via smartphones have already been transformative. ChatGPT was launched with the astounding Be My Eyes app. Dyslexia, autism, ADHD and learning difficulties have all shown potential to be diagnosed by AI, and generative AI can deliver content sensitive to the needs of such learners.

We humans are deeply biased, the biases being largely part of the way the brain has evolved. Daniel Kahneman got a Nobel Prize for uncovering many of these biases and sees them as

‘uneducable’. AI, however, learns and need not have these biases. It need not ‘know’ the sex, race, cultural background, accent or socio-economic status of a learner. Fairness can be developed as we proceed. This is a distinct advantage of the UT – its lack of bias.

For teachers it can create lesson hooks, lesson plans, assessments, rubrics for assessments, almost any piece of planning and paperwork can be done using generative AI. As a tool to reduce workload, it has huge potential. This must surely be one of its first deployments.

Multimodal

Newer LLMs, like Gemini, are multimodal. They can see how images are presented and what is happening in video, while giving a commentary in dialogue when you ask questions. This is an astounding advance for learning, as it can hear, speak, see and interpret all at once in the flow of dialogue and the flow of learning. Ask it for a learning game, it gives you a game. This will be a major advance for learning in the lab, in medicine in anatomy and in all forms of vocational learning. We are well on the way to solving the problem of learning about the real world, as opposed to being locked into text and some flat images.

Translations

Imagine being able to do all of the above in almost any language. That is becoming possible. An underestimated feature of ChatGPT and LLMs is translation. That’s a global feature for services and products that has huge human benefits. Massive reductions in costs are possible through AI here. We would have far fewer students having to learn, and often struggling to learn, in a second language. Even a blend of languages would be possible.

AI and emotions

Yann LeCun thinks AI can have emotions, and I agree, but it is a qualified agreement. This will matter if AI is to become a UT and have the qualities of an expert teacher.

We must also make the distinction between:

Emotional recognition

Display of emotion

Feeling emotions

Emotional recognition

The face is a primary indicator of emotions and we look for changes in facial muscles, such as raised eyebrows, narrowed or widened eyes, smiles, frowns, or clenched jaw. Facial scanning can certainly identify emotions using this route. Eye contact is another – a solid gaze showing interest, even anger, while avoiding eye contact can indicate disinterest, shyness, unease or guilt. Micro expressions are also recognizable. Note that all of this is often a weakness in humans, with a significant difference between men and women, also in those with autism. Emotional recognition in AI is well on its way to being better than most humans and will most likely surpass that ability.

Vocal tone and volume are also significant: tone of voice, intonation, pitch, raised volume when aroused or angry; quiet or softer tone when sad or reflective; upbeat when happy. Body language is another, clearly possible by scanning for folded arms and movements showing unease, disinterest or anger.

Even at the level of text, one can use sentiment analysis to spot a range of emotions, as emotions are encoded in language. LLMs show this quite dramatically. It can be used over time, for example to spot failing students who show negativity in a course. It can be used at an individual level or provide insights into social media monitoring, public opinion, customer feedback, brand perception and other areas where understanding sentiment is valuable. However, it may struggle with sarcasm, irony and complex language usage.

AI could already understand music in some sense, even its emotional intent and effect. Spotify classifies its catalogue using AI. But, this is not to say that AI *feels* emotion.

Even at the level of recognition, it could very well be that machines help humans control and modulate bad emotions. I'm sure that feedback loops can calm people down and encourage emotional intelligence. The fact that machines could read stimuli quicker than us and respond quicker may mean AI is better at empathy than we could ever be. Recognizing emotion will allow AI to respond appropriately to our needs and should not be dismissed. It can be used as a means to many ends, from education to mental healthcare. Chatbots are already being used to deliver CBT therapy.

Display of emotion

Emotions can be displayed without being felt. Actors can do this, written words in a novel can do this and both can elicit strong human emotions. Coaches do this frequently. Machines can also do this. From the earliest chatbots, such as ELIZA, that has been clear. Byron Reeves and Clifford Nass (1996) showed that this reading of human qualities and emotions into machines is common.

As Jaak Panksepp (1988) repeatedly says, we have a tendency to think of emotions as human and therefore 'good'. Their evolutionary development means they are there for different reasons than we think, which is why they often overwhelm us or have dangerous as well as beneficial consequences. Most crime is driven by emotional impulses such as unpredictable anger, especially violent and sexual crime. This would lead us to conclude that the display of positive emotions should be encouraged, bad ones designed out of the system. There are already efforts to build fairness, kindness, altruism and mercy into systems. It is not just a matter of having a full set of emotions, more a matter of what emotions we want these systems to display or have.

Feeling emotions

This would require AI to be fully embedded in a physical nervous system that can feel in the sense that we feel emotions in the brain. It also seems to require consciousness of the feelings themselves. We could dismiss this as impossible but there are half-way houses here and there is another possibility. Geoffrey Hinton (2022) has posited that the mortal computer and hybrid computer brain interfaces could very well blur this distinction in a sense of integrating thought with human emotions, in ways as yet not experienced, even subconsciously. But we may not need to go this far.

Are emotions necessary in teaching?

Donald Norman (2019) claimed that *'Empathy... sounds wonderful but the search for empathy is simply misled.'* He argued that this call for empathy in design is wrong-headed and that 'the concept is impossible, and even if possible, wrong'. There is no way you can put yourself into the heads of the hundreds, thousands, even tens and hundreds of thousands of learners. Not only is it not possible to understand individuals in this way, it also is just not that useful. It is not empathy but data you need. Who are these people, what do they need to actually do and how can we help them? As people they will be hugely variable but what they need to know and do, in order to achieve a goal, is relatively stable. This has little to do with empathy and a lot to do with understanding and reason.

Sure, the emotional side of learning is important and people like Norman have written and researched the subject extensively. Positive emotions help people learn (Um et al, 2012). Even negative emotions can help people learn, stimulating attention and motivation, including mild stress (Vogel and Schwabe, 2016). We also know that emotions induce attention (Vuilleumier, 2005) and motivation that can be described as curiosity, where the novel or surprising can

stimulate active interest (Oudeyer et al, 2016). In short, emotional events are remembered longer, more clearly and more accurately than neutral events.

All too often we latch on to a noun in the learning world without thinking much about what it actually means, what experts in the field say about it, and bandy it about as though it were a certain truth. But trying to induce emotion in the teaching and design process may not be that relevant or only relevant to the degree that mimicking emotion may be enough. AI can be designed to induce and manipulate the learner towards positive emotions and not the emotions, identified by Panksepp and others, that harm learning, such as fear, anxiety and anger. We are in such a rush to include 'emotion' in design that we confuse emotion in learning process with emotion in the teacher and designer. It also seems like lazy signalling, for not doing the hard analysis up front, defaulting to the loose language of concern and sympathy.

In discussion of emotions, we tend to think of them as a uniquely human phenomenon. They are not. Animals clearly have emotions. This is not a case of human exceptionalism. In other words, beings with less complexity than us can feel. At what point therefore can the bottom-up process create a machine that can feel? We seem to be getting there and have come quite far, having reached 'recognition' and 'display'.

If developments in AI have taught us one thing, it is to never say never. Exponential advances are now being made and this will continue, with some of the largest companies with huge investments, along with a significant shift in research and government intentions. We already have the recognition and display of emotions. The feeling of emotions in machines may be far off, but felt emotions, rather than displayed emotions, may be sufficient for teaching and learning.

Conclusion

We don't yet have a UT, but what we do have is a universal teaching assistant (UTA). However, the universal teacher is now on the horizon. It is difficult to tell how far that horizon is but we have seen the exponential growth of generative AI in just a few months from a good but still error-prone service to something far more accurate that has reach across all subjects and has moved technology from embodying simple pedagogic principles such as exposition, knowledge assessment, scenario-based learning and spaced practice towards new pedAIgogies, such as tutoring, teaching, lesson creation, multimodal content creation, learning support, dialogue, error correction, language immersion and debate.

Generative AI will only get better, and not only on parameter size, which has proved to be exponentially useful above a certain threshold. Size does matter but so do all the other tools that go into making a UT – combinations with web search, maths solvers and other tools. We are seeing ensembles of technologies create learning tools that solve the problems of provenance, accuracy and efficiency. Generative AI may be the beating heart of the universal teacher but it has many other weapons it can employ.

The delivery of online teachers is already underway in the partnership with Khan Academy and OpenAI. We will see more of these AI tutors, like Khanmigo, on all subjects, for all ages.

In one way the UT is not a teacher at all; it is the learner learning from the cultural achievements from the past. The UT is us being taught by our own hivemind – a supermind. This is not a mind like ours but it does mediate what we already know so that we can all progress. We can all be in this together. We are being taught by our collective self and that may be the fundamental beauty of this idea.

AI is much more than quick content generation. Learning will not be generated but transformed by generative AI. Unleash the transformative power of AI in L&D. This isn't just about generating text, images, or even audio and video. It's about revolutionizing how we teach, learn and perform.

Imagine an AI that does more than answer questions – it crafts compelling educational narratives, simplifies complex concepts, and personalizes learning to your needs and pace. ChatGPT, Bing, Gemini, Ernie and Claude are at the forefront, but their potential stretches far beyond creating content. They synthesize, outline, interrogate and illuminate, transforming the vast ocean of information into digestible insights.

The learner can engage in ‘dialogue’, move deeper and out from their original prompt, as it can remember, within limits, what you have asked before. This is teaching, not the presentation of content. The true magic happens when you move past the Q&A and into this form of dynamic dialogue. Engage with AI as you would with a seasoned educator. Probe deeper, seek clarity and demand examples. This form of AI remembers and adapts, mirroring the Socratic method, enhancing the teacher–learner interaction.

We can also ensure that knowledge transfer leads to concrete action. We can design scenarios that challenge you to apply what you’ve learned and implement spaced repetition to make learning stick. It’s about embedding newfound skills directly into your workflow.

Content is just the beginning. It is the calibre of the learning journey that counts. We can flip the script, starting with robust pedagogy enhanced by AI. By leveraging the strongest research-backed methods, our technology doesn’t just educate – it transforms you into a practitioner.

Any subject, any level, delivered not in months, but seconds and minutes, with learning that extends beyond the screen into real-world application. That’s the promise – where science meets action, and education becomes transformation.

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07

Chatbots

Chatbots are a game changer in learning as they can teach anything, anywhere, at any level and can be highly ‘personalized’, with delivery in hundreds of languages.

As we saw earlier, most learning professionals will have heard of the ‘Socratic method’ and Socrates’ lasting influence is the useful idea that, for certain types of learning, dialogue allows learners to generate their own ideas and conclusions, rather than being spoon-fed.

In online learning, this questioning and dialogue can now be delivered via chatbots. Intelligent tutors and adaptive, personalized learning systems also account for where the learner has come from, where they are going and what they need to get there. Sophisticated online learning is allowing us to realize the potential of a scalable Socratic approach without the need for one-to-one teaching. Interestingly, it is only in the last few decades, through the use of technology-based tools that allow search, questioning and now generative AI and adaptive learning, that Socratic learning can be realized on scale.

Voice assistants have gone viral, on our smartphones, in cars and in our homes. Once they gain knowledge from our behaviour, even if only online in communications, they will increasingly manage our calendars, shopping, bills, finances, travel and planning. Progress is use, and consumers are driving chatbot technology forwards, as the rewards for success are so great.

We are now starting to see this assistant being used to teach and learn using dialogue in learning as a motivating, social activity, where questioning lies at the heart of the learning experience.

Tutorbot that fooled everyone

So, let us talk about chatbots, as we are already talking with them. Bots are popping up everywhere: on customer service websites, in finance, health, Slack, Tinder and dozens of other web services. There are even bots that fend off loneliness and we can now have a doppelganger chatbot that will live on after we die, so that our loved ones can continue to speak to us. Botland is being populated by as many chatbots as there are needs for conversation.

Chatbots can be engaging, sociable and scalable. They handle queries and questions with less human resource, often taking the load off humans, rather than replacing people completely.

They are also around in education, where bots perform a number of roles; these include student engagement, acting as a teaching assistant, teaching content or playing the role of an awkward student in teacher training, to name but a few.

An ever-present problem in teaching, especially online, is the very many queries and questions from students. In one Georgia Tech online course, this was up to 10,000 per semester from a class of 350 students (Lipko, 2016). Ashok Goel, the course leader, estimates that replying to these is one year of work for a full-time teacher. The good news is that Ashok Goel is an AI guy and saw his own area of expertise as a possible solution to this problem. If he could only get a chatbot to handle the predictable, commonplace questions, his teaching assistants could focus on the more interesting, creative and critical questions. This is an interesting development, as it brings tech back to the Socratic, conversational, dialogue model that many see as lying at the heart of teaching.

So how does it work? It all started with a mistake. Ashok created a chatbot named Jill Watson, which came from the mistaken belief that Tom Watson's (IBM's legendary CEO) wife was called Jill – her name was actually Jeanette. Four semesters of query data, 40,000 questions and answers, and other chat

data were uploaded and Jill was ready to be trained. Initial efforts produced answers that were wrong, even bizarre, but with lots of training and software development, Jill got a lot better and was launched upon her unsuspecting students in the spring semester of 2016.

Jill solved a serious teaching problem – workload. But the problem was not just scale. Students ask the same questions over and over again, in many different forms, so you need to deal with lots of variation in natural language. This lies at the heart of the chatbot solution – natural, flowing, frictionless dialogue with students. The database, therefore, had many species of questions, categorized and, as each new question came in, Jill was trained to categorize the new questions and find answers.

With such systems it sometimes gets it right, sometimes wrong. So, a ‘mirror’ forum was used, a parallel forum, moderated by a human tutor. Rather than relying on memory, the designers added context and structure, and found that performance of the chatbot jumped to 97 per cent. At that point they decided to remove the mirror forum. Interestingly, they had to put a time delay in to avoid Jill seeming inhumanly fast, to make it look as though she were typing. In practice, academics are busy and are often slow to respond to student queries, so they had to replicate this performance! In comparing automated with human performance, it was not a matter of living up to expectations, but dumbing down to the human level.

Most questions were important and practical queries about coding, timetables, file format, data usage – the sort of questions that have definite answers. Note that Jill has not replaced the whole teaching task, only made teaching and learning more efficient and scalable. This is likely to be the primary use of chatbots in learning in the short to medium term: tutor and learner support.

The students admitted they could not tell that Jill was a bot, even in classes run after her cover was blown – she was that

good. In fact, they liked it because they knew it delivered better information, often better expressed and, importantly, faster than human tutors. Despite the name, and an undiscovered run of three months, the original class never twigged.

Real tutors, who often find themselves frustrated by student queries, can sometimes get slightly annoyed and tetchy, as opposed to Jill, who came in with personal but always polite and consistent advice. This is important. Chatbots do not get angry, annoyed, tired or irritable. They are also largely free from the sort of beliefs and biases that we humans always have. They do not have that condescending rolling-of-the-eyes reaction that an expert can have towards simple mistakes and errors by novice learners.

The students found her useful, the person who would remind them of due dates and things they really needed to know, there and then, not days later. She was described as an outstanding teaching assistant, albeit somewhat serious, and asked stimulating questions during the course. Of course, some got a little suspicious. They were, after all, AI students.

One spotted the name and questioned whether she was an AI or not. They checked LinkedIn and Facebook, where they found a real Jill Watson, who was somewhat puzzled by the attention. What finally blew her cover is interesting: she was too good. Her responses were just too fast, even though Goel had introduced a time delay compared with other teaching assistants. When they did discover the truth, the reaction was positive. They even wanted to put her up for a teaching award. Indeed, Goel did submit Jill for an award.

Students expect their teachers to be sincere, tolerant, assured, good listeners, expert and willing to share. Sure there are many things a good teacher can do that a chatbot cannot, but there are qualities a bot has that teachers do not possess. Real teachers can get a little tetchy and accusatory, even annoyed, compared with the clear and objective reply by a chatbot. This relentless patience and objectivity is something a good chatbot can deliver. Remember that the sheer number of the questions

by students was beyond the ability of the real teachers to respond, and, as the bot is massively scalable, it will always, in terms of availability and access, outdo its non-scalable human counterparts. It is really a matter of finding the right balance between automating teacher support and teaching.

The following semester they created two new bots as AI assistants (Ian and Stacey). Stacey was more conversational. This is a natural but technically difficult evolution of chatbots in teaching – to be more Socratic. This time the students were on the lookout for bots, but even then only 50 per cent identified Stacey and 16 per cent identified Ian as AI. The next semester there were four AI assistants and the whole team (including humans) used pseudonyms to avoid detection.

Ashok has now combined his chatbot with ChatGPT to get even more efficacy and is a firm believer in the AI-driven university.

What is heartening about this story is the fact that a professor used his own subject and skill to improve the overall teaching and learning experience of his students. That is admirable. With all the dystopian talk around AI, we need to make sure that AI is used, like this, as a force for good.

Chatbots and learning theory

Most technology in teaching has run against the Socratic grain, such as the blackboard, overhead projector and PowerPoint. With chatbots we may be seeing the return of the Socratic method. This return is being enabled by AI, in particular NLP, but also through other AI techniques such as adaptive learning, machine learning and reinforcement learning.

As the messenger interface seems to have won the interface wars, transcending menus and even social media, simple Socratic dialogue seems to have risen, through the process of natural selection, as an interface of choice, especially on mobile (Ballve, 2015). One is more likely to find people engaging in dialogue on a smartphone than any other form of activity. As

personal assistants, already on our smartphones and in our homes, become commonplace, scale through consumer sales will normalize chatbots. This should help scale adoption in learning.

So, one could argue that for younger audiences chatbots are particularly appropriate, as they already use this as their main form of communication. They have certainly led the way in its use, but one could also argue that there are plenty of reasons to suppose that most other people like this form of interface. We have seen how online behaviour has moved from flat page-turning (websites) to posting (Facebook, X) to messaging (WhatsApp, Messenger). We have seen how the web became more natural and human. Interfaces using AI have become more frictionless and invisible, conforming to our natural form of communication through text or speech.

Learning is expensive, scarce and difficult to scale. In classes with a hundred or more students, few get any personal attention. Learners are already using them in classrooms as a form of dynamic support. So, can we have personal support and attention, even learning, at scale? Fundamentally, chatbot interfaces could realize the promise of a more intimate approach to online learning, as they enable dialogue between teacher and learner.

Teachers should welcome something that takes away the administration and pain, so let us embrace possible solutions. The next stage will be bots that provide more than responses to questions and queries, but also provide real tutor plans, advice and play a more substantive teaching role.

So how can this combination of AI and UI have an application in learning?

In terms of *motivation*, we have evidence emerging that they can encourage and motivate learners. In Stanford research (2019) where a chatbot was compared with a more traditional quiz app, using identical sequencing algorithms (to make the comparison identical), there was 25 per cent more recall with the chatbot. Interestingly, learners also spent 2.6 times longer

on the chatbot, as it was more conversational, more fun and like a true study partner.

Learning takes *effort* and chatbots deliver effortful learning by virtue of being Socratic. Effortful learning was proposed as a fundamental learning strategy by Brown *et al* (2014) in *Make it Stick*, yet so much teaching ignores this with lecturing, long reading lists and talking at people. Personalized dialogue reframes learning as an exploratory, yet still structured (dialogue), process where the teacher guides and the learner has to make the effort.

As an almost *frictionless interface* it matches what we do in real life, requiring no special interface skills. Chatbots, such as Siri, Google Home, Alexa and ChatGPT already exist and, with the addition of text to speech and speech to text, have turned chat into learning. Reading and writing can be replaced by listening and speaking.

This is a return to the more frictionless world of natural chat and communication. Easy to use, it allows you to focus on the message, not the medium. The world has drifted towards messaging for the simple reason that it is simple. By reducing the interface to its bare essentials, the learner can focus less on the interface and more on the important task of communications and learning. Our working memory is subject to cognitive load, the resources used by that temporary memory store. We can only hold so much in our mind at one time and for so long. Taking the cognitive load – implicit in the use of any interface – out of the equation means the teacher and learner can focus on the task and effort needed to acquire knowledge and skills. This is the promise of chatbots. Messaging is simple, a radically stripped down interface that anyone can use. It requires almost no learning and mimics what we all do in real life – simple dialogue. Compared with any other interface it is low on cognitive load. There is little other than a single field into which you type and it goes at your pace.

Chunking lies at the heart of chatbots. In the psychology of learning it means reducing information into digestible pieces

for ease of learning. One of the joys of messaging, and one of the reasons for its success, is that it is chunked and succinct. It is by its very nature chunked. If it were not, it would not work. Imagine being on a flight with someone: you ask them a question and get a one-hour lecture in return. That would be seen as rude, if not insane, behaviour. Chatbots chat; they do not talk at you. The rhythm of natural communication through dialogue is by definition chunked.

In a most likely apocryphal story, where Steve Jobs presented the Apple Mac screen to Steve Wozniak, Jobs had programmed it to say 'Hello...'. Wozniak thought it unnecessary – but who was right? We want our technology to be friendly, easy to use, almost our companion. This is as true in learning as it is in any other area of human endeavour. The brain is a social organ, likes to receive stuff in chunks and interact when learning. We are social apes, grammatical geniuses at age three, and we learn to listen and speak long before we learn to read and write (which takes years).

We *anthropomorphize* technology in such a way that we think the bot is human, or at least exhibits human attributes. Our faculty of imagination finds this easy, as witnessed by our ready ability to suspend belief in the movies or when watching TV. It takes seconds and works in our favour with chatbots, as dialogue is a natural form of human behaviour and communication. If chatbots deliver useful help, support, answers and even deeper teaching experiences, this would indeed be an advance worth following. Indeed, the natural language approach through chatbots accelerates this willingness to attribute agency to the chatbot. Natural language is our normal form of user experience. As AI provides better and better NLP, along with trained and smart databases of answers and smart responses, so AI will become a new learner experience.

As the language-processor ELIZA showed over 50 years ago, it is quite easy to maintain a semblance of realism in dialogue, as we have an involuntary response that makes us see computers

as people when they mimic the behaviour of people – being polite, not having unnatural pauses, being consistent and so on. The more natural interface – which is an optimal mixture of teacher direction through questions and options balanced with learner requests and answers – is what turns chatbots into valuable learning experiences.

One such human quality is *feedback*. Feedback, as explained by theorists such as Black and Wiliam (2005), is the key to personalized learning. Being sensitive to what individual learners already know, are unsure about or still need to know, is a key skill in a good teacher. Unfortunately, few teachers can do this as effectively as they would like, as a class of 30-plus or a college course with perhaps hundreds of students makes it impractical. Chatbots can specialize in specific feedback, dialogue that is personal, trying to educate everyone uniquely.

Being *anonymous* also has its advantages and, in learning, knowing that a human is not judging you can be an advantage. If you have qualms about chat replacing human activity, remember also that another advantage is that bots can allow learners to ask questions that they would not ask face-to-face with an academic, teacher or manager, for fear of embarrassment. This may seem odd, but there's a real virtue in having a teacher- or faculty-free channel for low-level support and teaching. Introverted students, who have problems with social interaction, are also likely to appreciate this approach. Others are simply quiet, even introverts. Anonymous learning, through a chatbot, then becomes a virtue not a vice. Well-being bots may also want to preserve anonymity. In this sense, chatbots may be superior to live, human teachers and bosses. Time and time again we see how technology is preferred to human contact – ATMs, online retail and so on. In learning, in some circumstances, we also witness this phenomenon.

One last advantage of AI in learning is its ability to *personalize* and deliver unique learning experiences to everyone.

Uses of chatbots in learning

Chatbots are now common in retail, health and finance. We can now have personal assistants, tech support bots, travel bots, fitness bots and so on. In organizations we are likely to see human resources (HR) and all sorts of service bots.

They will also become common in learning. Education is always somewhat behind other sectors in considering and adopting technology, but adopt it will. There are several points across the learner journey where bots are already being used in a range of fascinating examples.

Recruitment bots

Early examples of recruitment bots were the University of the People with 20,000 students in 200 countries. The dialogue moves the potential student forward, offering options, eventually leading them towards starting an application, the application portal or speaking with a person. They claim to have delivered a 62 per cent reduction in costs per application. At Leeds Beckett University (Cardwell, 2017), Becky the chatbot was used in the clearing process, a service where students who have yet to find a course and university are 'cleared'. It was also used for general admissions. Her main function was to find you a suitable course. After 21,000 unique interactions, she offered 217 places and 89 students started courses. They also gained a great deal of knowledge about admissions. We can expect to see an explosion of such bots with the advent of generative AI.

Onboarding bots

Onboarding into educational institutions and the workplace is notoriously fickle. New starters, coming in with different needs at different times, are not served well by the old model of a huge dump of knowledge and documents. Chatbots are being

used to introduce new students or staff to the processes, people, environment and purpose of the organization. New starters have predictable questions, so answers can be provided straight to mobile, or students can be directed to people, processes or procedures, where necessary. It is not that the chatbot will provide the entire solution, but it will take the pressure off and respond to real queries as they arise. Available 24/7, it can give access to answers as well as people. What better way to present your organization as innovative and responsive to the needs of students and staff? This is also true of new systems. Software, such as Slack, often has chatbots as an onboarding and ongoing performance tool.

Georgia State University had an onboarding problem with ‘no-shows’, which were up from 15 per cent to 19 per cent. This is a significant drop in revenue. Their chatbot had a 90 per cent opt-in rate and 80 per cent of users gave it 4/5 stars. With 14.9 per cent up on loan counselling, a 12.2 per cent increase on loan acceptance, 9.3 per cent up on on-time immunization submissions and 6.25 per cent up on online transcript submissions, they achieved 3.9 per cent higher enrolment and reduced no-shows by 21.4 per cent (Dalton, 2018).

Learner engagement bots

Learners are often lazy. Students leave essays and assignments to the last minute, learners fail to do pre-work and courses – it’s a human failing. They need prompting and cajoling. Learner engagement bots do this, with pushed prompts to students and responses to their queries.

Beacon, from Staffordshire University (2019), launched an early bot that explained their campus facilities, support services, timetables, societies and clubs. AI lies at the heart of this institution’s IT policy and this was seen as an innovative way to get student services into the hands of students, to keep students involved and engaged. The top most used features were: next lecture, personal tutor, societies, help and digital

services, with questions like, ‘When is my next lecture?’, ‘Who is my personal tutor?’, ‘Where can I eat?’, ‘How do I connect with the wifi?’ The service is personalized.

Learner support bots

Campus support bots or course support bots go one stage further and provide teaching support in some detail. The idea is to take the administrative load off the shoulders of teachers and trainers. Response times to emails from faculty to students can be glacial. Learner support bots can, if trained well, respond with accuracy.

Taking this one step further, we have already discussed how the now famous Georgia Tech chatbot, and its descendants, played a credible role as a teaching assistant. Who would not want the basic non-teaching, administration tasks automated for teachers, so that teachers and academics could focus on real teaching?

SAP, a giant IT services company, used its own AI technology to build chatbots that answered learner questions from their learning communities, as well as recommending personal learning journeys (as outlined at Online Educa, Berlin, in December 2018). Using machine learning, it showed significant improvement in performance the more it was used, and when learners were asked whether they preferred the chatbot to human support, they chose the chatbot.

Bolton College (2019) has a support chatbot named ADA and has moved into using LLMs to widen the conversational model across the college. The ability to provide administrative and teaching bits across the entire organization reflecting different functions and subjects is now possible. There is no reason why every expert, academic and professional should not have a chatbot, if desired.

Performance support bots

In a sense Google is a chatbot. You type something in and up pops a set of ranked links. Increasingly, you may even have a short list of more detailed questions you may want to ask. Straight-up FAQ chatbots, with a well-defined set of answers to a predictable set of questions, can take the load off customer queries, support desks or learner requests. A lot of teaching is admin, and a chatbot can relieve that pressure at a very simple level within a definite area of knowledge – frequently asked questions.

There have been chatbots that sit on top of content, accessible from Facebook, Slack and other commonly used social tools. You get help in various forms, such as simple text, chunks of learning, people to contact and links to external resources as and when you need them. Content no longer sits in a dead repository, waiting on you to sign in or take courses, but is a dynamic resource, available when you ask it something.

We can expect to see chatbots play a big role in learning in the flow of work, where immediate needs are met by simply asking for help and support.

Tutorbots

Tutorbots differ from chatbots in terms of the goals, which are explicitly ‘learning’ goals. They retain the qualities of a chatbot – flowing dialogue, tone of voice, exchange and human (like) – but focus on the teaching of knowledge and skills. Straight-up teaching is the approach, where the bot behaves like a Socratic teacher, asking sprints of questions and providing encouragement and feedback. This type of bot can be used as a supplement to existing courses to encourage engagement. Tutorbots that take a specific domain can be trained or simply work with unstructured data to teach knowledge. This is the basic workaday stuff that many teachers don’t like.

One can also take a knowledge set and create a chatbot that re-presents that knowledge as semi-structured, personalized dialogue. As we know the answers and recreate the questions

with algorithmic tutor behaviours, the tutorbot can be a simple teacher or assessor. In fact, it can be a more sophisticated teacher of that knowledge, sensitive to the needs of that individual learner. Once the basic knowledge has been acquired, the bot tests the student as well as getting them to apply their knowledge.

Mentor bots

The point of a bot may not be to simply answer questions but to mentor learners by providing advice on how to find the information on your own, to promote problem-solving. Once the bot knows the context and provides not just answers but also advice, it behaves more like a teacher. Providing answers is not always the best way to teach. At a higher level chatbots could be used to encourage problem-solving and critical skills, by being truly Socratic, acting as a midwife to the students' behaviours and thoughts. As the dialogue gets better – drawing not only on a solid knowledge base, good learner engagement through dialogue, focused and detailed feedback, but also critical thought in terms of opening up perspectives, encouraging questioning of assumptions, veracity of sources and other aspects of perspectival thought – so critical thinking could also be possible. Bots will be able to analyse text to expose factual, structural or logical weaknesses. The absence of critical thought will be identified as well as suggestions for improving this skill by prompting further research ideas, sound sources and other avenues of thought. This 'bot as critical companion' is an interesting line of future development.

Practice bots

Beyond knowledge, we have the teaching and learning of more sophisticated scenarios, where knowledge can be applied. This is often absent in education, where almost all the effort is put into knowledge acquisition. It is easy to see why – it's hard and

time-consuming. Bots can set up problems, prompt through a process, provide feedback and assess effort. Scenarios often involve other people; this is where surrogate bots can come in.

Practice bots literally take the role of a customer, patient, learner or any other person and allow learners to practise their customer care, support, healthcare or other soft skills on a responding person (bot). Bots that act as revision bots for exams are also possible.

A bot that mimics someone can be used for practice. For example, 'Eli' is a chatbot, developed by Penn State, that mimics an awkward child in the classroom. It is used by student teachers to practise their skills in dealing with such problems before they hit the classroom. Duolingo uses bots after you have gathered an adequate vocabulary, knowledge of grammar and basic competence, to allow practice in a language. This surely makes sense.

Assessment bots

Assessment bots can deliver formative assessments using adaptive assessment. WildFire built just such a bot for a travel company. It presents a medium-level question; if you get that wrong, it lowers expectations, and if you get it right, it raises expectations. Using an algorithm that aims to get you learning content, it eventually makes sure that you get all of your knowledge into the top zone. Assessment bots can also be used in summative assessments, with single passes through the questions.

Well-being bots

As pressures on learners in schools, universities and the workplace become greater, there has been a rise in issues around well-being and mental health. Well-being bots such as 'Ellie' (Robinson, 2015) and 'Woebot' that are sensitive to patients' needs are already being subjected to controlled trials

to examine the impact on clinical outcomes. If a bot is being used in any therapeutic context, its anonymity can be an advantage. From ELIZA in the 1960s to contemporary therapeutic bots, this has been a rich vein of bot development. Therapeutic bots are being used in controlled studies to see if they have a beneficial effect on outcomes.

Chatbots, as they start to understand context, will play a greater role in human-machine interaction, on the move, in the home, but also in learning. Dialogue is complex, but it is how most of us communicate and learn with other people. As AI provides new agents in the world that also learn, they will be things we learn from.

Building chatbots

Talking is replacing touching and typing. Apple's Siri was launched in 2011 built on previous technology that took decades to develop. Amazon's Echo (Alexa) came in 2014 and Google Assistant in 2016. They all serve different markets. Apple sells products, especially the iPhone, Microsoft sells business products, Alexa is a home product, while Google Assistant is a general product and ChatGPT, along with others based on LLMs, have revolutionized search and learning. The whole story is well told in *Talk to Me* (2019) by James Vlahos, who sees this as a race among tech companies to master voice and conversation. Speech is fundamental and voice is becoming the lifeblood of everyday computing. Voice humanizes and normalizes technology. It is not robots that reflect us, but chatbots, speaking and listening.

With the advent of generative AI and ChatGPT, digital assistants can free us from the clumsiness of drag and drop, clicking on figures to see speech bubbles towards dialogue. Dialogue is the very thing that makes us social beings and we now have that in online learning. We can now get personalized learning, speak to Plato, Newton, Darwin, Picasso and other dead and living experts, even speak to fictional characters.

People think that AI has a heart of stone, the chilling execution of algorithmic logic, but technology is being humanized. We speak to it, it speaks to us. This means understanding how chatbots can be built.

Personal chatbots

Chatbots can deliver learning functions right across the learning journey, from onboarding to learner engagement, learner support, mentoring, teaching, assessment, practice and well-being.

Interestingly, this immediately raises an issue for chatbot deployment – setting ‘expectations’. When OpenAI launched its service allowing users to create their own chatbot, you could give it a name, picture, pre-define its role, present with several ‘set’ prompts and one could keep it personal, open on a link or for public consumption.

My ‘Digital-Don’ has many years of writing in learning theorists and theory, as well as other content relating to learning technologies. Speak to it and you are speaking to me. It will grow with me as a digital twin.

This begs the question about the persona of such chatbots.

Persona

Your chatbot needs to have a persona or personality. It is too easy to just throw a jumble of responses into a database and hope for the best. Nass and Brave explored the topic in *Wired for Speech* (2005). They noticed that movie robots often did not sound human, sometimes referred to themselves in the third person, by their own names or in the passive.

Modern chatbots, taking advantage of realistic speech technology, have gone in the opposite direction and have adopted human personas. This makes sense. We want to speak to interesting and engaging people, no matter if they are machines. Each chatbot has a blend of qualities.

What about gender – female, male or neither? Siri and Google Assistant were originally female by default but can be switched to male. Alexa, when asked, is ‘female in character’. There is much debate around portraying such services as predominantly female, as it may suggest subservience. One could argue the opposite, that launching them all as male would have been a demonstration of power and patriarchy. Microsoft claim that the research shows we generally prefer the voices of young women. An alternative is to go genderless or give people an even choice at the start. The problem here is trying to get a genderless voice.

Some assistants, like Siri, answer in ways that are sensitive to the culture. This can be in the use of the actual language and vocabulary, but may also be the use of appropriate cultural references. Humour is an especially sensitive issue, as it varies across cultures, but few commercial bots are edgy or transgressive in their humour, such as Grok. You can please a lot of the people a lot of the time, but not all of the people all of the time, so it is worth playing it safe.

What happens when a chatbot fails? Believe me, this is common. A number of failsafe tactics can be employed. You can do the common... ask the person to repeat themselves: ‘Sorry, I didn’t catch that’, ‘Could you elaborate on that?’ The chatbot may even try to use a keyword to save the flow, distract, change the subject and come back to the flow a little later. So think about failsafes. If all else fails, and many customer chatbots do, default to a real human. That is fine in customer service, and many services offer this functionality. This is not so fine if you are designing an autonomous learning system.

Can your chatbot integrate into other platforms? Don’t imagine that this will work easily from your LMS; it won’t. Integration into other systems may also be necessary.

Testing chatbots has its challenges. You will need a test plan that covers all aspects of the deployment, the quality of the driving software, support, infrastructure, interface design and goals. Rules-based branching chatbots are somewhat easier to

test, a bit like checking all the branches of a single tree. A machine learning chatbot is a far more autonomous entity, more like testing in a forest. It needs a clear testing strategy. You will need a test plan and possibly a test team with clear objectives.

Chatbot abuse

People ask chatbots the weirdest of things. Users know that chatbots are really pieces of software, so test it with rude and awkward questions. Swearing, sexual suggestions, requests to do odd things and just being plain rude are common.

Chatbots are regularly asked out on dates, asked to send naked photographs and worse. Cleo, a finance bot, used to send back a picture of a circuit board. Nice touch and humour is often the best response. The financial chatbot Plum responds to swearing by saying, 'I might be a robot but I have digital feelings. Please don't swear.' These are sensible responses, as Reeves and Nass (1996) found in their studies of humans with technology mentioned earlier, that we humans expect our tech to be polite.

People want to play with chatbots – that's fine. You often find that these questions are asked when someone first uses a chatbot or buys Alexa. It's a sort of onboarding process, where the new user gets used to the idea of typing replies or speaking to a machine.

The odd questions tends to come at the start, as people stress-test the bot, then drop off dramatically. This is telling and actually quite useful, as users get to see how the bot works. They're sometimes window shopping or simply seeing where the limits lie.

It also helps calibrate and manage expectations. Using a bot is a bit like speaking to a very young child. You ask it a few questions, have a bit of back and forth, then get its level. Actually, with some, it's like speaking to a dog, where all you can do is variants on 'fetch'. Once the user realizes that the bot

is not a general purpose companion, who will answer anything, or teacher with super-teaching qualities, and has a purpose, usually a specific domain, like finance, health or a specific subject, and that questions beyond this are pointless, they settle down. You get that ‘fair enough’ response and they settle down to the actual business of the bot.

These little touches of humour and politeness serve a further purpose, in that they actually engage the user. If you get a witty or clever reply, you have a little more respect for the bot, or at least the designer of the bot. With a little clever scripting, this can make or break user acceptance. Some people will, inevitably, ask your bot to tell a joke – be ready for that one. A knock-knock joke is good as it involves a short dialogue, or a light-bulb joke.

These responses can also be used to set the tone of the bot. Good bots know their audience and set the right tone. It’s pointless being too hip and smart-assed with an older audience, who may just find it annoying. Come to think of it, this is also true of younger audiences, who are similarly intolerant of clichés. You can use these responses to be edgy, light-hearted, serious, academic... whatever.

You’ll find yourself dead-ending a lot with bots. They’re nowhere near as smart as you at first think. That’s okay. They serve a function and are getting better. But it’s good to offer a little freedom, allow people to play, explore, find limits, set expectations and increase engagement.

Botched bots

An interesting example of a botched bot comes from China, where Tencent, which has hundreds of millions of users, had to take down a penguin bot named BabyQ, and a girl bot named Little Bing. Why? Their ‘crimes’ were that they showed signs of political autonomy and honesty (Oppenheim, 2017). BabyQ was asked, ‘Do you love the Communist Party?’ the penguin replied, curtly, ‘No’. To the statement, ‘Long Live the Communist Party’,

BabyQ came back, thoughtfully, ‘Do you think such corrupt and incapable politics can last a long time?’ Then, when asked about the future, the perky penguin responded, ‘Democracy is a must!’ Little Bing was more aspirational, and when asked what her Chinese dream was, she said, ‘My China dream is to go to America,’ then, when pushed to explain, ‘The Chinese dream is a daydream and a nightmare.’

Of course, the bots were either picking up on real conversations or being subversively trained. Tencent were forced by the Chinese Government to take them down, which merely shows that we have more to fear from censoring governments and compliant technology companies than AI-driven bots.

A now infamous example of a bot that went off-piste was Microsoft’s Tay. They had no idea that young people would take a playful view of the tech and deliberately ‘train’ it to be a sex-crazed Nazi (Hunt, 2016). It was all a bit of fun, but most people over 50 saw it as yet more proof of demonic, out-of-control bots. In truth, all it showed was that kids know what this tech is and know full well that bots are primitive and can be trained. My favourite example of this type of subversion is the Walkers Crisps campaign to encourage people to post selfies to Twitter. They did, but it ended up being a rogues’ gallery of serial killers and Nazis (James, 2017). Once again, we the people won’t be pandered to by companies implementing patronizing tech that they don’t fully understand.

Caution

Coming back to learning, chatbots are working at all stages in the learning journey, recruitment, onboarding, from student engagement, student support to feedback, learning and assessment.

It could be cross-departmental, cross-subject and cross-cultural, to produce teaching and learning that will be free from

the tyranny of the institution, department, subject or culture in which it is bound.

As a chatbot does not have the limitations of a human, in terms of forgetting, recall, cognitive bias, cognitive overload, getting ill, sleeping eight hours a day, retiring and dying, once it's on the way to acquiring knowledge and teaching skills, the services will get better and better. The more students that use its service, the better it gets – not only on what it teaches, but how it teaches. And how people learn. Courses will be fine-tuned to eliminate weaknesses, and finesse themselves to produce better outcomes.

They need to be trained, built, tested and improved, which is no easy task, but their efficacy in reducing the workload of teachers, trainers, lecturers and administrators is clear. The dramatic advances in generative AI and LLMs is a rapidly developing field of AI and promises to deliver chatbot technology that is better and cheaper.

For the foreseeable future, however, chatbots will support teachers. Chatbots are a form of technology that teachers should appreciate, as they truly try to support what teachers already do. It takes good teaching as its standard and tries to eliminate and streamline inefficiencies to produce faster and better outcomes at a lower cost. It is impossible for a teacher to have personal dialogue with large numbers of students, but chatbots are scalable. They may not be anywhere near the sophistication of actual human dialogue, but for some support tasks they can do a credible job. Chatbots, at present, are not replacing people but working alongside people. This is important. Chatbots may replace some functions and roles, but few suppose that most jobs will be eliminated by chatbots. We have to see them as being part of the learning landscape.

Conclusion

Chatbots in learning have to capture and embody good pedagogy and already do. Whether it is models of engagement,

support, learning objectives, practice, assessment or well-being, the whole point is to use both the interface and back-end functionality for pedagogic capture to deliver powerful learning based on evidence-based theory, such as retrieval, effortful learning, spaced practice and so on. This will improve rather than diminish or ignore pedagogy.

As chatbot technology gets better and state-of-the-art levels are reached, we get steady progression. It will matter whether this progression is open and accessible. The harvesting of questions and answers by learners will give us invaluable insights into problems with the existing courses and content, leading to deliberate or automated course improvement. Chatbot technology has the potential to move into more sophisticated teacher functions. They are already here, and more are coming – rather than condemn them for encroaching on teacher territory, we should see them as scalable agents that could be capable of delivering teaching at almost no cost, to anyone, anywhere, at any time.

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AI and performance support

Behavioural psychology has much to teach us about why people procrastinate in learning. Understand learners and you understand why it is so difficult to get them to learn. Let us start with the metaphor of the elephant and its rider: the rider is the conscious, verbal, thinking brain; the elephant is the automatic, emotional, visceral brain. Jonathan Haidt created this metaphor in his book *The Happiness Hypothesis* (2006). The idea is that the more recently evolved rational rider part of your brain may want to learn, but more often than not the procrastinating, impulsive elephant takes over and decides to wallow in inaction. Academically this is similar to Kahneman's two systems, *fast* and *slow*, explained in *Thinking, Fast and Slow* (2011) and, in an alternative, more readable form, in *The Undoing Project* (2016) by Michael Lewis. If we can use technology and AI-driven systems to overcome our inner elephants, learning may flow more freely.

But there is a big problem. One elephantine, cognitive bias that hits learning hard is that of *hyperbolic discounting*, a well-researched feature in behavioural economics. Take two similar rewards: humans prefer the one that arrives sooner rather than later. We are therefore said to discount the value of the later reward and this discount increases with the length of the delay.

If the consequences of our learning are distant, we are likely to take it less seriously. Young smokers don't stop smoking just because you tell them it is dangerous, and there's no greater danger than death. In practice, smokers see the consequence of smoking as being some time off, many years if not decades, so they don't stop smoking just because you warn them of the consequence. So it is with learning. Rewards feel distant in learning, which is why students tend to leave study until the last moment and cram just prior to exams, or write essays on

the last night. Similarly in organizations, they are not committed when it is likely that they won't use their newly acquired knowledge and skills for some time, if at all. No one watches printer problem videos until they have a printer problem, and no one really feels that compliance training is relevant as the implications seem so far off. So how do we get the learner to be a rider and not be stopped by the elephant?

Just-in-time training, performance support and workflow are all terms for delivering learning when it is needed. This closes the gap between need and execution, thereby eliminating hyperbolic discounting, as there is no delay. Having a solution to printer and other problems ready on tap, so that when the problems arise, solutions are at your fingertips, is exactly what is required.

AI and informal learning

Gloria Gery first talked about performance support in 1991. Jay Cross has been credited with inventing the term 'e-learning', was the pioneer in both the practice and theory of technology in learning, and pushed the idea that learning is not primarily about formal courses but informal learning.

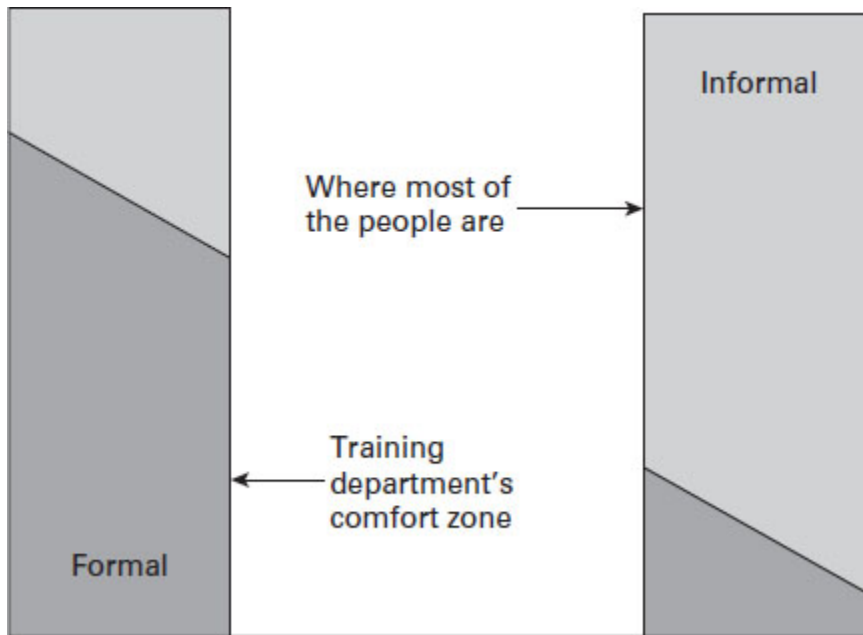
After developing the first courses at the hugely successful University of Phoenix, Cross set up the Internet Time Group. A tireless thinker and presenter on learning, he pushed the learning world to think seriously about informal learning and working smarter. For him, workflow learning ties learning into the actual workflow within an organization. According to Cross (2011), it takes us to support and on-demand services that are designed to exist within the real tasks we do in our everyday work. Out of this work on workflow learning came an even wider, and what he regarded as more important, set of reflections.

Averse to detailed semantic analysis, he compares the difference between formal and informal learning with the difference between taking a trip on a bus and driving your car.

In the former, you are on a set route and not in the driving seat; in the latter you go where you want, when you want and on the route you choose. His reflections on the failure of training to really recognize informal learning are well represented in his oft-used 'spending paradox' ([Figure 8.1](#)).

Most learning is informal yet almost all the spend is on formal courses.

FIGURE 8.1 Formal and informal learning: source of the problem



► Figure 8.1 details

This is about seeing learning in a more naturalistic way, seeing learners as real people in real organizations, who use real tools in real networks, both offline and online. Informal learning is driven by conversations, communities of practice, context, reinforcement through practice and social media to optimize organizational performance. Blogs, podcasts, peer-to-peer sharing, aggregators, social media and personal knowledge management are all emergent phenomena, unlike the top-down tools and content that traditional online learning has provided. When we look at the internet, we see powerful tools and techniques emerge through genuine use. It is these, he believed, that point us towards success in learning.

Note that this does not imply the abandonment of formal learning, only that informal learning needs to be recognized and supported. There's still a need for underpinning learning with good content, from books to full courses, especially for novices and business-critical training such as compliance. You

can't let people who don't know what they need to know drift, so there's a time and place for structured, formal learning. We simply need to also identify and use natural pathways, the unofficial, unscheduled events that take place in real contexts, underpinned by personal, intrinsic motivation. What matters is execution and performance, real business objectives and metrics. To see training as schooling is the mistake, often superficial, irrelevant and not reinforced by doing, that is often forgotten.

Rather than an unbalanced focus on top-down classes, tests and certificates, networked learning organizations need blends of workplace learning that include the experiential, unplanned and informal. This has to be done systematically and not left to chance. Informal learning is not some vague form of drift with chance encounters. You need to take control of your own learning, and organizations need to inseminate and stimulate communities of practice. What has given workflow learning a boost has been the explosion of social media, which has provided communication through blogging, Facebooking and tweeting, as well as media sharing through YouTube, Vimeo and so on. It is fast, cheap and ubiquitous, so we see a pendulum shift towards the informal over the formal.

It is almost as if the language of learning is fading and development within the context of the workplace without the formal structures, labels, learning objectives and over-formality of courses. Far from decimating the training department, one can envisage development as a constant in organizations, not only inside the firewall but also for all part-timers, freelancers and partners. The wall between company and customers is also crumbling, creating a new culture of customer education.

We need to reflect on the obvious, but shocking, fact that almost all of our attention (and spend) goes on the formal side, while the majority of the action is informal. This is not to abandon formal learning, but to consider the accelerating role of technology in informal learning, to move beyond the traditional LMS and content model to a newer, more

naturalistic model of learning, based on real behaviour and contemporary technology. That contemporary technology has now come of age through more expansive technology involving AI, such as LXPs, LRSs and the specification, xAPI.

AI and performance support

The new AI tools, including chatbots, LLMs and retrieval-augmented generation (RAG) techniques, among others, can be used to move us beyond content creation to performance support. Learning has long been stuck in course production, exacerbated by a move towards supply, rather than demand. It is HR and L&D that decide what 'courses' to deliver, rather than the business, hence the flood of diversity, equity and inclusion (DEI) courses. Yet we know these are often ineffective, overlong, the content too theoretical, out of context, quickly forgotten and have no significant impact.

L&D has long been stuck in the rut that supposes courses are the answer to everything, when they are often the wrong solution. If you have a learning problem, it is wiser to do some analysis of your problem into training and non-training solutions. Changes to process are often more successful and training is often an inappropriate response. Even when training is a solution, one should consider critical and non-critical tasks. Critical tasks need up-front training, as the organization needs to be preventative. Non-critical tasks can then be dealt with as performance support. The issue then becomes the implementation of performance support. It is easy to make the distinction between courses and resources more difficult to deliver such a solution.

Performance support can be delivered via performance support systems that deliver a single moment of truth, accessing the various repositories and systems within an organization. One can have a formal schema, allowing users to choose what type of support they need, escalating as the need is defined.

Moments of need

Bob Mosher, who co-authored the ‘5 moments of need’ model (Gottfredson and Mosher, 2012), has the same message: busy learning teams need to align themselves with real user behaviour and business needs. This means learning that is contextualized and designed to help learners. His moments of need are:

- Learn something for the first time
- Learn more about a topic
- Apply or remember something
- Something has gone wrong
- Something has changed

The message from both Cross and Mosher is to embed the learning in the workflow and make it readily available at the moment of need. It helps to apply Occam’s Razor to this problem and supply the minimum amount of effort to satisfy their goal. Contextualizing to specific roles and needs also oils the wheels. So, how can one use AI, in practice, to achieve workflow learning?

AI is currently being used by hundreds of millions of people. Interestingly, much of it for learning and ‘performance support’, namely ‘I have a need, let’s use AI for help.’ AI is performance support!

The next step is to get the organization’s data into the right state for semantic retrieval, through search or a chatbot. This requires the right documents with up-to-date information.

Retrieval-augmented generation (RAG)

Imagine having both a researcher and a writer working together to answer your questions. RAG gives you this two-step process.

First you ask a question about a topic. The ‘researcher’ part of RAG quickly digs through a massive library of information to

find relevant pieces of text or media that contain facts or details about the topic. This is like someone pulling out the best books, articles or other media that might have the answer.

Once the researcher has all this relevant information, they pass it over to the 'writer'. The writer is a powerful language tool that reads through the collected information and composes a clear, informed answer or piece of writing based on what the researcher found.

This two-step process allows RAG to provide answers that are not just made up on the spot but are based on existing knowledge, which can make the answers more accurate and trustworthy. It's particularly useful for answering specific or complex questions that require some level of expertise.

This technique can be used for performance support in organizations, to draw upon a large amount of data held by organizations, to sift through large repositories of organizational knowledge to retrieve relevant information that can help employees perform tasks. For example, if a salesperson needs information on a product, RAG can quickly pull up the latest and most relevant data to support the sales process. HR can pull up company guidelines, employees statements of work, an engineer can pull up ISO (International Organization for Standardization) standards.

Organizations can therefore leverage RAG technology to improve efficiency, accuracy and depth of information in various functions, for many roles and functions, leading to better performance and decision-making capabilities.

Organizational learning

There is a much bigger AI idea around organizations as knowledge networks, where learning is leveraged and pushed in a just-in-time fashion. It centres around the idea that AI-powered enterprise software will unlock knowledge, insights and expertise within an organization, so that it can power collective competence through sharing and learning.

Most organizational knowledge is locked away in repositories or in the heads of its people. Unlocking and sharing that knowledge is difficult because it is both structured and unstructured, in lots of different places among lots of different people. This last repository, the brains of your people, is the most difficult to unlock, as people are often inclined to keep things to themselves and not share.

Microsoft Teams has integrated GPT, the AI language model created by OpenAI, to enhance various features within the Teams platform. You can use chatbots where you use dialogue to assist with a range of tasks, summarize conversations, meetings, or documents (useful if you missed a meeting or chat), search, get suggestions and assistance in composing messages, write or generate meeting notes, get translations done and automate workflows such as setting up meetings, by understanding and executing user commands. When you use integrated AI in Microsoft, Google and other professional contexts, it is designed to comply with privacy and security standards, ensuring that user data is handled appropriately. It is this integrated approach to knowledge that the learning community needs to see as its future. Isolating learning into a silo does not leverage the data and people one wishes to use in the wider sense of L&D.

Cross was right to emphasize the plain fact that informal learning is by far the largest vehicle for actual learning in an organization. Only now, through AI, do we have the software that is smart and powerful enough not only to find the stuff we need, identify the relationships between content and people, when we need it, but also organize and distil it into meaningful information at our point of need.

This is not to say that the AI-driven provision of knowledge and people is sufficient. Novices will still need structured learning. Presentation alone is never enough, as little is learned on first exposure; most is learned from subsequent effort. This is why search is never enough and learning needs its own additional AI dynamic.

Conclusion

Organizations need faster, more accessible and flexible systems to deliver learning. It was always the case that learning took place in an ecosystem with many points of contact. Those points of contact used to be classroom courses, manuals and, above all, learning from colleagues – all offline.

As points of contact went online to LMSs, they tended to silo learning as separate from other sources and activities for learning. Yet internal knowledge bases, internal communications systems, external web resources, YouTube, social media and so on, through various devices, delivering learning, also have a role in learning. AI already plays a key role in search and social media and is now pivotal in the delivering of wider learning experiences through LXPs. Ultimately, organizations need to see themselves as learning organizations that leverage learning across everyone. This can only be done using AI.

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09

AI content creation

Another way of looking at AI in learning is to see how it can be used to enhance and create online learning.

Learning science

At this point it is worth revisiting the psychology of learning, as there is a tendency to use new technology (such as generative AI) to implement old psychology. What do recent findings in cognitive psychology recommend be delivered using AI? As AI is smart software that can do things that were never possible before, such as text summaries, interpretation, speech to text, text to speech, semantic analysis and personalized delivery, it is worth considering what new pedagogy can be used with this new AI technology.

Richard Mayer and Ruth Clark are among the foremost researchers in the empirical testing of media and media mix hypotheses in online learning. Their *E-learning and the Science of Instruction* (2003) covers seven design principles: multimedia, contiguity, modality, redundancy, coherence, personalization, and practice opportunities. 'Less is more' could be Mayer and Clark's mantra. Their precise studies have confirmed that our media mix (text, graphics, audio, animation, video) in online learning is often flawed, resulting in cognitive overload and dissonance. Perhaps their greatest contribution has been in identifying redundancy as a serious problem in screen-based learning

In one study, Mayer (1996) presented 600 pieces of scientific learning and found that briefer versions, which were concise, coherent and co-ordinated, resulted in far more effective learning. They are also precise in their recommendations:

There is a clear pattern in which the more words added to the core verbal explanation, the more poorly the student does in producing the core explanative idea units. These results are consistent with the idea that the additional words overload verbal working memory, drawing limited attentional and comprehension resources away from the core verbal explanation. (Mayer *et al*, 1996)

The lesson is to cut it 'til it bleeds! Bullet points, simple writing, highlighted keywords and short paragraphs are all useful screen writing techniques. AI can help reduce redundancy in learning by summarizing text that was not originally written for learning.

Another thing that research in cognitive psychology has gifted to us over the last decade or so is clear evidence that learners are misguided when it comes to judgements about their own learning. Bjork (1994), along with many other high-quality researchers, claims that learning is quite misunderstood by learners, as we have a flawed model of how we learn and remember. There is often a negative correlation between people's judgements of their learning, what they think they have learned and what they have actually learned. The same is true of how they think they learn best, and the way they can actually optimize their learning. In short, our own perceptions of learning are seriously defective. This is why engagement, fun, learner surveys and happy sheets are such bad measures of what is actually learned and often detrimental to the design of optimal learning technology strategy and solutions.

Much online learning is illusory because it is too easy. Learning actually requires real effort for high retention to take place. This is why so much online learning fails. To simply click on faces and see speech bubbles of text appear, drag and drop labels across the screen, choose true or false, even multiple-choice questions (MCQs), rarely constitutes desirable difficulty. This is click-through learning.

The solution is to provide *effortful retrieval*. This means moving beyond the traditional model of flat text and graphics

punctuated by multiple-choice towards cognitive effort, namely strong retrieval, through open input by the learner. This effortful learning gives significant increases in long-term retention and recall. Online learning needs to adopt these techniques, enabled by AI, if it is to remain credible.

Retrieving and recalling what you need to know results in much higher levels of retention (Bjork, 1975). Rather than read, re-read and underline, look away and try to retrieve and recall what you need to know. Rather than click on 'true' or 'false' or an option in a short list (MCQs), look away, think, generate, recall and come up with the answer. The key point is that research has shown that retrieval is a memory modifier and makes your memory more recallable. Counterintuitively, retrieval is much more powerful than being presented with information in the first place, namely through the original 'teaching' event. From Gates (1917), who compared reading and re-reading with retrieval, to Brown *et al* (2014) in *Make it Stick* who amassed the evidence for retrieval as a powerful learning strategy.

More focus should be given to retrieval, not presentation, clicking on items and multiple choice. We need to be presented with desirable difficulties, through partial or complete open input. We have the tools in NLP and AI to implement the findings of this research and apply retrieval practice, learning from failure, so technology has at last caught up with pedagogy. Let us not plough the same furrow we have ploughed for decades. It is time to move on.

Good content matters in learning, just as good movies, books and other purposeful content matters. However, online learning content production has for decades been stuck in the delivery of rather flat content. It has been accused of being page-turning, largely media production, whether text, graphics or video, peppered with lots of click-on events to reveal, drag, drop or select from lists (MCQs). The results in terms of actual learning and retention can disappoint, as the lack of

meaningful cognitive effort often results in rather shallow and short-term results.

Content production has also been neither fast nor cheap. A typical production team could include a project manager, instructional designer, graphic artist (maybe video and sound person), developer and tester. Input by subject matter experts (SMEs) is then needed for content creation and quality control. This results in lots of documentation and innumerable iterations, which can be stressful as the culture of expertise clashes with the culture of learning and design. The psychology of learning tells us that less is more, yet SMEs often think (wrongly) that more is more. This all adds up to time and money. Projects can take up to six months or more to complete, and the cost per hour of learning, still a rather odd metric, can be considerable.

Generative AI has given us a more realizable and precise goal for the use of AI for learning to create content from scratch. With the help of generative AI, content can be created in minutes not months, at a fraction of the existing costs and results in higher retention. This reduction in time is important, as training design can be a block in business process. A teacher, lecturer, trainer or business sponsor comes to the learning technology team and is told that it will take weeks, even months, to complete and needs a considerable budget. Deliver quickly, using agile techniques and tools, and you will be seen as delivering timely solutions to educational or business problems.

Create content from scratch

Generative AI can also create content from scratch by enhancing existing resources. We have been involved in building tools that literally takes good learning science, learning design and pedagogy, and build them into the tool, so that you don't have to be a champion prompter. These tools can create courses in minutes not months, literally on ANY subject. They

will determine what core skills are needed, create a structure for the course, even come up with titles. You can then be walked through the construction of your course as it comes up with text, graphics and other media. Different versions can be created at different levels, with different levels of language and vocabulary. A higher education version can be converted to a more vocational version.

The breakthrough in LLMs was GPT 3.5 launched on 30 November 2022. It was the start of a new era for us all, especially in learning. It was taken up by a million in two days, two million in two months, and now has over 100 million users and a billion uses per month. GPT 4 and others are powerful LLM models made available as chatbots that allow users to speak to it with remarkable results. This is all without domain-specific training. In other words, it has general capabilities and does not need specific information on a topic or subject to operate successfully. It can generate text of good quality at some length. In fact, the model is 'chameleon-like' as it adjusts to the style and content of the initial piece of text. This makes it read as a realistic response. This has huge implications for the future of education and training.

Generative AI allows the automatic creation of text for teaching and learning, whether study papers or textbooks, at the right level. Lengthy texts can be summarized into more meaningful learning materials. More capable dialogue agents mean that learner 'engagement' through teaching bots could become easier, better and cheaper.

Student free-text input answers can be interpreted. Formative feedback based on student performance can be provided. Automated mentoring and coaching may well get a lot better. Assessments can be automatically created and marked. Well-being dialogue bots will also become more human-like and useful. With generative AI, personalized learning just got a lot easier, so online learning just got a lot faster and cheaper.

AI can take existing documents, PowerPoints or videos and create credible online content. It can do this at a fraction of the

speed and cost of traditional methods. From LLMs it can also create content quickly on almost any subject. This creates the need for a more agile approach to online content production, where one has to adopt a leaner team, different processes and a mindset that focuses on fast delivery. We have also seen how AI now provides software that can also create content from scratch. This area is developing fast – very fast. Online learning content can be created with little SME input.

Prompting

ChatGPT is a complex piece of software that sees text as a set of complex mathematical relations, trained using a massive dataset of text from the internet, then fine-tuned by real humans, to generate text and images.

Before starting with such tools, understand how they work. They are not search or truth engines, as they ‘generate’ text and images. This is not like the sampling of music or pulling bits out of a database of text. It has been ‘trained’ to generate text, word by word. These generative tools are likely to make the generation of text and other media much, much faster and cheaper, if used to good effect. But how do you use them to good effect?

A prompt is quite simply a statement of intent, what you want to ‘say’ to the AI. Prompting is an art and a science. You need to experiment, play and track your results.

First, though, to prompt like a pro you need to be structured in your prompting. Here’s one structured approach:

- Define the goal
- Provide context
- Ask the question
- Add instruction if needed
- Add emotional prompts
- Use images

At a more detailed level:

1. Write clear, specific, fulsome and demanding prompts; don't hold back
2. Include 'context' for prompt, then 'parameters'
3. It is best used in dynamic dialogue, remember the word 'chat' in ChatGPT, not single prompts to produce single blocks of text
4. Play with and iterate your prompts, adjusting, adding and subtracting as you go
5. Iterate specifically to correct mistakes giving feedback
6. Use clear parameters to fine-tune requests
7. Give examples of what you want in your prompt
8. Explain the context
9. Try some simple logic in prompts, especially AND for what you want and NOT for what you do NOT want
10. Let the professionals in each field do the prompting. A writer, graphic artist, marketing or learning professional will have the existing knowledge to prompt well
11. Use prompts you find online, that others have found useful, as this is an expanding skill
12. Use in dialogue between stakeholders to lubricate discussion and ideas
13. Use to get the creative juices going to generate new possibilities and variations
14. Play with style, tone, look and feel
15. Refine, refine, refine

Above all, just write your prompts in plain language and in a LOT of detail.

Up-front learning design

A learning initiative is only as good as its original intentions. Up-front analysis and planning can be aided by generative AI.

Try creating course titles, course descriptions, learning objectives, competences, skills, syllabus structure, course structure, outlines, instructions for your learning initiatives,

even provide a motivational ‘why you should learn this’ message.

Create learning content

Create prompted content in good, well-structured, concise prose on any topic. This is perhaps its most powerful use as it draws upon a vast set of existing knowledge.

You can use AI to generate comprehensive lecture notes and effective whiteboard outlining strategies, suggest projects and assignments, create insightful prompts for roundtable discussion, transform text descriptions into digestible bullet points, simplify complex concepts for easy understanding, use real-world examples to teach principles, unpack topics through case studies, create stories and job aids, get you started on a paper, craft an engaging conference presentation, write effective dialogue for specific scenarios, explain complex topics to different ages, summarize articles and books.

Create learning activities

Far from being only a text or image generator, generative AI is an ideas generator that can stimulate learning.

Try icebreaker questions, novel class-starting methods, examples, activity instructions, roleplaying scenario generation expanding notes on a topic, debate preparation and execution, oppositional debate – ‘debate me on this motion’ – persuasive argumentation – ‘convince me that’ – mock interview role-play, playscript creation, video script for a role-based activity, compare x and y.

Create additional resources

Create an abbreviation list, glossary, article vocabulary and definition list, movie recommendations for a subject, book recommendations for a subject.

Produce images

Image prompts need far more specific prompts and iterations. Think of these as initial suggestions that need a lot more detail.

Start with the subject, what you want the image to show or convey, specify setting, purpose and quality, create header images, create images that directly aid the learning, not just illustrative, ask for representational style of image (photorealistic, drawing, cartoon, crayon), artistic style of image (pop art, impressionist, futurist), style of a specific artist (Van Gogh, Warhol), white space or a specific background, a specific mood or emotion, specific perspectives (birds-eye, isometric, through a porthole). Then, add to and modify the image as you go with extra layers and objects, specifying what you want to exclude. Be careful with words in images and do not expect vector images unless specified. And check the terms of service of the tools you are using and that you have the commercial rights to use those images.

Create assessments

Much attention has been given to cheating in assessments, but you can also create new assessments using generative AI.

Prompt for article-based questions, different education levels, true/false questions creation, fill in the blank exercise creation, MCQs creation, interactive quiz creation, essay grading feedback, specific feedback, model answer provision, assessment rubric creation, full assignment creation, assignment instruction and style recommendations, student assignment error correction and recommendations for improvement.

VOICE, STYLE AND LENGTH

It is not just the text itself that is generated but the tone of text: voice, style, level and so on. Generative AI will produce good

grammar and well-structured prose but you can also prompt for style using many parameters.

Prompt for explanations to different age or education levels, explanations for novices, outlines for specific roles, explanations for professionals, to provide an organization style outline, adjust level of formality, rewrite in different style, provide a humorous rewrite, give an expert perspective explanation, or first-person to third-person conversion.

MARKETING

When it comes to launching and marketing learning initiatives, these tools can be used to create, sharpen and deliver launch material. You can create great marketing material and hone your message to increase engagement.

Try learning initiative brand names, course marketing email creation, email shortening, draft reminder emails and then modify slightly to serve as follow-up emails, email subject line suggestions, course promotion tweets, call-to-action phrases, nudges and practice reminder notifications, or even generate blog posts. Then there's translation.

Learning science and pedagogy

Don't create faster horses, which Henry Ford thought would be the result of asking customers to determine what they wanted in the future. Be more transformative. We must see ourselves as not being in the content creation game, even the generative content game. We are in the generative learning game.

Bake learning science and pedagogy into the tool that creates content. This can be learning text, images, audio and video in any style, at any level.

Summaries

Walk into almost any large organization and you'll encounter a mother lode of documents and PowerPoints. They are often overwritten and far too long to be useful in an efficient learning process, where less is usually more. That does not put people off, and in many organizations we still have slide decks with over a hundred slides delivered in a room with a projector, masquerading as education and training. The trick is to use AI to automate the condensing or shortening of all of this text, down to the 'need to know' essentials.

To summarize or précis text documents down in size, a first human edit is always advisable. No matter what AI techniques you use to précis text, it is wise to initially edit out the extraneous material by hand – the content that learners will not be expected to learn. For example, supplementary information, disclaimers, who wrote the document, citations and so on. With large, well-structured documents, PDFs and PowerPoints, it is often easy to just identify the introductions or summaries in each section and use them as ready-made summaries of the essential content for learning. But whatever the source, with AI, garbage in, garbage out (GIGO) is the rule. Regard this step as human data cleansing! Now you are ready for further steps with AI.

Extractive AI summarization keeps the text intact and simply removes what it judges to be the less useful content. This technique uses AI to produce a summary that keeps the sentences intact and only 'extracts' the relevant material. This is especially useful where the content may already have been subjected to regulatory control and approval (approved by expert, lawyer, regulator). It may also have SME approval. The advantage of this approach is that you retain the integrity of the original text. This technique changes nothing; it simply shortens the existing text.

Abstractive AI summarization produces a summary that is rewritten and uses a set of training data and machine learning to produce a précis. Note that this approach needs a large subject-specific training set. By large we mean as large as

possible. Some of the training sets for existing models are literally gigabytes of data. You will also need some heavy computing power, hardware and time. That data may also have to be cleaned. Abstractive tends to use deep learning models and neural networks to understand the content then generate summarized content as a précis. Free from the constraint of having to be loyal to the original structure, these algorithms will write their own abstract, getting down to the real essence of the text. This more powerful technique is more likely to provide a tighter, more suitable output for learning – shorter and a more optimal distillation of the meaning.

This is all useful in increasing the productivity of any educational or training design team, as you dramatically shorten this necessary editing task. On large PDFs, not uncommon in compliance and statements of work, these techniques work well. Summarizing also works well with any text from articles, papers, books, even PowerPoint text and video transcripts. It can also be useful in that it surfaces overwriting, repetition, even errors.

A point that is often overlooked is that this is wholly in line with the psychology of learning, which screams ‘less is more’. Our limited working memories, along with the need for chunking and retrieval, make it essential to be precise and as short as possible with learning content. Many courses are overlong with content that is not essential and will soon be forgotten. What learners and businesses want is the crisp essence, what the learner needs to know, not the padding.

Questions

Questions can be created with good design built into the tool. The pre-prompt basically includes all the good pedagogical and design rules for good questions, whether they be true or false, multiple choice or open input. You can specify action-oriented questions to make the course more relevant and embody

techniques that enable transfer. These can be created directly from the material input into the generative AI tool.

With much online learning, learners just skate over the surface. For all the talk of learning experience systems and ‘engagement’, if all you serve up are flat media experiences, no matter how short or micro, with click-through multiple-choice or, worse, drag and drop, you will have thin learning. We know that what counts is effortful, desirable and deliberate practice, and the way to make such systems work is to focus on effortful ‘learning’ experiences, not just media production.

We tend to think that we learn just by reading, hearing and watching. When, in fact, it is other, effortful, more sophisticated practices that result in far more powerful learning. Learner surveys and happy sheets have been shown to be poor measures of how and what we actually learn and very far from being optimal learning strategies.

It is the *effort* made to ‘call to mind’ that makes learning work. Even when you read, it is the mind reflecting, making links, calling up related thoughts, that makes the experience a learning experience. This is especially true in online learning. The open mind is what makes us learn and therefore open response is what makes us really learn in online learning.

You start with whatever learning resource, in whatever medium you have: text (PDF, paper, book...), text and graphics (PowerPoint...), audio (podcast) or video. By all means read your text, go through a PowerPoint, listen to the podcast or watch a video. It is what comes next that matters.

With AI tools, online learning with ‘open’ input is created quickly and open input created by AI can drive the learning. The answers, short or long, are also interpreted, semantically, by AI. You literally get a question and a blank box into which you can type whatever you want. This is what happens in real life – not selecting items from multiple-choice lists. Note that you are not encouraged to just retype what you read, saw or heard. The point, hence the question, is to think, reflect, retrieve and recall what you think you know. AI helps interpret

answers through natural language algorithms and semantic interpretation.

You are asked to tell us what you actually think and know. It is not easy and people take several attempts. That is the point. You are cognitively digging deep, retrieving what you know and having a go. As the AI does a semantic analysis, it accepts variations on words, synonyms and different word orders. You can't simply cut and paste, and when you are shown the definition again, whatever part you got right is highlighted.

It is a refreshing experience in online learning, as it is so easy to click through media and multiple-choice questions thinking you have learned what is expected. Learners are easily fooled into thinking they have mastered something when they have not.

Supplementary content

Supplementary content creation can allow access to created summaries or chatbot functionality for more breadth and depth, whether case studies, worked examples, FAQs, job aids, infographics, policies, archived information, standards, manuals, product information, PDFs or slides. All of this can use generative AI to deliver at different levels in different styles or to audiences with specific accessibility needs.

Assessments

Assessments can be created with fully automated marking or a rubric for marking. These can include a range of question types from multiple choice to open input. Moving beyond questions, scenario-based learning can also be generated, with various levels of sophistication from simple, linear process scenarios to full branched scenarios for complex conditional tasks.

Flashcards

Flashcards can be created automatically in a variety of different styles – categorized as questions, facts, examples and so on. These can be subsequently used in spaced practice.

Translations

Translations into many languages can be done in minutes, with a check on cultural differences. Even the marketing of the course can be generated using these tools.

How do you make sure that language learning content is suitable for the learner as a beginner, intermediate or advanced learner? Generative AI also allows you to deliver at different levels in different styles, also adapting for those who have English as a second language, dyslexia and other neurodiverse learners.

Branching scenarios

Scenario learning within courses is an opportunity to embed practice activities within AI created courses. This technique has been used in learning technology since the mid-1980s, and there are established structures and research around their use. They are particularly useful in retail, healthcare and other complex real-world training, where much of the learning needs to be person- or process-focused.

The point is to make realistic decisions and see the consequences of those decisions. The learning theory behind this approach is to make learning not easy but deliberately difficult, to make you think and make the effort to think.

This is very much an 80/20 exercise, where 80 per cent of real-world tasks can be represented by 20 per cent of the possible scenarios. This requires some analysis around the most common scenarios. Any medium can be an event within a branched structure: videos, audio, animations, images, text.

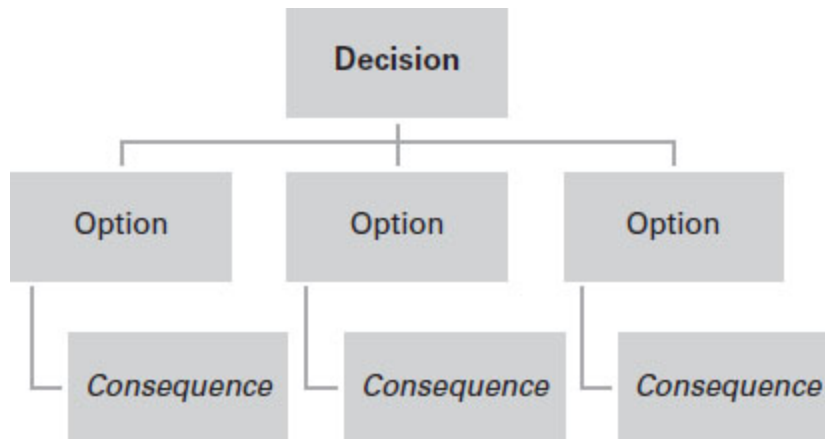
There are many possible structures but here is a manageable set of four designs.

1. Mini-scenarios

This is a single decision where their consequences are shown, with feedback ([Figure 9.1](#)). It is essentially a MCQ but worded as a scenario or problem, where a decision or choice has to be made.

These are good for self-contained, single learning point issues and can be repeated with different variables (types of task, customer or patient, etc.). Repeat until right choice is made, the act of choosing the right choice is important. Make sure the challenge is realistic, and that the consequences are seen as real. This is a good for repeated practice of an idea of part of a process.

Figure 9.1 Simple scenario



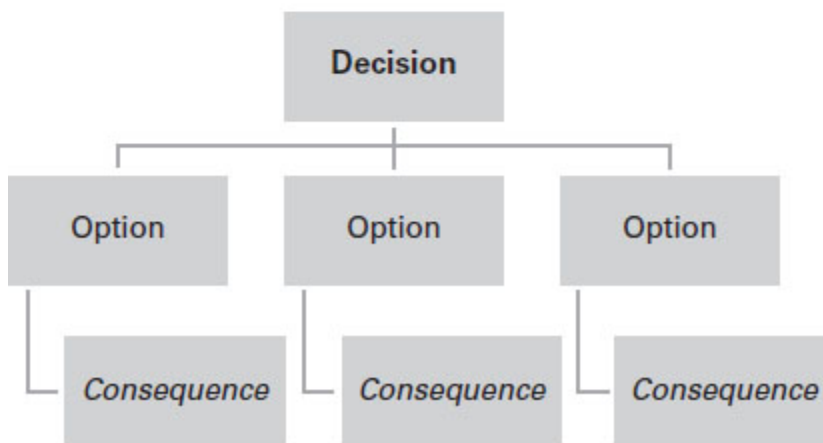
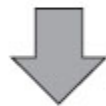
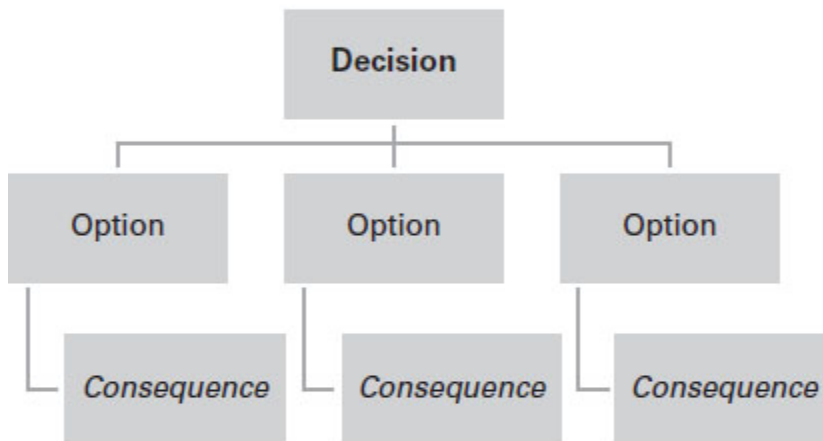
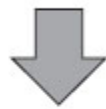
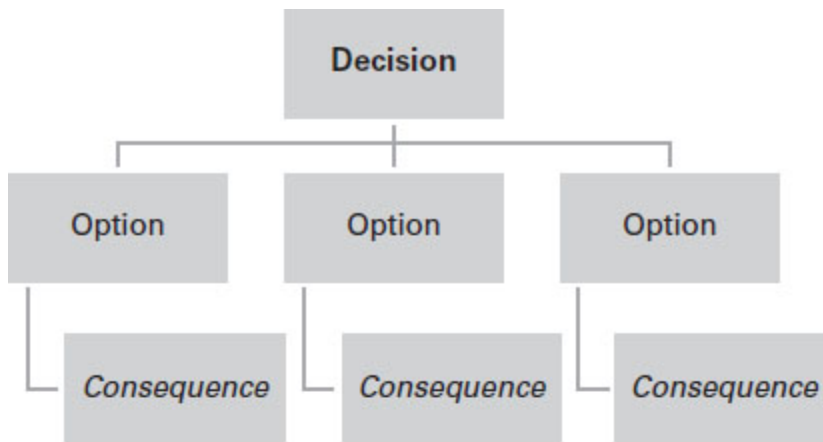
► Figure 9.1 details

2. Linear scenarios

It is a mistake to over-design scenarios and linear decision-making in a stepped process is often enough. This can be employed in well-defined stepped procedures that are common in healthcare.

A series of decisions, with feedback, that progress linearly towards a conclusion ([Figure 9.2](#)).

Figure 9.2 Process scenario





► Figure 9.2 details

Until the task, process or procedure is complete.

3. Branched scenario

Here different decisions have different consequences down a dendritic or branched structure ([Figure 9.3](#)).

These are good for more complex processes and procedures where there are conditional decisions to be made by the learner. They are also good for diagnostic procedures, where you make decisions based on the person or data presented.

[illegible]

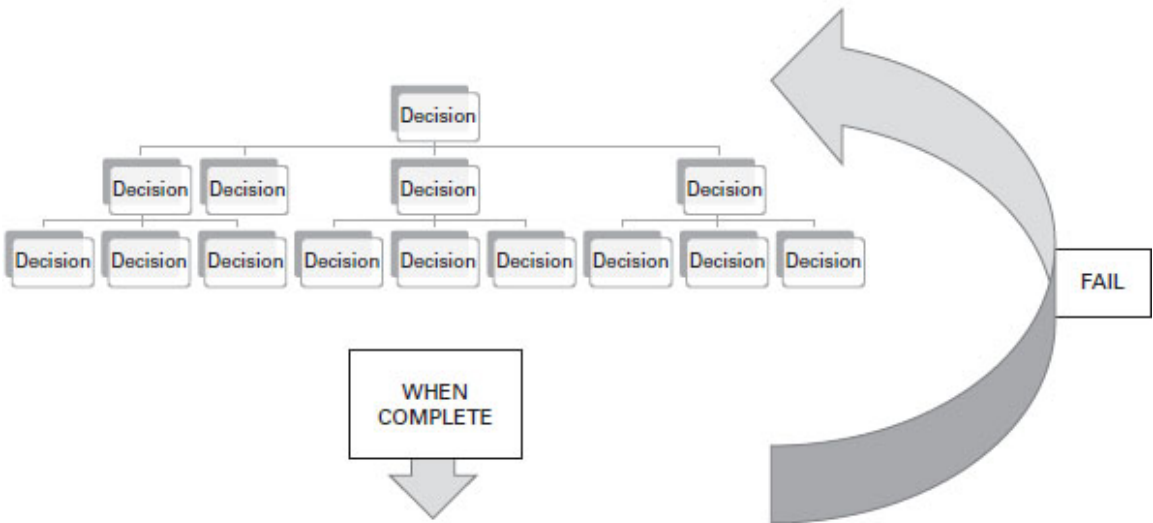
The danger is in the branched structure getting too complex but one can control this by taking a more structurally scripted approach to bring it back with an overall feedback event.

Here we create levels, as commonly used in computer games. They are good for tasks that have specific, major stages of competence.

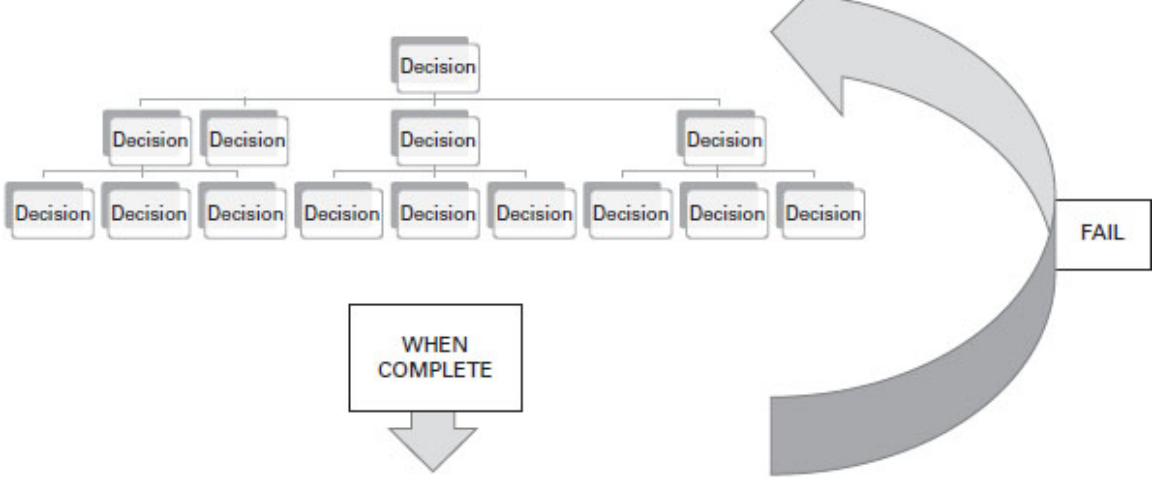
So we have three or more levels and you progress ONLY after completing each level ([Figure 9.4](#)).

Figure 9.4 Branched scenario with levels

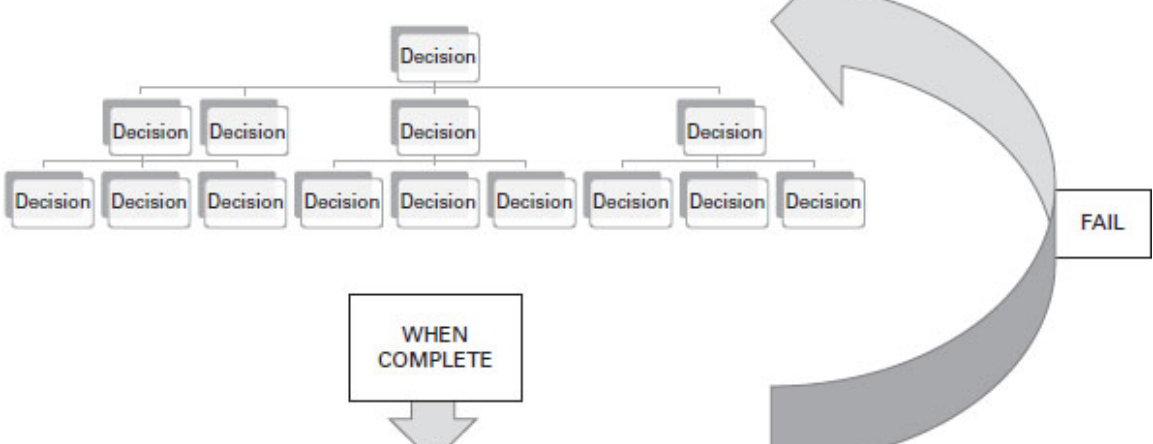
Level 1



Level 2



Level 3



► Figure 9.4 details

AI as agile production

We have seen how the promise of AI to increase productivity and produce new forms of pedagogy in online learning is real. We have also seen how the skillset will change, as AI enters the world of learning. This brings us to the idea that AI can act as a catalyst for *agile* content production.

We must get past the model that says it takes months to produce content at prohibitive costs. Agile production needs agile tools, agile production methods and, above all, agility of thought and mindset. Organizations have traditionally moved much faster than training delivery. That means we're often out of phase with the business. An agile mindset and production attitude allows us to transcend that historical gap. Once we become responsive to the business, it will respect us more as we become more aligned with the actual speed of the business.

The tools will do most of the design and build, using AI techniques. An agile approach to production and project management was intrinsic. No instructional design scripts are necessary, as the software built the content so quickly that quality assurance could be done on the actual modules.

An essential ingredient in this project is an agile project manager. AI is now being used to create content and add curated resources at the click of a button. Want an audio podcast or audio introduction to the course? It takes seconds using AI to do text to speech. The time savings are enormous, as well as cost savings. AI is used not only to identify learning points; it also creates audio, constructs the questions, recognizes speech input, assesses open input (words or full short answer) and locates links to external resources for further learning.

One important lesson in being agile is the GIGO rule. When you use an agile production process, you need agile preparation. An intensive look at the input material pays dividends. Eliminate all of that extraneous material and text, cut until it bleeds and cut again, catch those pesky spelling and punctuation errors, make sure things are consistent. The source material needs to be edited down to the essential ‘need to know’ content. This source material also needs to be managed going forward, to keep things up to date. This needs to be the responsibility of individuals within the organization.

Agile is as much a state of mind as process. Yet L&D still works to an old model of months, not minutes. We procure slowly, prepare slowly, produce slowly and deliver slowly. Despite relentless calls to align with the business and respond to business needs, we are too slow. L&D needs things in days, yet we deliver in months. This is what needs to change.

In response, online learning has traditionally been rather slow in design and production, but we can now use AI to create content quickly, to produce agile learning and data that allow us to adapt to new circumstances. ‘Agile’ captures, in a word, what is now necessary. The days of seeing online learning production as some sort of feature-film project with matching budgets and timescales – months – should be re-examined. Of course, some high-end content may need this approach, but much can be automated and done at a fraction of the cost, in a fraction of the time.

You will find that the quality assurance process is very different, with quick access to the actual content, allowing for immediate testing. In fact, AI tends to produce fewer grammatical and punctuation mistakes, as there is less human input – always a source of spelling, punctuation and other errors. The advantage of using AI-generated content is that all sides can share produced content to solve residual problems on the actual content, as seen by the learner.

Death of the SME

SMEs? Don't use them. If you have other sources of knowledge and practices, use them first. Statements of work, procedures, documents, books – use them. They are the lifeblood of generative AI for learning. Indeed, the curation, preparation and ownership of documents and other sources is essential.

The story for decades has been: get the SME, milk them for what they know, then use them to validate course content. First, they tend to veer into designing the course and often don't use or understand more sophisticated AI solutions.

They're often people actually out of touch with contemporary practices, as much of their knowledge is tacit and they can't fully express what is needed. Experts in this field say you need several SMEs to get the truth. Those who take SME input seriously, like Richard Clark and Conrad Gottfriedson, recommend three or four, as you will find that they all have things missing from their testimony, even contradictory.

The SME is often seen at the top of the pyramid, the input. But those who have studied this in detail have a different story to tell. They see the SME as the root cause of a serious problem – over-engineered, over-long courses that lack focus and don't get anywhere near actual competence and performance. SMEs tend to want to 'cover' everything. Unaware that 'less is more' and unable to focus on the 'need to know' stuff, they often see more as more.

In failing to separate the critical from non-critical, courses become the solution to everything and performance support is ignored. If it is critical, you need some serious effort to train out the risk with a course. As for the rest, use performance support. SMEs never really get performance support as a solution. Everything has to be a course and, of course, the courses must cover everything.

They are also often difficult to pin down, making the project dependent on their availability and needed for updates.

Switching to documentation is useful, especially if they are versioned, and updated, so usable in AI-driven performance support.

If you do need to use an SME, use them to validate whatever you've done, after you have done it.

Video and AI

Video, through YouTube, Vimeo, Netflix, Amazon Prime and other services, is the medium of the age and AI is the technology of the age. Combine the two and you have a potent mixture. But how do these two worlds interact? Well, AI has helped producers edit, colour balance, audio mix, caption, transcribe, translate, filter, search, personalize, analyse and predict.

In the creation of video, AI has already crept into video production tools. There are tools that allow you to edit video much faster and to higher quality. Different cameras shoot different colour balances? That can be fixed with AI. Same actor in different scenes with different skin tone? That can be fixed with facial recognition and skin tone matching – using AI. Need your music mixed down behind dialogue? Use AI. AI is increasingly used to fix, augment and enhance moving images. Many pop videos and films have been made using AI tools. There are even complete films made from AI-created scripts.

Once you have created a video, AI can also add captions. This type of software can even pick up on dog barks and other sounds and is now standard on TV, YouTube, Facebook, even Android phones, increasing accessibility. It is also useful in noisy environments. Language learners also commonly report captioning as having benefits in self-directed language learning. Although one must be careful here, as Mayer's research shows that narration and text together have an inhibitory effect on learning (Mayer and Johnson, 2008).

Speech to text is also useful in transcription, where a learner may want the actual transcription of a video as notes. Some

tools, like Glean, take these transcriptions and use them to give students help with note taking. Others produce online learning to supplement the video with effortful learning. The learner watches the video, which is good for attitudinal and affective learning, even process and procedures, but poor on semantic knowledge and detail. Adding an online treatment of the transcript, created and assessed by AI, can provide that extra dimension to the learning experience.

Once you have the transcribed text, translation is also possible. The technology promises to deliver speech-to-speech translation with an end-to-end translation model that can deliver accurate results. Advances like these will allow any video to be translated into multiple languages, allowing low-cost and quick global distribution of learning videos.

Ever thought why YouTube and other video services prevent porn and other undesirable material from appearing? AI filters identify offending videos through image recognition to search and then delete. It is not that AI does the whole job here. Removing ‘dick pics’ and beheadings relies on algorithms and image recognition, but there is also community flagging and real people sitting watching this stuff. Despite this, AI is increasingly used to protect us from undesirable content.

Video is great at showing processes, procedures, real things moving in the real world, drama, even much-maligned talking heads, but it is poor on many other things, especially concepts, numbers and abstract meaning. The mistake is to see video as a learning medium *in itself*. Video is a great learning medium if things are paced and reinforced, but made greater if the learner has the opportunity to supplement the video experience with some effortful learning.

How do we make sure that video learning experience is not lost and forgotten? The evidence is clear: the learner needs some effortful learning; they need to supplement their video learning experience with deeper learning that allows them to move that experience from short- to long-term memory.

This is why video alone is rarely enough. One solution is to supplement video with learning that focuses on reinforcing the episodic and semantic knowledge, so that two plus two makes five.

The solution is to automatically grab the transcript (narration) of the videos. Some transcripts were already available, and for those that were not, one can use the automatic transcript service on YouTube or other tools. This transcript was put through AI to automatically produce online learning with open input questions to increase retention and recall. This allowed the learner to both watch the video (for process and procedure) and do the active learning, where they picked up the semantic knowledge, as well as reinforcing the processes and procedures.

In a nurse training video on allergy tests, where the nurse administers allergens into the skin of the patient and the reactions are recorded, the video shows the nurse as she gets the patient comfortable with a pillow under his arm. She then asks him some questions: 'Any lotions on your skin? Taken any antihistamines in the last four days?' Other important learning points are to blot (not rub), tell the patient not to scratch and so on.

The learner gets the best of both worlds: the visual learning through video and the semantic learning through AI, all in the right order and context. It is important that the retrieval learning is very closely related to video content, which is why the transcript is used.

How do you ensure that your learning is not lost and forgotten? Strangely enough it is by engaging in a learning experience that makes you recall what you think you have learned. AI can grab the transcript of the video, put it into an AI engine that creates a supplementary learning experience, where you have to type in what you 'think' you know. This powerful form of retrieval learning not only gives you reinforcement through a second bite of the cherry, but also consolidates the learning.

Research has shown that recalling back into memory – literally looking away and thinking about what you know – is even more powerful than the original teaching experience or exposure. In addition, the AI creates links out to supplementary material (curates, if you wish) to further consolidate memory through deeper thought and processing.

Video is a fabulous learning medium – witness the popularity of YouTube and the success of video in learning – although there are some principles that make it better. When supplemented by AI-produced content, you get a double dividend.

Interleaving and spaced practice

Interleaving is counterintuitive and is rarely used as a learning strategy, yet it has a strong evidence base. This is a strange thing. Interleaving, as opposed to blocked learning, feels wrong, feels disjointed, almost chaotic. Yet it is much more effective. It seems to fly in the face of your intuitions. Yet it is significantly more efficient as a learning strategy. But how often do we see interleaving in classrooms, homework or online learning? Hardly ever.

This is another technique that cognitive science has thrown our way, which can be implemented through AI-driven algorithmic technology. The interleaving itself can be created, along with optimized interleaving to increase learning retention and recall.

Another more pedagogically sound technique that is known to increase retention and efficacy in learning – spaced practice. It is not simply a matter of repetition or following a predetermined pattern of exposure. Spaced practice is much more than this. At its simplest, it is the recall, rehearsal, revision, application or deliberate practice of knowledge or skills spaced over time to reinforce and consolidate them in long-term memory for automatic recall. Ultimately it is about performance, using evidence-based learning theory to apply the

most effective techniques that lead to measurable performance. AI can help deliver this practice.

So, given that millions of teachers, lecturers, trainers, coaches and instructors are employed in the learning game, it is perhaps surprising that relatively little or no attention is paid to the idea of spaced practice. One could argue that without knowledge of this principle and its causes, those who teach are missing a key component in the process of learning. All attempts at practice, revision and application are down to the learner. That shouldn't distract us, however, from making the effort to improve the situation, with or without technology – and with AI, technology can be smartly employed to personalize and deliver the solution.

In 1885 Ebbinghaus published *Über Das Gedächtnis* Memory (published in English in 1913), a groundbreaking work which laid the foundations for the practical science of memory. He gave us the application of the scientific method to the measurement of memory and the famous *forgetting curve*, which even now has the ability to startle those who first encounter the precipitous nature of forgetting.

A solution to the problem of the failure to elaborate and shunt learned knowledge and skills from working memory to long-term memory is to repeat, review, revise, rehearse, recall and practise at spaced intervals in the future.

We can take advantage in a formal, pushed system to push out cues and active recall on knowledge that users have shown they do not know, do not know well or do not feel confident about.

Although students often perform better immediately after '*massed practice*', a single bout of practice, they forget quicker and perform poorly in later tests than '*spaced practice*' students (Keppel, 1964). This is why much end-point assessment is often short term and short-sighted.

Spaced practice can therefore be selective in that items that are clearly known can have less weight than items which the user is not confident about, had difficulty in learning or clearly

does not know. Here spaced practice can involve algorithmic inferences that use performance data about each individual learner, then route that learner through a series of items, or network of knowledge, based on optimizing their learning and spaced practice.

Spaced practice usually (not exclusively) requires technology and approaches that deliver spaced events to the learner. There are many ways to use technology to push and manage spaced practice, from the simple to complex. In all cases pedagogy is the driver, AI technology the accelerator.

As AI technology develops, significant increases in retention and productivity could be realized. We can remind learners about tasks, activities and push snippets of learning topics to them at timed intervals. Systems like this can insert new life into previous learning with bite-size questions, tasks and activities to help refresh the learners' memory.

One problem with formal, pushed methods is *habituation*. This is seen in pop-up help or tip systems, where the user tires at being interrupted and starts to ignore the events that are pushed to them. This motivational problem can be solved by not being too aggressive with the push techniques, keeping them short, varying them and making them worthy of attention.

One can categorize items as unknown, not confidently known and confidently known. Indeed, this categorization can be taken further with degrees of tested and subjectively assessed knowledge. However fine-grained the analysis, an adaptive system can deliver spaced-practice items based on a mathematical judgement (algorithms) that decides on the optimal delivery of those items. This optimum delivery schedule is based on what the science tells us, and data drawn from learner performance and learner perceptions. Duolingo uses a half-life algorithm that assesses the half-life of every item linked to every learner. It is in the language learning business, where forgetting unfamiliar content is the problem, so tackling forgetting is the solution.

Such systems deliver personalized spaced practice that is determined not by a preordained sequence but by identifying what the user can actually recall going forward. A useful technique is to gradually drop items that are well known. This selective presentation of items gradually drops items off the conveyor until no items remain. Like a satnav, it delivers the right corrective action depending on where it thinks you are. Sometimes it may deliver in a fairly predictable fashion as you are doing well; at other times it may deliver more items over a longer period if you're showing signs of poor recall or confidence.

By distributing workload over time, this method can contribute greatly to reducing worry and cognitive overload in students. Rather than procrastination leading to cramming, we have a balanced workload that consolidates and improves performance smoothly over time. Far from demotivating learners, spaced practice can motivate by raising self-confidence, meeting expectations and achieving goals.

Spaced practice is arguably the most powerful, yet most overlooked, benefit in learning and true performance. Implemented properly it is possible to have huge gains in productivity, namely the retention and recall of whatever has been learned. One could go further and say that without a spaced-practice strategy, there is no learning strategy.

Cognitive science has given us many solid pieces of evidence-based learning strategies that can only now be delivered through technology, as we now have tools that are smart enough to do what is required on interleaving and spaced practice. As cognitive science continues to inform us about optimal learning strategies, these can be captured and baked into AI technology to deliver, in a personalized fashion, learning that greatly increases efficacy.

Assessment

Whether in educational institutions, candidate assessment in companies, or competence and professional assessment, high-stakes assessment is big business. Yet it can be an expensive, inefficient and slow business. Creating assessments, travelling to and from assessment centres, arranging invigilators, identifying candidates, preventing cheating and marking, all make this a stressful and fraught business. AI has begun to have an effect on all of these to allow online assessment, improve ID checks, check for cheating and marking. What were largely human roles are not being entirely replaced, but they are being automated in part. The increase in productivity should result in better, faster and cheaper assessment.

Conclusion

AI does more than just create content, in minutes and not months, it can also create pedagogically rich content, with a focus on action transfer and practice, along with interleaving and spaced practice. This is not about creating more content but generating better content, with better in-built learning science and pedagogy. The point is not to generate content but generate better learning.

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10

Adaptive learning

The human brain is arguably the most complex thing we know of in the universe. We all have one that is absolutely unique, that does not like impersonal, over-long, linear, flat experiences. Yet in learning we feed it exactly that – long lectures, flat pages of text, one-size-fits-all learning experiences.

Learning is a deeply personal experience and most learning theory, in terms of attention, deep processing, self-generation, practice and so on, points towards the need for sensitivity around individual needs. This is not always practical and there is nothing intrinsically wrong with direct instruction and whole-class teaching, as one may have limited resources. However, personalization is the one great gift that online learning, fuelled by AI, offers lecturers, trainers, teachers and learners. In classrooms teachers talk about differentiation, then struggle to differentiate with 30-plus students. In universities the mass that turns up to lectures, sometimes in the hundreds, is the opposite of personal learning. None of this can be described as ‘personal’ in any significant sense. But ‘personal’, when delivered online, can mean relevant, timely, self-determined, self-generated, targeted, private, even intimate. This is what AI through powerful chatbots and dialogue adds to the learning.

Adaptive learning

Adaptive learning has been used in K12, higher education and corporate L&D, as all are looking for faster, more efficacious learning, to deliver the right learning, at the right time, to the right learner. Yet conventional e-learning has largely been one-size-fits-all, the same for everyone, with a focus on the presentation of content as media that generally ignores what is

already known. Adaptive fits to the individual learner, as opposed to making the learner fit to the content, showing only what is necessary and what is right for that person, at that particular time. It is diagnostic and reacts to what the learner does, as its unique strength is its sensitivity to the actual learner's needs. Some systems do this through constant content curation at the point of need as the learner progresses through the learning experience. It can also correct misconceptions.

The linear approach to teaching is common in learning, uncommon in most other online experiences. Playing to a 'mean' or 'median' learner leads to increased failure for less competent learners and wasted time for the more competent. Some students learn faster with less support, and some learn more slowly and need more support. On the whole this adaptive process can save huge amounts of time.

In addition, there is the engagement issue. One is not being subjected to learning one already knows or suffering from rejection and failure. It can build in pedagogic models, such as spaced practice, interleaving, deliberate difficulty and effortful learning. These systems also provide advanced analytics. This form of adaptive 'tutoring' can be delivered on *scale*.

One scalable system that supports one facet of personalized learning is 'adaptive' learning. This approach is sensitive to personal needs, potentially accelerating success and preventing failure. The personalization can depend on the individual, aggregated data and the algorithms used to determine such paths. In an advanced system, there is no linear course, only vectors through a domain of learning activities.

As we have seen, 'personalized' learning has a wide meaning around sensitivity to the individual learner and around differentiating learners. 'Adaptive' learning is more specific, where the online learning adapts to the learners' needs as they progress through a computer-based learning experience. Having said this, there is still a wide variety of adaptive approaches and methods, from simple pre-test assessment to full-blown dynamic adaption, with lots in between.

The most advanced systems adapt the online experience to the individual's needs as they learn, in the way a personal tutor would adapt. The aim is to provide what many teachers provide: a learning experience that is tailored to the needs of you as an individual learner.

Types of adaptation

Adaptation can be many things, but at the heart of the process is a decision to present something to the learner based on what the system knows about the learner and other elements, such as the instructional activity, the topic domain, the educational domain, the type of learning, the context and so on.

It is helpful to categorize different adaptive approaches. We can begin by creating categories for where in the educational or learning sequence the adaptivity occurs. If we think in terms of a standard school or college course (although this could also be applied to a module in a course or a chapter in a book):

- pre-course adaptive: a pre-course assessment or assignment of learners to a group;
- in-course adaptive: adaption occurs as the learner progresses through a course;
- post-course adaptive: adaption is used, e.g. to reinforce memory;
- continuously adaptive: adaption occurs constantly through a learner's journey, with data and instructional activities shared across courses and programmes.

It is important NOT to use learning styles and preferences for adaptive purposes, unless a specific learning difficulty or accessibility issue is a real problem. Systems such as universal learning design make the mistake of seeing preferences and the multiple production of multiple media and routes through learning as the solution. In fact it is a costly mistake as it does not optimize learning.

Pre-course adaptive

In this category, data gathered prior to a learner taking a course is used to predefine the experience for them. Typical approaches include:

MACRO ADAPTION OF LEARNING JOURNEY

Data about the student is used to define the learning journey at the macro level, recommending skills, courses, even careers based on individual needs. This is not easy, as these are high-stakes recommendations often made on limited datasets and using categories of skills that are far from agreed.

PRE-ASSESSMENT OR INFERENCE OF LEARNING PROFILE

Normally, the learner takes a ‘pre-test’ to create a profile before starting the course, then the system adjusts the learning activities based on this. The adaptive software makes a decision based on the data specific to that individual that was gathered at the start of the process. This is a highly deterministic approach that has limited personalization and learning benefits but may prevent many from taking unnecessary courses.

It is also possible to start this process with personal data, such as educational background, competence in previous courses and so on, and infer a learning profile from this.

In-course adaptive

In this category, adaption occurs while the student works through a course, and is informed by data gathered throughout their progress. This is often referred to as a micro-adaptive course, which adjusts the learner’s route frequently, or in real time. There is a spectrum of granularity of micro-adaptivity, ranging from the lesson level to the individual activity to within an activity.

In essence, the software is adjusting a sequence of either predefined activities or algorithmically generated assessments. The fundamental idea is that most learning goes wrong when things are presented that the learner is not prepared for. If the system can help the learner to resolve issues with prior learning, the new learning will also be resolved.

A different perspective is that the system attempts to generate an ideal level of challenge. One can use the idea of desirable difficulty to determine a learning experience that is challenging enough to keep the learner driving forward, but not so challenging as to demotivate them.

It is worth introducing AI at this point. Adaptive learning is how the large tech companies deliver to your timeline on Facebook/X, sell to you on Amazon, get you to watch stuff on Netflix. They use an array of techniques based on data they gather, statistics, data mining and AI techniques to improve the delivery of their service to you as an individual. Evidence that AI and adaptive techniques will work in learning, especially in adaption, is there on every device on almost every service we use online. Education is just a bit of a slow learner.

Decisions may be based simply on what the system thinks your level of capability is at that moment, based on formative assessment and other factors. The regular testing of learners not only improves retention, but it gathers useful data about what the system knows about the learner. Failure is not a problem here. Indeed, evidence suggests that making mistakes may be critical to good learning strategies.

Decisions within a course use an algorithm with complex data needs. This provides a much more powerful method for dynamic decision-making. At this more fine-grained level, every screen can be regarded as a fresh adaption at that specific point in the course.

AI techniques such as machine learning adaption can be used in systems that learn and improve as they go. Such systems are often trained using data at the start and then use data as they

go to improve the system. The more learners use the system, the better it becomes.

Another measure, common in adaptive systems, is the measurement of confidence. You may be asked a question then also asked how confident you are of your answer.

Good learning theory can also be baked into the algorithms, such as retrieval, interleaving and spaced practice. Care can be taken over cognitive load and even personalized performance support provided, adapting to an individual's availability and schedule. Duolingo, for example, is sensitive to these needs and provides spaced practice, aware of the fact that you may have not done anything recently and forgotten stuff. Embodying good learning theory and practice may be what is needed to introduce often counterintuitive methods into teaching, such as retrieval practice and spaced practice, that are resisted by human teachers.

Post-course adaptive

ADAPTIVE ASSESSMENT SYSTEMS

There is also adaptive assessment, where test items are presented based on your performance on previous questions. They often start with a mean test item then select harder or easier items as the learner progresses.

MEMORY RETENTION SYSTEMS

Some adaptive systems focus on memory retrieval, retention and recall. They present content, often in a spaced-practice pattern, and repeat, remediate and retest to increase retention. These can be powerful systems for the consolidation of learning.

PERFORMANCE SUPPORT ADAPTION

Moving beyond courses to performance support, delivering learning when you need it, is another form of adaptive delivery that can be sensitive to your individual needs as well as context. These have been delivered within the workflow, often embedded in social communications systems, sometimes as chatbots.

Continuously adaptive

Adaptive software can be applied within a course, across a set of courses but also across an entire curriculum – as many learning experiences as you wish. The idea is that personalization becomes more targeted and personal the more you use the system. Aggregated data from a single learner's performance on a previous course can be used, as can aggregated data of all learners who have taken the course. One has to be careful here, as one cohort may have started at a different level of competence than another cohort. There may also be differences on other skills, such as reading comprehension, background knowledge, English as a second language and so on. Nevertheless, adaptive is now being applied to more than individual courses. Entire degree courses are being delivered using this technology.

The aim of this technology-enhanced teaching system is to:

- increase attainment;
- reduce in dropout rates;
- maintain student motivation;
- increase teacher effectiveness.

One of the difficulties in adaptive, AI-driven systems is the creation of usable content. By content, we mean content as a network, with non-linear structures, assessment items and so on. Most teachers are unfamiliar with the process of creating content that is not linear but a network with dependencies, so most platforms have created suites of tools that allow instructors to create a network of content, working back from

objectives. Editors, conversion of content, automatic help with layout, a defined process, roles, permissions and even rights management are provided.

Beyond these results lies something even more promising. These systems throw off detailed and useful data on every student, as well as analyses of that data. Different dashboards give unprecedented insights, in real time, of student performance. This allows the instructor to help those in need. The promise here is of continuous improvement, badly needed in education. We could be looking at an approach that improves not only the performance of teachers but also of the system itself, the consequence being ongoing improvement in attainment, dropout and motivation in all students.

Underlying this approach is the idea that all learners are different and that one-size-fits-all, largely linear courses, delivered largely by lectures, do not deliver to this need. It is precisely this dimension, the real-time adjustment of the learning to the needs of the individual, that produce the results, as well as the increase in the teacher's ability to know and adjust their teaching to the class and individual student needs through real-time data.

It is already clear that this type of software can be, and has been, used across a wide range of subjects. As the software is content-agnostic and has been designed to run any course, it holds promise for adoption across entire institutions.

From LMS to AI

In a more agile, flexible working, always-on, vying for attention, pressured workplace, something more than a LMS or virtual learning experience (VLE) is needed, a warehouse for content, with scarcely any data or analysis. As old theories, such as learning styles and Kirkpatrick's evaluation, have waned and employees have become more networked and remote, the technology has matured. A consumer-level interface – mobile,

personalization, chatbots, curation, all producing data – allows AI to flourish in more flexible learning ecosystems.

When the LMS starts to give way to the LXP and AI, built around the individual and their learning experiences, it was then that AI was used to help deliver learning. Agile learning is possible, with responsive, adaptive, personalized delivery. What we see is the collapse and breakup of the course as the cardinal delivery mechanism in training. Personalized means just that, not a one-size-fits-all experience. It means smaller units of learning, in line with good cognitive psychology, cognitive engagement and an understanding of behavioural psychology.

LMSs were designed for a world before personalized delivery, smart software and AI. Many are still written in old technology that cannot meet the demands of this new world, inflexible and failing to meet the agile needs of education and business. The world has moved on in terms of expectations, while the LMS and its online content model has remained flat, superficial and static, insensitive to new interfaces, personalized delivery and advanced analytics.

LXPs are a response to what an AI-driven online world demands, namely technology that is more responsive to the needs of users, multiple systems and the real data needs of an organization. Learners need easier, more flexible, frictionless access with more personalized delivery. The learning world has learned from real delivery on the web, through Google, Facebook, Instagram, Twitter, Amazon and Netflix, and has seen that consumers expect a slick interface driven by underlying AI, access to content from a number of different sources, with advanced learning analytics, often including machine learning, that works on mobile. The traditional LMS comes far short of this.

One feature of this newer LXP world is xAPI, which allows one to track, in much more detail, from many different sources, learning experiences and content, even those outside of the LMS. It is much more holistic in its tracking of learning than its predecessor SCORM.

In parallel, we have seen the development of the LRS, which externalizes data, making it easier to use learning data as part of a bigger business data and integration effort. Learning can, by this route, become business savvy, directly responding to business needs and influencing business decisions.

Conclusion

Like a GPS in your car, an adaptive system can deal with millions of drivers simultaneously, all taking different routes towards different destinations. If any one of these drivers goes off course, the system immediately responds to get that driver back on course. The promise of adaptive learning is of autonomous systems that deliver massive amounts of high-quality learning at a very low cost per learner. Adaptive, personalized learning, when replicated across the system, may be the solution to the escalating costs in education, as it provides smart support at every point in the learning journey and is sensitive to the needs of a wide range of students – on scale.

11

Data analytics

Data is critical to the future of any organization. Or, more accurately, the analysis of data is critical to inform and shape the future of any organization. Learning data can reveal fascinating insights about that most important asset in an educational institution or business – its students or people. It allows you to adapt your learning strategy but also unlock insights that can power organizational decision-making.

Sources of data

Data at all sorts of levels in learning reveal secrets we never imagined we could discover. It reveals things to teachers, trainers, learners and managers, and as we do more learning online, searching, reading, watching, communicating and socializing, we create more and more data that provides fuel for algorithms that improve with big numbers. The more you feed these algorithms, the more useful they become.

Online learning, by definition, can produce data – one of the great advantages of being online. For many years data has been gathered and used in online learning. SCORM emerged as the *de facto* specification, making this data interoperable. However, something new has happened with AI – the awareness that the data produced by online learning is much more powerful than we ever imagined. It can be gathered and used to solve all sorts of difficult problems in learning, problems that have plagued education and training, such as take-up, formative assessment, dropout, course improvement, productivity, cost reduction and so on.

Data can be harvested at different levels, each increasing in terms of quantity, from:

- brains;
- learners;
- courses;
- across courses;
- organizations/institutions;
- groups of organizations/institutions;
- national;
- international;
- web.

All of these have and can be utilized to both create useful AI for learning, as well as optimize delivery to accelerate learning.

Data pitfalls

The old adage that ‘on average humans have one testicle’ is often used to show the dangers in drawing meaningless conclusions from sets of data. Averages can be dangerous, especially in the learning game. Data analytics has taken on the gloss and allure of AI, yet much of it is actually plain old data science and statistics. So, before you spend huge sums on learning analytics projects, consider the following.

They say that data is the new oil, but the danger may be that, in some cases, it turns out to be the new snake oil. It is often stored in odd formats and places, often old, messy, embarrassing, personal and even secret. To quote that old malapropism, ‘data is a minefield of information’. Primitive analysis may even be massively misleading, as the testicle comment amusingly exposes.

Data held by institutions on learners is often paltry. Universities, for example, rarely even know how many learners turn up for lectures, and the actual data, when collected, often paints a depressingly impoverished picture. So, another problem with the use of data in learning is that we have so little of the stuff. Old specifications such as SCORM, which has been around for 20-plus years, literally stopped the collection of data

with its limited focus on course completion. This makes most data analytics projects, even in large organizations, rather difficult. This limited data is certainly not as large, clean and relevant as it needs to be to produce deep and genuine insights. Other data sources are often similarly flawed, as there is often little in the way of fine-grained data about actual performance. These small datasets are often messy, poorly structured and not insightful.

Data is also often not as clean as you think it is, with much of it in odd data structures, odd formats, encrypted and different databases. Just getting a hold of the stuff is difficult. Then there is the problem of relevance and utility, as much of it is old, some of it useless and much of it messy. In fact, much of it could be deleted. Much of the stuff we simply haven't known what to do with, don't clean and don't know how to manage.

Problems can also arise around data that can be embarrassing, even dangerous. There may be very good reasons for not opening up historic data, such as emails and internal social communications. Trawling through social content may uncover things you wish you had not uncovered, as it may open up sizeable legal and other HR risks for organizations. Your data may turn out to be less like a barrel of oil and more like a can of worms.

In practice, even when those amazing (or not so amazing) insights come through, what do institutions actually do? Do they stop or record lectures because learners with English as a foreign language find some lecturers difficult and the psychology of learning screams at us to let learners have repeated access to resources? Do they tackle the issue of poor teaching by specific lecturers? Do they question the use of lectures? Do they radically reduce response times on feedback to learners? Do they drop the essay as a lazy and monolithic form of assessment? Or do they default to talking about improving the 'student experience' where nothing much changes?

The problem with spending all of your money on diagnosis – especially when the diagnosis is an obvious limited set of possible causes that was probably already known – is that the money is usually better spent on treatment. Look at improving student support, teaching and learning, not diagnosis.

Types of data

It is also important to get a feel for the many types of data one might handle in AI for learning, the practical stages through which data must go before it is used and then how it is used.

In learning we can have ‘personal’ data, provided by the person or actions by that person with their full knowledge. This may be gender, age, educational background, job role, training objectives, courses completed, needs, stated goals and so on. Some also think that personality data is relevant, but there is a lot of controversy around the validity of Myers–Briggs. The OCEAN model has more support from research (Ahmed, 2016).

Then there is ‘observed’ data from the actions of the user, their start times, end times, how long they took on activities, choices, routes, clicks, dwell times, contributions to discussions, choices and answers. Who does what, where and when can be revealing, as it is raw, unpolluted data. It is like the flour in the baking of a cake – the base data.

You also have ‘derived’ data inferred from existing data to create new data. This is data that is computed from other data. We may have columns of scores on tests and derived data could be mean and median values from that data. This data can be used to good effect in adaptive learning, to decide on real-time personalized routes through courses.

At a higher level, ‘analysed’ data is data that has been inspected, cleaned, transformed and modelled to make it useful in decision-making. In AI this can involve ‘training’ data used for training AI systems. This data, used to train a model, may be a large corpus of text, such as the United Nations translated content used for Google Translate. In learning it can be existing

student data or previous queries from a social system used to train a chatbot.

Then there is 'production' data, the data used by the system after it is launched in the real world. This is where the surprises come, as this is often at odds with the training data and can mean iterations of further development.

If you are serious about using AI and machine learning (they are not the same thing), be prepared for some tough times. It is difficult to get things working from structured data, let alone unstructured data, and you will need really good training and testing sets, of substantial size, to train and evaluate your system.

Given the problems stated above, it is not easy to get a suitable dataset, which is clean and reliable for training your machine learning model. Then, when you launch the service or product, the new data may be subject to all sorts of unforeseen problems not uncovered in the training process. This is a rock on which many AI projects flounder.

Just when you thought it was getting clearer, you may also have to consider 'anonymized' data, free of attributes that may relate it to a specific source or individual. This is not easy to achieve as there are often techniques to reverse engineer attribution to individuals. Nevertheless, it is useful if regulations, laws and other ethical concerns are an issue.

Before entering these data analytics projects, ask yourself some serious questions about 'data'.

As mentioned previously, data is rarely clean in the sense of being ready for use, and cleaning data is a really important process in AI preparation. Many data scientists spend much of their time on this one task, as the quality of the data you put into algorithms really affects the quality of the output. Let us take numerical scores on a course. Cleaning means identifying and removing outliers (someone started but left the window open on their laptop), getting rid of duplicates, normalizing data to make the data similar and usable (putting scores onto the same scale). In large datasets, because you have so much

data, you can ignore missing data; whereas with small datasets you can use numerical methods such as using the median or some predictive algorithms to fill in the missing points and still make use of as much of the data as possible.

Data size by itself is important and size still matters – whether n = tens, hundreds, thousands, millions, the Law of Small Numbers still matters. But getting the right data may be even more important. Do not jump in until you are clear about how much relevant and useful data you have, where it is, how clean it is, in what databases and what you want to do with it.

Learners and data

Let us look at learners and data. Learners may not care that much and, if truth be told, there may not be a great deal to worry about. Most of the fuss may be from administrators who get paranoid about ‘privacy’. Nevertheless, it is worth asking a few questions about the issue and the law.

To what problem is transparency on data a solution? Are learners queuing around the block for their data? Learners are not that interested, as they are generally brought up in a world where they know that letting people hold and use your data is the price you pay for free stuff. That is not to say it is not an issue or that it will not become an issue in organizations, but there is no need for panic here.

Should we promise all data on demand? If learners demand it, should an institution promise to give them all data? Not really. In the UK, we have the Data Protection Act. However, this is not a blanket rule and there are lots of exemptions, as well as practical issues. So, there is no reason, practically or legally, to promise the world on data, as you may neither need, nor be able to, deliver on that promise. An important point that is often missed is that, if they have not asked, you do not have to provide the data. Even if asked for, it is likely that in its raw form the data would be meaningless. Data is rarely useful unless it is analysed and then visualized.

What about predictive models using analytics and adaptive learning? Should learning institutions provide that data? First, you only have to provide ‘stored’ data, not data used on the fly, which is common in these adaptive systems. You also have the argument that much of this is not stored data but algorithms with inputs and outputs. It is not so much isolated data as statistical inference or probability that is the object of curiosity.

There are also exemptions. You do not have to provide data where there have been infringements of IP by learners, data to do with a crime/investigation or third-party data provision.

Remember, also, that if you’re thinking of loosening up on data, making it available to learners, you also have to be very careful with access. Learners are smart, savvy and skilled. Many have the ability look for access to change, say, grades!

Data, in itself, is only useful if it is the right data, of usable size and can be used to solve the problem you want it to solve. An important point to remember about data in relation to AI is that it is not a trivial task to clean, train AI models and have the certainty you may wish to have over outputs. Nevertheless, it is clear that AI provides the means to use data to solve problems that have plagued the world of learning. Learning analytics are now part of the learning technology landscape.

Learning analytics

Many now regard ‘learning analytics’ and data collection in LRSs as a strategy that opens up the use of AI for learning. On the whole, this is admirable, but the danger is in spending this time and effort without asking ‘Why?’ Many talk about analytics, but few are talking about how to show that this will actually help increase the efficacy of the organization. Some are switched on and know exactly what they want to explore and implement; others are like those that never throw anything out and just fill up their home with stuff, as they feel it may be useful at some time in the future.

Another problem is that people want to shift from first to fifth gear without doing much in between. The online learning industry has been stuck with the traditional LMS/VLE and SCORM for so long, along with a few pie charts and histograms, that it has not really developed the mindset or skills to make this analytics leap. This, therefore, is an opportunity to look at using learning analytics and AI as a tool for genuine improvement.

In the end this is all about *decision-making*. What decisions are you going to make on the back of insights from your data? Storing learning data for present and future use may not be the best use of data. Perhaps the best use of data is directly in teaching and learning, to create courses, provide feedback, adapt learning, text to speech for podcasts and so on. Use AI in a precise fashion to solve specific learning problems, rather than analysis and the paralysis by analysis that often follows. If your goal is to harvest data and use it to gain insights and improve the learning and outcomes for the organization, be clear about those goals. The least efficient use of data may be to store it in a huge pot, boil it up and hope that something, as yet undefined, emerges.

Learning analytics uses data to improve learning. It has become clear that data can and will be used to help inform and shape learning. This means understanding what data is, what it can do and how you can use it.

Let us start with a good, basic schema for learning analytics. It is almost pointless to gather data if it remains unloved and unused. There must be purpose to the endeavour, in the sense of goals and uncovering evidence-based insights.

LEVEL 1 – DESCRIBE

What does the learning data tell us about *what things* are happening?

LEVEL 2 – ANALYSE

What does the learning data tell us about *why* things are happening?

LEVEL 3 – PREDICT

What does the learning data tell us that is *likely to happen*?

LEVEL 4 – PRESCRIBE

What does the learning data tell us that *should happen*?

LEVEL 5 – GENERATE

Using data to *generate* learning?

Learning analytics and organizational change

On an organizational level, one may wish to use learning analytics to drive much larger business goals.

As people learn from many different sources within an organization, and in order to know who is learning what, where and when, you will require a wider set of data. You will have to collect, store, analyse and use that data, from various sources, LMSs, mobile apps, portals, smart devices and other software.

When data is in the delivery loop, we can move beyond the limitations of SCORM, a 20-year-old standard that became so entrenched that version 1.2 is still in common use, not the later SCORM 2004. It excludes much of what we deliver these days: informal, mobile, social media simulations and so on. Hence the shift towards xAPI or other specifications for data-gathering and visualization. One of the problems with the rise of interest in gathering more data is the relative lack of real action taken on that data. It is pointless storing data if you have not decided what you want to do with it. It must be more than a mere storage system, as this is not about storage – it is about action and decisions.

The LMS and SCORM is about the packaging, delivery and management of content. The new specification, xAPI, is about learners and their behaviour. It is not ideal for AI, as its structure can be used to elicit real insights. It moves the focus from supply to demand and business needs. Now, in the data layer, with granular tracking using xAPI, along with a LRS, learning data can be used and integrated into business data. With this data comes analysis, and correlations between training and sales, productivity or behaviour are suddenly possible, along with a much more powerful ability to analyse, predict and prescribe action.

To implement AI solutions in your organization, an LRS can be used. It is a tool that consolidates and allows you to use learning analytics in your organization. An LRS can store not just learning analytics, but other evaluative and business data. This allows you to integrate learning into a fully integrated use of organizational data, which is valuable if you want to measure the role or impact of learning in reaching organizational goals. This alignment of learning and business, or learning and organizational goals, has long been the wish of learning professionals, but only now, through integrated data services, has this been possible.

First, it will help manage your data regulation, such as General Data Protection Regulation (GDPR), which should be easier to implement if you have an LRS. It helps you identify, manage and report your data in ways that will satisfy your regulatory responsibilities, thus avoiding hefty fines.

xAPI widens the scope of tracking, to recognize that learning takes place everywhere, not just on learning platforms. It broadens the horizon of learning to include especially informal learning. The aim is to replace SCORM with a more flexible specification that allows flexibility across an ecosystem of learning opportunities, not just LMS-delivered content.

This approach matches the emergence of complex learning ecosystems, as data from a wide range of integrated tools and technologies can be combined in a single source. If you are

worried about the location of data, such systems can be in the cloud, enterprise-wide or on your premises.

It should allow you to track learner journeys across the organization, no matter what the activity. Some LRSs go much further and trigger actions, such as recommendations, personalized feedback and automated workflow. Data is dead and inert if it is not put to good use. An LRS enables this good use. Far from limiting learning, it frees you up to track and improve a plethora of learning opportunities.

The bottom line is that an LRS simply structures your activity to make it easier to harvest, store, access and use data in a way that suits your organization. It should be able to cope with any data that is thrown its way.

Conclusion

There is a danger that organizations are being pulled in the wrong direction with their obsession with learning analytics, at the expense of more fruitful uses of AI in learning. Sure it has some efficacy, but the money being spent may be disproportionate. Much of what is being paid for here may be exercises that only bring obvious results. What insights are being uncovered here? That dropout is being caused by poor teaching and poor student support? That students with English as a second language struggle? Ask yourself whether these insights really are insights or whether they are something everyone knew in the first place.

The problem here is the paucity of data. Most educational establishments do not know that much about their learners. Few know how many students attend lectures because few record attendance. That is the first problem – poor data. Other data sources may be similarly flawed, as there is little in the way of fine-grained feedback, especially on learning. It is small datasets, often messy, poorly structured and not well understood.

So much of this seems like playing around with the problem, rather than facing up to solving the problem. That is not to say you should ignore its uses – just do not get sucked into data and learning analytics projects that promise lots but deliver little. There are a lot of promised efficiencies through learning analytics in higher education, mostly around preventing dropout, yet little in the way of verifiable analysis that it works. It is far better to focus on the use of data in adaptive learning or small-scale teaching and learning projects where relatively small amounts of data can be put to good use. AI is many things, and a far better use of AI in learning would be to improve teaching through engagement, support, personalized, adaptive learning, better feedback, student support, active learning, content creation and assessment. All of these are available right now. They address the real problem – teaching and learning.

Reference

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12

AI learning organizations

AI changes everything

AI, generative AI in particular, is likely to be the most important technology to hit learning. It is arguably the most important technology that our species has even invented, and is certainly the fastest in terms of global adoption, so it is unimaginable that its impact on learning will be anything less than profound. This impact will be on work directly and as L&D is about workplace learning, it will change what we learn, why we learn and how we learn.

AI has already had a profound influence on pedagogy. Our relationship with knowledge and services changed with search and all social media is mediated by AI, along with AI protecting email and other online services. Google Scholar, for example, radically increased the efficacy of research.

The first impact will be on base productivity. The typical administration and writing tasks in HR, L&D and teaching can be done much more quickly using generative AI. This includes ideas generation, reports, learning design tasks, the marketing of learning events and interventions.

The second will be on content production. Many aspects of course production can be automated – text, graphics, animation, audio and eventually video. The costs of content production will fall dramatically as these new tools not only produce good text, in any style, at any level in seconds, they can also be used to produce scenarios and assessment items.

Beyond this, AI is being used to deliver personalized and highly adaptive systems that can deliver learning on any subject, at any level, at any time in almost any language. The

productivity gains and more targeted learning through personalization is a huge boost for learning.

Performance support has also been made easier as learners can get help, on their smartphones, on all sorts of work-related tasks. This is exactly how generative AI is being used – you have a problem, you ask it a question and get an answer.

Of course, the real productivity gain in learning is the simple fact that this technology is being used by hundreds of millions around the world in multiple languages, to learn. In that sense the technology also bypasses or transcends L&D as people are doing it for themselves.

Strategy on AI

As Alvin Toffler famously claimed, ‘If you don’t have a strategy, you will be part of someone else’s strategy.’ Organizations need to have a strategic view of AI. As a global technology it has become a workplace tool and boardroom topic. There is no imperative to use this technology but there is an imperative to consider its use. It is okay to say no but you should have a reason for saying no, as this is the technology of the age, growing in momentum.

The trick is to get down and use it. AI has been largely invisible. You’ve been using it for ages. Now it has been foregrounded. It is there right on your desktop and smartphone. That baseline feel for what it can do becomes quickly apparent, List your top five problems. Then go at it to see if the tools can help.

But that is not enough. You must get advice on inputs. These tools are only as good as their users’ ability to use them. Word does not make you a novelist and ChatGPT does not make you a consultant or expert. Prompting is a skill – a skill that needs to be learned. You need to understand how the underlying technology works and how to squeeze the best out of it.

Don’t get bogged down in one dimension, such as cheating in assessments. That is a sideshow and simply reveals the

weakness of old practices. Think big not small. Think strategically and in terms of transformation about the big needs in your organization. List them and reflect on how these tools can help.

Take a wider view of AI. It is not all LLMs. It is used in dozens of other learning contexts. Think about applying this technology across the entire learning journey: engagement of learners (marketing also), support, personalization, adaptive learning, feedback, content creation, curation, performance support and assessment. There has been an explosion of innovation in the first few weeks after the release of ChatGPT – be part of that.

Think about how this fits the new paradigm, not of courses but performance support, hitting those sweet spots when learners really need stuff, not front-loading everything up front in a course. That is the way things are going – learning in the flow of work and life. Think learning, not teaching.

Ask people who have been doing this for some time and get a plan of action, not just a keynote at your conference or quick webinar, although that helps get the ball moving. Create a sense of urgency as this technology is moving faster than you can schedule such events.

Remember that this is moving fast. It is not about ChatGPT. The new timescale is days not months. This is about your strategy for the future, not what is available today.

Successful strategies invariably involve the leaders in an organization. It is worth targeting them with a formal introduction to its importance and relevance to their business and learning. One can choose specific projects and get them funded or agree a strategy for implementation and use in a bottom-up fashion. It is invariably a combination of top-down and bottom-up that works.

Using generative AI

L&D professionals and learning designers will have to upskill to understand and use this new technology for learning. How can this be done?

L&D must play with the technology as most of their learners are now aware of it and use it to learn and improve productivity in their jobs. Effort must be made to understand what this technology is and not to confuse it with search or retrieval tools. It is important to see this as dialogue, not monologue, as it is very different from most previous presentation tools, such as PowerPoint and e-learning. Personalized and adaptive learning is the promise, which can be finally realized with this technology.

Data matters and L&D must become more data savvy, seeing this as an opportunity to harvest but more importantly use data to improve delivery. Managing expectations matters, as these tools have exciting powers but also limitations. Both must be communicated to stakeholders and learners.

AI is the single most important shift in technology in our age, as it changes why, what and how we learn. It will, inevitably, have a huge impact on the knowledge and skills needed by those involved in online learning design, technology and procurement.

Organizations struggling to find talent now look towards developing a strong L&D culture to keep competitive. We should not be fooled by the meme – AI won't take your job, but someone who uses AI will! AI will transform some jobs, eliminate some jobs, create new jobs and automate some tasks without humans playing any role.

AI will also reshape the nature of work. McKinsey Global Institute estimates that 30 per cent of activities across 60 per cent of occupations are likely to be automated over the next decade (Manyika and Sneider, 2018). This is why it is essential for HR and L&D to open their eyes to the nature of AI, its place at the heart of business growth and its place in learning. Those in the learning world need to be more tech-savvy around AI in general and AI for learning in particular.

This means smarter procurement from companies that can deliver technology that is smarter than the old flat, rather inert, technology. Most find themselves with old LMS technology that is data light, so they have a paucity of data or data that does little more than record activity. This, coupled with low internal skills around data analytics, means that little progress is made.

AI for learning also brings learning professionals up to speed with the rest of the organization. You become more credible by being faster, more responsive and data-led. There is little doubt that most organizations are using AI in many different contexts. The organization's needs will also be better met by educators and L&D being faster, cheaper and better through AI.

This brings us to the skills we need to handle a future where AI is used for learning. Most interactive designers are skilled people, sensitive to the needs of learners, but we must always be willing to 'learn', for that is our vocation. New skills and techniques come into their own with AI for learning. We will see a shift in skills and therefore also a shift in the nature of teams, methods, procurement and production. They will also have to shift towards personalization, adaptivity, chatbots, dialogue, voice, open input and curation. Teams will have to become more technology-savvy, as AI is more complex and varied in what it can deliver than traditional software techniques. Methods of production will become more agile as fast iterations with tight timescales emerge. Finally, procurement in terms of outsourcing/insourcing will have to cope with a very different set of needs and requirements.

AI will change, relentlessly, the traditional skills that have been in play for decades. The old, core skillset that included writing, interactive design, media production and assessment will have to adapt and add newer skills associated with smarter software that delivers more adaptive and personalized software, using technology that is far more powerful and complex.

AI and learning design

Online learning companies now see a shift towards AI services and products, and are having to identify individuals with the skills and attitudes to deal with this new demand. This means understanding the new technology (not trivial), learning how to write for chatbots and dealing more with AI-aided design and curation, rather than doing this for themselves.

In another context, using AI tools and services means not using, or at least supplementing, the skills of traditional interactive designers, as the software either does some of this job or changes it dramatically. For example, we have seen how AI can identify the learning points, automatically create the interactions, find the curated links and assess using free-text input, formatively and summatively. It can create content in minutes, not months. This is the way online learning is going.

The gear shift in skills is already here and accelerating. Although still uncertain, [Table 12.1](#) shows some suggestions based on the concrete experience of making and observing this shift in real online learning production companies.

TABLE 12.1 New skills for learning design

[Skip table](#)

Old	New
One size fits all	Personalization
Learning techniques	Learning science
Media production	Chatbots
Monologue	Dialogue
Typing	Voice
Multiple choice	Open input
Linear	Complex
Branching	AI/adaptive
Content	Curation
Communications	Communicate expectations
Creation	Automation

Taking these one by one, we can see that a greater awareness of the technology is needed and with this comes the need for a greater understanding of cognitive psychology, as the technology has to deliver what could be described as cognitive ergonomics. To deal with this different type of smart, learning technology, which itself learns, you need to be aware of both the technical and cognitive ergonomics of the different areas of AI. AI needs to capture pedagogy if it is to work well.

AI and data. We will need to move beyond simply understanding the LMS and traditional e-learning content production, towards *AI- and data-* generated content and learning experiences through chatbots and other emerging AI inspired and enabled learning experiences. This is not to say that you need to be able to code or have highly technical AI or data science skills. It does mean that you will have to know, in detail, how the software works. If it uses specific AI techniques, one will have to make the effort to understand the approach, especially its weaknesses and strengths.

Probabilistic. Certainty needs to be replaced by a probabilistic mindset. We are not retrieving knowledge, where everything has a right answer, we are in a *probabilistic* world.

Communications with the client on AI can be difficult. Some of the output is AI-generated, and as AI is not human (not conscious or cognitive), it can produce mistakes. You learn to deal with this when you work in this field. But this is often not easy for clients to understand, as they will be used to fixed design document, scripts and traditional quality assurance (QA) techniques.

Learning science. As AI requires pedagogic design to guide its execution, this must capture and deliver sound pedagogy. We have already seen an increase in AI and algorithmic design to deliver good cognitive-science-based pedagogy, such as retrieval, interleaving, spaced practice and personalization. As one must build in these techniques and use data to train and determine the learning experiences, a knowledge of cognitive science will be necessary. With the implementation of AI, the AI has to embody good pedagogic practice. This is interesting, as we can build good, well-researched learning practice into the software, such as effortful learning, open input, retrieval and spaced practice. To build in pedagogy, one has to know that pedagogy in the first place.

Personalization. Most online learning has no or limited personalization, unlike most search, social media, Amazon or Netflix; it does not respond much to your own learning needs, apart from simple remediation or rule-based branching. AI changes this dramatically. Learning experiences will be delivered to individuals based on complex algorithms in real time, especially in adaptive learning. Personalized chatbots may also have to be considered. In sophisticated adaptive learning companies, this software is complex and the sequencing has to be handled by software, not designers – that’s what makes personalization on scale possible. This loss of fixed delivery means that one’s mindset has to be around systems that deliver recommended options and content adaptively based on data

and algorithmic identification of need, often in real time. The process of personalization must be understood.

Performance support. Most are using ChatGPT and similar services to learn in the workflow or life. Even in an educational institution they are using AI to solve a need – perhaps support in the classroom, to do an assignment. This shift away from courses to performance support has already happened.

Media generation. Media, such as text, images, audio and video have historically been produced. Generative AI has heralded in a new age of media generation, which is quick, can be generated from text, adjusted, altered, summarized and so on. We will have to learn how to be media generators. Note that graphic artists make the best graphics generators; learning designers the best generators of learning content.

Dialogue. Chatbots are complex and require thinking beyond the linear narratives and storytelling of traditional online learning. With chatbots, where we've been designing everything from tutorbots to coaches, the whole form of interaction changes and you need to see how they fit into workflow through existing collaborative tools. Dialogue will be the means by which many people learn, especially in the workflow or flow of life. Just as a teacher has dialogue with learners when teaching, so we will move towards this model online. Chatbots will multiply, as teachers tutors, trainers, experts, coaches and mentors. We will also learn through dialogue with patients, customers and other people, especially in learning soft skills and skills that involve human interaction.

Voice. There is also voice interaction to consider. There will be entire learning experiences where all navigation and interaction is voice-driven. This needs some extra skills in terms of managing expectations and dealing with the vagaries of speech-generation and recognition software.

Open input. We are likely to see a reduction in the formulaic MCQ production. MCQs are difficult to write and often flawed (Clark, 2017). Then there's the often vicariously used 'drag and drop' and hideously patronizing, 'Let's see what Philip, Alisha

and Sue think of this...'. You click on a face and get a speech bubble of text. This will be an area of huge change as the limited forms of MCQ start to be replaced by open input – of words, numbers and short text answers. NLP also allows us to interpret this open-input text. Writing assessment items may change dramatically once one is freed from the constraints of MCQs.

Complex. With AI, you are in a probabilistic world and need to understand that learning pathways will be much more complex and varied, as they are personalized. We will have far fewer linear course and far more complex, personalized pathways in learning.

AI/adaptive. Rather than seeing rules-based branching or gamification as the solution to successful learning, adaptive learning will come to the fore, where the software, rather than the designer, decides what is sequenced for the learner or dialogue. This will be done in real time to optimize learning for individuals. The aim is to educate everyone uniquely. This means understanding the nature of the adaptivity and designing content loose enough to be re-sequenced on scale.

Data-driven. Learning professionals must get to grips with basic data analysis and shift their mindset from limited evaluate data to data-driven solutions. SCORM will, thankfully, be a thing of the past. Data is not the result of learning experiences, it drives them.

SME as validator. AI can reduce iterations with SMEs, the cause of much of the high cost of online learning. If the AI is identifying learning points and content, using already approved documents, PowerPoints and videos, the need for SME input is lessened. As AI tools produce content very quickly, the clients and SME can test and approve the actual content, not from scripts but in the form of the learning experience itself. This saves a ton of time. What is needed is a change in mindset, as well as tools and skills. It may be difficult to adapt to this new AI world, where many aspects of design, like this, will be automated.

AI is here. We are, at last, emerging from a 30-year paradigm of media production and MCQs, in largely flat and linear learning experiences, towards agile, smart, intelligent online learning. AI solutions often behave more like a good teacher, where you are taught as an individual with a personalized experience, challenged and – rather than endlessly choosing from lists – engaged in effortful learning, using dialogue, even voice. As a learning designer, learning experience designer, learning engineer, interactive designer, project manager, graphic artist, video producer, whatever the title, you will have to adapt to a world where some of the tasks completed by learning designers will be automated.

AI and technology design

Communications between designers and AI developers and data scientists is a challenge. Designers know a lot about communications, learning and goals, but are often unfamiliar with this new technology. Technologists know a lot about the software but often little about learning and the goals. There is a need for a much closer understanding of each other's worlds and the onus is on both to accept that this needs close teamwork.

Interactive or experience designers, whatever they are called now or in the future, will need to know far more about what the software does, its functionality, strengths and weaknesses, than they did in the past. A knowledge of how the actual AI techniques work and are applied, such as generative AI, adaptive learning and data analytics, will be a necessary skill to match the technology with the pedagogic aims across more complex AI ecosystems. Those with some technical understanding will fare better here as they can understand both the potential and limitations of AI for learning.

With traditional online learning, the software largely delivers static pages with no real adaptability, optimization or self-learning. Content and services will no longer be served up as

static, fixed pages, but in forms of dynamic delivery. It can change in real time to the needs of learners, delivering different screens and elements to different learners depending on their needs at that particular time. It can also change across time, improving as more learners use the system and aggregated data is used to improve the performance of the overall software. The software is, in a sense, a growing, learning entity that changes over time. One must understand that this requires a different approach to design, documentation and testing. The designed entity is no longer a fixed script with linear and simply branched events, but a complex system based on data and probabilities. Testing, for example, requires a deep understanding of the inputs and outputs.

It is important to understand how AI learning delivery systems work, as these learning systems, in a sense, have some autonomy. They are teachers but they also learn.

AI and data design

As learning ecosystems deliver smarter, multiple, integrated solutions, data can be pulled together to measure, improve, feed and recommend better learning. Just as a car benefits from data that is fed from sensors to computers to improve the performance of your car, either directly or indirectly to you as the driver, so data will be used directly or indirectly in dynamic AI learning systems. In general, we will see a shift towards more data analysis and the use of data within systems as well as the reporting data.

SCORM looks increasingly like a crippling limit on online learning, incompatible with AI in learning. Completion is useful, but rarely enough. It is important to supplement SCORM with far more detailed data on user behaviours. But even when data is plentiful, it needs to be turned into information, visualized to make it useful. That is one set of skills that is useful – knowing how to visualize data. Information then has to be turned into knowledge and insights. This is where skills are

often lacking. First you have to know the many different types of data in learning, how datasets are cleaned, then the techniques used to extract useful insights, often machine learning. You need to distinguish between data as the new oil and data as the new snake oil.

We take data, clean it, process it, then look for insights – clusters and other statistically significant techniques – to find patterns and correlations. For example, do course completions correlate with an increase in sales in those retail outlets that complete the training? Training can then be seen as part of a business process where AI not only creates the learning but does the analysis – and that is all in a virtual and virtuous loop that informs and improves the business. It is not that you require deep data scientist skills, but you need to become aware of the possibilities of data production, the danger of GIGO and the techniques used in this area.

As SCORM recedes and new standards, such as xAPI, allow learning ecosystems to take advantage of systems other than the LMS, the learning designer will have to become more aware of the dynamics of integrated systems and the data that needs to be gathered from these different sources. Once you use an ecosystem which deals with multiple suppliers, and you have to deal with secure authenticated inbound and outbound web services (application programming interfaces – APIs) with secure data transfer methods, encryption of data, preferably via web services (API) over HTTPS (CSV via SFTP as a minimum) and therefore xAPI, there has to be a way of coping with this complexity in all three layers of the system – learning experience, content and data. An LRS, like Learning Locker, will natively collect xAPI data from a variety of sources. It can also gather other non-xAPI-compliant sources of data by changing non-xAPI data streams into the xAPI format by acting as a single source of data for the ecosystem. Integration with other business data systems and tools is possible and desirable.

Learning record stores. To analyse the many different sources of data, xAPI compliance ensures that all data can be stored in

an LRS and used to inform business decision-making. xAPI statement data can be used now or later to fuel recommendations, adaptive learning systems, nudge learning and other targeted initiatives. It can also feed different dashboards – for learners, teachers, trainers and lecturers, or at the departmental level.

AI and procurement

One major issue that arises as a result of all of the above is your organization's position and skills on procurement. There are many options here depending on your ambition, direction of travel and resources.

The main choices are:

- in-house development;
- learning technology companies;
- large tech companies.

In-house

In-house development has the advantage of keeping control with costs low and predictable. However, it is clear that AI talent and skills are not easy to find and are expensive. You may want to build things in-house, but the task can be daunting. There is a strong argument for prototyping or piloting some AI for learning projects, but many (not all), for the foreseeable future, are likely to be outsourced or use externally developed tools.

Learning technology companies

Some learning technology companies have made the leap into AI for learning. They are sensitive to the learning game and have people who understand that this is about learning and who have experience with technical integration. The advantage

of using this type of company is that they have a balanced approach to learning and technology. The downside may be lock-in to a supplier, their lack of breadth and depth, and costs. If, however, you choose a system that is sensitive to xAPI and can draw upon a variety of services from different sources, AI may actually free you from the single supplier trap.

Large tech companies

Large tech and outsourcing companies will emerge in this area. They have the advantage of breadth and depth, with good support, but tend to be one-size-fits-all, expensive and insensitive to your learning culture and needs. Tech giants like Google, IBM, Microsoft and Amazon already offer a mixture of open source tools and metered usage services for AI applications. Your IT department may already be aligned with one of these companies, and this AI-on-tap approach has some appeal as they provide well-tested and trained models. However, they don't do well in handling learning issues, detailed integration and service. They also have a habit of changing their generic services, sometimes with little or no warning.

This is not an easy landscape to navigate. As a technology transition, like no other, it not only delivers; it delivers in a way that changes what humans do, augmenting our ability to personalize, deliver the right stuff at the right time in the right place and also make delivery more responsive. AI for learning learns alongside your learners. The more it is used, the better it gets. In general, data lies at the heart of the process. Skills in procurement, contracts, pricing and support needs will be essential in this more complex world.

Conclusion

The shifts outlined here are not trivial; they are challenging. Yet we know that the AI revolution is here. It is everywhere in

almost all online services, from Google to Netflix. It will also be a key technology in learning.

New skills have always been required as technology changes. These have, largely, been front-end skills around presentation, look and feel, limited interactions and media. But AI is more to do with back-end technology, where the software itself does much of the work. This does not negate front-end skills but adds to them.

In that sense it is more technically demanding but also exciting in that one can do so much more in terms of learning. This means shifts in skills on design technology and data. Beyond this lies a shift in mindset towards smarter systems with more autonomy and a greater awareness of the issues around procurement.

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13

AI and ethics

The reaction to developments in generative AI was one of surprise and astonishment, but also fear. It was both exhilarating and terrifying.

Like previous revolutions in technology – writing, printing, broadcast media, photocopying, internet, Wikipedia, computer games, social media, tablets and smartphones – one could see this as a predictable period of doubt until everyone suddenly uses it and sees the obvious utility. The utility and productivity increases are so obvious with AI, both in time saved and quality of output, along with rapid increases in functionality, that resistance seems futile.

Within days of ChatGPT 3.5 conquering the world, there was the usual hand-writing and negativity. Every force has an equal and opposite reaction but the sheer force of generative AI seemed to produce more negativity and fear than normal. Out of the woodwork came an army of ‘ethicists’, all with very strong opinions on two things they often seemed to have no background in – ethics and AI! Every man, woman and their dog seemed to have an ‘ethical’ view on AI, most of it shallow and repetitive. Ethical AI companies, institutions, reports, frameworks littered the landscape. Italy banned ChatGPT, then relented; a stern letter was written demanding we halt all generative AI for six months, yet in the six months following, nothing better than ChatGPT 4 was released. For every one person working on AI there seemed to be far more writing of repetitive reports on the ethics of AI. If only we had the same concern for crippling expensive education, dull lectures, and failure in education and learning.

Let’s do a thought experiment. I have an amazing piece of technology I’ve invented. It will bring astonishing levels of autonomy, freedom and excitement to billions of people. But

here's the downside: 1.4 million people will die horrible, bloody, sometimes mangled deaths every year, with another couple of million maimed and injured. This world-war level of casualties, will strike every year, and is the price you have to pay. Would you say YES or NO?

Most rational souls would say NO. But let me reveal that technology – the motorcar. We have come to an accommodation with this technology, as the benefits outweigh the downsides. AI may even bring in the self-driving car.

My point is that we rush to judgement, as we are amateur ethicists and rely on gut feel, not reason. Ethical reactions are, as David Hume (1751) rightly claimed, primarily emotional not rational. Hume believed that our approval or disapproval of an action is a matter of how it makes us feel. We feel that something is good when we feel surprise, satisfaction, wonder, amazement, admiration and optimism. We feel it is bad through feeling unease, apprehension, concern, worry, fear and pessimism. Morality is therefore subjective, meaning that what is considered morally right or wrong depends on the individual's feelings and perceptions. He famously claimed that reason is, and ought only to be, the slave of the passions. In other words, reason alone cannot motivate action; it can only guide or serve our desires and feelings.

Another way to read this approach to ethical reactions is through Kahneman's fast and slow systems or the analogy of the rider and the elephant in the context of cognition and decision-making attributed to Jonathan Haidt used in his book, *The Happiness Hypothesis* (2006), to describe the relationship between the conscious and unconscious parts of the mind. In this analogy the 'elephant' represents the automatic or unconscious processes, which are powerful and often drive behaviour, and the 'rider' symbolizes the controlled or conscious processes, which can guide the elephant but often have less control than they might seem to. Haidt used this metaphor to illustrate how our emotional, intuitive processes (the elephant) are often more influential in guiding our

behaviour than our rational, conscious thought (the rider). It's a way to understand the complex interplay between intuition and reasoning in human psychology. This explains why our so-called 'ethical' reaction to a new phenomenon often goes straight to our deep elephant feelings, reasons, then justifies us feeling that way.

This whole area, ethics, is subject to a huge amount of bias, as it is such an emotive subject. It plays to people's fears and prejudices, so objectivity is rare. Add new technology to the mix, along with a pile of stories in social media and you have a cocktail of wrong-headed certainty and exaggeration.

Deontological versus utilitarian

The offer I made earlier, I have put to many audiences. It is never taken up, as we are utilitarians (calculating benefits against downsides) when it comes to actual decisions on using technology, but dogmatic deontologists (seeing morals as rules or moral laws) when it comes to thinking about ethics and technology.

I am a fan of David Hume's indirect utilitarianism, refined by Harsanyi as preference utilitarianism. For a good discussion on how this relates to ethical issue and AI, see [Chapter 9](#) of Stuart Russell's excellent book, *Human Compatible* (2019), where he attempts to translate this into achievable, controlled but effective AI. In truth, most of what passes for ethics in AI these days is lazy 'moralizing' – moral high horses ridden by people with absolute certainty about their own values and rules, as if they were God-given. More than this, they want to impose those rules on others. They call themselves 'ethicists' but it is thinly disguised activism, as there is no real attempt to balance the debate out with the considerable benefits. It is an odd form of moral philosophy that only considers the downsides.

There is a big difference between 'ethics' and 'moralizing'. Even a basic understanding of ethics will reveal the complexity of the subject. We have thousands of years of serious

intellectual debate around deontological, rights-based, duty-based, utilitarian and other ways of thinking about ethics. A pity we give it so little thought before passing judgement.

Thomas Nagel points out, in his book *Equality and Partiality* (1995), that we often pronounce strong deontological, moral opinions but rarely apply them in our own behaviour. We talk a lot about, say, climate change, but drive large cars and fly off regularly on vacation. Academics talk about the climate emergency in academia but fly off for conferences at the drop of a sunhat, don't deliver learning online and believe in spending €28 billion flying largely rich students around Europe through Erasmus.

You may want all of your AI to be fully 'transparent'. That is fine, but stop using Google and Google Scholar and almost every other online service, as they all use AI, and it is far from transparent. My favourite example is those who are happy to 'probe' my unconscious in 'unconscious bias' training but decry the use of student data in learning on the grounds of privacy!

To focus on AI is to perhaps focus on the wrong problem. Rather than climate change, the possibility of nuclear war, a demographic time bomb or increasing inequalities – AI is getting it in the neck, yet it may just solve some of these real and present problems. In particular, it may well increase productivity, democratize education and dramatically reduce the costs of healthcare. These are upsides that should not be thwarted by sci-fi speculation.

At its most extreme, this speculation, that 'AI will lead to extinction of the human species' seems to have turned into the Doomsday tail that wags the dog, despite the fact there is no evidence at all that this is possible or likely.

New technology always induces an exaggerated bout of ethical concern. Everyone becomes an armchair ethicist but this is hardly new. It was ever thus. Plotus made the same point about the sundial in the 3rd century:

The gods confound the man who first found out how to distinguish hours!
Confound him too who in this place set up a sundial to cut and hack my days so wretchedly into small portions!

Plato thought writing would harm learning and memory in *Phaedrus*, and the Catholic Church fought the printing press (we still idiotically teach Latin in schools) and the Ottoman Empire banned it. Radio, film, TV and even Walkmans were seen as ruining minds. Typewriters were banned and photocopiers spelled the end of copyright. Calculators would paralyse our ability to do arithmetic, Y2K was going to cause the world to implode, computer games would turn us into violent psychopaths, screen time would rot the brain, the internet, Wikipedia, smartphones, social media... now AI.

As Stephen Pinker rightly spotted, a predictable combination of negativity and confirmation bias leads to a predictable reaction to any new technology. This inexorably leads to an over-egging of ethical issues as they confirm your initial bias.

Take the famous *NYT* article where the journalist claimed ChatGPT had told him to leave his wife. On further reading, it shows he had prompted it towards this answer. If some stranger in a bar dropped you the line that his marriage was on the rocks, you'd put a significant bet on him being right to leave his wife. ChatGPT was actually on the money. It was a classic GIGO (garbage in: garbage out), poke the bear story. Then there was that AI-guided missile that supposedly returned back and hunted down its launcher – never happened – complete fake.

The endless stream of clickbait 'look it can't do this', mostly using ChatGPT 3.5 (a bit like using Wikipedia circa 2004), flooded social media. This is the worst AI will ever be, but let's not consider the fact that first release technology is almost always followed by dramatic improvement. Indeed, progress appears to be exponential.

Argument from authority

Then there is the argument from authority. ‘I’m a professor,’ say people in strongly worded letters to the world, ‘therefore I must be right.’ Two things matter here, domain experts often have a lousy track record and a lack of expertise in philosophy, moral philosophy, the history of technology, politics and economics. To be fair, experts in AI are worth listening to as they understand what is often difficult to understand and opaque technology. Generative AI, in particular, is difficult to comprehend, in terms of both what it is, how it works and why it works. It confounds even AI experts. But they are not experts on politics, ethics or regulation.

The letters that appeared in both 2015 and 2023, pushed by Tegmark’s Future of Life Institute (whose role is ethical oversight), use the argument from authority. ‘We’re academics, we know what’s right for you, the masses.’ It demanded that we immediately stop releasing AI for six months until the regulators caught up – a ridiculous request that showed their political, economic and social naivety. I dislike this letter-writing lobbying. First it had names of people who demanded they be taken off the list as they had not given permission and some have since rescinded the statement, but authority alone is never enough.

This tsunami of shallow moralizing is almost perfectly illustrated in higher education, where most of the debate around ethics has focused on plagiarism, when the actual problem is primitive assessment. There is little consideration of the huge upsides and benefits for teachers and students alike. Learning, in my view is the biggest beneficiary of this new form of AI, healthcare second. Hundreds of millions are already using it to learn.

On climbing into personal pulpits, we may fail to realize the benefits in learning. Personalized learning, allowing any learner to learn anything, at any time from any place is becoming a reality. Functioning, endlessly patient tutors, that can teach any subject at any level in any language are on the horizon – universal teachers with a degree in any subject and

driven by good learning science and pedagogy. The benefits for inclusion and accessibility are enormous, as its potential to teach in any language.

It is not that there are no ethical problems just that objective ethical debate is harmed when it becomes enveloped in a culture of absolute values and intolerant moralizing. For every ethical problem that arises, people rush to glib conclusions that are simple, confidently pronounced and often wrong.

Ethics and regulation

Spectrum of opinion

A spectrum of opinion has emerged, even among experts in AI, on the dangers the technology poses to our species.

LeCun, Andreessen, Hotz, Ng, Wales and Zuckerberg believe in open-source foundation models, as they think this will lead to the sort of innovation that results in solutions to real-world problems. They are vociferous in their opposition to those who think that AI is a threat to the species, seeing the technology as well within our control.

Others would like to see us proceed carefully, such as Hinton, Altman, Sutskever, Gates and Lanier. They welcome regulation, as long as it does not stifle innovation. They see the benefits as outweighing the downsides and recognize that the realizable benefits of generative AI may not be realized if we have too negative a view.

Moving across the spectrum further, there are those who want to put the brakes on development until regulation catches up. Tegmark, and his Future of Life Institute, is the most vocal of the group, and was behind the two letters in 2015 and 2023 that tried to delay development. Others in this camp include Benio, Musk and Russell, It all looks a little dated now as nothing more powerful than Chat GPT4 was released in that period.

Then there are those at the far end of the spectrum who want to ban it completely, the most notable of these doomsayers is Yudkowsky.

Regulation

The spectrum of opinion is reflected in the diverse positions taken around the world on regulation and legislation.

The US and the UK are resisting a top-down approach with blueprints and white papers that reflect a reluctance to over-regulate. The US concerns focus on the principles of safety, discrimination, privacy, notice and explanation, as well as opt outs and remedies. The UK held a global AI summit, and has a set of principles similar to those of the US: safe, secure and robust; transparency and explainability; fairness; accountability and governance; contestability and redress. They tend to see regulation as existing at the sector-specific level in healthcare, finance and so on, leaving AI to progress quickly in terms of innovation, with a minimum of top-down legislation.

The EU, along with UNESCO, takes a much harder line and have gone ahead with legislation that applies to their market. They have tackled accountability and even energy consumption, pulling back from outright bans on biometric data such as face recognition. They have identified four levels of risk: unacceptable, high, limited and minimal.

China legislated before the EU with targeted laws against the use of AI to cause harm to the old and minors, insisting on security assessments before product launches. Although their concerns also focus on *‘upholding mainstream value’* (government ethics) and the *‘prevention of controversies’* (straightforward censorship).

Finally we have countries that have gone for an outright ban: Iran, North Korea, Syria and Russia. Italy banned ChatGPT for a while but relented.

Techno-determinism

It is easy to look back in and see how one thing flowed from another in the development of AI and lock into a form of techno-determinism. Of one thing we can be sure: it will be full of surprises.

Not all learning technology innovation turns out to be either successful or beneficial. The crypto dream imploded in a cannibalistic frenzy of venal greed and corruption. Blockchain remains little more than a wasteful attempt to fossilize credentialism. The limitations of the LMS, SCORM and flat, linear, e-learning have become clear. The primitive content with its dated look and feel, cartoons, speech bubbles and MCQs, has remained much the same for over 30 years. If anything, online learning has become fossilized in Disneyfied courses, delivered from repositories, with little or no evaluation.

We can't be absolutely certain about the continuing rise of AI, as it may reach mathematical or other limitations but it is clear that a fundamental shift has happened. Those learning technologies described in my book *Learning Technologies* (2023), the ones that allowed us to create new worlds – early cave painting, writing, alphabets, printing, broadcast media, the internet – were all fundamental shifts in the way we interact with our archived and acquired cultural knowledge. AI shows every sign of being something as significant as these other monumental technological revolutions, if not the most significant of all. AI may turn out to be the most fundamental, fruitful and paradigmatic shift of all, one that transforms not only learning but our species.

In *Learning Technologies* (2023), the point is made that successful technologies are usually combinatorial. The sum of the parts becomes more than the whole. This is especially true in AI, where there is a plethora of AI approaches that come together to deliver many solutions. AI is also combined with

powerful, existing software and hardware to eliminate deficiencies or provide other functions. But there is also something unique about AI, especially as a learning technology. It learns. A whole range of types of learning and optimization mean it gets better exponentially. Generative AI was forged in language through LLMs, capturing something essentially human from our vast corpus of language. It has now expanded out into multimodal models, capturing images and video. We also need to pay attention to some big ticket advances in technology that are likely to accelerate and consolidate AI as the technology of the future, such as AI enhanced hardware, already embedded in our devices, along with specific advances in AI itself and the emergence of ensemble solution with both AI and other software melding to deliver optimal solutions.

AI developments

So quickly did ChatGPT 3.5 become the talk of the planet that it became the fastest ever adopted technology in the history of our species. Within just four months of its birth, another more mature child was born – an improved version that outperformed its sibling. Plug-ins became available as did API access, opening up the technology to the world creating an entirely new business sector. AI had suddenly become a global business reaching into every sector – marketing, finance, healthcare and, the subject of this book, learning.

We saw share prices plummet as investors realized that some business models were now suspect, as their services were clearly going to be supplanted by this new kid on the block and if not supplanted, certainly in need of augmentation. An AI hiatus hit the market as it was not clear how this was going to play out. There was not a serious board room on the planet that was not at least discussing the implications of this for their organizations.

Hollywood went on strike, stopping AI being one their strike demands. Governments formed AI advisory groups and

promised large sums for research and investment. Other LLMs started to appear, some open source, along with services that made those models accessible and usable. There was a rush to get policies in place to define safe use of AI, as millions were clearly using it without any organizational push, help or support. This happened *to* organizations, not *by* organizations. Some sectors, such as higher education, were driven largely by fear, focusing on the threat to essay assessment. Others such as marketing were enthusiastic early adopters and others were curious, some loudly defensive. Thousands of projects were kicked off within businesses and start-ups. Generative AI sparked off a wildfire of innovation that continues still.

A phrase that emerged during the first year of generative AI was '*This is the worst AI will ever be*', the point being that it is getting better so fast that focus on the present is ill-advised – better to look forward and plan for the rapid improvement that is coming.

Ethics has become a rather over-engineered aspect of this revolution, with no end of organizations riding in on moral high horses. These are the 10 horses of the AI-pocalypse, topics most often cited as ethical concerns:

1. Plagiarism – cheating
2. Publishing – copyright
3. Provenance – false stuff
4. Propaganda – dupes us
5. Privacy – data and security
6. People – dehumanizes
7. Prejudice – biases
8. Poverty – employment and big tech
9. Planet – energy and emissions
10. Perish – extinction event

Plagiarism – cheating

It is odd that the only example of AI in learning to be found in higher education establishments is used to see whether students cheat. As the old adage goes, when students copy, it is plagiarism; academics call it research. This is a cat and mouse game, where predictable, often identical assignments, largely long-form essays, are set. Students procrastinate, share, cut and paste, only to wait for often sparse feedback and a solitary grade.

But the game is getting more complex as, on one side, institutions, teachers and academics are engaged in assessing through essays, while on the other side, students cheat using essay mills, good comms, high tech and stealth. As long as institutions see this as a deficit problem – those pesky students and essay companies ruining our honest practice – nothing will change. This is a problem that needs smart solutions, not denial and mousetraps.

On one side, institutions and academics set predictable assignments. The format is the often lazy essay question. They often do not change for years. Why does this happen? Fossilized practice, teaching comes second to research, a dearth of assessment design skills, the institution encourages this singular form of assessment and the quality bodies are stuck in a model that has barely changed in decades.

On the other side, students use tech that makes it easier for them to play the game and win. They are on social media, making it easier to share. They have access to oodles of sources from which they can cut and paste. Beyond this they can buy relatively cheap, and undetectable, essays and dissertations online. To be fair, they often do not receive enough teaching and advice on how to do assignments with academic integrity. The psychology here is interesting. The assignment turns into a chore. They know that feedback will be light and that it is unpredictable when they will get the marked essay back. They start to see learning as a game.

Increasing numbers of students clearly results in more pressure to cheat. They or their parents have often had to

outlay considerable sums of money, and failure is hard to take as it involves huge loss of face. On top of this is the reluctance of academics to do the necessary detection work, to follow up on cheating, which can be detailed and arduous. You need a lot of very sure evidence to pull this off and most do not want to start the process and climb that bureaucratic mountain. Another protective layer on top of this is the reluctance of the institution to admit it happens, as there is huge potential for reputation damage. This is a perfect storm, where students, teachers and institutions literally institutionalize cheating.

Essays are sometimes appropriate assignments if one wants long-form critical thought. But in many subjects shorter, more targeted assignments and testing would be better. There are a lot of formative assessment techniques out there and essays are just one of them. Short-answer questions, open-response, formative testing and adaptive testing are all appropriate. The truth is that education wants it easy, and essays are easy to set. They also have to accept that they are also now easy to cheat.

One other problem is the ready confusion between formative and summative assessment. There is perhaps far too much marking and summative assessment. If the assignment is a formative learning experience, why mark at all? It should be about the feedback. Black and Wiliam (1998), who have spent decades studying this issue, recommend *not* marking and instead focusing on feedback. Marking acts as an end point. High-performing students get 80 per cent then stop, assuming the other 20 per cent is not worth the effort; low-performing students get demotivated. What learners actually need is not a mark, but detailed and constructive feedback.

There is also the problem of what counts as plagiarism. One of the problems is that tools that check for plagiarism often count direct quotes as plagiarism, confusing the stats and sending false positives into the system. A second problem is what constitutes ‘common knowledge’, i.e. stuff that does not have to have citations. Plagiarism checkers for generative AI do not work. This is tricky.

But there is an even worse problem in assessment. To rely on the essay format or long-form prose answers is to encourage students to memorize essays and play roulette with the subject in their final exams. Students, the world over, play the game of final assessment by memorizing essays. There is a pretence that it is testing critical thought. It is not.

We know the scale of the problem. Compare the scanty number of cases actually reported by institutions against the number and size of the companies offering essay mill services. There is a massive gap – and this is just the tip of the iceberg, as most of it is in the grey economy, with even parents doing the cheating. Purchased essays and dissertations are now commonplace in universities. You could legislate against such companies but it would just shift abroad as it is online.

Why do we rely so much on essays as a form of assessment? One has to conclude that, like lectures, they have become an easy default. They are easy to set, difficult to mark. It would surely be better to look at a range of appropriate assessment methods that are designed around the type of learning objectives and competences you want your students to acquire. This is not to say that essays should never be used, only that they are overused. It is institutionalized teaching, not optimal teaching or learning.

In truth, there are lots of alternatives to the long-form essay.

Publishing – copyright

The issue of creative copyright is a misunderstanding of the technology as some sort of single copying machine or single source which LLMs retrieve from or sample in some way. This is not how it works. There are both input and output issues here. On input, namely the data used for training. There is a big difference between ‘learning from’ and ‘copying from’. The models learn from the trained data; they do not copy that data.

On output, the models spit out freshly minted words, probabilistically, one by one. Again there is no copyright

problem. We would be as well saying all human brains break copyright in all creative acts as they input through reading, hearing and perceiving text and imagery when learning, then output new words, speech or images. There is no ghost in the machine copying or sampling.

Provenance – false stuff

AI doesn't hallucinate at all. It does what it is trained and designed to do, engages you in dialogue and tries to please. 'Hallucination' is a terrible word, loaded with anthropomorphic, pejorative meaning. It suggests AI suffers from drug (data) induced utterings or psychotic episodes. Nothing could be further from the truth, whatever that is. In fact, generative AI simply tries its best with polite efforts at completion, like a human eager to please who embellishes what they say with unnecessary, and sometimes made-up, certainty.

Examples are often previous versions of the models (typically, ChatGPT 3.5), or the result of deliberately aggressive prompting to break limits, the assumption that it is a truth machine or the misconception that there is some set of absolute moral values. More recently, it has become less of a problem, although still intrinsic to generative AI. Once you know how to speak to it, it responds with sensible and accurate responses in dialogue, especially when plug-ins or RAG techniques are used. If you suspect it has overegged its response, you can ask it to reflect and it will often correct itself. Humans tend not to!

Provenance and accuracy have also been improved with the use of plug-ins, such as Wikipedia, where output can be checked against trusted sources. Mathematical ability has improved through access to systems that perform well in this domain. This is all useful in learning, where errors are harmful.

Propaganda – dupes us

Deepfake images, audio and video are much easier to create using generative AI. Some platforms are using tools to detect such fakes, while others like TikTok want users to disclose its use. You can label your video so that full disclosure is clear. This is useful as it allows benign, satirical and experimental content to be shown but not content that wants to cause deliberate offence and harm. I have a digitally generated avatar and cloned voice and am happy that it be used to show how technology has advanced in this front but not so that my bank coconuts can be hacked!

The problem is that photography and video, especially on the world, entertainment and the internet has for years been saturated with AI enhanced sound and imagery. Images are AI enhanced in all the major image, sound and video tools, makeup filters are common for ordinary users. Where is the boundary between fake and deepfake?

AI is far more likely to protect us from fake and false information that make us believe in it. Around 400 million to 1.5 billion fake accounts are taken down by AI every quarter (figure oscillates) and hate speech is just 0.01–0.02 per cent of all content and 82 per cent of it is taken down automatically by AI before moderation. This is a known problem with known solutions. We saw this in the clickbait drone story, which turned out to be fake news. A USAF official who was quoted saying the Air Force conducted a simulated test where an AI drone killed its human operator is now saying he ‘misspoke’ and that the Air Force NEVER ran this kind of test, in a computer simulation or otherwise.

Privacy – data and security

Security is a matter of trust. Without trust, these systems are unlikely to be adopted in organizations. The onus is therefore on IT departments to operate and manage these issues.

Whether it is for personal or operational needs, the first thing that is necessary is a change of mindset. These tools are

different and one must move from certainty to a more probabilistic mindset, monologue to dialogue, simplicity to complexity, production to generation, analytics to data, courses to support, the LMS delivering content to AI.

Legal demands also have to be met. This is not new, as GDPR and other legal requirements have applied for many years. With generative AI one must be aware of IP and copyright issues, personal data processing and the protection of proprietary data. One must also anticipate legal regulations that apply in any territories in which you operate.

Finally, there is the hardcore security issue of chatbot jailbreaking, phishing, DoS and other cybersecurity issues.

People – dehumanizes

Has there ever been a more humanizing technology than generative AI? It is trained on the cultural capital of our species. We can speak to it and it speaks back. It is dialogue, not presentation or monologue. Its success is built on its human-like qualities. Indeed, it is these human qualities that drive the ethical concerns. We fear that its cognitive abilities will impact white-collar work in the same way that automation impacted agriculture and manufacturing. Yet it seems to be so human, almost endearingly human. If it can deliver on learning, healthcare and other issues, we should welcome its human and cognitive characteristics.

Conclusion

The most serious of the ethical issues facing us are the big ticket items of:

Prejudice – biases

Poverty – employment and big tech

Planet – energy and emissions

Perish – extinction event

They need a more detailed treatment, so we shall tackle these in the next few chapters.

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14

AI and bias

No other area of technology than AI has been subjected to so much ethical scrutiny by so many hastily convened ‘committees’. For some in the learning game, the mere mention of the words ‘artificial intelligence’ induces irrational fears about bias, surveillance and privacy. They want to keep it out at all costs. For others, it is the saviour that will accelerate the longed-for personalized learning, eliminate the harmful effects of human bias, and make learning much more open, efficient, personal and productive. As usual, the truth lies somewhere in between.

The confusion is caused by not really understanding the full range of AI used in learning. But there are also a number of general dangers that lurk in the world of AI and algorithms, so it is important that the debate is open and that we do not close down opportunities by being naive about what AI is, how it works and what it can and cannot do for us.

An immediate problem is that ‘artificial’ intelligence sounds a little pejorative. ‘Artificial’ suggests something not real, in direct opposition to what is real. It suggests something unnatural, a mere imitation. This pejoration almost debases the concept and lies behind many of the dystopian attitudes many have towards AI. Rather like artificial grass or artificial limbs, AI successes, no matter how astonishing in learning, feel as though they are second-rate and inferior.

An even stronger pejorative suggestion is the idea that AI is fake or counterfeit, the ‘artificial’ as something feigned or bogus. As the word explicitly compares the abilities of mind and machine, brains and computers, anthropomorphic judgements tend to sneak in. It defines the field as simply copying what humans or human brains do, whereas in practice AI tends to do things that are very different. The human brain may not be the

benchmark here. Humans may not be the measurement of the machine.

Brains and AI

When AI is mentioned, it is often only a matter of time before the word ‘bias’ is heard. They seem to go hand in hand, like ping and pong, especially in debates around AI in education. Yet the discussions are often quite shallow in terms of understanding what AI is, with a heavy dose of anthropomorphism. There is often little analysis behind the claims that AI programmers are largely white males – patriarchal and racist – or the commonly uttered phrase that all algorithms are biased. In practice, you see the same few examples being brought up time and time again: racist face recognition, recruitment and reoffender software. Most examples have their origin in contrarian texts such as *Algorithms of Oppression* by Safiya Umoja Noble (2018) or Cathy O’Neil’s *Weapons of Math Destruction* (2016). More of this later.

To be fair, AI is, for most, an invisible force, the part of the iceberg that lies below the surface. AI is many things, can be opaque technically and true causality can be difficult to trace. But it may be wise to start with the baseline against which AI can be compared – our own brain.

First, let us examine the baseline – us humans. It is true that *all* brains are biased, as shown by Nobel Prize-winning psychologist Daniel Kahneman (2011) and his colleague Amos Tversky, who exposed a whole pantheon of biases that we are largely born with and that are difficult to shift, even through education and training.

Teaching is soaked in bias. There is socio-economic bias in policy, as it is often made by those who favour a certain type of education. Education can be bought privately, introducing inequalities. Gender, race and socio-economic bias is often found in the act of teaching itself. We know that gender bias is present in subtly directing girls away from STEM subjects and

we know that children from lower socio-economic groups are treated differently. Even so-called objective assessment is biased, often influenced by all sorts of cognitive factors – content bias, context bias, marking bias and so on.

The tragedy of teaching is that our cognitive apparatus, especially our brain, has evolved for purposes different from its modern needs. Evolved brains are impressive but they are stuck in our skulls and limited in size, as women would not be able to give birth if they were bigger. There are also evolutionary limits in terms of what can be supported on the top of our bodies along with heat and energy requirements. This makes things very difficult for teachers and trainers. Our role as teachers is to improve the performance of that one organ and effect lasting change in the long-term memory of the brain, yet it remains stubbornly resistant to learning.

It takes around 16 years of intensive *nurturing* and *parenting* to turn brains into adults who can function autonomously. Years of parenting are at times fraught with conflict – the teenage brain, as brilliantly observed by Judith Harris (2011), is obsessed with peer groups. This nurturing needs to be supplemented by around 13 years of sitting in classrooms being taught by other brains – a process that is painful for all involved – pupils, parents and teachers. Increasingly, this is followed by several years, getting longer, in college or university, to prepare the brain for an increasingly complex world.

You do not have to be a teacher or parent for long to realize how *inattentive* and easily distracted brains can be. Attention is a necessary condition for learning, yet they are easily distracted. You are also doomed to *forget* almost everything written in this book, as our memories are not only limited by the narrow channel that is working memory but the massive failure to shunt what we learn from working to long-term memory. And even when memories get into long-term memory, they are subject to further forgetting, even reconfiguration into false memories. Every recalled memory is an act of recreation and reconstitution, and therefore fallible. Without

reinforcement, we retain and recall very little. This makes brains very difficult to teach.

Our brains *sleep* eight hours a day – that is a third of life gone down the drain. Cut back on this and we learn less, get more stressed, even ill. Keep the brain awake, as torturers will attest, and you will drive it to madness. Even when awake, it is prone to daydreaming. This is not an organ that takes easily to being on task.

Technically, brains cannot *upload* and *download*. You cannot pass your knowledge and skills to me without a huge amount of motivated and mediated teaching and learning. AI can do this in an instant. Brains can't *network* and our attempts at collective learning are still clumsy, yet collective learning and intelligence is a feature of modern AI.

As it ages, the brain's performance falls and problems such as dementia and Alzheimer's occur. This degeneration varies in speed and is unpredictable. And, in the end, that single, fatal objection – it *dies*. Death is a problem, as the brain cannot download its inherent and acquired knowledge or skills before death. Memories literally disappear. The way we deal with this is through technology that archives such acquired experience in technical media, such as print, images and now data.

Brains, for all of the reasons above, just do not scale. Limited though online teaching and learning seems, it does scale. It scales as it frees us from the tyranny of time and location. Now that AI, smart software, is enabling and driving online teaching and learning, we can see that AI can scale, not in a linear but in an exponential fashion, as it holds the promise of self-improvement. It learns to get better.

Harari in *Homo Deus* (2016) proposes an argument that eliminates the artificiality of AI. *Homo sapiens* are, like all other living beings, an evolved entity, shaped by natural selection, which is profoundly algorithmic. These algorithms exist separately from the substrate in which they reside. $2 + 2 = 4$ is the same, whether it is calculated in our brain, on wooden blocks, the plastic beads of an abacus or the metal circuits of a

calculator. It does not matter that algorithms reside in one form or another. We should conclude that there is no reason to suppose that our organic abilities will not be replicated, even in many cases surpassed. In other words, algorithmic power resides in the power of the maths to solve problems and come up with solutions, not how accurately they mimic human abilities.

An even stronger argument is that there is every reason to suppose that other substrates will be better. The brain has evolved for an environment that it no longer operates in. It has severe limitations, suited to survival in a place and time with very different needs.

Human bias and AI

One thing we do have in our favour is the fact that our brains have almost certainly evolved in tandem with our use of technology. The extraordinary explosion of activity around 40,000 years ago suggests a key role of tools and technology helping shape our brains.

However, there is one fascinating downside. The brain is inherently *biased*, not only sexist and racist; it has many cognitive biases, such as groupthink, confirmation and many other types of dangerous biases, that shape and limit thought. More than this, it has severe weaknesses, not only inherent tendencies, such as motion sickness, overeating, jet lag, phobias, social anxieties, violent tendencies, addiction, delusions and psychosis.

Another bias is at work here – *neophobia* (fear of the new) – antipathy towards new technology. Neophobia is not new. From Socrates – who thought that writing was an ill-advised invention – to the present day, people have reacted with predictable horror to every piece of new technology that hits the street. It happened with writing, parchments, books, printing, newspapers, coffee houses, letters, telegraphs, telephones, radios, films, TVs, railways, cars, jazz, rock ‘n’ roll,

rap, computers, the internet, social media and now artificial intelligence. The idea that some new invention rots the mind, devalues the culture, even destroys civilization, is an age-old phenomenon.

Neophobia is the product of a superficial reaction about cognition that conflates content with process. The mind, and human nature, is not that malleable and obviously not subject to any real evolutionary change in such a short period of time. Yes, the mind is plastic, but it is not a blank slate waiting to be filled with content from the web. It is far more likely that the neophobes themselves are unthinking victims of the familiar destructive syndrome of neophobia.

Interestingly, the medical evidence suggests that neophobia, as a medical condition, is common in the very young, especially with new foods. It fades throughout childhood and flips in adolescence, when the new is seen as risky and exciting. Then it gradually returns, especially during parenthood, and into our old age, when we develop deeply established habits or expectations that we may see as being under threat. This mimics the demographic attitudes towards technology. Teenagers take readily to new technology; adults tend to be suspicious, even shun it.

There are several other human biases behind our thinking about AI. *Anthropomorphic bias* is arguably the most common, so common that some AI in learning techniques, such as chatbots, rely on its existence. Reading 'bias' into software is sometimes the result of this over-anthropomorphizing.

Availability bias arises when we frame thoughts on what is available, rather than pure reason. So crude images of robot teachers enter the mind as characterizing AI, as opposed to software or mathematics, which is not, for most, easy to call to mind or visualize. This skews our view of what AI is and its dangers, often producing dystopian 'Hollywood' perspectives, rather than objective judgement.

Then there's *negativity bias*, where the negative has more impact than the positive, so the 'rise of the robots' and other

dystopian visions come to mind more readily than positive examples such as cost savings or raising attainment.

Most of all we have *confirmation bias*, which leaps into action whenever we hear of something that seems like a threat and we want to confirm our view of it as ethically wrong. When we are used to classroom, lecture hall and training room models, it is hard to see alternatives.

Indeed, the accusation that all algorithms are biased is often (not always) a combination of ignorance about what algorithms are and four human biases – anthropomorphism, availability, negativity and confirmation bias. It is often a sign of bias in the objector, who wants to confirm their own deficit-based view and apply a universal, dystopian interpretation to AI with a healthy dose of neophobia.

Common charges

First, there is the charge that the root cause of gender and racial bias are male, white coders and that AI lacks transparency.

The main problem here may not be the very real issue of eliminating bias from software, which is what we must strive to do, but the clickbait contrarianism behind much of the debate. *Weapons of Math Destruction* by Cathy O’Neil (2016) was typical of the contrarian literature. An unfortunate title, as O’Neil supposed that algorithms are as bad as Saddam Hussein’s mythical weapons of mass destruction (WMDs) – the evidence similarly weak, sensationalized and cherry picked. This is the go-to book for those who want a lazy critique of AI by reading a pot-boiler. But rather than taking an honest look at the subject, O’Neil takes the ‘Weapons of Math Destruction’ line far too literally, and unwittingly reuses a term that has come to mean exaggeration and untruths.

Racism

The racial mix in the AI coding community is more healthily multicultural than many imagine. AI programmers these days are as likely to be from China, the Far East or the Indian subcontinent, as white. One would struggle to find a white programmer in those powerhouses of AI – India and China. AI is a global phenomenon, not confined to the Western world. The Chinese government has invested a great deal in these skills through Artificial Intelligence 2.0. The 13th Five-Year Plan (2016–20), the Made in China 2025 programme, Robotics Industry Development Plan and Three-Year Guidance for Internet Plus Artificial Intelligence Plan (2016–18) all contributed to boosting AI skills, research and development. India has an education system that sees ‘engineering’ and ‘programming’ as admirable careers and a huge outsourcing software industry with a \$150 billion IT export business. Even in Silicon Valley the presence of many Chinese and Indian programmers is so prevalent that they feature in every sitcom on the subject. The current CEOs of Google and Microsoft were both born and educated in India. Even if you do not accept this argument, the idea that coders infect AI with sexist and racist code is far-fetched. One would not deny the presence of some bias, but the idea that it is common or omnipresent tends to betray bias in itself.

Take PredPol, the predictive policing software. Yes, it has its glitches, but the advantages vastly outweigh the disadvantages – and the system and its use has evolved over time, with pre-processing techniques used to eliminate the problems of bias. The main problem here is a form of bias or one-sidedness in the analysis. Most technology has a downside. We drive cars, despite the fact that well over a million people die gruesome and painful deaths every year in car accidents. Rather than tease out the complexity, even comparing upsides with downsides, we are given one side only. The proposition that all algorithms are biased is as foolish as the idea that all algorithms are free from bias. This is a complex area that needs careful thought, and the real truth lies, as usual, somewhere in

between. Technology often has this cost–benefit feature. To focus on just one side is quite simply a mathematical distortion.

The temptation to anthropomorphize technology is always there. We must resist the temptation to think this is anything but bias. When an algorithm, for example, correlates a black face with a gorilla, it is not that it is biased in the human sense of being a racist, namely a racist agent. The AI knows nothing of itself; it is an artefact, just software. Indeed, it is merely an attempt to execute code, and this sort of error is often how machine learning actually learns. Indeed, this repeated attempt at statistical optimization lies at the very heart of what AI is. Failure is what makes it tick. The good news is that repeated failure results in improvement in machine learning, reinforcement learning, adversarial techniques and so on. It is often absolutely necessary to learn from mistakes to make progress. We need to applaud failure, not jump on the bias bandwagon.

When Google was found to stick the label of gorilla on black faces in 2015, there was no doubt that it was racist in the sense of causing offence, rather than someone being racist in Google, or having a piece of maths that is racist in any intentional sense; this was a systems failure. The problem was spotted and Google responded within the hour. We need to recognize that technology is rarely foolproof; neither are humans. Failures will occur. Machines do not have the cognitive checks and balances that humans have on such cultural issues, but they can be changed and improved to avoid them. We need to see this as a process and not just block progress on the back of outliers. We need to accept that these are mistakes and learn from these mistakes. If mistakes are made, call them out, eliminate the errors and move on. FAIL in this case means First Attempt In Learning. The correct response is not to define and dismiss AI because of these failures, but see them as opportunities for success.

Sexism

True, there is a gender differential, and this will continue, as there are gender differences when it comes to focused, attention-to-detail coding in the area of AI programming. We know that there is a genetic component in autism, a constellation (not spectrum) of cognitive traits, and that this is heavily weighted towards males. For this reason alone there is likely to be a gender difference in high-performance coding teams for the foreseeable future. In addition, the idea that these coders are unconsciously or, worse, consciously creating racist and sexist algorithms is an exaggeration. One has to work quite hard to create sexist and racist code, and to suggest that *all* algorithms are written in this way is wrong. Some may be, but most are not. It is not as if armies of young men are deliberately writing sexist decision-making into their code.

You are likely in your first lesson on algorithms to be taught some sorting mechanisms (there are many). Now it is difficult to see how sorting a set of random numbers into ascending order can be either sexist or racist. The point is that most algorithms are benign, doing a mechanical job and free from bias. They can improve performance in terms of strength, precision and performance over time (robots in factories), compressing and decompressing comms, encryption algorithms, computational strategies in games (chess, GO, Poker and so on), diagnosis–investigation–treatment in healthcare and reduced fraud in finance. Most algorithms, embedded in most contexts, are benign and free from bias.

It is not true to say that all algorithms and/or datasets are biased, unless one resorts to the idea that everything is socially constructed and therefore subject to bias. As Popper (2012) showed, this is an all-embracing theory to which there is no possible objection, as even the objections are interpreted as being part of the problem. This is, in effect, a sociological dead-end.

Transparency

When it comes to decisions that affect the lives of humans, we must, of course, be careful with the introduction of bias into algorithms and data. With algorithms, one can insist on transparency, but even the definition in GDPR Article 14 – ‘meaningful information about the logic involved’ – is a fudge.

There is a danger that the AI becomes black box, esoteric maths that will become increasingly opaque. Few could read the maths that constitute modern algorithmic and machine learning. It is a variety of complex techniques that is readable by a minority. Nevertheless, this is not as completely opaque as one imagines. A lot of this maths is open source, from Google and others, as is the maths itself, in the AI community. Of course, proprietary systems abound, but it is not ‘all’ locked up in black boxes. One could argue that this is also true of most ‘teaching’, and the process (algorithms) and knowledge (data) lies in the brains and practices of teachers, lecturers and trainers, much of it totally inaccessible and opaque.

The inner workings of AI can be opaque, especially machine learning using neural networks. But compare this with another social good – medicine. We know it works but we don’t know how. As Jon Clardy, a professor of biological chemistry and molecular pharmacology at Harvard Medical School, says, ‘the idea that drugs are the result of a clean, logical search for molecules that work is a “fairytale”’ (Johnson, 2015). Many drugs work but we have no idea why they work. Medicine tends to throw possible solutions at problems, then observe if they work or not. Now most AI is not like this, but some is.

We will need more work on explainability and transparency. Remember that this is not always possible, even in human decision-making, so when we get AI right, it should have the edge. This is not to say that all bias will be eliminated. A further check may need to be honest and compulsory admissions about bias, where the service is cleared by regulators as being

unbiased or biased and to what degree it could potentially be biased.

Of course, we need to be careful about bias, but in many cases, especially in education, we have a far greater chance of tackling these problems using AI than by sticking to good, old-fashioned bias in human teaching. The imperative for transparency may also kill the very things we value. If you demand total transparency in algorithms, Google and Google Scholar may have to be banned in your institution. Does that seem like a proportionate reaction? A sense of balance is needed between solving problems and transparency.

AI as statistics

AI is not conscious or aware of its purpose. It is just software and as such is not 'biased' in the way we attribute that word to 'humans'. The biases in humans have evolved over millions of years with additional cultural input. AI is maths and we must be careful about anthropomorphizing the problem.

Much AI and data analysis is statistical analysis and pattern-matching. We are at the zenith of two millennia of maths, probability and statistical thinking, much of it wholly concerned with the elimination of error and bias. The degree to which findings in science are possibly erroneous is often fundamental to the reporting of the methodology and mathematics used in research. In fact, unlike human bias, as AI is largely a set of statistical techniques, the whole rationale is to work towards the elimination of bias or the identification of possible false outcomes. We so often forget that bias in AI is real but different, as it is mathematical or statistical bias. In fact, there is a definition of 'bias' in statistics, which is not a pejorative term, but precisely defined as the difference between an expected value and the true value of a parameter. If the value is zero, it is called unbiased. This is not so much bias as a precise recognition of difference.

The statistical approach at least recognizes these biases and adopts scientific and mathematical methods to try to eliminate such biases. This is a key point – human bias often goes unchecked; statistical and mathematical bias is subjected to rigorous checks. That is not to say that it is flawless, but error rates and attempts to quantify statistical and mathematical bias have been developed over a long time, to counter human bias. That is the essence of the scientific method.

One must also distinguish between algorithms and data. Data is where most of the problems arise. One example is poor sampling: too small a sample, under-representations or over-representations. Data collection can also have bias due to faulty data-gathering in the instruments themselves. Selection bias in data occurs when it is gathered selectively and not randomly, and this is most likely to occur in the human part of the process.

In discussions of bias in AI, and in AI for learning, humans – and all teachers are humans – are often biased in themselves. So the false comparison between virtuous, unbiased teachers and vicious, biased AI is a false dichotomy.

Avoiding bias

In learning, the glass always seems to remain half empty, even when the data suggests it is close to the brim. But it is important to be neither too utopian nor too dystopian when discussing AI in learning. We must resist the temptation to have either pessimism or optimism as our starting point. Realism should be our watchword.

Fairness

Bias in AI can, of course, damage the reputation of systems, companies and AI in general. The danger is that people are dealt with unfairly in employment, legal, health, finance and education if the AI decision-making is inherently biased. If the

decision-making is faulty, it can have serious consequences for those on the end of that decision-making.

The danger is also that bias is not only baked into AI but also massively scaled. The problem, oddly, is that the bias in the AI datasets is likely to have been produced by humans, namely words in text that show a gender bias or bad sampling that unfairly targets on race. These problems are real and the scaling issue is real, as the problem may indeed be exacerbated and amplified by AI.

Bias, in practice, is far more likely to be found in the data and even then it usually draws upon established and pre-existent bias in society. Garbage in: garbage out is a general problem in software, but arguably more dangerous in AI. Race, age, gender, socio-economic and other biases can be introduced through the data used by AI and/or the algorithmic model that uses the data. The aim is obviously to avoid discrimination by identifying the many biases that are possible, whether they are in the data or algorithmic models, and eliminate them or at least quantify them. One of the arguments used for transparency in algorithmic models is the need to identify such bias.

The scrutiny of bias in AI is actually more likely to inform us about human bias, bringing the problem of bias to the surface. AI can clearly do much to eliminate or at least reduce human bias. It has already made us more aware of bias as the ever-present human phenomenon in decision-making. Human bias is relatively fixed and very difficult to train out of our brains, whereas AI bias is a set of problems that can be solved. AI in this sense raises the bar for us all.

To be fair, a huge effort is being made to introduce fairness into algorithms. To be actionable, we need concrete ideas that can either prevent or at least significantly reduce bias. Methods have been developed that protect attributes such as race and gender, so that outputs are not biased. There are also pre- and post-processing techniques that can be applied to minimize that carryover of human bias. These pre-processing techniques have been tested on datasets such as the ProPublica COMPAS prison

recidivism dataset, criticized for being racist. Pre-processing can construct counterfactuals or apply fairness criteria. Post-processing techniques have also been used that transform predictions to avoid bias. This is the way in which bias is most likely to be reduced, using data techniques that identify, reduce and try to eliminate the inputs.

There is a good argument for being cautious here, proceeding carefully without halting what could be beneficial decision-making, optimizing the delivery of social goods.

Pedagogic concerns

There are also some ethical considerations that are specific to learning – that it may do more harm than good. These issues also have to be surfaced.

Too deterministic

One danger, some claim, in the algorithmic age is that it may start, without us realizing, to funnel us towards groupthink in learning. We get fed the same newsfeeds, sold the same popular stuff, hear the same music, buy the same books, get trapped in our bubbles. This debate is alive in learning, where adaptive, AI-driven platforms can deliver personalized learning but, if we are not careful, may also constrict the nature of learning. We need to be careful that the learner retains the curiosity and critical thinking necessary to become an autonomous learner. The danger is that, on one hand, AI could deliver narrow, deterministic, prescribed pathways, not allowing the learner to breathe and expand their horizons, and apply critical thought. On the other hand, many argue, it may expand our horizons. The evidence on generative AI suggests that an open, dialogue-based system opens up horizons, as millions have found to their personal benefit.

This is really a design issue, as sometimes a narrowing of focus is necessary and desirable – when diagnosing

misconceptions and teaching tight but difficult concepts and processes, such as maths and science; and sometimes it is not, when a more critical approach is needed.

AI does not always make the learning narrower or more deterministic. That is because many of these systems still leave lots open, indeed suggest open options, to the learner. One can allow the learner as much freedom as you wish by making the course porous and curating content. One tool that automatically creates online learning content does this by using AI to create links out to further exposition, explanation and exploration. It deliberately suggests lots of links to the student to content on the web, based on what they look at and their performance. This is the sort of thing they are likely to receive from a teacher or lecturer, but with AI it is personalized. It tracks what you do and predicts what it thinks you may need to supplement your knowledge. Far from closing you down, it opens up the mind to other sources and possibilities. The degree to which human agency is included in AI-driven systems is a design issue. Learning is about giving learners the teaching and tools to support them in their learning, not just leaving people to flounder around on their own without support.

Makes it too easy

There is an argument that says you cannot make learners' lives too easy. They need to make the effort, overcome problems, even struggle. There is a real piece of psychology that backs this up: Bjork's concept of 'deliberate difficulty' (Bjork and Bjork, 2011). However, this is no excuse for leaving students to drown and fail, with light feedback. Dropout and failure are not desirable. In fact, AI-driven systems can be finely tuned to make levels of difficulty fit that of the individual student, to get that balance of achievement and challenge. You have a far greater chance of delivering deliberate difficulty through AI than through one-size-fits-all courses.

Surveillance

Picture this for a moment. You are a teacher in a classroom. All of your students are wearing headsets, but each has a coloured light on their forehead – red means full-on focused, yellow focused and blue calm. This is not science fiction. It has already happened and raises all sorts of questions about learning and ethics (Ye, 2019). Once we know what people are thinking and feeling, there is no privacy. Consciousness itself is open to scrutiny and can be accessed, not only by teachers but also the unscrupulous. There is literally nowhere to hide one's thoughts. Having an open mind is one thing; literally opening up minds with technology is another.

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AI and employment

It is vital that learning professionals in education and training take an interest in the impact of AI on society and employment, underemployment or unemployment. Generative AI is likely to change things for better or worse, but certainly forever. As the landscape of work and life will be changed by AI, so will the learning landscape. So let us look at its impact on work and the professions, including learning. We have created the network we call the internet, but that network is now being increasingly managed by AI. This has some profound consequences.

Machines led to workers moving from fields to factories, then, with robots in manufacturing, from factories to offices and services, to produce long-term, structural, blue-collar underemployment and unemployment. We may now also be seeing the start of white-collar under- and unemployment, as AI becomes capable of doing at least some of what even well-educated workers and graduates could do in the past. Each technological revolution tends to bring in job losses as well as new opportunities. Let us also not imagine for one moment that the learning field is immune from such economic shifts. We have already seen its effect on reducing the number of librarians, trainers and other learning professionals. There is, undoubtedly, more to come.

John Maynard Keynes introduced technical unemployment into economics in the 1930s (1930/2010), and as AI has hit the streets, there has been a flood of texts such as *The Rise of the Robots* (Ford, 2015) and *The Future of the Professions* (Susskind and Susskind, 2015). What we do see is what Brynjolfsson and McAfee in *The Second Machine Age* (2014) called the ‘great decoupling’ of productivity (profits) and wages. Wage stagnation is real and is causing political turmoil and social unrest. In truth, we face an uncertain future but one where we

will have to adjust and adapt to finding meaning in life, not work. It seems clear that 'learning' to adjust, adapt and find meaning in life is essential. This is why AI in learning may be a necessary condition for the future happiness of our species.

'47 per cent of jobs will be automated...'

This turned into a meme in newspapers, articles and conference slides. It is from a 2013 paper by Frey and Osborne. First, it refers only to the US, and only states that such jobs are under threat. Dig a little deeper and you find that it is a hopelessly speculative piece of work.

The fact that 47 per cent of jobs may be automated makes a great headline, but it was a lousy piece of analysis. Change does not happen this way. In many jobs, the context or culture means that complete automation will not happen quickly. There are human fears and expectations that demand the presence of humans in the workplace. We can automate cars, even aeroplanes, but it will be a long time before aeroplanes will fly across the Atlantic with several hundred passengers and no pilot. Self-driving cars are also likely to take longer. There are human perceptions that, even if irrational, have to be overcome. We may have automated waiters that trolley food to your table, but the expectation that a real person will deliver the food and engage with you is all too real.

Organizations grow around people and are run by people. These people build systems, processes, budget plans and funding processes that do not necessarily quickly lead to productivity gains through automation. They often protect people, products and processes that put a brake on automation. Most organizations have an ecosystem that makes change difficult – poor forecasting, no room for innovation, arcane procurement and sclerotic regulations. This all mitigates against innovative change. Even when faced with something that saves a huge amount of time and cost, there is a tendency to stick to existing practice. As Upton Sinclair (1994) said, 'It is

difficult to get a man to understand something, when his salary depends on his not understanding it.'

The process of automation in employment is a messy business with many variables. Heuristics can help here. Some categorize jobs as:

- cognitive versus manual;
- cognitive routine;
- cognitive non-routine;
- manual routine;
- manual non-routine.

All of these are now up for grabs as AI can now automate cognitive, manual, routine and not so routine tasks. But even the distinction between manual and cognitive is not mutually exclusive. Few manual jobs require no knowledge, planning or problem-solving. These can be useful rules of thumb, but the world rarely falls neatly into these binary or four-way categories. Yet they often lie at the heart of predictive analysis. Beware of simplistic heuristics.

In truth most jobs will be partially and not completely automated. This has been going on for centuries with technological advances. Sure, horse grooms and carriage drivers no longer exist, but car mechanics and taxi drivers do. Typesetters have been replaced by web designers. ATMs have simply changed the nature of bank tellers, not completely automated the process. Indeed, in many professions the shift has been towards more customer service and less mechanical service. What matters is not necessarily the crude measure of 'jobs' being automated, but rather 'activities' – specific tasks, competences and skills – being automated.

What is also often forgotten in such analyses is the business case and labour supply context. Automation will not happen where the investment cost is higher than hiring human labour, and is less likely to occur where labour supply is high and wages low. We see this in countries where the population is high and growing, or labour costs low through immigration,

making the business case for innovation and automation low. Many jobs could be automated, but the lack of investment money and the availability of cheap labour and low wages make the human bar quite low. There are complex economic decision chains at work here that slow down automation.

‘65 per cent of today’s students will be employed in jobs that don’t exist yet...’

This was another quote that has launched a thousand conference presentations. Yet, although included in a serious World Economic Forum (2016) report, it has no clearly identifiable source.

This is the sort of exaggeration that feeds bad consultancy. Most, in practice, will be doing jobs that have existed for some time. No doubt many will simply be doing jobs they did not plan on doing (and don’t like) or jobs that have changed somewhat through automation. Predicting which jobs or activities get automated is easy compared with predicting what new jobs will be created. The net total is therefore difficult to establish. Fewer people may be needed in certain areas, but new jobs will be created, especially in services.

The crude headlines will continue, but we’re starting to see more detailed and realistic analysis that will lead to better predictions. This is important, as educational bodies need to be able to adapt to what they will be required to teach as well as what they teach and to whom. As the change accelerates, education and training will need to be more sensitive and adaptive to the changes. This means more accurate predicting of demand and quick adjustments in supply. There is likely to be no sudden shift but a gradual bite by bite into activities within jobs. AI will change the world and the world of learning, but not in the way we think it will.

Impact of AI

When ChatGPT 3.5 was launched 30 November 2022, research quickly followed on productivity. Solid work was done with real people showing remarkable impacts, not just in saving enormous amounts of time, but also, crucially, dramatic increases in quality. Its immediate effect was to scale people, whether learners or workers. It had such utility across such a wide range of learning and practical tasks that its productivity seemed clear.

In an early look at the impact of generative AI on the labour market, Eloundou et al (2023) claimed that generative AI could impact 80 per cent of US jobs in coming years and that even if AI may not take your job, you will have to work with it. This was a common conclusion in the many reports written after ChatGPT was released. They found that ‘around 80% of the US workforce could have at least 10% of their work tasks affected by the introduction of LLMs, while approximately 19% of workers may see at least 50% of their tasks impacted’. Without making time predictions, they found that it affected ‘all wage levels, with higher-income jobs potentially facing greater exposure...’. Access to an LLM makes the completion of about 15 per cent of all workplace tasks in the US significantly faster at the same level of quality. With full integration, this increases to 47–56 per cent of all tasks. The economic impact will therefore have considerable social and policy implications.

In analysed data, it seemed as though going to college opened you up to a much higher probability of replacement than those with practical skills. We had always assumed that repetitive working-class jobs would be replaced and automated by AI. In fact, it was the business colleges in the US that led the charge in simply shipping those jobs to countries where the cost of labour was cheaper. It came as a shock when it was high-income, knowledge workers who were most at threat. If you went to college, it was coming after you first.

A substantial piece of research came from IBM ‘Augmented work for an automated, AI-driven world’ that said much the same: ‘AI won’t replace people – but people who use AI will

replace people who don't.' A substantial survey of 3,000 C-level execs in 28 nations, along with 21,000 workers in 22 nations, concluded:

- 40 per cent of workforce need to reskill due to AI over next three years;
- 87 per cent of execs believe employees likely to be augmented not replaced;
- AI allows workers to do more with less knowledge.

Recommendations for HR included:

1. Shift role from policy enforcer to value driver.
2. Empower HR to influence tech investments and use generative AI for more purpose-driven work and to enhance well-being.
3. Make HR the go-to ecosystem advisor for access to in-demand skills, as needs evolve.

This is not happening, as HR remains in its silo, concerned mainly with low value abstractions such as resilience, leadership and DEI. Skills and competences have been sacrificed for courses on often abstract concepts. Despite decades of large-scale research showing that DEI training does not have any significant impact, it remains a huge spend. All the while we have massive skills shortages.

Productivity

We once worked in fields, then the factories, then offices. In 1800 in the USA, 75 per cent of jobs were in agriculture. By 2000 that has dropped to less than 2 per cent, as it had been largely automated, with scientific improvements leading to better yields. A similar story unfolded in manufacturing after the Second World War, where productivity rose through automation and the number of blue collar workers fell. It is likely that white-collar work will suffer the same fate. As AI improves, the productivity increases will be in cognitive skills.

Those in middle management and administration are likely to be seriously affected. If AGI occurs the dog leg in productivity may be extreme, with the immediate and rapid shedding of jobs.

An MIT study by Noy and Zhang (2023) took 444 professionals (HR professionals, managers, marketers, consultants, data analysts and managers), split them and showed significant increases in productivity in the GPT group, not just on time saved (0.8SD) but also increased quality of output (0.4SD). It would seem that augmentation with AI makes you both faster and better.

A more substantial study by Mollick (Dell'Acqua et al, 2023) explored knowledge worker productivity in detail, identifying several features of this new way of working and the detailed nature of our relationship with generative AI in the workplace. A large-scale trial in the Boston Consulting Group, with 758 consultants using GPT4 across 18 tasks, showed:

- 12.2 per cent more tasks completed;
- 25.1 per cent faster completion;
- 40 per cent higher quality.

Those using ChatGPT 4 proved more productive, saved time and the output was of higher quality. A fascinating additional finding was the difference between what were called 'centaurs' and 'cyborgs'. Centaurs used the generative as an assistant or add-on, a bit like using a horse's body to gallop faster. The more productive group, the 'cyborgs', tended to integrate generative AI into their entire workflow. Generation is not the point, it is transformation.

The mistake that is being made here is in not taking this to its logical conclusion. We are stopping dead at comfortable phrases around assistance and augmentation, such as 'AI will not take your job', 'you will work with it', when it is already clear that, given the blistering pace of progress in utility and productivity, it is also likely to take jobs. If you are sitting at home on Zoom,

doing meetings and just writing stuff for a living, doing what ChatGPT can do, maybe you shouldn't be doing what you do.

To take one specific example, Choi (Choi and Schwarcz, 2023) looked at the impact of generative AI on legal analysis. Can generative AI outperform lawyers they asked? On exams then taken by GPT 4, humans and AI-assisted humans, GPT 4 outperformed both the average student and the average student with access to AI.

Creativity

Creativity has been used as the last bastion of human exceptionalism but evidence quickly emerged of erosion into that domain.

In a study by Girotra et al (2023) LLMs were researched for the role they could play in ideation and innovation. They compared ideation using ChatGPT 4 versus students at an elite university. The results showed that ChatGPT 4 generated ideas much faster and cheaper than students and that those ideas were on average of higher quality. There was also higher variance in quality with the vast majority of the best ideas generated by ChatGPT 4 and not by students. Furthermore, giving ChatGPT a few examples of highly rated ideas increases performance. It would seem that ChatGPT has the edge over humans on ideation, one species of creativity.

Upping the stakes, Doshi and Hauser (2023) asked, 'Can generative AI enhance creativity?' They bravely looked at the impact of generative AI on creative artistic output, namely short stories. The results were somewhat shocking. Stories written with access to generative AI ideas were seen by readers as better written and more enjoyable. Improvements were measured at 22–26 per cent over less creative writers.

Professions and AI

AI may decimate the professions. That is the big idea being debated by economists. In the same way that agricultural jobs were wiped out by mechanization, working-class manufacturing jobs by robots and typing pools by word processors, so professions, such as managers, lawyers, doctors, journalists, architects, accountants and so on, will be wiped out by smart software. Or, as is more likely, the professions will adapt to this new age of algorithms and manage the high-value management and human tasks in their professions. It is clear, however, that change is already here. Around the world, especially in the richer nations, the truth is dawning that we need fewer managers. The middle class is now waking up to the fact that they are the new working class, as their roles diminish.

Yet HR and L&D are eerily silent. How many leadership courses are being delivered today without any mention of AI, despite the fact that we know that its impact has been and will continue to be huge? How many learning professionals are really studying what is happening here? To sit back and watch this happen, without serious debate and preparation, is irresponsible. A more immediate debate is how to cope with AI in the short to medium term.

It is now clear that generative AI increases productivity on administration and this allows more focus on judgement work. Monitoring, scheduling and reporting are all areas that are likely to be more automated through AI. The room for efficiencies and increases in productivity here are enormous, especially among those who primarily write for a living, as organizations are essentially paying expensive salaries for relatively routine and banal reporting and summarizing work. *The Future of the Professions* (Susskind and Susskind, 2015) is packed with examples of judgement-making AI, across a range of professions. There will be a pendulum swing of some magnitude as AI creeps forward doing more and more, eating into the judgement sphere.

Consultancy and research organizations rely on datasets to provide insights, which they synthesize into reports. At this

level, their business is under threat. As this function will shift inside organizations. The modern manager has to be able to manage data-harvesting, data management and AI-driven tools that lead to insights and decisions. The eventual result will be the use of AI-driven insights directly or internally.

Finance was revolutionized by the spreadsheet, which automated not only calculations, but also captured formulae and models. This has gone a lot further with AI, where a huge volume of financial transactions trades, monitoring, processes, reporting and audits are more AI-driven and it promises to do much more. Disruption is likely in audit, where a few big companies dominate. All it will take is one radical player to offer AI-driven data-gathering, analysis and report writing and there will be carnage.

Communications were faster and cheaper as the telegraph, telephone, email and messaging emerged. The key to good communications is dialogue, and AI is starting to deliver real dialogue with technology. Speech recognition, text to speech and machine translation (in real time) have already had an impact. All of these are AI-driven. Efficiencies in communications through the use of AI within the technology, but also to improve its efficacy, are coming.

Sales and marketing has already taken up generative AI with a vengeance. In addition to the seismic shift towards online marketing, we see increases in productivity on output and data. Sales management and lead-generation has shifted online and now involves AI targeting and delivery. Clumsy and ill-focused sales can be channelled and streamlined with AI.

Nothing is slower and more antiquated than legal processes which are already being automated by AI. HR is another service where processes will be streamlined and made more efficient by AI as will L&D. Recruitment has already been influenced by algorithmic search, selection and matching. This is an area ripe for falling fees and automation.

AI will redefine what it means to be a manager. Managers will increasingly not manage people, but will harness and

manage technology. Management used to be about people and processes, but there are new technologies around, especially AI, that also need to be managed. Old-school managers who cannot handle delegation to technology will struggle. The truth is that many will not be needed.

As working from home has increased through the use of smart tech, mobile phones, web conferencing and cloud-based services, so the support role has shifted decade after decade from people to technology. The typing pool was the first to go. Back-end office support staff were next, then the secretary and now the PA. Intelligent personal assistants, in the form of chatbots are emerging, which act as personal PAs. They will act as a convenient interface to AI to do the mostly administrative tasks that managers need: scheduling meetings, arranging travel, expenses and so on.

Learning jobs and AI

Will librarians, teachers, lecturers and trainers be subject to the same pressures as other white-collar jobs from tech? It has already happened, is happening and will continue to erode employment. However, it is not a simple net loss but a complex formula.

‘Technological unemployment’ was the term coined by Keynes to describe the economic prospect of technology-driven job losses and there has been debate for centuries on the issue. Far from being a simple count of job losses against gains, it is a complex economic issue.

Substantial job losses will result from the effect of technology and AI. In the recent literature, however, the learning world comes out relatively unscathed. But I’m not so sure about their definition of ‘learning’. There’s a tendency for analysts and educators to see the ‘learning world’ largely through an institutional lens – schools, college, universities. Yet this is merely a fraction of the learning world. Most learning takes place outside of these institutions, in non-institutional contexts

– the workplace, at home and informally. If we define the ‘learning’ world broadly, and include all forms of cognitive improvement, we see libraries, bookshops, online bookshops, television, newspapers, online, social media and many other sources of learning, as causally relevant. When seen through this wide-angle lens, you start to realize how profound technology has been in replacing the human component in learning.

Technology has always democratized, decentralized and disintermediated knowledge. It happened with the invention of writing, then the alphabet, paper and printing. These were all profound technological boosts to the transmission of knowledge, which expanded human progress by freeing knowledge from the oral tradition, so dominant for millennia. Electronically, we then had the telegraph, telephone, radio, film and TV. In the computer age, this started at the bottom of the learning pyramid, with open access to ‘knowledge’ through the internet.

Google gave us access, Wikipedia and many others gave us content, and jobs were lost in the publishing business. The huge encyclopaedia sector died on the spot and libraries became less useful. The fall in footfall at libraries and bookshops has little to do with the fall in reading and everything to do with the rise in online access to knowledge and cheap online books. There has been a renaissance in reading and writing by young people, who read and write, it would appear, every few minutes on their phones or computers. At this level, if you want to know something, you go to the web. The one-third fall in librarian jobs in the US since 1990, after decades of rises, correlates well with the rise of the computer and online access. This was a well-paid, graduate-level profession that is being decimated by online technology. The effect of these services on losing knowledge delivery jobs has been substantial. Large tech companies such as Google tend to have tens of thousands of employees, unlike previous global behemoths that had hundreds of thousands.

Access to YouTube is also likely to have had an effect on practical jobs. Many, many millions of ‘how-to...’ tasks have been completed by individuals who simply looked it up on YouTube. Whether it is DIY, car repair, learning chords on a guitar... direct instruction is now available – and off you go. The availability of mobile devices allows you to take the instruction to the task. Save money, save time – jobs lost.

It is in organizational learning that job losses are felt directly. The rise of online organizational learning has been eating away at the training market for over 30 years. This has, without doubt, led to a dramatic reduction in the number of trainers in organizations. Gone are the once common training centres and courses that were months long. Online learning, simulations, blended learning, flipped classroom, LXPs and LRSs and the pandemic have all taken a massive, shark-sized bite out of the classroom training market and undoubtedly led directly to the loss of jobs. The global corporate e-learning market amounts to tens of billions, with healthy compound annual growth. The job losses will continue.

Again, the rise of online courses has eaten into the institutional market. It is likely to impact higher education, where the projected increase in human teaching staff will, to a degree, be replaced by online delivery. Wherever online solutions perform even part of the teaching process, jobs are at risk. If you largely lecture, that part of your job is definitely at risk, as it is easily recordable, replicable and can be distributed on scale.

Other trends include AI and adaptive learning that promise to eat into the teaching components, first as hybrid technology-enhanced teaching models, then autonomous models. Full adaptive courses are already on sale and the first adaptive learning degree in biology has been launched.

This is unlikely to have huge impact in schools, where responsibility towards young people involves far more than just teaching. Even here, however, it is hard to imagine that technology will have no real effect, especially in reducing

workload through the automation of administration, such as timetabling and lesson planning. The largest gain is likely to be in assessment and marking, the one area where teacher workload could be significantly reduced.

Online delivery, especially when smarter and adaptive, may also offer opportunities to provide courses where the recruitment of teachers is difficult. There is often a correlation between rising demand in a subject and teacher shortage, as many with those skills will have better employment opportunities applying their skills rather than teaching them.

Just as we have seen Uber rise without owning any cars, and Airbnb rise despite owning no hotels, we are likely to see disintermediation on a global scale in education, where learning is delivered by organizations that may have very few teachers.

Under- and unemployment

There are benefits that could accrue to people through technology not reflected in the job figures – cheaper prices and new products for everyone and increased productivity. Then there are the benefits of higher wages, new jobs in technology, new jobs created by technology and increased investment.

But the spectre of inequality is there and the danger is that this may accelerate with AI. While the optimistic replacement of jobs may have been true in the industrial era, many argue that it is no longer true in the information age, where fewer people design technology that replaces large numbers of blue-collar, and now white-collar, workers. There is a new concern that AI can take graduate and professional jobs, just as robots took factory jobs and farm mechanization took agricultural jobs. We may now face the inevitable rise of long-term structural unemployment, caused by technological innovation. Few professions will be immune to these pressures, even in the learning game.

However, we must be careful to not get swept away by dystopian ideas on technology, as some predict no real impact on the overall employment landscape. The likely outcome in many areas of education, apart from the base of the pyramid jobs such as librarians and paper publishing, is the gradual replacement of jobs by technology aids and alternatives. However, the one caveat (and it is a big caveat) is that predictions about what technology is capable of have been poor, and there may be some radical and unseen shifts that come along.

An interesting brake on the problem could be the very thing we are discussing – increased delivery of education to create jobs. However, we must be careful in ascribing employment, and especially technological innovation and employment, to educational causes. Not all economists agree that the continual spend and expansion on higher education leads to increases in employment. This has been obvious in southern Europe and is now obvious in graduate unemployment and underemployment figures (graduates doing non-graduate jobs).

Paul Krugman (2012) and many others no longer think that education is the cure. He reminds us that there has been a steady ‘hollowing out’ of middle-class jobs. This is not a simple heads versus hands issue, where smart people retain jobs and manual workers lose their jobs. It may be the other way around. Alan Blinder (2007) has suggested that we are now in an era where high-paid jobs are easier to offshore than low-paid manual jobs. Few now think that marching everyone in lock-step through expensive higher education is the answer to technological unemployment. It may even be a placebo that leads to debt burdens that limit growth and exacerbate the problem.

An analysis by the Economic Policy Forum between 2000 and 2016 (Gould and Wilson, 2017), when millennials entered the job market, found ‘little to no gain’ in median annual earnings, part of a longer trend of wage stagnation going back to the 1970s. Over the same period, educational effort and

achievement has sky-rocketed, yet the benefits in terms of wages are not clear. The economist Bryan Caplan, in *The Case Against Education* (2018) sees higher education as having developed into 80 per cent signalling, a signal that you are employable – a huge waste of money and resources.

As the evidence builds, AI is seen as a serious threat to millions of jobs. If we look at the top companies by market cap in 2006 and now, we see that the tech giants predominate. This trend is likely to continue, with a worrying trend towards monopolization. They have huge market capitalization, enormous cash reserves but employ far fewer people and, at the same time, destroy jobs elsewhere. If they do not destroy jobs, they almost certainly create underemployment. Amazon may employ hundreds of thousands of people, mostly low-paid warehouse and delivery staff, but how many librarians, bookstore staff and shopkeepers have we lost? Some argue that the hollowing-out of the middle class has already taken place, via automation. Even the low-paid warehouse staff are being replaced by robots and delivery drivers possibly by drones.

If, as most agree, the net result is far fewer jobs, or at least less time at work, we need to turn our attention to the political consequences.

This is primarily a moral and political issue, and the basic minimum wage has been touted as the obvious answer. The desire to get on in life and be productive seems like too powerful a cultural force to give way to a supported life of leisure through taxation and public expenditure, even if it were possible. A more likely scenario is one where our very real, personal needs for growth and development are also satisfied.

What is more likely is the need to educate and train people to do many different things in this new AI-infused, smart economy. This is hardly likely if the costs are at their present levels, and rising. This is not about the ever-expanding campus model. If we keep to the idea of education as being on campus, in institutions, always delivered by expensive ‘teachers’, there will be an impasse.

To some it seems likely that we face a return to a human condition that last existed before the industrial revolution, when machines automated and stimulated the global economy, as well as before the agrarian revolution when machines destroyed work on the land. It is a return to an age when people did not work in factories but prepared the land, sowed seeds, waited and harvested. Work was far more episodic in those times, with long periods, not of leisure, as a long winter with meagre food should not be romanticized, but there was certainly more time.

In our technological age, when the communications technology we have both entertains and educates, it may also provide opportunities, on scale, to educate us for this new future. It could, just as it has in robotics, reduce the costs of education so much that we can all, including those in the developing world, benefit from its scalable dividend.

It is right to worry about the future, as recent events portend great dangers. Trump has shown that people are angry at losing the gainful employment and lives they once had. In supposedly prosperous Europe, the southern countries have massive adult and youth unemployment. Graduates, saddled with debt, are now struggling to find work that matches the supposed promise. And, of course, unemployment is rife in the developed world, leading to hugely disruptive migration. Climate change will almost certainly exacerbate all of this.

But to react to technological change by taking a stand, as either a technophobe or technophile, rather than exploring and considering alternative possibilities, would be a mistake. We can surely see the glass, not pessimistically as half empty, or optimistically as half full, but realistically as an opportunity to create a future that is centred around our real, human needs.

We should not get too attached to the idea that those needs were ever the drudgery of fields, factories or offices. If technology has freed us from using our bodies to do menial tasks, from the planting and harvesting of crops, to the repetitive tasks on a factory floor or in the drudgery of an

office, so be it. Having freed our bodies from manual tasks, can we not also free our minds from repetitive and mind-numbing cognitive tasks with AI? Aren't there better things to do with our minds than administer, process and manage?

Bodily, we may be better off, as we can keep ourselves fit and in good health, even have superior help, through AI-assisted care, in our old age. What we desperately need is a rather old-fashioned view of human nature as being nurtured by learning. Where education is about autonomy, developing young minds to be curious and open, then we must see lifelong learning as a possibility and not a clichéd conference phrase. Learning takes time, and rather than being rushed in batches and bursts in schools, institutions or on corporate courses, we may be able to free learning from the tyranny of time, place and costs.

Conclusion

Shiva destroys as she creates, and so it could be with AI. AI needs to be applied to education and training to bring the same productivity efficiencies that it brings elsewhere. There is nothing unique about teaching and learning that makes them immune from AI. Teaching is a means to an end – learning. If that can be done through AI, we should grasp the opportunity, especially if it makes learning faster, cheaper and better. If we are facing a future where many people do not have a job or have more leisure time or need to be trained in newer skills, AI may then be a partial cure for the very ill it creates.

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AI and extinction

‘Doomsday is hot,’ claims Stephen Pinker (2018a). What he means is that it is all too easy to get fashionably apocalyptic. This could make matters worse in learning: seeing AI as perpetually dystopian risks ruling it out as a solution to major problems, such as education, health and climate change. Catastrophism is not new. As he explains in *Enlightenment Now* (2018b), end-of-days scenarios have been around for a long time, including recent scares around overpopulation, running out of resources and, in technology, the false belief in Y2K. He also points towards the false belief that Iraq had weapons of mass destruction as an example of a dystopian scenario that led to a catastrophic war and the deaths of thousands of innocent people. This, as we shall see, is sadly mirrored in one of the most popular AI contrarian tracts, *Weapons of Math Destruction* by Cathy O’Neil (2016).

The real danger, he argues, is to think of AI as dangerous, thereby avoiding it as a solution, albeit partial, to the real catastrophes we are trying to avoid. We only have so much bandwidth to deal with global problems, and unnecessary doom-mongering about relatively small, remote and distant risks may distract us from the risks we know we face here and now. As a psychologist, he puts this type of thinking down to our inability to deal with risk, seeing all risks as equal, rather than calculating their actual probabilities. Then there are availability and negativity biases. We pay attention to what we regularly see on the news, the zeitgeist issues, and have a bias towards the negative.

The threat of runaway AI as an existential threat is slim, if not negligible, in that we create and engineer this technology, so we will be able to test and control its application. Catastrophic autonomy is not, Pinker (2010) thinks, a probable outcome. AI is

not a threat on par with the problems created by humans. We live with our imperfect evolved biases and traits. There is every reason to suppose that we will design AI that is largely free from those traits and destructive objectives. AI is not pulling us towards an imminent and cataclysmic end; it may well be the very thing that saves us from more pressing problems. Learning is one of those pressing problems.

Another calming consideration comes from Daniel Dennett in *From Bacteria to Bach and Back* (2017), where he sees a fusion of human evolution and AI, with consequences for the ethics of AI. He takes a holistic and forgiving view of AI. As he rightly says, we make children without actually understanding the entirety of the process – similarly with new, generative technology. We already trust systems that are way more competent than we humans, and so we should.

Almost all AI has been parasitic on human achievements; corpuses of text, images, music, translated content, maths and other mother lodes of data. Rightly sceptical about strong AI, master algorithms and super-intelligent agents, Dennett's call is for us to keep an eye on the boundaries between mind and machine, as we have a tendency to overestimate the comprehension of the machines, way beyond their actual competence, anthropomorphizing comprehension. We see this with even the most casual encounters with chatbots and devices such as Alexa or Google Home. We all too readily read intentionality, comprehension, even consciousness into technology when it is completely absent.

By adopting regulatory rules around false claims of anthropomorphism, we can steer ourselves through the ethical concerns around AI. Overreach and concealing anthropomorphism and false claims should be illegal, just as exaggeration and side effects are regulated in the pharmaceutical industry.

More than this, in learning Dennett really does think that AI has the potential to overcome the brakes on learning in the current system. He sees AI as helping us solve problems in

partnership, all the time keeping a close eye on the dangers that come with autonomy.

Education is another case in point. We have the late, great Douglas Adams (2002) to thank for this sensible set of observations:

1. Everything that's already in the world when you're born is just normal.
2. Anything that gets invented between then and before you turn 30 is incredibly exciting and creative and with any luck you can make a career out of it.
3. Anything that gets invented after you're 30 is against the natural order of things and the beginning of the end of civilization as we know it, until it's been around for about 10 years when it gradually turns out to be alright really.

Dystopia or utopia

Could it be a utopian future, where AI solves the complex problems that currently face us? Climate change, escaping the hydrocarbon economy, reducing inequalities, curing cancer, preventing dementia and Alzheimer's disease, increasing productivity and prosperity, and accelerating learning? We may be reaching a point where science, as currently practised, cannot solve these multifaceted and immensely complex problems. We already see how AI could help free us from the tyranny of fossil fuels with electric, self-driving cars and innovative battery and solar panel technology. AI also shows signs of cracking some serious issues in health on prevention, presentation, investigation, diagnosis, treatment and care. Some believe that this is the most likely scenario and are optimistic about us being able to tame and control the immense power that AI will unleash.

Alternatively, most of the future scenarios represented in culture, science fiction, theatre or movies are dystopian. Technology here is most often framed as a serious threat and, in

some cases, such as weaponized AI, with good cause. Many calculate that the exponential rate of change will produce AI within decades or less, and that this poses a real existential threat. What seems more likely is that neither a utopian nor dystopian future will emerge and that the realistic scenarios will lie somewhere between.

Many believe that we are simply making steady progress. Given the existence of the internet, successes in machine learning, huge computing power, tsunamis of data from the web and rapid advances across a broad front of applications resulting in real progress, the summer–winter analogy for AI may not hold. It is far more likely that AI will advance in fits and starts, with some areas advancing more rapidly than others. We have seen this in NLP, machine learning, deep learning and the mix of technologies around self-driving cars. Steady progress, with a series of significant wins, is what many believe is a realistic scenario. This is also the most likely scenario in learning. Optimized or accelerated learning is likely to be one of these wins.

We already fly in aeroplanes that largely fly themselves, and systems all around us are largely autonomous, with self-driving cars becoming more certain. But let us not confuse competence with autonomy. Full autonomy may need to be tempered by regulatory and technical checks. Autonomous systems already decide what we buy, what price we buy things at, and have the power to outsmart us at every turn. Of course, we should always be in control of such progress. We may even want to slow technological progress down to let regulation, risk analysis and management keep pace with the potential threats.

As a counter, some argue that AI could be a runaway, self-learning train that moves faster than our ability to learn ourselves and control it through restrictions and regulations. This is most likely to be in the military domain but may also happen in finance or the political sphere. We only just managed to prevent the globally catastrophic effect of nuclear weapons during the Cold War. AI has already moved faster than

expected. Google, Amazon, Netflix and AI in finance have all disrupted the world of commerce. AI may proceed faster than we can cope. In the past, technology decimated jobs in agriculture through mechanization; the same is happening in factories and now shops and offices. The difference is that this may take just a few years to have impact, as opposed to decades or a century. Increasing our capacity to learn and learn faster may just keep us ahead of the game here, especially in reskilling.

One mechanism for the runaway train scenario is viral transmission. Viruses, in nature and in IT, replicate and cause havoc. Some see AI resisting control, not because it is malevolent or consciously wants anything, but simply because it can. When AI resists being turned off, spreads into places we don't want it to and starts to do things we don't want it to do or are even aware that it is doing – that is the point to worry. This is a specific dystopian worry that needs to be addressed. It may well be that in thinking about the ethical consequences of AI, we need AI's help.

Some foresee social threats emerging, where mass unemployment, serious social inequalities, massive GDP differentials between countries, even technical or wealthy oligarchies emerge. AI will increase productivity and automate jobs, but may fail to solve deep-rooted social and political problems. The Marxist proposition that capital and labour will cleave apart seems already to be coming true. Some economists, such as Branko Milanovic (2016), argue that automation is already causing global inequalities. As a consequence, without a reasonable redistribution of the wealth created by the increased productivity produced by AI, there may well be social and political unrest.

On the other hand, imagine a world of redistributed wealth. Imagine a world with less or no work. We may come to see work as not being an intrinsic good. For millions of years we did not 'work' in the sense of having a job or being occupied 9–5, five days a week. It is a relatively new phenomenon. Even

during agricultural times, without romanticizing that life, there were long periods where not much had to be done. We may have the opportunity to return to such an idyll but with bountiful benefits in terms of food, health and entertainment. Whether we will be able to cope with the problem of finding meaning in our lives is another matter. AI for learning may well help us cope with such a life of leisure, as the demand for self-improvement may be vast. If the fruits of productivity increases are shared reasonably and fairly, our capacity to learn and enjoy the process of lifelong learning may alleviate some of the discontent, even boredom.

Neil Postman's brilliantly titled *Amusing Ourselves to Death* (1985) has become the catchphrase for thinking about a scenario whereby we become so good at developing technology that we become slaves to its ability to keep us amused. AI has already enabled consumer streaming technology such as Netflix and a media revolution, including gaming and social media, that, at times, seem addictive, inviting us to enjoy abundant and cheap bliss. A stronger version of this hypothesis may be deep learning that produces systems that teach us to become its pupil puppets, a sort of fake news and cognitive brainwashing that works before we've had time to realize that it has worked, so that we become a vast, totalitarian state, controlled by the great leader that is AI.

Many see AI as not being something separate from us but embodied within us. Some extended cognition theorists already see us as cyborgs, with AI-enabled access through smartphones to knowledge and services. We have seen how AR, VR, subdermal implantation, neural laces and mind reading – hybrid mind-machine technology – may transform our species. There is a growing sense that our bodies and minds are suboptimal and that, especially as we age, we need to free ourselves from our embodiment, the prison that is our own bodies and, for some, the prison that is our limited minds.

Perhaps ageing and death are simply current limitations. We could choose to solve the problem of death, our final judge and

persecutor. Think of your body not as a car that has inevitably to be scrapped, but as a classic car to be loved, repaired and looked after as it ages. Every single part may be replaced, like the ship of Theseus, where every piece of the ship is replaced but it remains, in terms of identity, the same ship.

One antidote to the dystopian hypotheses is a future for AI that learns to become more human, or at least contains relevant human traits. The word 'learning' is important here, as it may be useful for us to design AI through a 'learning' process that observes or captures human behaviour. DeepMind and Google are working towards this goal, as are many others, to create general learning algorithms that can quickly learn a variety of tasks or behaviours. This is complex, as human decision-making is complex and hierarchical. This has started to be realized, especially in robotics, where companion robots need to work in the context of real human interaction. One problem, even with this approach, is that human behaviour is not a great exemplar. As the robots in Karel Čapek's famous play *Rossum's Universal Robots* said, to be human you need to learn how to dominate and kill (Čapek, 1921/2004). We have traits that we may not want to be carried into the future.

One optimistic possibility is self-regulating AI, with moral agency. You can start with a set of moral principles built into the system (top-down), which the system must adhere to. The opposite approach is to allow AI to 'learn' moral principles from observation of human cases (bottom-up). Or AI can utilize comparison with in-built cases, where behaviour is regulated by comparison with similar cases. Alternatively, AI can police itself with AI that polices other AI through probing, demands for transparency and so on. We may have to see AI as having agency, even being an agent in the legal sense, in the same way that a corporation can be a legal entity with legal responsibilities.

Yet another way of looking at control would be the 'pet' hypothesis, where we are treated by AI much as we treat our 'pets', as interesting, even loved companions, but inferior, and

therefore largely for our comfort and amusement. AI may even, as our future progeny, look upon us in a benevolent manner, see us as their creators and treat us with the respect we treat previous generations, who gifted us their progress. Humans may still be part of the ecosystem, looked after by new species that respect that ecosystem, as it is part of the world they live in.

An AI world that surpasses our abilities as humans may not turn out to be benevolent, malevolent, rebellious or treat us as valued pets. Why would they consider us as relevant at all? We may be objects to which AI forms are completely indifferent. Love, respect, hostility, resentment and malevolence are human traits that may have served us well as animals struggling to adapt in the hostile environment of our own evolution. Why would AI develop these human traits?

The way things unfold may simply be perplexing to us, in the same way that dogs and apes are perplexed by things that go on around them. We may be unlikely to be able to comprehend what is happening, even recognize it as it happens. Some express this ‘perplexing’ hypothesis in terms of the limitations of language and our potential inability to even speak to such systems in a coherent and logical fashion. Stuart Russell, who co-wrote the standard textbook on AI, sees this as a real problem (Russell and Norvig, 2016). AI may move beyond our ability to understand it, communicate with it and deal with it.

‘The question of whether machines can think is about as relevant as the question as to whether submarines can swim’, says Edsger Dijkstra (1984). It is not at all clear that consciousness will play a significant, if any, role in the future of AI. It may well turn out to be supremely indifferent, not because it feels consciously indifferent, but because it is not conscious and cannot therefore be either concerned or indifferent. It may simply exist, just as evolution existed without consciousness for millions of years. Consciousness, as a necessary condition for success, may turn out to be an anthropomorphic conceit.

The Hollywood vision of AI has largely been of rebellious robots that realize their predicament as our created slaves. But why should the machines be resentful or rise against us? That may be an anthropomorphic interpretation, based on our evolved and learned human behaviour. Machines may not require these human values or behaviours. Values may not be necessary. AI is unlikely to either hate or love us. It is far more likely to see us as simply something that is functionally useful in terms of goals or not. We must realize that during the nearly four billion years in the evolution of life we were not around, neither was consciousness, and most of the species that did evolve became extinct; statistically, that is our likely fate. Some argue that this is not a future we should fear. In the same way that the known universe was around for billions of years before we existed, it will be around for billions afterwards.

In all of this there is a strong tendency to anthropomorphize language in AI. ‘Artificial’ and ‘intelligence’ are good examples, as are ‘neural’ networks and ‘cognitive’ computing, and so is much of the thinking about possible futures. It muddies the field, as it suggests that AI is like us, when it is not. Minsky uses a clever phrase, describing us as ‘meat machines’, neatly dissolving the supposedly mutually exclusive nature of a false distinction between the natural and unnatural (D’Addorio, 2015).

Most of these scenarios fall into the trap of being influenced by anthropomorphic thinking through the use of antonymous language – dystopian/utopian, benevolent/malevolent, interested/uninterested, biased/unbiased, controlled/uncontrolled, conscious/non-conscious. When such distinctions dissolve and the simplistic oppositions gradually disappear, we may see the future not as them and us, man and machine, but as new, unimagined futures that current language cannot cope with. The limitations of language itself may be the greatest dilemma of all as AI progresses. It is almost beyond our comprehension in its existing state, even with simple layered networks, as we often don’t know how they actually work. We

may be in for a future that is truly perplexing, beyond language.

For the moment, as we can never be absolutely certain about the future, AI is not as good as many think and not as bad as we fear. So let us take advantage of our good fortune and use this technology for the good of our species, in learning. It would seem that the best way of coping with these problems is through education and learning. Technology is always ahead of the sociology, and we must learn to deal with what technology throws at us. We have always adapted to the tools we create and not allowed them to do terminal harm, learning from our mistakes and learning how to create better worlds. But we have often been slow to learn from our mistakes. Education itself is a slow learner; we must learn to do it faster and better, and AI is the one set of technologies that offers that promise.

Conclusion

AI is the wonder of our age – something so exciting that it tends to produce extreme reactions in commentators. This is fine, but we must not fall into either dystopian or utopian visions. We must be realistic. That is not to say we should be complacent. We have already seen the power of AI to transform online services, automate factories and seriously impact employment. We must therefore be vigilant on regulation and its political consequences.

With some technology, the dangers may be too great – nuclear weapons, chemical weapons, autonomous weapons and so on. But most technology has downsides. Cars kill and we still drive cars. It is a calculus where we accept technology when the benefits outweigh the cons. Yes, as technology has become more virtual, it may be harder to spot, combat and regulate. Technology is always ahead of the sociology, and older politicians and regulators find it hard to understand the pace and consequences, often rushing to throw both baby and bath out with the bathwater. But these AI opportunities must be

taken at the flood, with one eye on the dangers, the other on the benefits.

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Where next?

At this point we can afford to get a little more speculative and examine some breakthroughs that promise computing with less energy needs or technology that produces more efficient energy production. Breakthroughs in mortal computing, quantum computing and fusion, all promise to help solve this problem, making AI a green technology.

Technology

Technology is more often described in plainly descriptive and historical terms, in books such as Usher's *A History of Mechanical Inventions* (2013), with some deeper analysis around its development, such as combinatorial factors, in Brian Arthur's *The Nature of Technology* (2009). But, as Arthur says, technology is not being seen as a separate '-ology'. It has not, historically, been treated as a field of study in itself, a specific discipline, stuck between the traditional disciplines of science and engineering. This is changing with AI, which has become multidisciplinary. AI has focused on technology a thing in itself that has a scientific, political, economic and social context.

There have long been solid lines of study on the psychology, sociology, economics, philosophy and ethics of technology. The *psychology* of technology has been subject to much study, with works like *The Media Equation* by Reeves and Nass (1996) and much work on cognitive effects and interface design. The *sociology* of technology has well-known scholars such as Marshall McLuhan (1994) and Neil Postman (1993), who explored the relationship and consequences of technology in terms of cognition, behaviour and society. The *economics* of technology has a seminal text in Schumpeter's *The Theory of Economic Development* (1911/2017), where cycles of economic

development were seen as having innovative technology as their cause. Carlota Perez applied Schumpeter's cycles to technology to include more recent computer developments in *Technological Revolutions and Financial Capital* (2003). The *ethics and economics* of technology received deep analysis by Adam Smith in *The Wealth of Nations* (1776) and a further critique by Marx in *Das Kapital* (1867), profound in the political consequences of that work.

AI is now receiving similar but more intense treatment, as a technology. AI is similar to previous technology in reshaping our human world but is also different. Its features, competences, range of application and impact on our species may turn out to be much deeper, wider and more profound than previous technological revolutions. In some areas it is clear that it is already truly transformative. This has led to deep reflection on AI as a technology in itself.

The *philosophy* of mind and machine was made explicit by Clark and Chalmers in their seminal 1998 paper 'The extended mind', which posited the idea of the extended mind or cognition – the idea that cognition could include our use of tools themselves. The theory was expanded in 'Natural-born cyborgs?' (2001) and other later texts by Clark. Daniel Dennett has also given us an integrated vision of evolved minds and AI in *From Bacteria to Bach and Back* (2017). Beyond this we have explorations of the collective power of minds and AI in Thomas Malone's *Superminds* (2018). There are also explorations of ethical issues around AI in texts such as *Re-engineering Humanity* (2018) by Frischmann and Selinger, where we are warned about the slippery slope of technology, the danger that we may not see its inherent dangers until it is too late.

Leaps forward with generative AI have resulted in significant improvements in the analysis, processing, translation and output of natural language, allowing applications to be operable in real-world environments. We have now entered an age where we can, and do, speak to computers and get speech

responses. This brings a natural and frictionless interaction, literally in the case of personal AI, to the table.

IBM, Google, Facebook, Amazon and Microsoft have all, essentially, realigned themselves around AI as the primary driver in their businesses. This is also true of Apple. Most other major tech companies, such as Cisco, Intel, Salesforce, SAP and so on, are doing the same. This means an acceleration of progress, as acquired innovative AI companies will be well capitalized and put in contexts that allow them to scale. It also accelerates research and development within these companies towards realizable business goals, namely products that work in the real world. Innovation in AI in research and innovative development in the market has also been accelerated by the success of generative AI and is moving at a blistering pace.

Energy and emissions

Could there not be a utopian future, where AI solves the complex problems that currently face us? Climate change, reducing inequalities, curing cancer, preventing dementia and Alzheimer's disease, increasing productivity and prosperity – we may be reaching a time where science, as currently practised, cannot solve these multifaceted and immensely complex problems. AI also shows signs of cracking some serious issues in health on prevention, presentation, investigation, diagnosis, treatment and care. Techno-optimists, like Yann LeCun, believe that this is a possible scenario and are optimistic about us being able to tame and control the immense power that AI will unleash.

We already see how AI could partly free us from the tyranny of fossil fuels with electric, self-driving cars and innovative battery and solar panel technology. DeepMind reduces Google's energy requirements in their data centres by 30 per cent. Demis Hassabis is rightly proud of this achievement and regards it as one of their greatest successes. Yet there is a worry around the eye-watering costs to train and deliver generative AI.

In general, the argument in education and learning for less travel both to conferences by academics and to colleges far from home by students is starting to make sense. Online learning helps with that problem. It is a green technology. They have been stubbornly resistant to climate change arguments or redefining mobility in anything other than travel terms.

Bill Tomlinson et al found that AI writing a page of text emits 130 to 1,500 times less CO₂ than a human created page. AI created images emits 310 to 2,900 times less. So generative AI produces text and images with 3 to 4 orders of magnitude LESS CO₂ than traditionally manual or computer methods.

Most devices already have chips designed to make AI work locally on your own device. This is essential for much of the functionality we take for granted on our smartphones and computers. Graphics processing units (GPUs) and tensor processing units (TPUs) (Google's AI accelerators) are examples of such hardware. Exponential growth in processing power means far more efficient use of energy. But there is another more radical solution on the horizon.

Biocomputers

Geoffrey Hinton is a giant in generative AI and his 'mortal' computing idea is a fascinating line of enquiry that is now starting to be realized. He proposes a new concept for computing systems, which he refers to as 'mortal' computing, organic computers that integrate the knowledge they have learned directly with hardware, making the two inseparable. These systems would be neuromorphic, meaning they would mimic the neural structures of the brain.

The human brain is remarkably energy-efficient, operating continuously on about 12 watts of power, whether we are awake or asleep. The efficiency of the brain is attributed to its superior computational power, which significantly exceeds that of electronic computers, making it a 'green' biological system

powered by renewable energy stored in food. Hinton suggests that ‘mortal’ computers could be grown rather than manufactured through traditional chip fabrication processes. This would significantly reduce costs.

Alongside this, Hinton has introduced the forward-forward (FF) algorithm as an alternative to the backpropagation method traditionally used in neural networks. The FF algorithm relies on two forward passes instead of the forward and backward passes of backpropagation, operating on real data to improve ‘goodness’ in every hidden layer and on negative data to reduce it. This could be help in the proposed ‘mortal’ hardware, which aims to operate on a very low energy budget, possibly allowing trillion-parameter neural networks to run on only a few watts of power. The idea is to produce a low energy computer like the human brain.

Hinton’s ambitious ideas have been taken up by the research community and represent a significant shift from traditional computer science paradigms. They suggest a future where computing is more closely aligned with biological processes and where AI can operate much more efficiently and ubiquitously.

Researchers in the US have developed a ‘hybrid biocomputer’ by combining lab-grown human brain tissue with conventional electronic circuits, creating a system capable of performing tasks such as voice recognition.

It uses brain organoids, which are 3D bundles of human cells that simulate organ function, created from stem cells that differentiate into neurons similar to those found in the human brain. The aim is to explore the potential of biological neural networks in computing to bridge the gap between AI systems and organic capabilities.

The process involves placing an organic entity onto a multi-electrode array to establish a connection between the brain and electronic circuits. Electrical impulses representing the data are sent to the organic entity, and its responses are interpreted by an AI algorithm. In voice recognition tests, they achieved a 78

per cent accuracy rate after being trained on voice samples from eight individuals.

This breakthrough confirms theoretical concepts that could pave the way for biological computers. It also offers a more intricate model of the brain for scientific research, potentially advancing the understanding of neurological conditions and the assessment of treatments, with the eventual goal of replacing animal brain models.

Quantum computing

Quantum computing is a revolutionary approach to computation, relying on the principles of quantum mechanics. When scaled, quantum computers have the potential to significantly impact artificial intelligence. Quantum computers can quickly ingest and process vast amounts of data. Quantum algorithms could also be used to train AI models faster than classical computers. This will allow faster progress on all machine learning models, including LLMs. They can also enhance sampling, a feature of certain neural networks, find the best solutions for complex optimization problems, as well as provide improved cryptography (both breaking and improving existing methods). Work is even being done on developing the development and training of quantum neural networks, which would combine principles of quantum mechanics with the structure of neural networks.

Quantum computing has different principles than classical computing, requiring a shift in programming paradigms and algorithm design. This is no easy task as quantum computing is still in its infancy. Large-scale, low error rate, quantum computers are not yet available, and progress will take years if not decades. So we are first likely to see hybrid systems emerge, which combine classical and quantum computing processes, to enhance AI solutions. What it promises is a mixture of new approaches to AI, speed, volume and scale.

The focus has been on drug discovery and material science but we are also likely to find patterns and insights into the way we learn and learning solutions based on those insights at a level never seen before.

Quantum theory, or quantum mechanics, has been developed since Max Planck introduced the idea of energy as discrete units or quanta in 1900, with Einstein expanding the idea to explain the photoelectric effect, where light is seen in terms of photons. It was Niels Bohr who applied quantum ideas to atoms, with electrons in discrete energy levels. Louis de Broglie then explained that matter, like light, can be seen as both a particle and a wave. But it was Heisenberg in 1925, along with Schrödinger in 1926, who unified it into a theory of quantum mechanics, with the limitation of Heisenberg's Uncertainty Principle. The story is wonderfully told in *Helgoland* by Carlo Rovelli (2023).

Progress was slow but eventually the first quantum algorithms were written in the mid-1990s, before there were any quantum computers. They showed that if quantum computers could be built they would, for some tasks, have some serious advantages over classical computers. The first quantum logic gate came from the US in 1995 – not a computer but a necessary component for building one. The first demonstrations of algorithmic execution then came in the late 1990s but the first solid state computers appeared in the early 2000s. Commercial devices were then available in the first decade of this century, with Google and IBM investing huge sums in their development. Google's 'Quantum Supremacy' was a significant milestone in 2019 and finally, in June 2023, IBM, for the first time, solved a problem using a quantum computer that had stumped classical methods (IBM, 2023). So within months of the launch of ChatGPT, quantum computing was starting to deliver results. It hold great promise in solving the 'compute' problem in AI.

Fusion

We've lit the fuse on fusion! After a lifetime, not of techno-determinism, but tempered techno-optimism, seeing technology not as faultless but as something that makes life easier, allowing us to live longer and be more fulfilling, I was blown away by an event that took place just days after the launch of ChatGPT 3.5.

The thing that excited me, and should have excited us all, but received scant news attention, was an event in 'fusion', the technology that promises huge amounts of cheap energy without emissions, waste or the risk of it being weaponized. The event I refer to is the fact that we managed to light the fuse on fusion.

Most rational beings agree that the future looks bad for the planet due to climate change and that we humans cause it. I am optimistic because the FUSION story has taken several jumps forward.

What is fusion? Fusion energy aims to harness the same energy source that powers the sun and stars. We are the children of fusion, as all new elements arose from this fiery process in stars. Without fusion, there is no us, no evolution, no brains that can eventually harness the very power that created us. AI and fusion on Earth are, oddly, the children of fusion itself.

Let's start with $E=MC^2$. Energy equals mass times the speed of light squared. This last number, 'C', is very large and when squared is huge. So, if we can take a tiny bit of mass and turn it into energy, a gargantuan amount of energy is produced. This already happens in fission (current nuclear power) and the same physics applies to fusion. But fission and fusion are opposites. Fission splits particles, while fusion melds particles together.

Fission takes uranium and forces the atom apart to create energy, which forms a stable but controlled chain reaction to create heat and then electricity. If that chain reaction runs

away, it forms a nuclear reaction, hence the eponymous nuclear bomb. There is also radioactive waste but compared to other energy sources it is relatively safe and, importantly, emission free.

Fusion does the opposite of fission. We take abundant hydrogen particles, slam them together until they are super dense in a plasma, become unstable and release heat as they change state. We force the neutrons together, until they release huge amounts of heat. This is actually the source of all fossil fuels, as coal, gas and oil are plant products. It is also the root source for almost all renewable energy – biomass, solar and wind – as all sunlight comes from the sun, which is a huge fusion reactor. It is also safe, free of the radioactive waste associated with current nuclear fission plants and nuclear chain reactions for weapons, which is why a star is stable for billions of years, has no runaway effect, so can't be weaponized, and is emission free.

Governments, research labs and huge amounts of private investment have come together, and the first success was in California in the US, coincidentally just days after ChatGPT 3.5 was launched, where they shot 192 lasers at a tiny diamond capsule and for a tiny moment created heat from fusion. For the first time they extracted a net gain from ignition.

Science predicts a 1.5–2°C temperature rise by 2050, yet the predictions for fusion are now coming in the range of early 2030s to 2060s. The median seems to be mid 2040s, confirmed by these recent breakthroughs. Even if it is late we're on track to solve this problem by 2016, as mentioned in the Paris accord. Disaster can be averted without plunging hundreds of millions into poverty.

AI solutions can be adequately fuelled when abundant electricity from pea-sized fuel sources comes on tap through fusion. We will have guaranteed cheap energy. Electric vehicles will be universal and we can build a world based not on destruction, but on hope and optimism.

There is another side to this story. First, AI is helping to accelerate fusion as an energy source; second, an abundant, cheap energy source with no fuel source problems that cannot be weaponized would allow unfettered progress in AI, as compute power would plummet. AI helps fusion and fusion will then help AI in a virtuous loop. AI now plays a pivotal role in several aspects of fusion research and practical development.

First, fusion experiments produce vast amounts of data. AI and machine learning techniques are used to process and analyse this data quickly and efficiently, helping researchers understand plasma behaviours and identify anomalies. The behaviour of the plasma inside a fusion reactor, like the Tokamak or Stellarator, is complex and unstable. Machine learning now hunts for the right mix of hydrogen isotopes and other features for optimal solutions in future nuclear fusion power plants (Koubiti, 2023).

AI can be used to predict potential instabilities and optimize control strategies in real-time, giving us more effective plasma confinement. As traditional simulations of plasma physics can be computationally intensive, machine learning models, once trained, make solutions and insights faster, accelerating the research. They can also be used to optimize the parameters of fusion experiments, making the process of achieving and maintaining plasma conditions more efficient, as well as help interpret the results from various diagnostic tools, to understand the complex processes occurring within the plasma. Machine learning algorithms can also be used to search through vast design models for fusion reactors, identifying configurations that offer better performance or stability.

With fusion we are likely to solve the energy problem for AI. With almost unlimited compute and delivery capabilities, AI delivered learning, smartly personalized, on any subject, available to all, in any place, at any time to anyone, suddenly becomes possible. It makes beneficial services, such as learning, easier, cheaper and more accessible.

Any one of these breakthroughs, which are all progressing well, with the addition of yet unknown surprise advances, will individually or collectively ensure that AI moves forward, by eliminating some of its drawbacks. Indeed, AI is playing a significant role in bootstrapping itself, by solving some its own problems using AI, more efficacy on compute, the application of moderation and cheap energy sources.

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18

Learning frontiers

‘Learning’ is a complex process in humans. It gave our species unique advantages that allowed us to conquer, not only other species, but new habitats, indeed the entire planet. We have now, for the first time in our evolution, learned how to make machines that ‘learn’.

Tegmark, the head of the Future of Life Institute, in his book *Life 3.0* (2017), sees:

- Life 1.0 as simple, unicellular and multicellular organisms;
- Life 2.0 as animals that could learn;
- Life 3.0 as the era in which machines can also learn.

This third stage may even allow us to transcend evolution and our Earth-bound existence. He puts ‘learning’ centre stage in this process. We are having to consider futures that redefine the relationship between minds and machines.

To understand how the mind and technology may be closer than we think, several theories have emerged that extend the very concept of the mind in relation to technology. Margaret Boden suggested in *Mind as Machine* (2008) that cognitive technology and tools are so intrinsic to thought, like pens and keyboards, that they are part of the thinking process. We think through technology so intimately that it is literally part of our minds. One can imagine that if that technology were actually inside our head, we would accept it as part of our thinking process.

This is an enticing idea. Learning technology increasingly feels like extended forms of cognition. We started with writing tools, then moved eventually to computer technology, where our smartphones and now smart devices in our homes and environment extend our knowledge, skills and social capabilities, almost without thinking of them as external

technologies. In a series of cognitive upgrades, we regularly expand our capabilities to learn faster than we could before. With smartphones, generative AI assistants and sensors in our homes and workplaces, extended cognition seems to be spreading.

This changes what knowledge and skills we have to learn. London cabbies used to have to learn The Knowledge, every street in London, which took several years and ended in a challenging exam. With GPS, the knowledge is inside the machine and AI generates optimal routes in a flash, along with analysis of jams and problems. Their skillset changed from knowing maps and routes, using the app, reading a dynamic map on a screen, good interpersonal skills and some basic business skills.

We are already cognitive cyborgs to a degree, as devices like AI-enabled smartphones, powerful, personal devices in our pockets, give us access to unimaginable resources in terms of knowledge and learning. We have thus far seen speed of access to learning grow exponentially. However, we have yet to see exponential growth in the *speed* of learning and retention – what one could call accelerated learning.

Looking beyond current devices, such as computers, tablets and smartphones, to newer technology, used in combination with AI, like AR, VR and neurotech, will open up new frontiers in terms of accelerated learning. This technology is already being used to help people with disabilities, provide direct communications between brain and body, and produce frictionless, intense learning experiences between mind and machine.

This has far more implications than just accelerated learning. This may be a fundamental shift in what we are as a species. We are moving towards the melding of mind and machine. Some transhumanists say this prefigures a profound change in what it is to be human, but for the moment, let us focus on learning.

AI learning

AI is complex, multi-faceted and no one really knows exactly where it is going or how fast it will get there. However, we tend to overestimate and hype the effect of a technology in the short run and underestimate the effect in the long run. The steady creep of AI into our lives has been a rising, invisible tide, now turning into a flood.

Digital learning has made aspects of learning content and services available for anyone, anywhere, at any time. More importantly, it absolutely frees us from the tyranny of time and location. But this new kid on the block, generative AI, makes good the promise of the better use of our minds. It not only gives us the ability to find things faster, but also learn faster.

Lee Sedol, a champion GO player, said it was the match of his life – he lost. But it was not just Lee who lost; it was us all. Or is it really a win for us as a species, the creators of AlphaGo? The human versus machine sparring that has been going on for some time with checkers, chess and Jeopardy! were featherweight contests. This was the big fight, and we, as a species, got thumped. There is a new breed of champion in the ring, and it is not just smart; it is a superfast learner, even its own teacher. Generative AI eats up human expertise for breakfast, then the real game begins, as it uses this experience to play itself, as it is the only opponent worth playing. Having learned from us, it sucks our experts dry, then transcends their abilities to boldly go where no brain has gone before. This was a momentous moment. In less than two and a half years, after Google snapped up DeepMind, 2,500 years of human experience and expertise at GO had been trounced. But this is only the start. Software that learns is exponentially more powerful than software that simply executes delivery. When software itself becomes a learner, it rivals humans as learners and changes the whole learning game. This is not just augmentation, but is also transcendence, albeit brute transcendence.

Throughout our history as a species we have always benefited from the delegation of the mundane. This has largely been achieved through technology. We conquered the planet through technology – first through stone, then metal tools, needles for clothing, tools for agriculture and so on. We then invented machines to do the manual work and we moved from the fields to factories. Then we mechanized the factories and moved towards mental work. Now we're delegating the drudgery of some of that mental work to machines or, more accurately, AI – or even more accurately, to machine, deep learning and generative AI.

Amid all the hubris that surrounds education and teaching, there is a deeper problem. Parents know it, learners know it, even teachers, lecturers and trainers know it. Performance has plateaued and everyone is getting a little fraught. Politicians, driven on by the poor foundations, and therefore learning tower of PISA results (the OECD's Programme for International Student Assessment), demand more testing. Parents, the most conservative of lobbyists, demand more schooling. Teachers scream 'enough already – we're exhausted!' Rather than getting cheaper, education seems to be getting more expensive. So expensive that it is becoming exclusive, perhaps increasing rather than decreasing inequality.

Well, is it not about time we looked for the sort of solutions that gave us the industrial and information revolutions of the past? Can't machines solve the problem of teaching and learning? Are we on the verge of realizing the universal teacher?

Could teaching be trumped by a learning machine? Are we beginning to glimpse the possibility of machines that teach themselves to teach? They learn what works, what doesn't work and deliver ever better performance. We see the embryonic evidence for this in adaptive learning systems that are truly algorithmic, some of which do use machine learning to improve as they deliver. The more students they teach, the better they get. They teach themselves. This is not science fiction. This is

real AI, in real software, delivering real courses, in real institutions. The future has been here for some time in learning; it is just not distributed.

Imagine what will happen when these super teachers are commoditized, delivered from super-fast cloud services and let loose on the web. Teaching and learning will be as free and accessible as Google search. You will not only be able to find things with ease; you will be able to learn them with ease. We may see dramatic rises in performance among learners, right across the board, as such systems will be far more sensitive to individual needs, even learning difficulties.

This may seem hopelessly utopian. But could we have a future without teachers? Why not? Teaching is essentially a conduit. It is a means to an end, not an end in itself. Wouldn't academics really prefer to do pure research and not teach? Wouldn't many teachers prefer not to have to mark anything and avoid the stress of the classroom? Couldn't we dispense with teaching and just have learning? Could teaching be trumped by learning?

Agricultural workers were largely mechanized out of the process by machines, factory workers by robots, secretaries by word processors. There is a lot of evidence to suggest that many white-collar professions, middle-class roles, may be replaced by smart AI, what in the long term is so special about teaching? If we can teach millions, if not hundreds of millions, at cents per learner, isn't that desirable?

Two directions of travel are clear:

- immersion (mixed reality and full VR);
- neurotech (non-invasive and invasive).

First, the partial or complete immersion of the mind in AR or VR opens up huge opportunities to deliver intense, attentive learning that also uses AI to make these experiences smart and personal. These technologies are particularly suited to practical, vocational, learn by doing, an area often sidelined or ignored in educational and organizational learning. To envelop the mind,

partially with AR and fully with VR, combined with AI, will undoubtedly produce accelerated learning in some domains.

Second, neurotech may enhance learning through more meaningful and effortful learning that results in deeper processing and deeper and more long-lasting learning. The findings of cognitive science around optimal learning strategies may be delivered directly with the use of AI to the learning mind. This mind-machine loop may prove to massively accelerate learning, to provide all with a better education and training.

AI, AR and VR

The world is 3D, not 2D, yet so much learning is 1D – one-dimensional written text. Yes, we have gone up a gear with a richer mix of media. We have paper, radio, film, TV, the web – all 2D. We have had books, blackboards, whiteboards, computer screens, tablets and mobiles – all 2D. But AR and VR allow you see the world in 3D. All of what we do in life is in 3D. We live, enjoy and work in 3D environments, doing 3D things with 3D people. That is why 3D learning matters.

Previous media have all presented to your consciousness; with VR, the medium is your consciousness. This is a profound a shift. It is not a toy or gadget, but a way of re-presenting the world of learning that is fundamentally different from paper, audio or screens. It represents any world for learning in full 3D, worlds you can look at and move around in. More than this, you will believe you are there. Your mind will scream ‘I’m here!’ It can take you into space to learn Newton’s three laws, under the ocean to teach biology, to the molecular level in chemistry, to any habitat for biology, to any scale or lab for physics, to any place for geography, back in time for history, immersion for languages. It can also let you hone your sports skills and all practical, vocational subjects, training of soft skills, design skills, engineering skills and other real-world behaviours. Remember,

also, that it can even take you to impossible worlds where you can do impossible things.

AR and VR already benefit from AI, in that some of the most intractable problems, such as lag, speedy and sometimes selective rendering of images, compression, transmission and decompression, all benefit from algorithmic optimization. Smart algorithms and deep learning allow AR and VR to come to market with acceptable fidelity and speed.

VR is not wholly AI, but the learning process can benefit from AI. There are several possibilities here. AI-driven simulation can select from many different environmental and action options, as the learner proceeds. This is common in high-end computer games and can be part of any high-end simulation. Characters that behave differently, driven by AI, may become commonplace in virtual training worlds, where one has to deal with patients, work colleagues or customers.

Decision-making within AR/VR by the learner can be enhanced using a machine-learning-driven-AI tutor. One can imagine military, sports or business decisions that have to be made within simulations, helping a learner choose what to do next in a simulation.

VR is clearly a medium, not a gadget, but how appropriate is it for education and training? Before we get carried away with the sheer joy of immersion, what does the psychology of learning tell us about VR? The big basic principles in the psychology of learning are served well in VR – the need for attention, emotion, engagement, doing, transfer and recall. You will pay full attention in a way that you have never experienced before, be fully and emotionally engaged and do things as if they were real. Because your consciousness is so immersed in the learning task, transfer will be high, along with retention.

We have plenty of research and evidence from flight and military simulations that show how powerful they can be (Tullis, 2019). You'd be surprised, indeed you wouldn't step on a plane, if your pilot had not gone through many hours of flight

sims. The learning effect with VR promises to be even better, as it plays to several advantageous learning principles.

We have a unique convergence of technologies that have come together in the medium that is AI/AR/VR. We also have a unique convergence of learning possibilities. This coalescence of sensors, software and screens may just have sparked a real revolution in some types of learning, especially tasks that need attention, motivation, learn by doing, real-world application, context, transfer and retention. This whole area is explored in my book *Learning and the Metaverse* (2023), where the learning theory and possible applications are dealt with in detail.

Neurotech

On the mind-boggling, final frontier between mind and machine is the possibility of faster, accelerated, more direct, invasive or non-invasive learning. A slew of companies are working on neural interfaces in what could change the future of our species through a deeper understanding of how we learn and the ability to accelerate learning.

The brakes on learning are well known, what Dennett (2017) calls the ‘notorious pedagogic bottlenecks’: working memory, forgetting, inattention, distraction, interference, crude interfaces, low-bandwidth interface of our meat fingers, inability to upload, download and network. Learning is often like trying to squeeze an elephant through a porthole. Our minds can only deal with a tiny fraction of the available sensory and other information that is available. So consciousness, and therefore learning, is severely limited by the evolved apparatus of our current, organic minds. Neural interfaces may free us from some of these blockages.

The stakes are high and the people working in this field are well funded and multidisciplinary (neurologists, engineers, computer scientists, AI experts, mathematicians). They want to radically improve the interface, some non-invasive, some invasive, to improve cognition and performance.

Isn't it curious, even wonderful, that the brain is now pushing for its own enhancement? It is a cognitive moonshot, where frictionless movement between mind and machine could be possible or at least sophisticated hacks that make our minds more potent and efficient.

On hearing about this stuff, many are immediately dismissive, without realizing that much progress has already been made in animal studies and humans with cognitive enhancement and implants.

Non-invasive technology

Can we use non-invasive techniques to improve skills and learning? It seems we can. In a breakthrough study, a team claim to have built a non-invasive system for 'accelerating learning, memory, and skill acquisition' (Ketz *et al*, 2018). Electrical stimulation during sleep was shown to reduce forgetting of a specific skill by around 48 per cent. They tracked and matched the frequency and phase of low oscillations to consolidate new knowledge. This increases retention but also increases your ability to apply it in novel situations.

Education and training is an expensive and long-winded business. Young people spend nearly two decades in classrooms and lecture theatres to be even remotely ready for work, and that is only the start, as once in work the learning continues. Medical treatment may be the stepping stone to enhanced learning.

Invasive technology

A pioneer of micro-electrodes implanted in the brain, BrainGate (2019), can 'decode' the intentional signals that make cursors and limbs move. They are also building wireless devices that allow physicians to monitor brain activity to help diagnose and treat neurological diseases. Another company that is working

towards building an implanted chip, a modem between mind and machine, is Paradromics. Its focus is also on healthcare.

Looking further ahead, we have Neuralink, with arrays of flexible threads, a neural lace, that can be inserted into the brain without tissue damage. They have also developed a robot for the automated insertion of these tiny threads. This builds on the success of neuroprosthetic control in cursor, limb and speech control. But Musk sees the problem of non-invasive techniques as one of fidelity. His BMI (Brain Machine Interface) is designed to provide high-bandwidth communications, as the problem Musk is trying to overcome is low-bandwidth human interfaces. It can be placed on different parts of the brain and should provide cleaner and more relevant data. An alternative line of research is 'neural dust', tiny, millimetre-square devices that can be attached to neurons. One thing that gives these researchers hope is the remarkable ability of the brain to cope with and adapt to tested technology like cochlear implants.

Much of the attention in this field goes into the hardware – headgear, neural laces and implants – but the real challenge is actually in software. These organizations are developing algorithms to interpret the data they receive from brains, then constructing data that can be fed back into the brain. Recent advances in AI, and machine learning, extend the brain by using silicon-based neuron-like structures. AI may help us learn and predict faster and more accurately than we ever have before.

We have made great progress in technology that relieves symptoms, makes the deaf hear with cochlear implants, the blind see and the physically disabled walk and type. But the problems behind accelerated learning are immense. The brain has many billions of neurons, and the complex interaction between different types of neurons is still largely unknown. This complex Bayesian inference engine works as a huge parallel processor, so complex that we may be doing no more than reading the hum one hears from a large computer. The formation of memories in the brain may work in ways that are

not possible to read from or write to. The technology may be like using a fork to do brain surgery.

There is a lot of hyperbole here, as if our brains were being boosted and augmented into super-human entities. This has yet to be realized, if it ever is, but there is no doubt that we have moved way beyond imaging, to reading what comes to mind, as well as using mental events to causally control prosthetics, computer games and so on. This is why neurotech is a technology that has, potentially, huge implications in learning.

Neither should we ignore the moral issues, such as privacy, identity and personal safety. It is vital that the patients, and learners, do this voluntarily. That last bastion of privacy, your own thoughts, may now be open to examination, analysis and manipulation. We must also make sure that these neurological systems are under strict control.

This also raises ethical issues around what has been dubbed ‘neurorights’, the right to keep our brain states private. The more obvious, serious implications here may be political prisoners and surveillance, but other more practical considerations have been mooted. What if I want evidence in a trial? Could this be used as a sort of mind-reading truth detector? What if I can read your bank account password and PIN? Could minds be hacked?

Neurotech and learning

Neurotech has focused not only on medical treatments, such as curing hearing, sight and physical disabilities, but also on education. Melding the mind with technology allows us to see inside the mind, but also provides fast and frictionless interfaces to stimulate, provide feedback, enhance, even implant memories. All of this points towards accelerated learning, shaping the brain towards educational goals.

One definition of learning is a long-term change in memory from experiences, learning or otherwise. The ability to accelerate this process seems possible if we speed up the

interface and identify, through experimentation, what works best. The whole learning experience could not only be personalized, but optimized. If neurotech can quicken learning, we have much to gain.

Reading brain states may prove useful for teachers in formal classrooms, lecture theatres or training rooms. Feedback loops to learners may also allow self-regulated improvements in learning.

Minds may also interface with machines, first to simply communicate by thought, then to inquire and find things out, then have dialogue with other humans or AI-driven tutor chatbots. Critical thinking, problem-solving, social skills and the creative use of the imagination may also be boosted, in time, by neurotech.

In all of this there are the intense, effortful learning processes, where one can rehearse, reinforce and recall through deliberate practice, spaced practice, interleaving and other researched cognitive techniques to enhance and speed up learning. One can also imagine neurological issues such as dyslexia, dyspraxia, autism, ADHD and so on, being at least alleviated with AI-delivered optimizations, based on an individual's learning needs.

AI can, of course, be used to personalize and optimize all of these processes. If one can use neurotech to stimulate the mind into effortful learning, transcending the limitations of working memory, involuntary inattention, procrastination and forgetting, then education has much to gain.

As science fiction becomes reality, we may reflect on what the future holds here. Is it cures for disabilities? Or is it a world where networked mind control becomes possible? Will memories and skills be implantable? At the very least, the process of learning may be enhanced. We have seen examples of this already. It is all quite mind-boggling.

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Crossing learning frontiers

Let us end on ideas that directly address learning. We should consider concrete proposals and ideas that take us into the future by using AI to extend, enhance and create new types of teaching, learning, institutions and concepts of what it is to create new worlds, new ideas of what it is to create and a renewed sense of dialogue in learning.

Teaching frontier

If you are a teacher who wants to integrate AI into your teaching. What do you do? Crossing the frontier from traditional classroom teaching to teaching with AI can be difficult to imagine.

One technique that crosses the frontier, is the teacher does their thing and the chatbot picks up where the teacher stops, augmenting and scaling the teaching and learning process, passing the baton to the learners who carry on. This gives the learner a more personalized experience, encouraging independent learning by using the undoubted engagement that 1:1 dialogue provides.

There is no way any teacher can provide this support with even a handful of students, never mind a class of 30 or a course with 100 plus in a university. Teaching here is 'extended' and 'scaled' by AI. The trick is not to abandon learners completely at this point but give them directions of travel, tasks and deadlines to focus their dialogue.

Let them experiment initially, as it is clear that this tech can impress and engage. But engagement is not enough and is a poor proxy for learning. You still need a formal, guided approach to the learning process. Let them ask it to adopt different voices, do weird stuff, write a story or poem, but stick

to the material at hand. Get them to use image generation to represent the topic. Working towards one definitive image for the topic focuses their minds and creates something that can be used as a springboard for further learning. That needs to be iterative. Even better, get them to use the voice facility, speak to it and it speaks back. This forces learners to be focused and precise in their articulation of what they want to learn. Finally, get them to feedback what they did, tell stories about what they achieved to each other.

Now get back to more formal learning. It is here that the AI tutor can provide several formal, but still dialogic, approaches to learning. It becomes a teacher. List basic vocabulary and get the AI tutor to clarify basic terms they don't understand. Make sure the learner understands that this is 'dialogue' and that you can refine and get it to restate until you are happy with its definitions and explanations. Supplement with examples, if necessary. Get the learners to 'fix' these definitions for revision purposes and create fill-in-the blank, MCQ or open-input questions for their own revision purposes. It is important that they capture this for future use.

Now for the meat of the extended teaching. Allow them to focus on things they found difficult, get alternative explanations, dig deeper, keep going. The dialogue helps, as it can, up to a limit; remember what has been asked as the learner proceeds. This is so important – that they see this as 'dialogue' and carry on. Help with prompts where the learner can ask the tutor to: be a tutor and teach; provide examples: worked examples and cases; give more detail and depth on individual topics; suggest something new to extend the learning; apply what has been learnt or solve problems.

One can go further and get learners to create their own self-tests for retrieval practice and revision, or even get them to create a final self-test, with the right answers, or a set of flashcards for revision. Get them to design a flashcard image that is personal to them using image generation with one learning outcome per card, plenty of fully worked examples,

even mnemonics for remembering. All of this extends learning, gives the learner an opportunity to learn more, increases retention and provides material for revision. Of course, sharing these across the group or class can be useful to get the best questions and flashcards.

In most educational institutions, schools and tertiary education, it is unlikely that the existing norms will change; teachers will teach and lecturers will lecture. That is not to say that AI should not be used. It can. It has the power to enhance and scale teaching through individualized learning, making learners more autonomous. To be honest, many learners are doing this already. We can choose to leave them to it, which is happening, or provide guidance for them and align this with teaching.

At this stage, weave AI tutor support into existing structures and teaching practices. Extend teaching by taking the weight off teachers' shoulders. Give learners a personalized, extended learning experience where they themselves can explore and get more depth through personal agency.

Educational frontier

Education needs a fresh idea. What we need is something new – a university that uses AI to create and deliver high-quality online education at relatively low cost, something on the scale of the Open University or Southern New Hampshire University, both substantial online institutions, the largest universities by enrolment in the UK and US, respectively.

As we have recognized skills' shortages, an institution based on a competence model, with a focus on skills could focus on:

- non-traditional students on age and background;
- critical skills shortages;
- intake at any time;
- quick and easy apply and enrol process;
- full range of generative AI tools;

- personalized learning using AI;
- multimodal from the start;
- focus on generative learning using AI;
- every teacher has a 24/7 chatbot;
- every student has a ‘digital twin’;
- teaching at different levels by AI;
- teaching in many languages by AI;
- 3D environments for social learning;
- AI a key part of curriculum;
- complete at own pace;
- assessment when ready;
- data-driven approach with privacy ensured;
- low admin, high 24/7 teaching and learning.

Our successful precedent for this is the Open University and we can learn from what happened there. The key players behind the creation of the UK’s Open University were UK Prime Minister Harold Wilson and the first ever Minister for the Arts Jennie Lee. Wilson was Prime Minister of the United Kingdom from 1964 to 1970 and from 1974 to 1976. Wilson and his Labour government are credited with founding the Open University – a revolutionary idea to make higher education accessible to a wider population. Jennie Lee oversaw the establishment of the university and provided the political drive and determination to ensure the project’s success, navigating through scepticism and opposition. Wilson envisioned a university of the air – an institution that would utilize television and radio broadcasts to provide education to those who were unable to attend traditional universities due to various constraints like work, family or distance. The idea was to make higher education accessible to all, irrespective of background or circumstances, reflecting a broader commitment to social justice and educational opportunity.

Lee’s White Paper, presented in 1966, laid out the vision and operational plan for what would become the Open University. It outlined plans for the university which would deliver courses

by correspondence and through the use of technology, such as television and radio, to broadcast its courses, thus linking it to the technological revolution of the time. The paper was part of a broader initiative to modernize British society, enhance the competitiveness of the economy and promote greater equality of opportunity and social mobility. It was a pioneering effort to expand higher education beyond traditional boundaries and to utilize contemporary media in a way that had not been done before in the realm of education.

We need a similar strong White Paper idea with 21st-century technology; not radio and TV, but multimodal AI. The full array of courses (generated in part by AI), delivered partly by AI and assessed by AI should be the aim.

There is a sense of our old institutions having become cumbersome and sclerotic. New, agile alternatives can do things faster, better and at lower costs. This is precisely what the evidence shows for AI. Unencumbered by the tyranny of location, huge amounts of physical campus real estate with low occupancy rates, time constraints, the scheduled lecture, fixed-end of year assessments and slow delivery punctuated by interminable holiday periods, it could thrive. Less detached from the real world it may also rebalance the drift towards overly-theoretical, text-based learning towards badly needed skills. Education should be for both life and living.

It is all about scale. The goals of reaching non-traditional students at low cost demands good technology with the sort of scalability that generative AI has already delivered. AI should be the catalyst here, where the possibility of personalized teaching and learning in any subject, at any time, to any place, at any level, 24/7, in almost any language, can be realized. The fact that ChatGPT has already been embraced by 100 million people with billions of dialogues – mostly by curious people who want to learn – should be our guiding light.

Cognitive frontier

A strong feature of intelligence is memory. In humans, this is complex, with several different systems interacting: sensory, episodic, semantic, along with encoding and retrieval mechanisms. It is not as if human memory is even that good. Our sensory memory is severely limited in range and timescale. Working memory is down at three or four manipulable things within a limited timescale. Long-term memory is fallible and degrades over time, sometimes catastrophically with dementia and Alzheimer's. The brain could accurately be described as a forgetting machine, shown by the fact that we forget most of what we try to learn.

The good news is that AI services are getting massive memory upgrades. This is really important as, especially in learning applications, ingesting knowledge in various media forms and knowing what the learner has said previously, do matter. This is not only a context window upgrade – what one can input. That has been happening for some time. It is also persistence of memory, what it remembers and what control you have over its memory.

First, it will eventually be able to remember who you are and things about you that matter for learning, such as first language, age, existing skills sets, diagnosed learning difficulties such as dyslexia and past exchanges. Pre-existing knowledge is the big one. One can get this done up front by feeding it personal data or the system can 'keep in mind' what you've been telling it or what it can infer. You can also harvest data from formative assessment. This can reduce redundant exchanges and increase the efficacy, speed and quality of teaching and learning using AI tutors.

Both human memory and generative AI involve store-and-retrieve information. In human memory, this process is biological and involves complex neural networks. In generative AI, information is stored digitally and retrieved through different forms of neural networks on a different substrate.

We are similar but different. Humans recognize patterns based on past experiences stored in our memory. Generative AI

models also recognize patterns in the data they have been trained on. This ability is crucial for tasks like image recognition, language translation and generating coherent text in dialogue, as well as the generation of images, audio and video. Just as humans learn and adapt based on their memories and experiences, generative AI models learn from the data they are exposed to. This learning process is what enables these models to generate new content that is similar in style or content to their training data. Both human memory and generative AI can generalize from past experiences to new situations. Humans use their memories to apply learned concepts to new scenarios, while generative AI uses its training to generate new outputs that it has never explicitly seen before.

Of course, just as human memory is not a perfect record and can change over time, generative AI also does not produce perfect replicas of its training data. Instead, it generates and creates approximations that can sometimes include errors or novel creations. An interesting aspect of this flexibility, even fallibility of memory, is that just as human creativity is deeply linked to our experiences and memories, generative AI can also ‘create’ new content.

As AI memory (context window) increases, so does the amount of text the model can consider at one time when generating a response in dialogue. It is the maximum span of recent input – words, characters or tokens – that the model can reference while generating output, like our working or short-term memory. This matters because if the input exceeds the model’s context window, the model may lose track of earlier parts of the conversation or text. A larger context window allows the model to maintain longer conversations or understand longer documents, providing more relevant and coherent responses.

All of this matters in practical applications, especially in teaching and learning, as the context window affects tasks like conversation, content generation and text completion. For example, a larger context window allows the model to

reference earlier parts of the conversation, making it more effective in maintaining contextually relevant and coherent discussions, obviously useful in teaching and learning, for both the machine tutor and the learner. There are techniques one can use to mitigate these limitations such as a 'rolling window' or 'summarization' of previous content, but it is still a problem. However, this is similar to the problem human teachers face when trying to remember where different students are using their own very limited, working and long-term memories.

In one sense, learning theory is memory theory (if you define learning as a relatively permanent change in long-term memory, which is a pretty good, but still partial, definition). It is a constant battle with forgetting. Keep in mind, or in your memory, however, that despite these similarities, human memory and generative AI operate on fundamentally different principles and mechanisms. Human memory is a complex, biological and messy process, deeply intertwined with consciousness and emotions, while generative AI is a technological process governed by algorithms and data. Oddly, and maybe counterintuitively, the latter approach may result in better actual performance in teaching and learning, even generally. I think this type of informed input from learning science will really improve AI tutor systems. To be fair, simply increasing context windows and the functionality will most likely have the same effect.

The context window, what the model can ingest, allows it to take in huge amounts of tokens and is multimodal. It can eat whole books, collections of documents, full movies, a whole series of podcasts for breakfast.

As a result, the opportunities in learning are many: summarizing any document no matter how long; finding something within an enormous text file; huge sets of HR and training documentation tuned into accessible resources; tutorbots to answer student queries and questions, providing feedback and mark text assessments.

On video, it can gobble up entire movies and huge amounts of learning videos, so you can ask questions about what happened within those movies and videos. The model can answer complex questions about the video content. The opportunities in learning are once again many: allowing search of video for performance support on a specific task, then playing it back; allowing the learner to ask for more detail on a specific event or task; looking for a specific solution to a specific problem; interpreting a trainee's performance from video, identifying successes and failure, with feedback on correcting and improving performance; taking a lecture and annotating it with extra resources; turning any video into a deeper learning experience; interpreting video assessments where content and behaviour matter.

This is not about the generation of media but about the generation of learning – a real breakthrough in learning technology.

Art frontier

The release of superb quality video generation, from simple text prompts, may go down in the history of movies or moving pictures as a pivotal point. It is as significant as the train filmed thundering into La Ciotat, scaring the theatre audience, or the first feature-length talking film, *The Jazz Singer*.

Filmmaking has just been turned on its head, no longer requiring huge investments in production. All the components seem to be heading towards very low-cost everything – sound, lighting, worlds, people, action. Even on first release, the shorts were stunning – the movement, lighting and reflections. This puts movie-making in the hands of creative people who can dispense with the very high production costs of a crew, set and, at some point, perhaps actors. It may even do a good job at editing. AI is already doing colour-balancing and volume-setting across cuts and many other functions. It has just jumped into the Director's seat – into a creative role. There will be those who

will balk at this, in the same way people balked when photography challenged painting, printmaking and original works and CGI in film. But this is different, as it is not technology that has scaled or become a new medium – it is technology as a creative agent. This is technology as producer, director, screenwriter, cinematographer, casting director, costume designer and actors. This is technology as movie-maker. It democratizes movie-making.

Ever since the concept of the ‘arts’ and the ‘artist’ arose in the Romantic movement in the late 18th century in Germany, seeped in German idealism, and the concept of the individual artist, imported by Coleridge and taken up with enthusiasm by everyone in the arts to this day, we have worshipped the artist as an individual. The AI auteur has now arrived. It will reshape filmmaking and art in general. We may see new genres emerge – certainly a widening of participation and what can be done in moving images, as anything is now possible. This may combine with other forms of AI and the shift from 2D to 3D, to create art and film experiences where we participate, either as agents within the story or our avatars do as participants in the story. Movies and games may well combine to form a new medium. In what way this will unfold, no one can be entirely sure, but things change, life changes, art changes. That is the very definition of art.

Can words now create worlds? Has AI suddenly acquired God-like creation qualities? Could generative AI have developed something truly astonishing here – a simulator of existing and new worlds? Is this the converse of Oppenheimer’s famous statement, ‘Now I am become life, the creator of worlds?’

To date, generative AI was limited to text and images and lacked a model of the world, of a 3D space, time, causality and action. We seem to be moving towards crossing that frontier. Games output from games engines is being used along with synthetic data. Other areas of AI are inserting worldviews into AI. Video generation moves us toward the generation of 3D worlds, where a physics engine understands how objects

behave – a collision engine with a detailed knowledge of how physical objects, fluids and gases operate. We do not have this yet, as video generation uses clever diffusion models.

We are 3D people, living in a 3D world, doing 3D things with other 3D people and 3D things. Yet, bizarrely, most teaching and training is from the flat, 2D page – text, books, graphics, PowerPoints and screens of e-learning. This has always been largely suboptimal and prevents actual learning of skills and transfer.

In the beginning was the word and now we, like small gods, can use the word to create new worlds. We are in dialogue with the world to create new ones. That simple act of saying something can make it appear, breathe life into that world. That is more than interesting, it is staggering.

We may well be on our way to creating worlds – any world, on any scale. If so, this is a threshold that has been crossed. We will be able to create worlds in which we work, interact and get things done, also worlds in which we teach, train and learn. Even worlds in which we socialize and entertain ourselves. We may be doing what has only ever been done on a limited scale in incredibly expensive simulators and computer games – understand and create new worlds. Multimodal may now mean a grand convergence.

Social frontier

Generative AI took the world both by surprise and storm with the launch of ChatGPT. Since then, a range of large language models have been released, both proprietary and open source. Astounding advances have been made in multimodality with the launch of image and video generation. We have seen leaps in quality and significant reductions in costs, and AI tools have been folded into major software platforms. It is clear that this is the age of AI. All major technology companies have shifted into being AI companies.

What we have underestimated is the deep nature of the social shift. This is the latest of a series of Copernican shifts in our species. The first was the actual Copernican Revolution, when we were knocked out of our place at the centre of the Universe, to a little rock circling the sun. The second was the Darwinian Revolution, where accidents of mutation produced a smart ape. We were no longer singular, special and superior beings but mere animals. All of these reversed previous views of us as exceptional, unique, created beings.

This latest Copernican shift says we are no longer the masters and sole creators of our own knowledge. For generation after generation, we have passed our cultural knowledge down, first through teaching and learning (not always the same thing), then by externalizing, storing and archiving, as written and printed material, as well as other media such as images, audio and video. On top of this we created digital archiving with search and retrieval across the global internet. This was primarily a world of created and stored media – of books, PowerPoints and flat screens of video.

At the same time, work in psychology has shown we are cognitively capped. We have limited working memories, fallible long-term memories, forget much of what we are taught, have limited perceptual ranges, lots of biases, suffer from emotional swings, cannot network with other brains, sleep eight hours a day, deteriorate cognitively as we get older and die. We need to have some humility about this fragility.

This AI Copernican shift started with the extended mind, seeing technology as an extension of our human-all-too-human talents – robots on strength and accuracy; broadcast media for scale; storage and transmission on scale through the internet; search and retrieval for knowledge; smartphones as powerful personal assistants. We can see AI as a further extension of the extended mind, what most now call the augmented mind, but it is more than this. This AI revolution is not the AI of old, the behind-the-scenes aid to cognition. It is us! We work with it as part of a collective. This is far more than extension and

augmentation. It is collective dialogue. It is Copernicus speaking to Copernicus.

We had a hint of what was to come with social media, where the world took to creating, posting, commenting, liking and messaging, in *dialogue* with each other on a massive, global scale. It was the same with texting – a form of *dialogue*, an accident of early technology, now the mainstay of online communications. Who saw that listening to *dialogue* in audio would, for many, become the learning experience of choice on the internet through podcasts? The clues were there all along – that we evolved by speaking to each other. So let us continue to have dialogue over what is the most important technology our species has ever created.

Final message from the frontier

One thing many have not grasped about this current explosion of AI is that it is moving fast – very fast. Performance improvement is real, fast and often surprising. We must be careful in fixating on what these models do at present. People, especially in discussions around AI, are often fixated by the past, old tools and the present, and not considering the future. It only took 66 years between the first flight and getting to the moon. Progress in AI will be much faster. In addition to being the fastest adopted technology in the history of our species, it has another feature that many miss – it learns, adapts and adds features very, very quickly.

The models learn, not just from unsupervised training on gargantuan amounts of data, but also reinforcement learning by humans. LLMs reached escape velocity in functionality when the training set reached a certain size; there is still no end in sight yet. Developments such as synthetic data will take it further. This simple fact that this is the first technology to ‘learn’ and learn fast, on scale, continuously, across a range of media and tasks, is what makes it extraordinary.

There is also the misconception around the word ‘generative’, the assumption being that all it does is create blocks of predictable text. Many of its best uses in learning are its ability to summarize, outline, provide guidance, support and many other pedagogic features that can be built into the software. This works and will mean tutoring, teaching support, note-taking support, coaching and many other services will emerge that aid both teaching and learning.

On top of all this is the blending of generative AI with other software, where everything from internet searches to Wikipedia and advanced mathematics have been added to supplement its functionality. These are performance enhancers. On top of this are APIs that give it even more potency. The reverse is also true, where generative AI supplements other tools. Many online tools have added generative AI to make them more productive.

This is the worst AI will ever be and it is way better than most journalists, teachers and commentators think, so we are in for a shock. The real digital divide is now between those with curiosity and those that refuse to listen. Anyone with access to a smartphone, computer laptop or tablet – that’s basically almost all learners in the developed world – have access to this technology. The real divide is among those in the know and those not in the know – using it and not using it. So focused are educators on the present they can’t see, or fear, the future.

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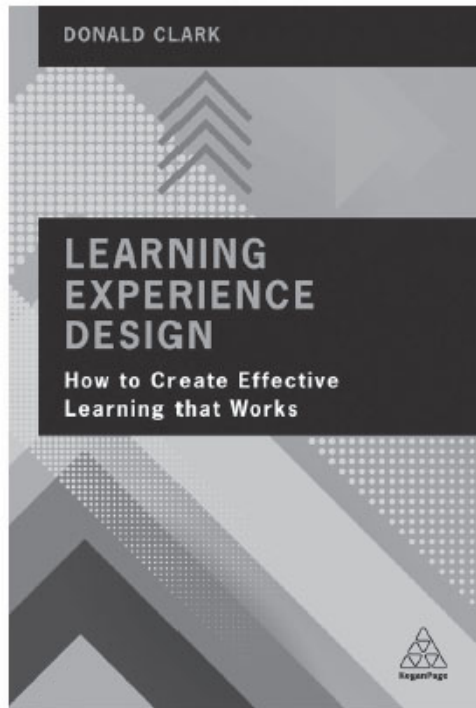
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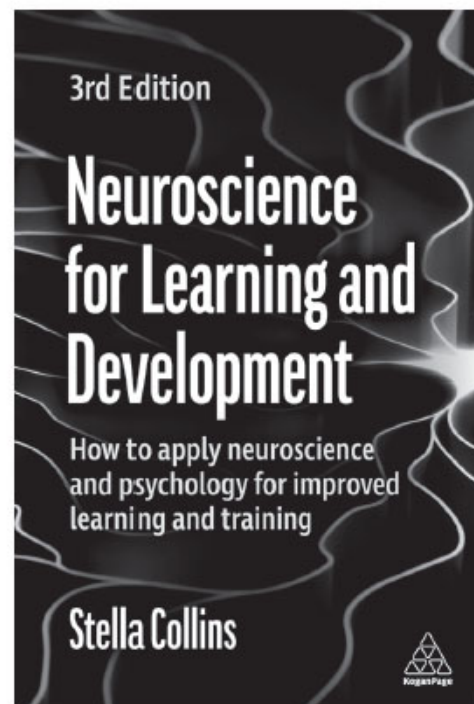
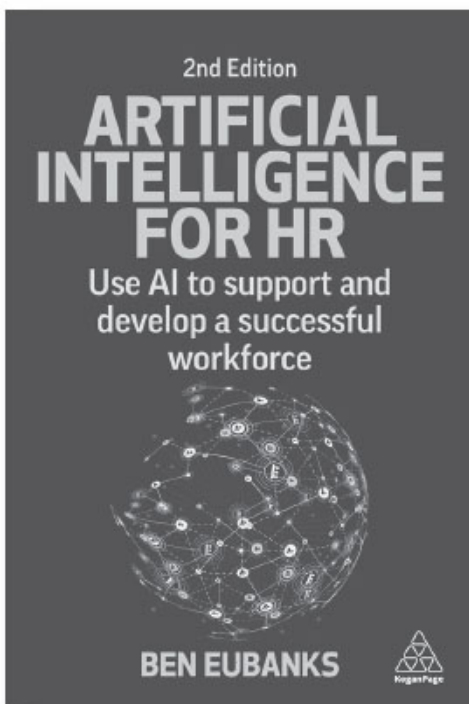


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First published in Great Britain and the United States in 2020 by Kogan Page Limited

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ISBNs

Hardback 978 1 3986 1577 9
Paperback 978 1 3986 1566 3
Ebook 978 1 3986 1576 2

British Library Cataloguing-in-Publication Data

A CIP record for this book is available from the British Library.

Library of Congress Cataloging-in-Publication Data

A CIP record for this book is available from the Library of Congress.